The electromagnetic sphere

Structure, matter, dynamics and causal structure of the split-signature Clifford algebra $\mathrm{Cl}(3,3)$ and its tower

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Abstract

We study the real Clifford algebra $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ of the split space $\mathbb{R}^{3,3}$, read as three temporal directions paired with three spatial ones, together with its graded tower $\mathrm{Cl}(3n,3n)$, and ask which ingredients of relativistic field theory the algebra fixes and which it merely admits. At one constituent the joint commutant of the interior Lorentz action and the internal bivectors determines the imaginary unit up to sign; charge conjugation is the surplus temporal plane, time reversal squares to $-\mathbf{1}$, and the kinetic and mass forms reduce the admissible gauge algebra to a single $\mathfrak{u}(1)$, which is also the entire internal centraliser. Along the tower the internal centraliser is $\mathfrak{so}(3n-1,3n-3)$. At three constituents an explicit axial projector of rank 64 carries the Pati–Salam algebra and one chiral family with a right-handed neutrino, whereas a Lorentz action taken diagonally across the constituents yields a compact internal algebra $\mathfrak{so}(6) \oplus \mathfrak{so}(3)$ forming a reductive dual pair with the Lorentz algebra, on a vector-like module. The anti-isometry pairing the two triples identifies $\mathbb{R}^{3,3}$ with the field-strength space $\Lambda^2\mathbb{R}^{1,3}$, on which it acts as the Hodge star; covariance then fixes the form of Maxwell's equations but not their sources. The interior dynamics binds only in neutral signature, and its force law closes in exactly one way, into a rotor whose angle is the duality angle of the momentum. On the 272-dimensional space of Lorentz-invariant mass kernels of the chiral block we classify the renormalizable potential of its $(\mathbf{10},\mathbf{1},\mathbf{3})$ component, give sufficient and, separately, necessary conditions for the Standard-Model orbit to be its global minimum, and compute the one-loop spectrum of its angular modes. In a curved arena both readings of $\mathbb{R}^{3,3}$ construct the Einstein tensor of a given metric. Finally, choosing a Lorentzian four-plane is equivalent to choosing the complex structure, no invariant scalar potential selects that choice cleanly, and the surplus-time constraints follow from a worldline gauge symmetry. Throughout, the algebra fixes forms and representations; it selects neither the projector, the hypercharge direction, the couplings nor the metric.

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1 Introduction

1.1 The question

Start from a flat real vector space $V = \mathbb{R}^{3,3}$ of split signature, with an orthonormal basis $e_1, \dots, e_6$ in which e_1, e_2, e_3 have square $+1$ and e_4, e_5, e_6 have square -1, and with the map $e_i \mapsto e_{i+3}$ that pairs each positive direction with a negative one. Read the positive triple as three times and the negative triple as three lengths. The space has six directions, the same number as an electromagnetic field strength, and the pairing looks like the correspondence between a magnetic and an electric component. This geometric picture, which we call the *electromagnetic sphere*, is the starting point of the paper.

The hypothesis we test is that as much of relativistic field theory as possible should be read off the internal structure of the Clifford algebra $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ and of its graded tower $\mathrm{Cl}(3n,3n) \cong M_{2^{3n}}(\mathbb{R})$, rather than added by hand. Split signature is the smallest setting in which three positive directions can be paired with three negative ones by an anti-isometry, and the tower is the natural way to enlarge the arena without changing its local structure. The classification of real Clifford algebras and their gradings is classical [75, 78, 110, 91], and several programmes have tried to extract Standard-Model structure from division and Clifford algebras [42, 54, 25]; the representation-theoretic limits of such embeddings are discussed in [33].

The question we ask is not whether the Standard Model fits inside this arena. It does, in more than one way, and fitting proves nothing. The question is which of the free choices of relativistic field theory the algebra removes, and which it leaves free. That question has sharp answers, and the answers are the content of this paper.

1.2 A deflationary principle

One theorem, proved in Section 4, governs everything else. Take the four interior gamma matrices $\gamma^\mu = (e_1, e_4, e_5, e_6)$ and the operator $J = e_2e_3$, which commutes with all four and squares to $-\mathbf{1}$. Then the algebra generated over $\mathbb{C}$, with J playing the role of i, by the four gamma matrices is the whole of $M_4(\mathbb{C})$. The real Dirac theory built on $\mathrm{Cl}(3,3)$ is therefore ordinary complex Dirac theory written in real variables, exactly and not in a limit. A change of variables predicts nothing, so a statement about the arena carries information only if it could have come out otherwise. We apply this test throughout. Facts that are true of Dirac theory in any notation, such as $g = 2$ at tree level, current conservation or the Ward identity, are *inherited*; they are reproduced here only where they are needed to state something else. The statements we report are of a different kind: which bilinears are admissible, which generators may be gauged, how large the internal centraliser is at each rung, what the faithful gauge algebra of a projected block is, which potentials can select a vacuum, and which constructions are impossible under stated hypotheses.

Every claim in this paper belongs to one of four classes, and we say in the text which one. A result is *fixed by the algebra* when it is an identity or a theorem that could have failed. It is *constructed* when the algebra admits it and we exhibit it, without anything selecting it among alternatives. It is *obstructed* when it is impossible under hypotheses that we state, and those hypotheses are part of the result. It is *open* when it is neither established nor refuted. The finite-dimensional statements are exact: every blade of the realisation used here is a signed permutation matrix, so products, commutators, traces and ranks are integer computations. How the statements were verified is described in Appendix B.

1.3 Summary of results

The arena (Sections 2 to 5). A bivector e_ae_b squares to $-\mathbf{1}$ when its two directions have the same sign and to $+\mathbf{1}$ when they have opposite signs; the first kind generates compact rotations that can be gauged, the second kind generates boosts that can only grade. The three *pole operators* e_ie_{i+3} are commuting involutions grading the algebra by $\mathbb{Z}_2^3$, and they span a maximally split Cartan subalgebra of $\mathfrak{so}(3,3)$. The pairing has two square roots: a para-complex structure $\mathcal{K}$ with $\mathcal{K}^2 = +\mathbf{1}$, which grades, and a complex structure G with $G^2 = -\mathbf{1}$, which is self-adjoint for η and, as we show later, binds the interior dynamics and acts as a Hodge star. A clock is a unit vector of the positive triple; the symmetric clock $\widehat{W} = (e_1 + e_2 + e_3)/\sqrt{3}$ is the unique permutation-invariant point and the entropy maximum of the clock simplex under the Fisher–Rao metric, and exactly that clock makes the conversion operator invisible in the equations of motion. The imaginary unit is selected rather than adjoined: inside the joint commutant of the interior Lorentz action and the internal bivectors, the only square roots of $-\mathbf{1}$ that commute with the gamma matrices are $\pm J$. Charge conjugation is the surplus plane $\langle e_2, e_3\rangle$ and coincides with the Majorana structure; time reversal is forced to square to $-\mathbf{1}$; the kinetic forms alone fix the massless gauge algebra $\mathfrak{u}(1) \oplus \mathfrak{su}(2)$, which the mass form cuts to the electromagnetic $\mathfrak{u}(1)$; and the quantum state space must be formed by complexifying from outside.

The tower and its matter (Sections 6 to 10). The bivectors commuting with the interior Lorentz action of $\mathrm{Cl}(3n, 3n)$ form $\mathfrak{so}(3n-1, 3n-3)$, nested along the tower, of dimensions $1, 28, 91, 190, \ldots$. At one constituent this centraliser is the single bivector J, so the photon is not one symmetry of the interior structure but the only one. At three constituents we construct an orthogonal projector

$$P_* = \tfrac{1}{8}\,(\mathbf{1} + \Gamma\chi_W)(\mathbf{1} - J\chi_C)(\mathbf{1} - Jq_A)$$

of rank 64 which commutes with the Lorentz action and with J, is axial in the sense that $P_*\gamma^\mu P_* = 0$, keeps the whole of $\mathfrak{su}(4)$ acting faithfully, and carries $(\mathbf{4}, \mathbf{2}, \mathbf{1})_L \oplus (\mathbf{4}, \mathbf{1}, \mathbf{2})_R$: one Pati–Salam family with a right-handed neutrino. Its two central abelian factors cannot be

gauged on this matter. Charges built from three commuting involutions are conjugately paired whenever they are anomaly free, so hypercharge cannot be read off from anomaly cancellation inside that class. When the Lorentz action is instead taken diagonally from all three constituents, the internal algebra becomes the entirely compact $\mathfrak{so}(6)\oplus\mathfrak{so}(3)$; the copy algebra and the Lorentz algebra form a reductive dual pair satisfying $C_{\mathrm{cop}} + C_{\Delta} = \frac{15}{2}$, and the module is vector-like. A single theorem gives the candidate gauge algebra of every Lorentz action on the tower, and with it a rung-by-rung census of the vector bosons the tower can carry.

Gauge fields and the electromagnetic reading (Sections 11 to 14). A varying projector induces a connection whose available curvature algebra is far too large, so a parent action would have to cut the projector's freedom rather than build a gauge field. The centraliser of the complex structure G in $\mathfrak{so}(3,3)$ is a Lorentz algebra, under which $\mathbb{R}^{3,3}$ *is* $\Lambda^2\mathbb{R}^{1,3}$: boosts act by $\delta\mathbf{E} = \mathbf{v}\times\mathbf{B}$, $\delta\mathbf{B} = -\mathbf{v}\times\mathbf{E}$, G is the Hodge star, and the two invariants $|\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\mathbf{E}\cdot\mathbf{B}$ are the real and imaginary parts of one holomorphic quadratic form. This reading excludes the interior four-plane on which the matter is built. Covariance fixes the form of Maxwell's equations, because the vector module occurs in $\Lambda^2\otimes V$ with multiplicity exactly two, and fixes nothing else. The photon's spin one appears in two places, related by the Klein correspondence $V \cong \Lambda^2(S^+)^*$, and at the second rung both can be coupled to a Dirac block in a single gauge theory. The orthogonality of $\mathbf{E}$ and $\mathbf{B}$ is the reality of the invariant $q = z\cdot z$; its failure breaks parity and time reversal, is never spontaneous, and does not generate mass.

Interior dynamics (Sections 15 to 17). The Lagrangian of a single shifted operator $\partial_\tau + \kappa g$ factorises exactly when the shift is symmetric, and then it describes an inverted oscillator; for every constant shift some solution is unbounded, and with a positive form no solution with nonzero momentum recurs. In the split form the same factorisation admits bound motion, and it does so exactly in neutral signature, with the complex structure G as the shift. The force law is a dual transport which is a Lie–Poisson equation only on an extension of the frame; that extension is unique, its rotor angle is the duality angle of the momentum, and its free motion is a pendulum that drifts the frame except on null data.

The vacuum (Sections 18 to 22). At one constituent we classify the vacuum of every coordinate three-plane in four senses. At three constituents the space of Lorentz-invariant mass kernels of the chiral block has dimension 272 and decomposes as $(\mathbf{10},\mathbf{3},\mathbf{1})\oplus(\mathbf{10},\mathbf{1},\mathbf{3})\oplus(\mathbf{15},\mathbf{2},\mathbf{2})\oplus 2(\mathbf{6},\mathbf{1},\mathbf{1})\oplus(\mathbf{1},\mathbf{2},\mathbf{2})$; the Standard-Model invariant kernels form one complex line, the right-handed neutrino Majorana kernel. We give the complete renormalizable potential of the $(\mathbf{10},\mathbf{1},\mathbf{3})$ component, including a holomorphic quartic that violates baryon number by two units, and sufficient conditions, together with separate necessary conditions, under which the Standard-Model orbit is its global minimum (the two do not coincide, and numerics locate the transition in the gap between them); we compute the radiatively generated spectrum of all 271 angular modes, the Goldstone content, the discrete symmetries of the vacuum, and an electroweak stage with the Georgi–Jarlskog factor and a seesaw.

Curvature and the photon (Sections 23 to 26). In a curved arena the pairing becomes an almost para-Hermitian structure. An explicit six-dimensional metric with a normal $U(1)$ has a closed-form Einstein tensor whose tangential block on the four-dimensional slice is the Einstein tensor of the slice, while the normal field strength gravitates with negative energy, as the signature forces. In the field-strength reading the curvature operator of a four-metric lives on the frame, the first Bianchi identity is the single condition $\mathrm{Tr}(G\mathcal{R}) = 0$, and the duality rotation of the curvature ends at the double dual, which contracts to the Einstein tensor. The projector current sources a massive normal vector, and a massless photon requires a vacuum that commutes with the charge; such a vacuum exists at the second rung and not at the first.

Causal structure (Sections 27 to 30). A field propagating freely in six dimensions of signature $(3,3)$ is ill-posed and has no ground state, and its time-like cone has no past and future. Choosing a Lorentzian four-plane inside $\mathbb{R}^{3,3}$ is the same as choosing a complex structure, and at one constituent the light cone of that plane is the set of chiral spinors squared with the help of J. No bounded-below invariant potential of degree below eight isolates that choice, and every invariant kinetic term for it has ghost directions. The surplus-time constraints, by contrast, follow from a gauge symmetry of the worldline.

1.4 Conventions

Spacetime signature is $(+,-,-,-)$, units are Heaviside–Lorentz with $\hbar = c = 1$ unless stated. A *constituent* is a tensor factor $\mathrm{Cl}(3,3)$ of the tower; the word never refers to a particle, and three constituents do not describe three particles. Anomaly coefficients are chiral traces per Weyl component with every field written left-handed. In the field-strength reading of $\mathbb{R}^{3,3}$ the vector $X = \sum_i (B_i e_i + E_i e_{i+3})$ is identified with the two-form F of components $F_{0i} = E_i$, $F_{jk} = \epsilon_{ijk} B_i$; with this orientation $\mathbf{B}$ is minus the textbook magnetic field, which we write $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$, while $\mathbf{E}$ is the textbook electric field. Formulas taken from the literature are written in textbook variables, and where a sign matters the variables are named. Compact Lie algebras are realised by skew operators and the corresponding charges by the symmetric operators $\hat{t} = -Jt$.

1.5 Organisation

Sections 2 to 5 develop one constituent. Sections 6 to 10 climb the tower and build the matter. Sections 11 to 14 treat gauge fields and the field-strength reading, Sections 15 to 17 the interior dynamics, and Sections 18 to 22 the vacuum. Sections 23 to 26 pass to a curved arena and to the photon, Sections 27 to 30 to causal structure, and Section 31 collects what is fixed, what is constructed, what is obstructed and what is open. Appendix A gives the explicit matrices, so that every finite-dimensional claim can be checked by hand, and Appendix B describes the verification.

Part I
The arena at one constituent

2 The split space and its Clifford algebra

2.1 A fixed realisation

Let $V = \mathbb{R}^{3,3}$ with orthonormal basis $e_1, \dots, e_6$, $e_a^2 = \eta_{aa}$ and $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$. We call $V^+ = \langle e_1, e_2, e_3 \rangle$ the temporal triple and $V^- = \langle e_4, e_5, e_6 \rangle$ the spatial triple. The Clifford algebra $\mathrm{Cl}(3,3)$ is generated by the e_a subject to $e_a e_b + e_b e_a = 2\eta_{ab}\mathbf{1}$; it is isomorphic to $M_8(\mathbb{R})$, and we fix once and for all its realisation on $S = \mathbb{R}^8 = (\mathbb{R}^2)^{\otimes 3}$,

$$\begin{aligned} e_1 &= \sigma_x \otimes \mathbf{1} \otimes \mathbf{1}, \quad & e_2 &= \sigma_z \otimes \sigma_x \otimes \mathbf{1}, \quad & e_3 &= \sigma_z \otimes \sigma_z \otimes \sigma_x, \\ e_4 &= \varepsilon \otimes \mathbf{1} \otimes \mathbf{1}, \quad & e_5 &= \sigma_z \otimes \varepsilon \otimes \mathbf{1}, \quad & e_6 &= \sigma_z \otimes \sigma_z \otimes \varepsilon, \end{aligned} \tag{1}$$

with $\sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\sigma_z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ and $\varepsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. All six matrices are signed permutation matrices. For a subset $I = \{a_1 < \cdots < a_r\}$ we write $e_I = e_{a_1} \cdots e_{a_r}$ for the corresponding blade of grade r; the 64 blades form a basis of $M_8(\mathbb{R})$.

The tower is built by graded tensor products. At n constituents the spinor module is $S_n = (\mathbb{R}^8)^{\otimes n}$ and the generators of the k-th constituent are

$$e_a^{(k)} = \underbrace{\omega \otimes \cdots \otimes \omega}_{k-1} \otimes e_a \otimes \mathbf{1} \otimes \cdots \otimes \mathbf{1}, \qquad \omega := e_1 e_2 \cdots e_6. \tag{2}$$

Since $\omega^2 = \mathbf{1}$ and ω anticommutes with every e_a, generators of different constituents anticommute, and the $6n$ matrices $e_a^{(k)}$ generate $\mathrm{Cl}(3n, 3n) \cong M_{8^n}(\mathbb{R})$. When we need a single index we write $E_{6(k-1)+a} = e_a^{(k)}$, so that at three constituents the eighteen generators are $E_1, \dots, E_{18}$ and the temporal ones are E_A with $A \in \{1, 2, 3, 7, 8, 9, 13, 14, 15\}$.

Two elementary rules carry most of the finite-dimensional arguments of this paper.

Lemma 2.1 (Blade rules)**.** *For blades e_I, e_J of grades $|I|, |J|$,*

$$e_I e_J = (-1)^{|I||J| - |I \cap J|}\, e_J e_I, \tag{3}$$

and for a blade B of grade r

$$B^\top = (-1)^{r(r-1)/2} \Big(\prod_{a \in I} \eta_{aa} \Big) B = B^2 B. \tag{4}$$

In particular a blade is antisymmetric if and only if it squares to $-\mathbf{1}$, and symmetric if and only if it squares to $+\mathbf{1}$.

Proof. For (3), move each generator of e_J past the generators of e_I: it anticommutes with the $|I|$ of them it differs from and commutes with itself when it occurs in I, which gives the exponent. For (4), the matrices σ_x, σ_z are symmetric and ε is antisymmetric, so each e_a satisfies $e_a^\top = \eta_{aa} e_a$; transposing a product reverses it, and restoring the order costs $(-1)^{r(r-1)/2}$. The same two factors give $B^2 = (-1)^{r(r-1)/2} \prod_{a \in I} \eta_{aa}\, \mathbf{1}$, so $B^\top = B^2 B$. □

Example 2.2. The bivector $e_2 e_3$ has $(e_2 e_3)^2 = -e_2^2 e_3^2 = -\mathbf{1}$, so it is antisymmetric and $\exp(\theta e_2 e_3) = \cos\theta\, \mathbf{1} + \sin\theta\, e_2 e_3$ is a rotation of period 2π. The bivector $e_1 e_4$ has $(e_1 e_4)^2 = -e_1^2 e_4^2 = +\mathbf{1}$, so it is symmetric and $\exp(\theta e_1 e_4) = \cosh\theta\, \mathbf{1} + \sinh\theta\, e_1 e_4$ is a boost, unbounded in θ. The two bivectors share the index 1 and $|I||J| - |I \cap J| = 4 - 1 = 3$, so they anticommute; $e_2 e_3$ and $e_5 e_6$ share nothing and commute.

2.2 The signature rule

Theorem 2.3 (Gauging versus grading)**.** *For $a \neq b$, $(e_a e_b)^2 = -e_a^2 e_b^2$. A bivector therefore squares to $-\mathbf{1}$ when its two directions have the same sign and to $+\mathbf{1}$ when they have opposite signs, and by Lemma 2.1 the first kind are exactly the antisymmetric bivectors. The three time–time and three space–space bivectors span the maximal compact subalgebra $\mathfrak{so}(3) \oplus \mathfrak{so}(3) \subset \mathfrak{so}(3, 3)$, and the nine mixed bivectors are boosts.*

Corollary 2.4 (No gauge boson is a boost)**.** *For a mixed bivector B, $\exp(\theta B) = \cosh\theta\, \mathbf{1} + \sinh\theta\, B$ generates $\mathbb{R}$ and not $U(1)$: it carries no phase and defines no compact charge.*

The criterion that matters for a gauge generator is antisymmetry with respect to the positive form that a kinetic term supplies, equivalently membership of the $+1$ eigenspace of the Cartan involution $\theta(X) = -X^\top$ [72, 62]. For a single blade this coincides with $X^2 = -\mathbf{1}$ by (4), which is why the rule is decisive for bivectors. For general elements it is not: $\mathrm{diag}(J_2, 2J_2)$, with J_2 the standard 2×2 rotation generator, generates a group of period 2π although its square is $\mathrm{diag}(-\mathbf{1}, -4\mathbf{1})$.

2.3 The pole operators and a multigrading

The mixed bivectors are not idle. The three *pole operators*

$$\omega_i := e_i e_{i+3} \qquad (i = 1, 2, 3)$$

are boosts, so $\omega_i^2 = +\mathbf{1}$, and they commute pairwise by (3), since any two share no index. Square $-\mathbf{1}$ gauges and square $+\mathbf{1}$ grades, and no bivector does both.

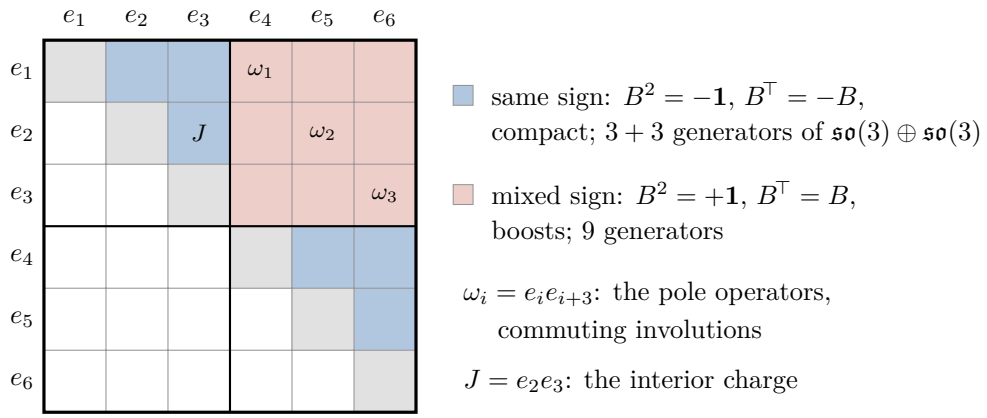


Figure 1: The fifteen bivectors $e_a e_b$, $a < b$, of $\mathrm{Cl}(3,3)$. Blue cells pair two directions of the same sign and generate compact rotations; red cells pair a temporal with a spatial direction and generate boosts (Theorem 2.3). The diagonal of the red block holds the three pole operators, which grade the algebra (Proposition 2.5); the cell $(2,3)$ holds the bivector J that will become the electromagnetic charge and the imaginary unit.

Proposition 2.5 (A $\mathbb{Z}_2^3$ grading of $\mathrm{Cl}(3,3)$)**.** *For a blade e_I put $\chi_i(e_I) = (-1)^{|I \cap \{i,i+3\}|}$. Then $\omega_i e_I \omega_i^{-1} = \chi_i(e_I)\, e_I$, and $\mathrm{Cl}(3,3) = \bigoplus_{\sigma \in \{\pm\}^3} \mathrm{Cl}_\sigma$ is a multiplicative grading with eight sectors of dimension eight.*

(i) The joint spectrum of $(\omega_1, \omega_2, \omega_3)$ on S consists of all eight sign triples, each on a line, so the pole operators cut $\mathbb{R}^8$ into eight one-dimensional eigenlines.

(ii) The vectors occupy the three weights with exactly one minus sign, two each (e_i and e_{i+3}); the bivectors occupy the four even weights with multiplicities $3,4,4,4$; the all-minus weight is spanned by eight trivectors, namely $e_1e_2e_3$, $e_4e_5e_6$ and six mixed ones.

(iii) Correspondingly $\mathfrak{so}(3,3) = \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_3$ is graded by the Klein four-group, of dimensions $3+4+4+4$, with $\mathfrak{g}_0 = \mathrm{span}\{\omega_1, \omega_2, \omega_3\}$ abelian, self-normalising and consisting of semisimple elements with real eigenvalues: a maximally split Cartan subalgebra of $\mathfrak{so}(3,3) \cong \mathfrak{sl}(4,\mathbb{R})$.

Proof. The conjugation sign is (3) with $|J| = 2$: $(-1)^{2|I| - |I \cap \{i,i+3\}|}$. The weight is multiplicative because $|I \triangle I'|$ and $|I| + |I'|$ have the same parity in each pair. The census in (ii) is a count of how many generators of each pair a blade contains. For (iii), a bivector inside one pair is a pole operator, of weight $(+,+,+)$; a bivector with one generator in each of two pairs has two minus signs, and each of the three such weights holds $2 \times 2 = 4$ bivectors. The pole operators commute, are symmetric with eigenvalues ± 1, and the centraliser of $\mathfrak{g}_0$ in $\mathfrak{so}(3,3)$ is $\mathfrak{g}_0$ itself, because every other bivector has a nontrivial weight; its dimension equals the real rank three of $\mathfrak{so}(3,3)$. □

The three characters are grading weights, not metric signatures: e_1 and e_4 carry the same weight and opposite squares, so nothing in the grading tells a temporal from a spatial direction. The decomposition in (iii) is the root decomposition of $\mathfrak{sl}(4,\mathbb{R})$ with respect to a split Cartan subalgebra, grouped by the root lattice modulo two; gradings of Clifford and classical Lie algebras by $\mathbb{Z}_2^n$ are well known [91], and we use this one because the pole operators will reappear in the field-strength reading as the six planes (e_i, e_{i+3}) that carry one complex component each.

Example 2.6 (The paired-mode projector)**.** Let $\mathsf{P} = \omega_1 + \omega_2 + \omega_3$ be the sum of the pole operators and $\mathcal{V} = e_1 e_2 \cdots e_6$. On the joint eigenline of signs $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$, $\mathsf{P} = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 \in \{3, 1, -1, -3\}$ with multiplicities $1, 3, 3, 1$. Reordering $\omega_1\omega_2\omega_3 = e_1e_4e_2e_5e_3e_6$ into $e_1 \cdots e_6$ costs

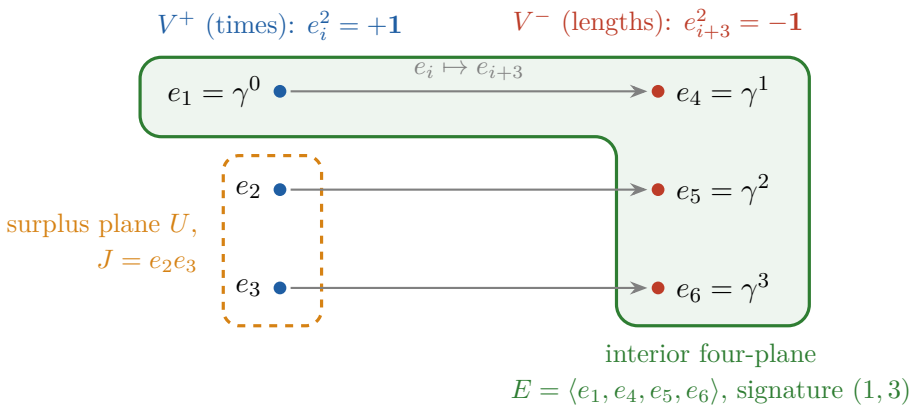


Figure 2: The arena. The positive triple V^+ (times) and the negative triple V^- (lengths) are paired by $e_i \mapsto e_{i+3}$; the pairing has two square roots, the grading $\mathcal{K}$ and the binding operator G (Proposition 3.3). The interior four-plane E, on which the Dirac sector is built, contains the clock e_1 and the whole spatial triple; its complement is the surplus plane $U = \langle e_2, e_3 \rangle$, whose rotation generator J is the only bivector commuting with the interior Lorentz algebra (Proposition 2.7).

three transpositions, so $\mathcal{V} = -\varepsilon_1\varepsilon_2\varepsilon_3$ on that line. Hence $\frac{1}{2}(\mathsf{P} + \mathcal{V})$ takes the values $1, 1, -1, -1$ on the four strata: it is the sign of P, and equals the polynomial $(13\mathsf{P} - \mathsf{P}^3)/12$. The operator

$$\Pi := \tfrac{1}{2}\mathbf{1} - \tfrac{1}{4}(\mathsf{P} + \mathcal{V}) \tag{5}$$

is therefore the orthogonal projector of rank four onto the strata with $\mathsf{P} < 0$. We call it the paired-mode projector; it is a natural candidate for an observation projector at one constituent and we test it against the photon in Section 26.

2.4 The interior four-plane and the surplus plane

The physical reading of the arena singles out a Lorentzian four-plane inside V. We take

$$E = \langle e_1, e_4, e_5, e_6 \rangle, \qquad \gamma^0 = e_1, \quad \gamma^a = e_{a+3} \ (a = 1, 2, 3),$$

of signature $(1, 3)$, and call the orthogonal complement $U = \langle e_2, e_3 \rangle$ the surplus plane. The interior Lorentz algebra $\mathfrak{l}_{\text{int}}$ is spanned by the six bivectors of E, with rotations $S_1 = \frac{1}{2}\gamma^2\gamma^3$ (cyclically) and boosts $T_a = \frac{1}{2}\gamma^0\gamma^a$. Its spin Casimir on S is $-\sum_a S_a^2 = \frac{3}{4}\mathbf{1}$, and a rotation through 2π is $-\mathbf{1}$ on S while it is $+\mathbf{1}$ on vectors (Section 13): the same rotors give spin one-half to spinors and spin one to vectors.

Proposition 2.7 (The interior charge is unique)**.** *As $\mathfrak{l}_{\text{int}}$-modules, $\Lambda^2 V = \Lambda^2 E \oplus (E \otimes U) \oplus \Lambda^2 U$, of dimensions $6 + 8 + 1$, is the adjoint representation, two copies of the vector representation, and the trivial representation. The centraliser of $\mathfrak{l}_{\text{int}}$ in $\mathfrak{so}(3,3)$ is therefore the line spanned by*

$$J := e_2e_3,$$

which satisfies $J^2 = -\mathbf{1}$, $J^\top = -J$, and commutes with each of the four gamma matrices.

Proof. The adjoint and vector representations of $\mathfrak{so}(1,3)$ contain no invariant, so only $\Lambda^2 U$ survives. That J commutes with e_1, e_4, e_5, e_6 is (3) with disjoint index sets and $|I||J| = 2$. □

Proposition 2.7 is the first instance of a pattern that recurs throughout: a structure is fixed once the right ambient algebra and the right conditions are named, and it ceases to be fixed when either is relaxed. Here the ambient space is the span of the bivectors and the condition is

commutation with $\mathfrak{l}_{\mathrm{int}}$. In Section 4 we enlarge the ambient algebra to all of $\mathrm{End}_{\mathbb{R}}(S)$ and find that a different condition, commutation with the internal bivectors, is what keeps the answer unique.

3 Time, conversion, and the two square roots of the pairing

3.1 One conversion constant

The split form is not dimensionally innocent. If V^+ carries times and V^- lengths, then $\eta(v,v) = \sum_i (t^i)^2 - \sum_a (x^a)^2$ adds squared times to squared lengths, and a conversion is needed, a priori one constant for each temporal axis.

Proposition 3.1 (A pairing alone does not give isotropy)**.** *A map exchanging V^+ and V^- cannot carry a single dimension; it converts times to lengths one way and lengths to times the other. In block form*

$$\mathcal{K}_C = \begin{pmatrix} 0 & C^{-1} \\ C & 0 \end{pmatrix}, \qquad C : V^+ \to V^-, \qquad [C] = LT^{-1}, \tag{6}$$

one has $\mathcal{K}_C^2 = \mathbf{1}$ for every invertible C, including $C = \mathrm{diag}(c_1, c_2, c_3)$ with distinct entries. An involutive pairing therefore does not force $c_1 = c_2 = c_3$.

Theorem 3.2 (One conversion constant)**.** *If a common $SO(3)$ acts on V^+ and on V^- by the vector representation and C intertwines the two actions, then $C = c\,\mathbf{1}$ with $[c] = LT^{-1}$.*

Proof. The vector representation of $SO(3)$ is irreducible of real type, so its commutant is $\mathbb{R}$ and Schur's lemma places C in $\mathbb{R}\,\mathbf{1}$ [53]. □

3.2 The two square roots of the pairing

Dimensional analysis fixes $[C]$ and leaves the sign of the lower-left block free. The two choices are different operators, and the rest of the paper turns on the difference.

Proposition 3.3 (The grading root and the binding root)**.** *In units with $c = 1$, and in block form on $V^+ \oplus V^-$, put*

$$\mathcal{K} = \begin{pmatrix} 0 & -\mathbf{1} \\ -\mathbf{1} & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix},$$

that is $\mathcal{K}e_i = -e_{i+3}$, $\mathcal{K}e_{i+3} = -e_i$, and $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$. Then:

	square	η-adjoint	$\eta(X\cdot, X\cdot)$	role
$\mathcal{K}$	$+\mathbf{1}$	$\mathcal{K}^\dagger = -\mathcal{K}$	$-\eta$	an element of $\mathfrak{so}(3,3)$; grades
G	$-\mathbf{1}$	$G^\dagger = +G$	$-\eta$	a complex structure outside $\mathfrak{so}(3,3)$; binds

with $X^\dagger := \eta^{-1} X^\top \eta$. Both are anti-isometries. The eigenspaces of $\mathcal{K}$ are the two totally null three-planes spanned by $e_i \mp e_{i+3}$.

Both roots can be built inside the algebra, and they are related by a single trivector.

Lemma 3.4 (Both roots from Clifford data)**.** *Let $\mathsf{P} = \sum_i e_i e_{i+3}$ and $T = e_1 e_2 e_3$, and define $\mathcal{K}(v) = \frac{1}{2}[\mathsf{P}, v]$ and $S(v) = TvT^{-1}$ on vectors. Then $\mathcal{K}(e_i) = -e_{i+3}$, $\mathcal{K}(e_{i+3}) = -e_i$, so $\mathcal{K}^2 = \mathbf{1}$; S fixes the temporal and negates the spatial generators, so $S = \eta$ as a map of V; and*

$$G = -\mathcal{K} \circ S.$$

Proof. $[e_i e_{i+3}, e_i] = e_i e_{i+3} e_i - e_i^2 e_{i+3} = -2e_{i+3}$ and $[e_i e_{i+3}, e_{i+3}] = e_i e_{i+3}^2 - e_{i+3} e_i e_{i+3} = -2e_i$, while the other pole operators commute with both by (3). T is a product of the three temporal generators, so it commutes with each of them up to the sign $(-1)^{3-1} = +1$ and anticommutes with each spatial generator. Finally $-\mathcal{K}(Se_i) = -\mathcal{K}(e_i) = e_{i+3} = Ge_i$ and $-\mathcal{K}(Se_{i+3}) = \mathcal{K}(e_{i+3}) = -e_i = Ge_{i+3}$. □

So the root that grades is $\frac{1}{2}\operatorname{ad}_{\mathsf{P}}$, the adjoint action of the sum of the pole operators, and the root that binds is the same map composed with the Cartan involution $S = \operatorname{Ad}_{e_1 e_2 e_3}$. That the two roots are distinguished exactly by the Cartan involution is the cleanest single statement the flat and the curved analyses below jointly support.

Shift operators. A clock coordinate λ has dimension T, the operator G is dimensionless, and their sum requires a scale. With a length ℓ,

$$\kappa = \frac{c}{\ell}, \qquad \alpha_\pm = \partial_\lambda \pm \kappa G. \tag{7}$$

The pointwise η-adjoint of κG is κG, while the formal adjoint of the differential operator, after an integration by parts with compatible boundary conditions, is $(\partial_\lambda + \kappa G)^* = -\partial_\lambda + \kappa G^\dagger = -\alpha_-$. The two adjoints must be kept apart: the factorisations below are insensitive to the overall sign, positivity statements are not.

3.3 The clock

A clock is a unit vector of V^+. Which one? Let $w = (w_1, w_2, w_3)$ be normalised non-negative weights on the three temporal axes, so that w ranges over the two-simplex Δ^2. The map $x_i = 2\sqrt{w_i}$ sends Δ^2 onto the part of the sphere $\sum_i x_i^2 = 4$ in the closed positive octant, and since $dx_i = dw_i/\sqrt{w_i}$ the Euclidean metric pulls back to

$$\sum_i dx_i^2 = \sum_i \frac{dw_i^2}{w_i},$$

the Fisher–Rao metric. That metric is not a modelling choice: by Čencov's theorem it is, up to scale, the only metric on a finite simplex invariant under Markov morphisms [23, 3]. The space of clocks is therefore an octant of a round sphere of radius two.

Proposition 3.5 (The symmetric clock)**.** *The barycentre $w = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is the unique point of Δ^2 fixed by the permutations of the three axes, and the unique maximum of the entropy $-\sum_i w_i \log w_i$, with value $\log 3$ and Hessian eigenvalues -9 and -3 in the chart (w_1, w_2). Its amplitude direction is the unit vector*

$$\widehat{W} = \frac{e_1 + e_2 + e_3}{\sqrt{3}}, \qquad \widehat{W}^2 = \mathbf{1}, \qquad \lambda := \eta(\widehat{W}, v) = \frac{t^1 + t^2 + t^3}{\sqrt{3}} = \sqrt{3}\,\bar{t}.$$

The permutation argument carries the weight here; the entropy argument is Jaynes' principle and is a hypothesis, recorded as such. What the proposition says is modest: if a clock is to treat the three temporal axes alike, it is $\widehat{W}$, and the physical time it reads is $\sqrt{3}$ times the mean of the three constituent times. It does not say that λ is a proper time, that it is invariant under a group, or that the directions orthogonal to $\widehat{W}$ are unphysical. Those directions, $\widehat{W}^\perp \cap V^+ = \{t^1 + t^2 + t^3 = 0\}$, spanned by $e_1 - e_2$ and $e_1 + e_2 - 2e_3$, are the *relative times*.

3.4 The spectrum of a conversion operator

An operator that converts times into lengths is a mixed bivector $B_C = \sum_{i,a} C_{ia}\, e_i e_{a+3}$, symmetric by Theorem 2.3. Its square controls the factorised interior dynamics of Section 15, where the equation of motion becomes $\ddot{X} = \kappa^2 B_C^2 X$, so its spectrum is worth knowing exactly.

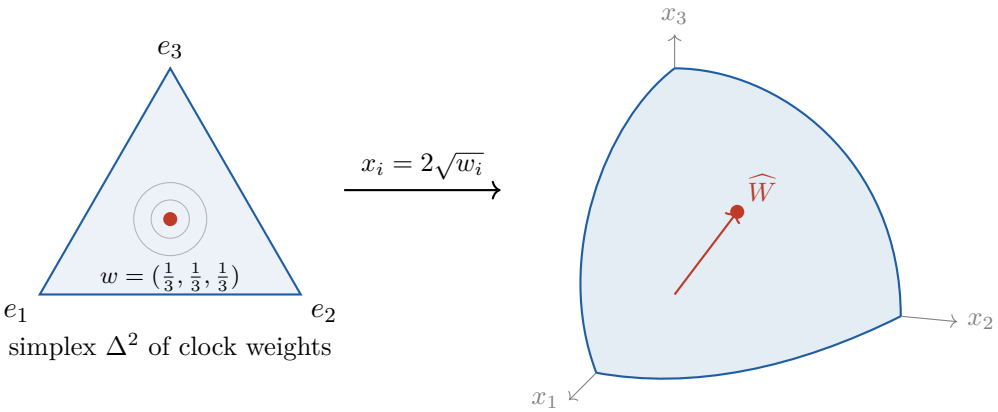


Figure 3: The space of clocks. Normalised weights on the three temporal axes form a simplex; the amplitude map carries it onto an octant of a round sphere, whose round metric is the Fisher–Rao metric. The barycentre, the only permutation-invariant point and the entropy maximum, corresponds to the symmetric clock $\widehat{W} = (e_1 + e_2 + e_3)/\sqrt{3}$ (Proposition 3.5).

Theorem 3.6 (Spectrum of a conversion operator)**.** *Write $C = \sum_k \sigma_k u_k v_k^\top$ in singular-value form and put $U_k = \sum_i u_k^i e_i$, $V_k = \sum_a v_k^a e_{a+3}$ and $B_k = U_k V_k$. Then $B_k^2 = \mathbf{1}$, the B_k commute, and*

$$B_C^2 = \|C\|_F^2\, \mathbf{1} + 2 \sum_{k<l} \sigma_k \sigma_l\, \Omega_{kl}, \qquad \Omega_{kl} = B_k B_l, \qquad \Omega_{12}\Omega_{13} = \Omega_{23}.$$

Consequently

$$\operatorname{spec}(B_C^2) = \{(\eta_1\sigma_1 + \eta_2\sigma_2 + \eta_3\sigma_3)^2 : \ \eta \in \{\pm 1\}^3\},$$

four values, each of multiplicity two on $\mathbb{R}^8$.

Proof. U_k and V_k are orthogonal unit vectors of square $+\mathbf{1}$ and $-\mathbf{1}$, so $B_k^2 = -U_k^2 V_k^2 = \mathbf{1}$; for $k \neq l$ the four vectors U_k, V_k, U_l, V_l are mutually orthogonal and B_k, B_l commute by (3). Expanding the square gives the formula. The seven nontrivial products of the commuting involutions B_k are, in a rotated orthonormal basis of V, blades of grade two, four or six, hence traceless; the joint eigenspace with signs η therefore has dimension $\frac{1}{8} \operatorname{Tr} \prod_k (\mathbf{1} + \eta_k B_k) = 1$, and on it $B_C = \eta \cdot \sigma$. □

Corollary 3.7 (Rank counts clocks)**.** *B_C^2 is a multiple of $\mathbf{1}$ if and only if $\operatorname{rank} C \leq 1$, and some eigenvalue of B_C^2 vanishes if and only if one singular value equals the sum of the other two. The rank of C is the number of independent temporal directions the conversion uses.*

Example 3.8 (The symmetric clock is dynamically blind)**.** The operators $G_a := \widehat{W} e_{a+3}$, which convert along the symmetric clock, have $C = \widehat{W} \hat{e}_a^\top$ of rank one and singular values $(1, 0, 0)$. Hence $G_a^2 = \mathbf{1}$, the factorised equation of motion is $\ddot{X} = \kappa^2 X$, and the conversion operator does not appear in it at all. The clock singled out by the symmetry is exactly the clock whose dynamics cannot see the Clifford structure. This is forced by the rank: if every left singular vector of C is proportional to $\widehat{W}$, then C has rank one. A conversion visible in the equations of motion therefore needs a temporal direction transverse to $\widehat{W}$, that is, a relative time. Raising the rank does not produce oscillation either: the eigenvalues $(\eta \cdot \sigma)^2$ are real and non-negative at every rank, so higher rank makes the instability anisotropic, with four rates $\kappa|\eta \cdot \sigma|$ instead of one, and can create flat directions, but never a bounded oscillation. Bounded motion has to come from elsewhere, and Section 16 shows where.

4 The imaginary unit and the Dirac structure

4.1 The complex structure is selected

At n constituents write $\Omega = \gamma^0\gamma^1\gamma^2\gamma^3 = e_1^{(1)}e_4^{(1)}e_5^{(1)}e_6^{(1)}$ for the volume element of the interior four-plane and ω_I for the product, in increasing order, of the remaining $6n-4$ generators, the *internal* directions. At $n=1$ these are e_2, e_3 and $\omega_I = e_2e_3 = J$.

Theorem 4.1 (Selection of the imaginary unit)**.** *Inside* $\mathrm{End}_{\mathbb{R}}(S_n)$ *the joint commutant of the interior Lorentz algebra and of all internal bivectors is* $\mathrm{span}\{\mathbf{1}, \Omega, \omega_I, \Omega\omega_I\} \cong \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C}$*, at every rung. Its solutions of* $x^2 = -\mathbf{1}$ *are the four isolated points* $\pm\Omega$, $\pm\omega_I$*. Of these only* $\pm\omega_I$ *commute with the interior gamma matrices, from which the kinetic operator is built. Hence the imaginary unit is* $i = \pm\omega_I$*, and the chirality is* $\gamma_5 = \mp\Omega\omega_I = \mp\Gamma$ *with*

$$\Gamma := \omega_I\,\Omega, \qquad \Gamma^2 = \mathbf{1}.$$

Proof. By (3), commuting with the six interior bivectors and with every internal bivector is a parity condition on each blade, and the blades that satisfy all of them are exactly the four listed, at every rung. A four-dimensional commutative real algebra with two commuting square roots of $-\mathbf{1}$ and no nilpotents is $\mathbb{C} \oplus \mathbb{C}$, whose square roots of -1 are those four points. Finally Ω is the product of the four interior generators, so it anticommutes with each of them, while ω_I contains no interior index and commutes with each. □

From now on J denotes ω_I; at one constituent this is e_2e_3, the generator of Proposition 2.7. The selection depends on its conditions, and it is worth saying exactly how.

Remark 4.2 (The ambient algebra is part of the statement)**.** If commutation with the internal bivectors is dropped, then at $n=1$ the commutant of the four gamma matrices in $M_8(\mathbb{R})$ is the quaternion algebra

$$\mathbb{H} = \mathrm{span}\{\mathbf{1},\ J,\ e_2\Omega,\ e_3\Omega\},$$

and its square roots of $-\mathbf{1}$ form a two-sphere. One further datum removes that freedom. The volume element of the temporal triple, $T := e_1e_2e_3$, depends only on the splitting $V = V^+ \oplus V^-$ and on an orientation of V^+; it commutes with J and anticommutes with $e_2\Omega$ and $e_3\Omega$ by (3), so

$$\{X \in \mathbb{H} :\ [X, T] = 0\} = \mathbb{R}\,\mathbf{1} \oplus \mathbb{R}\,J,$$

and $J = \gamma^0 T$ is recovered from the clock and the temporal orientation, up to the sign of that orientation. Higher up the tower the commutant of the gamma matrices alone acquires a matrix multiplicity algebra; at $n=3$ the Lorentz commutant is $M_{128}(\mathbb{C})$, whose own complex structure is Ω and not ω_I (Proposition 11.1). Selection tracks the conditions imposed, and every uniqueness statement in this paper is made with them.

That the imaginary unit is not adjoined but read off the algebra is the cleanest structural result at one constituent. It has an independent counterpart: on a real Hilbert space, Poincaré symmetry alone can force the complex structure [90].

4.2 Real Dirac theory is complex Dirac theory

Theorem 4.3 (Exactness of the realification)**.** *The* $\mathbb{C}$*-linear algebra generated by the interior gamma matrices on* $(S, J) \cong \mathbb{C}^4$ *is all of* $M_4(\mathbb{C})$*. Consequently the real Maxwell–Dirac system on* $\mathrm{Cl}(3,3)$ *is ordinary complex Maxwell–Dirac theory written in real variables.*

Proof. J commutes with the four γ^μ. The sixteen monomials in the γ^μ and their J-multiples are thirty-two linearly independent real matrices commuting with J, and the commutant of J in $M_8(\mathbb{R})$ is $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^4) = M_4(\mathbb{C})$, of real dimension thirty-two. □

Corollary 4.4. *Anything derivable from the real formulation alone was derivable from the complex one, and conversely. The gauged system is quantum electrodynamics in real variables, and no prediction can follow from the rewriting.*

This is the deflationary principle of the introduction. It does not make the arena uninteresting; it says where its content can lie. Everything below that depends only on the four gamma matrices and J is inherited. What is not inherited is what the arena adds: the selection of J, the surplus plane and its uses, the pairing, and the tower.

4.3 Admissible bilinears

A field ψ with values in S is real. If its components anticommute, a term $\psi^\top A\psi$ vanishes when A is symmetric, and $\psi^\top A\,\partial\psi = \frac{1}{2}\partial(\psi^\top A\psi)$ is a total derivative when A is antisymmetric. A mass term therefore needs an antisymmetric kernel, and a first-order kinetic term a symmetric one. With the clock $\gamma^0 = e_1$:

Proposition 4.5 (Kernels of fixed symmetry)**.** *The mass form $\gamma^0 J$ is antisymmetric. The kinetic kernels*

$$K^\mu := (\gamma^0 J)(J\gamma^\mu) = -\gamma^0\gamma^\mu, \qquad K^0 = -\mathbf{1},$$

are symmetric, while $(\gamma^0 J)\gamma^\mu$ is antisymmetric. The kinetic form must therefore carry the factor J: its J-free analogue defines a total derivative and constrains nothing a field equation sees.

The mass form $\gamma^0 J = e_1e_2e_3 = T$ is the temporal volume element of Remark 4.2. It is not the only admissible mass.

Proposition 4.6 (Scalar and pseudoscalar mass)**.** *The antisymmetric elements of $M_8(\mathbb{R})$ that commute with J and are invariant under the interior Lorentz algebra form the two-dimensional space* $\mathrm{span}\{T,\ T\Omega\}$*. Used as the kernel of the Dirac bilinear, the second contributes nothing to the kinetic term, because $T\Omega J\gamma^\mu$ is antisymmetric for every μ; the two differ only in the mass term. T is a scalar mass and $T\Omega$ a pseudoscalar one, since $\Omega^2 = -\mathbf{1}$ and Ω anticommutes with every γ^μ, so that Ω is the real form of $i\gamma_5$. The family $T(\cos\vartheta + \sin\vartheta\,\Omega) = Te^{\vartheta\Omega}$ is the chiral mass angle.*

So T is singled out among admissible masses by the requirement that the mass be scalar, and that requirement is an input. The central element Ω will reappear in Section 5 as the axial $U(1)$.

The Hamiltonian. The free Dirac Hamiltonian at momentum p is

$$H(p) = \gamma^0\gamma^a p_a + m\gamma^0, \qquad H(p)^2 = E^2\mathbf{1}, \qquad E^2 = |\mathbf{p}|^2 + m^2, \tag{8}$$

with spectral projectors $\Lambda_\pm(p) = \frac{1}{2}(\mathbf{1} \pm H(p)/E)$. Both $\gamma^0 = e_1$ and $\gamma^0\gamma^a = e_1e_{a+3}$ are symmetric, so $H(p)$ is a symmetric matrix that commutes with J; its eigenvalues are $\pm E$, each with real multiplicity four.

Example 4.7. At $\mathbf{p} = (0,0,3)$ and $m = 4$, $H = 3e_1e_6 + 4e_1$ satisfies $H^2 = (9+16)\mathbf{1} = 25\,\mathbf{1}$, because e_1e_6 and e_1 anticommute and $(e_1e_6)^2 = +\mathbf{1}$. The eigenvalues are ± 5; each eigenspace is a complex plane for J, and the helicity $\Sigma_3 = J\,\frac{1}{2}e_4e_5$ commutes with H and splits each of them into two real planes.

4.4 A first-order real form of free evolution

For a commuting real field the same data give a first-order symplectic system. Take

$$\mathcal{L}_1 = \tfrac{1}{2}\Psi^\top J\dot\Psi - \tfrac{1}{2}\Psi^\top H(p)\Psi.$$

Its Euler–Lagrange equation is $\dot\Psi = -JH(p)\Psi$. Since JH is antisymmetric with $(JH)^2 = -E^2\mathbf{1}$, the flow lies in $SO(8)$ and

$$\Psi(\lambda) = \cos(E\lambda)\,\Psi_0 - \frac{\sin(E\lambda)}{E}\,JH\Psi_0.$$

Reading J as i turns this into $i\dot\psi = h\psi$ with $h = \alpha\cdot\mathbf{p} + \beta m$, the free Dirac Hamiltonian at fixed momentum: an exact embedding of free Dirac evolution in a real symplectic form, with eight second-class constraints reducing a sixteen-dimensional phase space to eight. Three things norm conservation does not buy. The energy $\frac12\Psi^\top H\Psi$ is unbounded below, since $\operatorname{Tr} H = 0$ and $H^2 = E^2\mathbf{1}$ force the spectrum $\{\pm E\}$ with equal multiplicity. For anticommuting components the action collapses to a boundary term, $\Psi^\top H\Psi = 0$ because H is symmetric and $\Psi^\top J\dot\Psi = \frac12\frac{d}{d\lambda}(\Psi^\top J\Psi)$ because J is antisymmetric, which is why a fermionic Lagrangian must use the kernels of Proposition 4.5. And the bilinear $\Psi^\top J\dot\Psi$ is not preserved by the boosts. The real form of Dirac theory in which a bivector commuting with the gamma matrices plays the role of i is old [65]; what is specific here is that the bivector is the surplus rotation J, selected by Theorem 4.1.

5 Discrete symmetries, the massless gauge algebra, and the states

5.1 Charge conjugation, parity and time reversal

At one constituent write $U = \langle e_2, e_3\rangle$ for the surplus plane, $\Omega = \gamma^0\gamma^1\gamma^2\gamma^3$, and $\Gamma = J\Omega$. An operator X on S is *linear* when $XJ = JX$ and *antilinear* when $XJ = -JX$. It intertwines the free Dirac operator $J\gamma^\mu\partial_\mu - m$ when it commutes with every $J\gamma^\mu$. Parity is required to be linear and time reversal antilinear, and each must satisfy $XH(p)X^{-1} = H(-p)$ for every p and m.

Theorem 5.1 (Charge conjugation is the surplus plane)**.** *The antilinear operators that intertwine the Dirac operator form exactly the two-dimensional space U. Every unit vector $C \in U$ satisfies*

$$CH(p)C^{-1} = -H(-p), \qquad CJC^{-1} = -J, \qquad C\Gamma C^{-1} = -\Gamma, \qquad C\Lambda_+(p)C^{-1} = \Lambda_-(-p),$$

commutes with every interior Lorentz generator, and preserves the mass form and every kinetic kernel K^μ. Hence C is unique up to a gauge rotation $e^{\theta J}$, and the operator known as the Majorana real structure is charge conjugation.

Proof. If $XJ = -JX$, then $X(J\gamma^\mu) = (J\gamma^\mu)X$ is equivalent to $X\gamma^\mu = -\gamma^\mu X$. The conditions are linear, and conjugation by a blade sends every blade to plus or minus itself, so the solution space is spanned by the blades that anticommute with J and with e_1, e_4, e_5, e_6. By (3) a blade e_I anticommutes with a vector e_a exactly when $|I| - [a \in I]$ is odd. For odd $|I|$ this forces $I \subset \{2,3\}$, hence $I = \{2\}$ or $\{3\}$, and both anticommute with J. For even $|I|$ it forces $\{1,4,5,6\} \subset I$, hence $e_I = \Omega$ or $e_1\cdots e_6$, and both commute with J. The solutions are therefore e_2 and e_3. The listed identities follow from the same rule, since e_2, e_3 anticommute with every γ^μ, with J and with Γ, and commute with every interior bivector. □

Corollary 5.2 (Equal mass, equal spin, opposite charge)**.** *C commutes with every generator of the interior Lorentz algebra, so it preserves the Casimir $-\sum_a S_a^2 = \frac34\mathbf{1}$, and $(CH(p)C^{-1})^2 = E^2\mathbf{1}$. The Majorana condition $\Psi = C\Psi$ makes the current vanish: a particle that is its own antiparticle is neutral.*

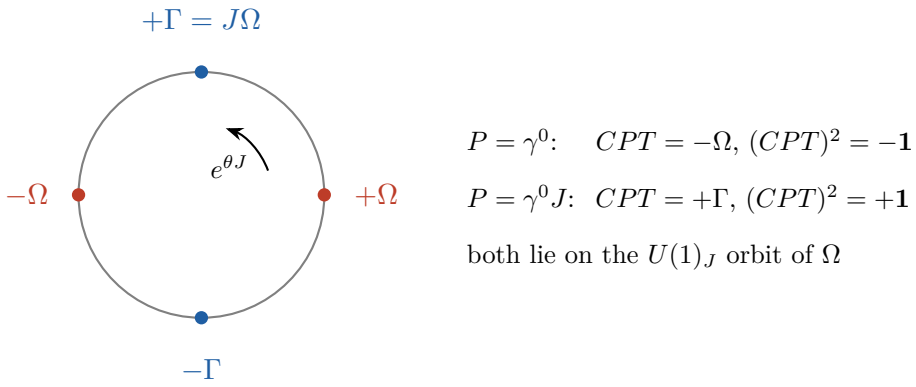


Figure 4: The CPT circle at one constituent (Theorem 5.4). The product of the three discrete operators is the spacetime pseudoscalar or the chirality according to the parity chosen; the two lie on one orbit of the gauge rotation $e^{\theta J}$.

Theorem 5.3 (Parity and time reversal)**.** *The admissible parities form the space $\langle \gamma^0, \gamma^0 J \rangle$, and $P^2 = +\mathbf{1}$ holds only at $P = \pm\gamma^0$, while $P = \pm\gamma^0 J$ gives $P^2 = -\mathbf{1}$. The admissible time reversals form $U \cdot \gamma^1\gamma^2\gamma^3$, so $T = C\gamma^1\gamma^2\gamma^3$ with*

$$T^2 = -\mathbf{1}.$$

The sign is not a convention: $B = \gamma^1\gamma^2\gamma^3$ has $B^2 = +\mathbf{1}$ and anticommutes with every vector C of the clock plane, so $T^2 = CBCB = -C^2B^2 = -\|C\|^2\mathbf{1}$ for every choice of clock in the positive triple.

Theorem 5.4 (CPT lies on a circle)**.** *The product of the three operators depends on the parity chosen:*

$$P = \gamma^0: \quad CPT = -\Omega,\ (CPT)^2 = -\mathbf{1}; \qquad\qquad P = \gamma^0 J: \quad CPT = +\Gamma,\ (CPT)^2 = +\mathbf{1}.$$

What is convention-free is the orbit of Ω under $U(1)_J$, $\theta \mapsto \cos\theta\,\Omega + \sin\theta\, J\Omega$, whose four real points $\pm\Omega, \pm\Gamma$ are the spacetime pseudoscalar and the chirality.

These are statements about explicitly constructed operators of the free theory; no CPT theorem is proved, and the interacting theory is not constructed.

5.2 The massless gauge algebra at one constituent

A gauge generator must leave the kinetic form invariant. Solving that condition over all of $\mathrm{End}_{\mathbb{R}}(S)$, with no ansatz, gives the following.

Proposition 5.5 (Gauge algebra from the kinetic forms)**.** *The solutions of $X^\top K^\mu + K^\mu X = 0$ for $\mu = 0, \ldots, 3$ form*

$$\mathfrak{g}_{\text{massless}} = \mathbb{R}\,\Omega \ \oplus\ \mathrm{span}\{J,\ e_2\Omega,\ e_3\Omega\} \ \cong\ \mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2).$$

The condition at $\mu = 0$ is $X^\top = -X$, and the conditions at $\mu = a$ then say that X commutes with the three boosts $\gamma^0\gamma^a$ and hence with the whole interior Lorentz algebra. The trace form is $-8\,\mathbf{1}$ on this basis, so the algebra is compact. Adding invariance of the mass form, $X^\top(\gamma^0 J)+(\gamma^0 J)X = 0$, leaves exactly $\mathbb{R}J$. The algebra meets the span of the pole operators in zero.

The $\mathfrak{su}(2)$ factor contains the fermion-number direction J and two generators that anticommute with J; it therefore mixes a field with its conjugate and is the Pauli–Gürsey algebra of a massless Dirac field [103, 56]. It is a symmetry of the free massless theory, not a gauge candidate

on this matter: its two Weyl components form a single doublet, and gauging $SU(2)$ on an odd number of doublets is forbidden by the global anomaly [148]. The central factor is the axial one. Since $\Gamma = J\Omega$, $\Omega = -J\Gamma$, the real form of $-i\gamma_5$; $U(1)_\Omega$ is broken by a Dirac mass and is anomalous on this matter content [15, 2]. The electromagnetic $\mathfrak{u}(1)$ is therefore singled out twice at one constituent: as the whole internal centraliser (Proposition 2.7) and as what survives of the massless gauge algebra once a mass is admitted.

5.3 The state space

Theorem 5.6 (The real module is not a state space)**.** *Eight* 16×16 *integer matrices* θ_i*, antisymmetric with* $\theta_i^2 = -\mathbf{1}$ *and pairwise anticommuting, realise the canonical anticommutation relations of the eight real field components exactly. Their commutant in* $M_{16}(\mathbb{R})$ *is* $\mathbb{R}\,\mathbf{1}$*, because the* 256 *Clifford monomials span the whole matrix algebra. Hence no internal complex structure commutes with all eight field operators, and the quadratic observables built from them are real antisymmetric matrices with imaginary spectra. The quantum state space has to be formed from outside, as* $\mathcal{F} = \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^{16}$*, with charge* $Q = -i\rho(J)$ *and Hamiltonian* $H = -i\rho(JH(p))$*, where* $\rho(A) = -\frac{1}{4}\sum_{ij} A_{ij}\theta_i\theta_j$ *lifts an antisymmetric* 8×8 *matrix to the module.*

It is the complex structure, not the reality, that is missing, which is the lesson of Remark 4.2 arriving one level up.

Theorem 5.7 (The free spectrum of one momentum)**.** *The lifted generators have characteristic polynomials*

$$\chi_{2\rho(J)}(x) = x^6(x^2+4)^4(x^2+16), \qquad \chi_{2\rho(JH)}(x) = x^6(x^2+4E^2)^4(x^2+16E^2),$$

so $\operatorname{spec} Q = \{0, \pm 1, \pm 2\}$ *and* $\operatorname{spec} H = \{0, \pm E, \pm 2E\}$*, with multiplicities* $6, 4, 4, 1, 1$*. The joint eigenspaces of* (H, Q) *are*

H	dimension	charges
$-2E$	1	$q = 0$
$-E$	4	$q = \pm 1$, twice each
0	6	$q = \pm 2$ once each, $q = 0$ four times
$+E$	4	$q = \pm 1$, twice each
$+2E$	1	$q = 0$

so $H + 2E$ *has spectrum* $0, E, 2E, 3E, 4E$ *with multiplicities* $1, 4, 6, 4, 1$*.*

The multiplicities $1, 4, 6, 4, 1$ are the binomial coefficients of a four-mode Fock space, as a single Dirac flavour with two particle and two antiparticle states must give. The ground state at $-2E$ is the zero-point energy $-\frac{1}{2}\sum E$ of the four modes; summed over momenta it is the familiar fermion term of the one-loop effective potential. A projector built from $\rho(JH)^2$ could not have produced the table: squaring loses the sign of the energy and conflates the vacuum with the filled state.

Proposition 5.8 (Conjugation on states)**.** $C \in SO(8)$ *lifts to* Spin(8) *as a product of four integral reflections. The lift is unitary on* $\mathcal{F}$*, anticommutes with* Q *and carries* $H(p)$ *to* $H(-p)$*, so* $\langle\psi|Q|\psi\rangle = 0$ *for every* C*-eigenstate. The stronger statement* $Q|\psi\rangle = 0$ *is false: the* $+1$ *eigenspace of* C *has real dimension eight and* Q *annihilates only a three-dimensional part of it.*

Remark 5.9 (Many-body chirality is a group lift)**.** ρ is a homomorphism of Lie algebras only, so the product $\rho(J)\rho(\Omega)$, of rank eight, is not a Spin(8) element and is not the chirality of the module. The correct object is $\Gamma_{\mathcal{F}} = \exp[\frac{\pi}{2}(\rho(J) + \rho(\Omega))]$, an involution of $\mathcal{F}$ whose eigenvalue on a joint eigenvector is $i^{q_J + q_\Omega}$; the sectors are labelled by the parity of $\frac{1}{2}(q_J + q_\Omega)$.

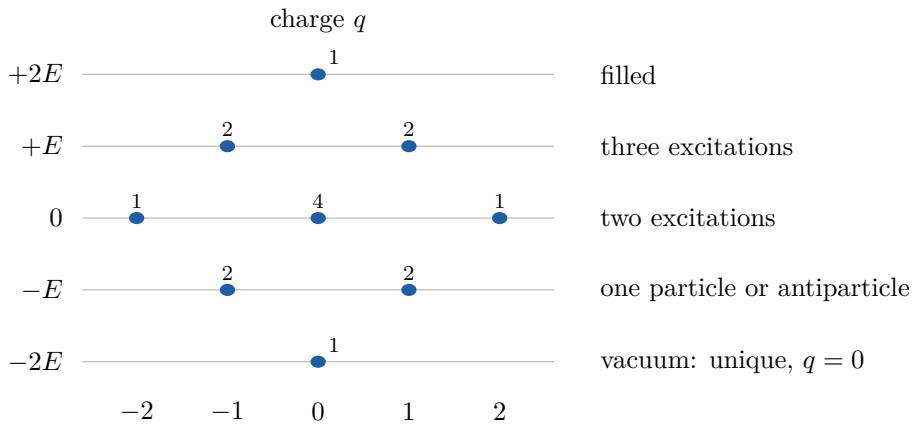


Figure 5: The Fock space of one momentum (Theorem 5.7). Each dot is a joint eigenspace of energy and charge, labelled by its dimension. The multiplicities $1, 4, 6, 4, 1$ are those of four fermionic modes: two particle and two antiparticle states of one Dirac flavour, read off rather than inferred.

Part II
The tower and its matter

6 The tower: centralisers, nesting, and carriers

Climbing the tower adds six directions per rung and leaves the local structure of each factor unchanged. The interior four-plane stays in the first factor; the question is what the added directions do to the internal symmetry.

Theorem 6.1 (Internal centraliser)**.** *In* $\mathrm{Cl}(3n, 3n)$ *the bivectors commuting with the interior Lorentz action form*

$$\mathfrak{g}_{\mathrm{int}}^{(n)} \cong \mathfrak{so}(3n-1, 3n-3), \qquad \dim \mathfrak{g}_{\mathrm{int}}^{(n)} = \tfrac{1}{2}(6n-4)(6n-5).$$

Proof. By (3) a bivector sharing exactly one index with an interior bivector anticommutes with it, and one sharing none or two commutes. A bivector commuting with all six interior bivectors therefore has either both indices in E, which gives $\mathfrak{l}_{\mathrm{int}}$ itself and is excluded by commutation with the boosts, or both indices in the Lorentz complement

$$W_n = \langle e_2^{(1)}, e_3^{(1)} \rangle \oplus \bigoplus_{k=2}^{n} \langle e_1^{(k)}, \dots, e_6^{(k)} \rangle,$$

which has $3n-1$ positive and $3n-3$ negative directions. All bivectors of W_n commute with $\mathfrak{l}_{\mathrm{int}}$, and they span $\mathfrak{so}(W_n)$. □

Corollary 6.2 (The photon at one constituent)**.** *At* $n = 1$, $W_1 = U$ *and* $\mathfrak{g}_{\mathrm{int}}^{(1)} = \mathbb{R}J$. *The electromagnetic* $\mathfrak{u}(1)$ *is not a symmetry of the interior structure but the symmetry, by exhaustion. In agreement with Proposition 5.5, no other charge direction gauges at* $n = 1$.

Proposition 6.3 (One frame, and nesting)**.** *The interior four-plane lies in the first tensor factor, so the six interior bivectors have the same index sets at every rung and identical structure constants. The tower is one Lorentz frame with an increasing family of modules over it. Moreover* $W_1 \subset W_2 \subset \cdots$, *hence*

$$\mathfrak{g}_{\mathrm{int}}^{(1)} \subset \mathfrak{g}_{\mathrm{int}}^{(2)} \subset \mathfrak{g}_{\mathrm{int}}^{(3)} \subset \cdots, \qquad \dim = 1,\ 28,\ 91,\ 190,\ 325, \dots,$$

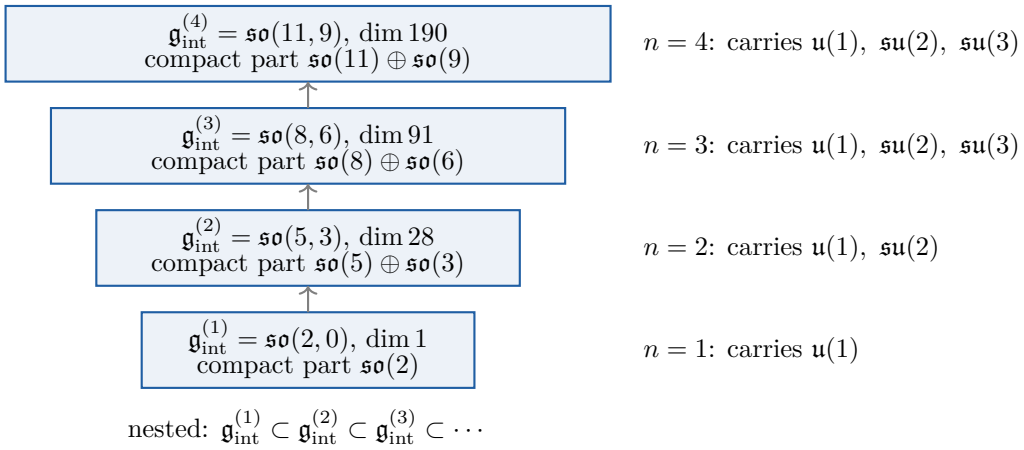


Figure 6: The tower of internal centralisers (Theorems 6.1 and 6.4). Each rung contains the previous one; the compact part decides which gauge factors can be carried. The electromagnetic $\mathfrak{u}(1)$ is the whole centraliser only at the first rung, the one rung with no room for a nonabelian factor.

with direct limit $\mathfrak{so}(\infty,\infty)$*. A connection valued at one rung is a connection at every higher rung with no matching condition. The inclusion runs in the wrong direction to justify a truncation to low rungs, which would be a separate postulate.*

A gauge algebra has to be compact, and every compact subalgebra of a reductive real Lie algebra is conjugate into a maximal compact one [72, 62], here $\mathfrak{so}(3n-1) \oplus \mathfrak{so}(3n-3)$. Since $\mathfrak{su}(2)$ needs a faithful real representation of dimension at least three and $\mathfrak{su}(3)$ one of dimension at least six [53], the rungs can be sorted by what they can carry.

Theorem 6.4 (Carrier census)**.**

n	$\mathfrak{g}_{\mathrm{int}}^{(n)}$	*maximal compact*	dim	*rank*	$\mathfrak{u}(1)$ *alone*	$\mathfrak{su}(2)$	$\mathfrak{su}(3)$
1	$\mathfrak{so}(2,0)$	$\mathfrak{so}(2)$	1	1	*yes*	*no*	*no*
2	$\mathfrak{so}(5,3)$	$\mathfrak{so}(5) \oplus \mathfrak{so}(3)$	13	3	*no*	*yes*	*no*
3	$\mathfrak{so}(8,6)$	$\mathfrak{so}(8) \oplus \mathfrak{so}(6)$	43	7	*no*	*yes*	*yes*
4	$\mathfrak{so}(11,9)$	$\mathfrak{so}(11) \oplus \mathfrak{so}(9)$	91	9	*no*	*yes*	*yes*

At $n = 2$ *the projection of a simple* $\mathfrak{su}(3)$ *to* $\mathfrak{so}(3)$ *must vanish and* $\mathfrak{so}(5)$ *has no proper subalgebra of dimension above six; at* $n = 3$ *colour enters through* $\mathfrak{su}(3) \subset \mathfrak{su}(4) \cong \mathfrak{so}(6)$*.*

Lemma 6.5 (Rank obstruction)**.** $\operatorname{rank}(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 4$*, while the maximal compact subalgebras above have ranks* $1, 3, 7, 9$*. The full Standard-Model algebra therefore embeds at no rung* $n \leq 2$*. At* $n = 2$ *the maximal compact subalgebra has dimension* $13 \geq 12$*, so dimension counting alone would admit it, and rank is the sharp invariant.*

Corollary 6.6 (The tower's trade-off)**.** *The* $\mathfrak{u}(1)$ *is the whole internal centraliser only at* $n = 1$*, the one rung with no room for either nonabelian carrier. Inside* $\mathfrak{g}_{\mathrm{int}}^{(n)}$ *the centraliser of the embedded one-rung charge* $e_2^{(1)} e_3^{(1)}$ *has dimensions* $1, 16, 67, 154, 277$ *for* $n = 1, \ldots, 5$*, since by* (3) *exactly* $12(n-1)$ *internal bivectors fail to commute with it. From* $n = 2$ *onwards the photon is one direction in a space that grows without bound; its uniqueness is lost to no dynamical process, only to the arrival of room.*

At $n \geq 2$ the embedded one-rung charge $e_2^{(1)} e_3^{(1)}$ must not be confused with the complex structure $J = \omega_I$ of Theorem 4.1, which commutes with every internal bivector and has all of $\mathfrak{g}_{\mathrm{int}}^{(n)}$ as centraliser; they agree only at $n = 1$.

Remark 6.7 (The gaugeable fraction does not improve)**.** Of the $\binom{6n}{2} = 3n(6n-1)$ bivectors of $\mathrm{Cl}(3n, 3n)$, $2\binom{3n}{2} = 3n(3n-1)$ pair directions of the same sign: the fractions 6/15, 30/66, 72/153 rise only to 1/2. Climbing adds boosts at essentially the rate at which it adds rotations. The census permits the chain $\mathfrak{u}(1) \subset \mathfrak{u}(1) \oplus \mathfrak{su}(2) \subset \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ at $n = 1, 2, 3$; it permits it, and nothing selects those subalgebras among others of the same rank.

7 Chirality, projectors, and colour

7.1 Unprojected matter is vector-like

With $i = J$ and $\gamma_5 = \Gamma$ as in Theorem 4.1, consider the part of the Lorentz commutant that preserves chirality. It splits into a *vector class* $\mathrm{Cl}^0(W)$, the even subalgebra of the internal directions, whose elements commute with the interior gamma matrices, and an *axial class* $\Omega\,\mathrm{Cl}^0(W)$, whose elements anticommute with them.

Proposition 7.1 (Unprojected matter is vector-like)**.** *Every grade-two internal algebra lies in the vector class. For a gauge algebra in the vector class, γ_μ is an invertible, complex-linear, gauge-equivariant map between the two chirality sectors, so the unprojected matter is vector-like at every rung.*

Proposition 7.2. *The two classes form a symmetric pair, $[\mathrm{a}, \mathrm{a}] \subseteq \mathrm{v}$, so the axial class is a Lie triple system and no nonabelian gauge algebra lies entirely inside it.*

This does not exclude generators with both parts. If $[\Gamma, t_a] = 0$ and $P_L = \frac{1}{2}(\mathbf{1} - \Gamma)$, then

$$[P_L t_a, P_L t_b] = P_L^2 [t_a, t_b] = \epsilon_{abc} P_L t_c,$$

a nonabelian algebra, half vector and half axial. Such *dressing* is available to nonabelian carriers exactly as to abelian ones; only the purely axial summand is excluded. For scalar dressings $f(\Gamma)t_a$ with $f = a + b\Gamma$ and unchanged structure constants, closure forces $f^2 = f$, that is $a^2 + b^2 = a$ and $2ab = b$, so $f \in \{0, \mathbf{1}, P_L, P_R\}$.

7.2 A family of projectors at three constituents

To obtain chiral matter one must project. At $n = 3$ write $J^{(k)} = e_2^{(k)} e_3^{(k)}$ for the surplus bivector and $\omega^{(k)} = e_1^{(k)} \cdots e_6^{(k)}$ for the volume of constituent k, and $K_{14}^{(3)} = e_1^{(3)} e_4^{(3)}$ for the pole bivector joining the first temporal and the first spatial direction of constituent three. The three operators

$$X_1 = J^{(1)} J^{(2)}, \qquad X_2 = \omega^{(3)}, \qquad X_3 = \omega^{(1)} K_{14}^{(3)}$$

are commuting symmetric traceless involutions, and they generate the orthogonal projectors $P_1 = \frac{1}{2}(\mathbf{1} + X_1)$ of rank 256, $P_{ij} = \frac{1}{4}(\mathbf{1} + X_i)(\mathbf{1} + X_j)$ of rank 128, and $P_{\mathrm{fam}} = \frac{1}{8}\prod_i (\mathbf{1} + X_i)$ of rank 64.

Definition 7.3 (Ambient and faithful algebra)**.** For an orthogonal projector P commuting with the Lorentz action let $\mathfrak{g}_{\mathrm{phys}}(P)$ be the internal bivectors commuting with P, $\mathfrak{k}(P)$ its compact part (the $+1$ eigenspace of $\theta(X) = -X^\top$), and

$$\mathfrak{k}_{\mathrm{eff}}(P) = \mathfrak{k}(P) / \{X \in \mathfrak{k}(P) : XP = 0\}.$$

The quotient is not a technicality: an element can preserve the block and annihilate every state in it.

Example 7.4 (An annihilator)**.** With $X_1 = J^{(1)}J^{(2)}$, $(J^{(1)})^2 = (J^{(2)})^2 = -\mathbf{1}$ and $[J^{(1)}, J^{(2)}] = 0$,

$$(J^{(1)} + J^{(2)})(\mathbf{1} + X_1) = J^{(1)} + J^{(2)} + (J^{(1)})^2 J^{(2)} + J^{(2)} J^{(1)} J^{(2)} = J^{(1)} + J^{(2)} - J^{(2)} - J^{(1)} = 0.$$

So $J^{(1)} + J^{(2)}$ is a compact element of $\mathfrak{k}(P_{\text{fam}})$ that annihilates the block of P_{fam}, although it does not annihilate the block of P_{23}. A table of gauge algebras that omits the quotient overcounts, here by three dimensions.

Theorem 7.5 (The projector family)**.** *Computed on the* 512*-dimensional module as the exact nullspace of* $X \mapsto [X, P]$ *over the* $\binom{14}{2} = 91$ *internal bivectors, with* $\mathfrak{k}_{\text{eff}}$ *decomposed on the image:*

P	rank	$\dim \mathfrak{g}_{\text{phys}}$	$\dim \mathfrak{k}$	$\dim \mathfrak{p}$	$\dim \ker$	$\mathfrak{k}_{\text{eff}}$	axial
P_{fam}	64	17	9	8	3	$\mathfrak{u}(1)^3 \oplus \mathfrak{su}(2)$	yes
P_{23}	128	23	9	14	0	$\mathfrak{u}(1)^3 \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(2)$	yes
P_{13}	128	33	17	16	3	$\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{so}(5)$	yes
P_{12}	128	27	15	12	3	$\mathfrak{su}(2)^{\oplus 4}$	no
P_1	256	51	27	24	3	$\mathfrak{so}(6) \oplus \mathfrak{su}(2)^{\oplus 3}$	no

Here a projector is axial when $P\gamma^\mu P = 0$*. Factors are identified by dimension and rank:* $(3,1) = \mathfrak{su}(2)$*,* $(10,2) = \mathfrak{so}(5) \cong \mathfrak{sp}(2)$*,* $(15,3) = \mathfrak{so}(6) \cong \mathfrak{su}(4)$*; a six-dimensional semisimple compact factor is necessarily* $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$*. The centres have dimensions* $3,3,1,0,0$ *and the derived algebras* $3,6,13,12,24$*.*

Corollary 7.6 (No colour on the axial members)**.** *The largest simple factor among the rank-*64 *and rank-*128 *blocks is* $\mathfrak{so}(5)$*, every proper subalgebra of which has dimension at most six. Since* $\mathfrak{su}(3)$ *is simple of dimension eight, it embeds in none of them. Dimension eight is necessary and not sufficient:* P_{13} *carries an* $\mathfrak{so}(5)$ *of dimension ten and still contains no* $\mathfrak{su}(3)$*, so the test is an embedding test.*

7.3 Colour at rank 256

Theorem 7.7 (Colour on P_1)**.** *Let* $f_1, \ldots, f_6$ *be the six spatial directions of the second and third constituents. The fifteen bivectors* $f_a f_b$ *square to* $-\mathbf{1}$*, are antisymmetric and close on* $\mathfrak{so}(6)$*; each commutes with* X_1*, because the index sets are disjoint and* $2 \cdot 4$ *is even in* (3)*, so all fifteen preserve* P_1 *and act faithfully on its image. Let* $\Omega_6 = f_1 f_4 + f_2 f_5 + f_3 f_6$ *generate the pairing* $f_k \mapsto f_{k+3}$*. Its centraliser in* $\mathfrak{so}(6)$ *has dimension nine and is* $\mathfrak{u}(3)$*; the derived algebra has dimension eight and rank two, is compact, closed and simple, hence* $\mathfrak{su}(3)$*, and its eight generators act faithfully on the block.*

Theorem 7.8 (P_1 is vector-like undressed)**.** X_1 *is a product of four generators, none of them interior, so* $|I||J| - |I \cap J| = 4$ *is even for every interior generator and* $[X_1, \gamma_\mu] = 0$*. Hence* P_1 *is not axial and its matter is vector-like by Proposition 7.1.*

The obstruction is to the undressed action only.

Theorem 7.9 (Colour with a chiral doublet on P_1)**.** *Keep the colour generators in the six negative directions of factors two and three, and in the three positive directions of factor three set*

$$t_1 = \tfrac{1}{2} e_1^{(3)} e_2^{(3)}, \qquad t_2 = \tfrac{1}{2} e_1^{(3)} e_3^{(3)}, \qquad t_3 = -\tfrac{1}{2} e_2^{(3)} e_3^{(3)}.$$

With $P_{1,L} = P_1 \tfrac{1}{2}(\mathbf{1} - \Gamma)$ *and* $L_i = P_{1,L} t_i$*,*

$$[L_i, L_j] = \epsilon_{ij}{}^k L_k, \qquad [C_A, L_i] = 0, \qquad -\sum_i L_i^2 = \tfrac{3}{4} P_{1,L}, \qquad L_i P_{1,R} = 0$$

for every colour generator C_A*. Vector colour and a commuting, purely left-handed doublet action therefore coexist on* P_1*.*

Proof. The t_i are half time–time bivectors of factor three, so each squares to $-\frac{1}{4}\mathbf{1}$ and they close on $\mathfrak{su}(2)$. The colour generators carry only negative indices of factors two and three, disjoint from $\{e_1^{(3)}, e_2^{(3)}, e_3^{(3)}\}$, and two even blades with disjoint supports commute. $P_{1,L}$ is an idempotent of rank 128 commuting with each t_i, so $L_i = P_{1,L}t_i$ inherits the bracket and $-\sum L_i^2 = P_{1,L}(-\sum t_i^2) = \frac{3}{4}P_{1,L}$, the Casimir of a doublet. $P_{1,L}P_{1,R} = 0$ gives the last identity. □

Ordering the family by $\dim \mathfrak{k}_{\text{eff}}$ and recording whether the projector is axial,

$$P_{\text{fam}}\,(6)\ \text{axial}, \quad P_{23}\,(9)\ \text{axial}, \quad P_{12}\,(12)\ \text{not}, \quad P_{13}\,(14)\ \text{axial}, \quad P_1\,(24)\ \text{not},$$

the two orderings are not reverses of one another, and what survives is a statement about these five projectors only: no axial one carries $\mathfrak{su}(3)$ in its effective bivector algebra, and $\mathfrak{su}(3)$ appears on exactly one of them, the vector-like P_1. The pattern is not a law of the arena. The next section exhibits a projector outside the family on which colour and chirality coexist.

8 A chiral Pati–Salam block

The five projectors of Section 7 are one family among many. Enlarging the class to all commuting symmetric blade involutions produces a projector on which colour and chirality coexist. The idea is to correlate spacetime chirality with an *internal* four-dimensional chirality, and to constrain the colour *volume* rather than any colour direction.

8.1 Construction

At three constituents take the interior gamma matrices $\gamma^\mu = (e_1^{(1)}, e_4^{(1)}, e_5^{(1)}, e_6^{(1)})$ and split the remaining fourteen directions into three groups:

group	directions	signs	generators E_A
weak W	$e_1^{(2)},\ e_1^{(3)},\ e_2^{(3)},\ e_3^{(3)}$	$++++$	$7, 13, 14, 15$
colour C	$e_4^{(2)}, e_5^{(2)}, e_6^{(2)},\ e_4^{(3)}, e_5^{(3)}, e_6^{(3)}$	$------$	$10, 11, 12, 16, 17, 18$
auxiliary A	$e_2^{(1)}, e_3^{(1)},\ e_2^{(2)}, e_3^{(2)}$	$++++$	$2, 3, 8, 9$

With $J = \omega_I$ the ordered product of the fourteen internal generators and $\Gamma = J\gamma^0\gamma^1\gamma^2\gamma^3$, define the three volumes

$$\chi_W = e_1^{(2)}e_1^{(3)}e_2^{(3)}e_3^{(3)}, \qquad \chi_C = e_4^{(2)}e_5^{(2)}e_6^{(2)}e_4^{(3)}e_5^{(3)}e_6^{(3)}, \qquad q_A = e_2^{(1)}e_3^{(1)}.$$

Theorem 8.1 (A chiral projector carrying colour)**.** *$X = \Gamma\chi_W$, $Z = -J\chi_C$ and $U = -Jq_A$ are commuting symmetric traceless involutions, and all seven nonidentity products X, Z, U, XZ, XU, ZU, XZU are traceless. Hence*

$$P_* = \tfrac{1}{8}(\mathbf{1} + \Gamma\chi_W)(\mathbf{1} - J\chi_C)(\mathbf{1} - Jq_A) \tag{9}$$

is an orthogonal projector with $\operatorname{Tr} P_* = 512/8 = 64$. It satisfies $[P_*, \Sigma^{\mu\nu}] = [P_*, J] = 0$ for the six Lorentz generators $\Sigma^{\mu\nu} = \frac{1}{2}\gamma^\mu\gamma^\nu$, and $P_*\gamma^\mu P_* = 0$, so it is axial; its two Weyl blocks have real rank 32 each, and $\chi_W P_* = \Gamma P_*$.

Rank is not automatic from the three factors being individually traceless; it is the products that decide, and here all of them are traceless. The statement $\chi_W P_* = \Gamma P_*$ is the correlation of spacetime chirality with the internal chirality of the weak directions.

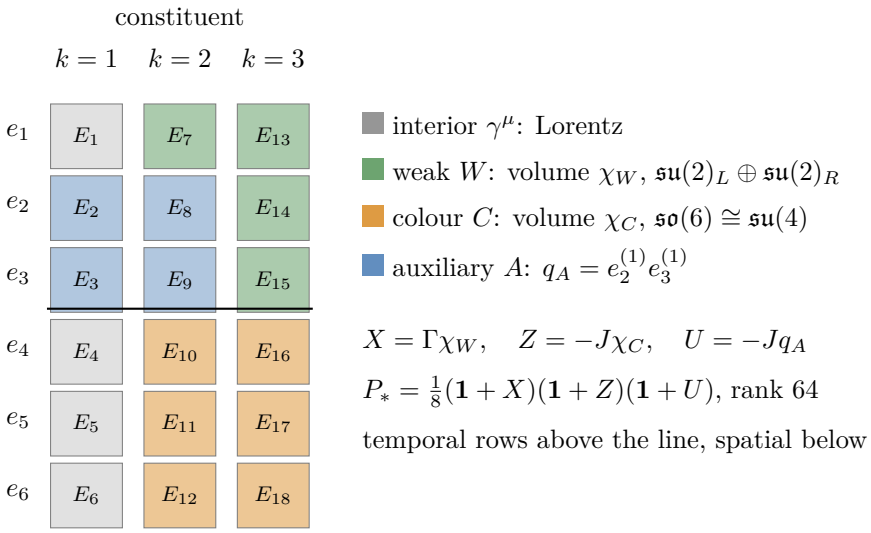


Figure 7: The eighteen generators of $\mathrm{Cl}(9,9)$ arranged by constituent, and the grouping that defines the Pati–Salam block. The interior four-plane carries the Lorentz action; the weak, colour and auxiliary groups enter P_* only through their volumes, which is why every rotation inside each group survives the projection (Theorems 8.1 and 8.2).

8.2 The effective algebra

Theorem 8.2 (The effective algebra and its Pati–Salam part)**.** *All fifteen colour bivectors $e_a e_b$, $a, b \in C$, commute with X, Z and U, so the whole compact $\mathfrak{so}(6)_C \cong \mathfrak{su}(4)_C$ survives and acts faithfully on the block; the colour volume χ_C is invariant under those rotations, so constraining it costs no colour generator. With $w_1, \dots, w_4$ the weak directions, $\tau_i = -\frac{1}{2} w_j w_k$ for cyclic (i,j,k), $t_i^{L,R} = \frac{1}{2}(\mathbf{1} \mp \chi_W)\tau_i$ and $T_i^{L,R} = P_* t_i^{L,R}$,*

$$[T_i^L, T_j^L] = \epsilon_{ijk} T_k^L, \qquad T_i^L P_R = 0, \qquad -\sum_i (T_i^L)^2 = \tfrac{3}{4} P_L,$$

and the mirror identities hold on the right. The $t_i^{L,R}$ are genuine bivectors, for example $t_3^L = -\frac{1}{4}(w_1 w_2 + w_3 w_4)$. The full effective compact bivector algebra is

$$\mathfrak{k}_{\mathrm{eff}}(P_*) = \mathfrak{su}(4) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R \oplus \mathfrak{u}(1)^2,$$

of dimension 23, entirely compact, with a derived algebra of dimension 21 and a two-dimensional centre. All 21 simple generators preserve the kinetic kernels $K^\mu = P_* \gamma^0 \gamma^\mu P_*$, and $K^0 = P_*$.

Proof. Each colour bivector has even overlap with each of the three involution masks, so all fifteen commute with X, Z, U and descend faithfully. For the weak part, $\chi_W^2 = \mathbf{1}$ and $[\chi_W, \tau_i] = 0$ make $\frac{1}{2}(\mathbf{1} \mp \chi_W)$ idempotents commuting with the τ_i; the τ_i close on $\mathfrak{su}(2)$ with $\tau_i^2 = -\frac{1}{4}\mathbf{1}$, so $-\sum_i (T_i^L)^2 = \frac{3}{4} P_L$, and $\chi_W P_* = \Gamma P_*$ places the two ideals on opposite chiralities. The dimension 23 is the full solution of $[B, P_*] = 0$ over the 91-dimensional space of internal bivectors: the nullspace is 23-dimensional, entirely compact, has trivial annihilator on the block, and has a derived algebra of dimension 21. □

Corollary 8.3 (One Pati–Salam family)**.** *The colour volume is not an involution: χ_C is a grade-six blade on six negative directions, so $\chi_C^2 = -\mathbf{1}$. The involution is*

$$\Gamma_C := J\chi_C, \qquad \Gamma_C^2 = \mathbf{1}, \qquad \Gamma_C P_* = -P_*, \quad \textit{equivalently} \quad \chi_C P_* = J P_*.$$

On the module with $J = +i$ each Weyl block then carries exactly four $\mathfrak{su}(4)$ weights, all of colour chirality $\Gamma_C = -1$, each with multiplicity $4 = 2 \times 2$, a weak doublet times a Weyl spinor, and

the two blocks carry the same colour weights. The content is

$$(\mathbf{4},\mathbf{2},\mathbf{1})_L \oplus (\mathbf{4},\mathbf{1},\mathbf{2})_R,$$

one Pati–Salam family including a right-handed neutrino [102]. Which of the two conjugate four-dimensional representations is called $\mathbf{4}$ *is a convention.*

8.3 Charges

The compact algebra is realised by skew operators; charges are the corresponding symmetric operators $\hat{t} = -Jt$. On the block the centre of $\mathfrak{k}_{\text{eff}}(P_*)$ is spanned by the skew operators JP_* and $J\Gamma P_*$, whose charges are P_* and ΓP_*. Concretely $q_A|_{V_*} = J$ and $E_8E_9|_{V_*} = J\Gamma$ on the image $V_* = \operatorname{im} P_*$.

Proposition 8.4 (Colour intertwiner, $B-L$, and the charge table)**.** $R = P_*\gamma^0 e_1^{(2)} e_2^{(2)}$ *satisfies* $R^2 = -P_*$*, exchanges the two Weyl blocks and commutes with every colour generator, so the colour representations of the two chiralities are equivalent and the content is* $(\mathbf{4},\mathbf{2},\mathbf{1})_L \oplus (\mathbf{4},\mathbf{1},\mathbf{2})_R$ *rather than a* $\mathbf{4}/\bar{\mathbf{4}}$ *pairing. With*

$$H = -(e_4^{(2)}e_4^{(3)} + e_5^{(2)}e_5^{(3)} + e_6^{(2)}e_6^{(3)}), \qquad Q_c = -JH,$$

the derived centraliser of H *inside colour* $\mathfrak{so}(6)$ *is* $\mathfrak{su}(3)$*, and* $(Q_c - \mathbf{1})(Q_c + 3\mathbf{1})P_* = 0$*: has eigenvalues* $+1$ *and* -3 *on the block, with real multiplicities* 48 *and* 16*, the triplet and the singlet. Setting* $B-L = Q_c/3$ *and using the weak charges* $-JT^3_{L,R}$*,*

$$Y = -J\,T^3_R + \tfrac{1}{2}(B-L), \qquad Q = -J\,T^3_L + Y,$$

gives one family:

field	$SU(3) \times SU(2)_L$	Y	internal Weyl components
Q_L	$(\mathbf{3},\mathbf{2})$	$\frac{1}{6}$	6
L_L	$(\mathbf{1},\mathbf{2})$	$-\frac{1}{2}$	2
u_R	$(\mathbf{3},\mathbf{1})$	$\frac{2}{3}$	3
d_R	$(\mathbf{3},\mathbf{1})$	$-\frac{1}{3}$	3
e_R	$(\mathbf{1},\mathbf{1})$	-1	1
ν_R	$(\mathbf{1},\mathbf{1})$	0	1

Four real spinor coordinates make one internal Weyl component, so rank 64 *is sixteen internal Weyl components, eight left and eight right, and not sixty-four particles.*

The $\mathfrak{su}(3)$ embedding, the right-handed Cartan direction and the hypercharge combination are chosen; what is established is their mutual compatibility. In the complex structure J every one of the sixteen Weyl fermions has charge one: J is fermion number, $F = 3B + L$, and the baryon number appears as $4B = F + (B-L)$.

Proposition 8.5 (The two central factors cannot be gauged)**.** *For a central charge* $a + b\Gamma$*, corresponding to the skew element* $aJ + bJ\Gamma$*, the anomaly coefficients on the block, per Weyl fermion with every field written left-handed, are*

$$SU(2)_L^2 :\ 2(a-b), \qquad SU(2)_R^2 :\ -2(a+b), \qquad SU(4)^2 :\ -2b,$$
$$\textit{cubic} :\ -16(3a^2b + b^3), \qquad \textit{gravitational} :\ -16b,$$

and the mixed anomalies with two central and one Pati–Salam generator vanish. All vanish only for $a = b = 0$*. The Pati–Salam algebra itself is anomaly free on the block: the* $SU(4)^3$ *anomaly cancels between two fundamentals and two antifundamentals in the all-left convention, and each weak* $SU(2)$ *sees four doublets, so the global anomaly is absent [148]. Hence, with the block as the matter content,* $\mathfrak{su}(4) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ *is the largest anomaly-free gauge algebra containing Pati–Salam, and the two central directions remain global symmetries.*

The obstruction is stated for this matter content; additional matter or another cancellation mechanism is not excluded. The two central symmetries are fermion number J, anomalous only through $SU(2)_L$ and $SU(2)_R$ and hence exact in perturbation theory and violated by their instantons [132], and the chiral charge $J\Gamma$.

Proposition 8.6 (Two rank-64 projectors, two orbits)**.** *Under the compact group of the connection sector, whose Lie algebra is the maximal compact* $\mathfrak{so}(8) \oplus \mathfrak{so}(6)$ *of dimension* 43*, the orbit through* P_{fam} *has dimension* $43 - 9 = 34$ *and the orbit through* P_* *has dimension* $43 - 23 = 20$*. Equal rank does not put two projectors on one orbit.*

Remark 8.7 (What the construction settles)**.** It settles an existence question: colour on a chiral block is not obstructed, and no counting argument based on the five-projector family can be a theorem. It does not select P_*; it does not derive the hypercharge embedding, which is compatible with anomaly cancellation but chosen; it gives one family and not three; and by Corollary 4.4 exhibiting a known representation content is a consistency result, not evidence. One further fact will matter in Section 21: γ^0 commutes with all 91 internal bivectors and with J and the rotations, and anticommutes with Γ and the boosts. It maps V_* onto an orthogonal rank-64 subspace on which the same internal algebra acts with Γ reversed. The chirality of the block is put in by the choice of P_*.

9 Charges and anomalies

The weak and colour algebras of a block are fixed by the projector; the abelian charges are not. This section asks how much the algebra says about them, first inside the arena and then in a layer that uses nothing from the arena at all.

9.1 A pairing obstruction

The family block of P_{fam} carries three abelian charges. The centre of $\mathfrak{k}_{\text{eff}}(P_{\text{fam}})$ is spanned by the block images of $X_a \in \{\frac{1}{2}J^{(1)}, \frac{1}{2}J^{(3)}, \frac{1}{2}e_5^{(3)}e_6^{(3)}\}$, and their charges $q_a := -J(2X_a)P_{\text{fam}}$ are commuting symmetric involutions of the block, with eigenvalues ± 1 and $\operatorname{Tr} q_a = 0$. The cubic anomaly of $u = aq_1 + bq_2 + cq_3$ is the chiral trace

$$\mathcal{A}(u^3) = \tfrac{1}{4}\big[\operatorname{Tr}(u^3 P_L) - \operatorname{Tr}(u^3 P_R)\big], \qquad P_{L,R} = P_{\text{fam}}\tfrac{1}{2}(\mathbf{1} \mp \Gamma),$$

per Weyl component, right-handed components entering with conjugated charge when every field is written left-handed. Since $q_a^2 = P_{\text{fam}}$ and the linear chiral traces vanish, only the mixed term survives:

$$\mathcal{A}(u^3) = 6abc\,\mathcal{A}(q_1q_2q_3) = 96\,abc,$$

while the plain trace $\operatorname{Tr}(u^3)$ vanishes for every u and carries no information about anomalies. The anomaly-free locus is the union of the three coordinate planes $abc = 0$, and on it there is nothing to select.

Theorem 9.1 (Conjugate pairing)**.** *Let a module carry three commuting involutions* q_a *with eigenvalues* ± 1 *and* $\operatorname{Tr} q_a = 0$*. For every* $u = aq_1 + bq_2 + cq_3$ *with* $abc = 0$ *the weight multiset of* u *is symmetric,* $\operatorname{mult}(w) = \operatorname{mult}(-w)$*.*

Proof. Take $c = 0$; only the joint multiplicities $M_{s_1 s_2}$ of (q_1, q_2) matter. The trace conditions give $M_{++} + M_{+-} - M_{-+} - M_{--} = 0$ and $M_{++} - M_{+-} + M_{-+} - M_{--} = 0$; adding and subtracting, $M_{++} = M_{--}$ and $M_{+-} = M_{-+}$. The weight of (s_1, s_2) is $as_1 + bs_2$, and $(s_1, s_2) \mapsto (-s_1, -s_2)$ negates it while preserving multiplicities. □

The proof uses only the three linear traces, with no reference to rank, rung or realisation, so within its hypotheses no enlargement of the block and no ascent of the tower repairs it. Charge pairing is all that is concluded: calling a full matter representation vector-like would also require the chirality convention and the other gauge factors. And the hypotheses are not decorative.

Example 9.2 (Vanishing traces do not imply pairing)**.** The multiset $\{(-3)^{\times 5}, 1^{\times 10}, 5^{\times 1}\}$ has $\sum q = -15+10+5 = 0$ and $\sum q^3 = -135+10+125 = 0$, yet it is not symmetric under $q \mapsto -q$. It is realised by three commuting involutions with joint signs and multiplicities $(+,+,+)\colon 1$, $(-,-,+)\colon 10$, $(+,-,-)\colon 5$: these have traces $(\operatorname{Tr} q_1, \operatorname{Tr} q_2, \operatorname{Tr} q_3) = (-4,-14,6)$, so the hypotheses of Theorem 9.1 fail, while $Y = q_1 + q_2 + 3q_3$ has spectrum $\{5^{\times 1}, 1^{\times 10}, (-3)^{\times 5}\}$ with $\operatorname{Tr} Y = \operatorname{Tr} Y^3 = 0$. Leaving the hypotheses therefore need not violate the gravitational condition, which constrains $\operatorname{Tr} Y$ for the gauged combination and not $\operatorname{Tr} q_a$ for each basis charge [2, 15].

9.2 Dressing, and a benchmark

The escape from Theorem 9.1 is to leave its hypotheses. On the family block let

$$t^1 = -J \cdot \tfrac{1}{2} e_5^{(2)} e_6^{(2)}, \qquad t^2 = -J \cdot \tfrac{1}{2} e_6^{(2)} e_4^{(2)}, \qquad t^3 = -J \cdot \tfrac{1}{2} e_4^{(2)} e_5^{(2)}$$

be the weak charges, whose skew counterparts, the rotations of the spatial directions of constituent two, span the derived $\mathfrak{su}(2)$ of $\mathfrak{k}_{\mathrm{eff}}(P_{\mathrm{fam}})$; let $\varepsilon = \pm 1$ be an orientation sign, and put $t^a_{L,R} = P_{L,R} t^a$,

$$B_0 = \frac{\varepsilon}{6} \Gamma(q_1 + q_2 + q_3), \qquad Y = B_0 + t^3_R, \qquad Q_{\mathrm{em}} = t^3_L + Y.$$

The cell sums $\varepsilon \sum_a q_a$ are $\{-1,-1,-1,3\}$ on the $\Gamma = -1$ half and $\{-3,1,1,1\}$ on the other, which gives $B_0|_{\Gamma=-1} = \{\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{2}\}$ and, after adding $t^3_R = \pm\frac{1}{2}$ on the right-handed cells, $Y \in \{0, -1, \frac{2}{3}, -\frac{1}{3}\}$ there. Over the sixteen left-handed Weyl components $\operatorname{Tr} Y = \operatorname{Tr} Y^3 = \operatorname{Tr} T_3^2 Y = 0$ with four doublets, so all four anomaly conditions hold, and $\operatorname{Tr} T_3^2 / \operatorname{Tr} Q^2 = 3/8$. This hypercharge is adopted, not derived.

Proposition 9.3 (The escape)**.** *Theorem 9.1 constrains undressed elements of $\mathfrak{u}(1)^3$, which lie in the vector class. The charge B_0 carries a factor Γ and is axial, not an element of $\mathfrak{u}(1)^3$, so the cubic-trace formula does not apply to it. Once chirality-dressed endomorphisms are admitted, which abelian charges can be gauged changes, and no-go results proved in the undressed grade-two setting do not carry across; a well-posed search must say which Clifford grades are permitted alongside $Y^\dagger = Y$, $[Y, t_{La}] = 0$ and $[Y, C_A] = 0$.*

Since $P_L + P_R = \mathbf{1}$, $Q_{\mathrm{em}} = t^3 + B_0$: the chirality dressing cancels identically in the electric charge when the same axis is used in both terms and is visible only in the split into T_3 and Y. The value $3/8$ is arithmetic on a charge table; it becomes $\sin^2\theta_W$ only after a common normalisation of the gauge kinetic terms and a matching scale, neither of which is supplied. Five choices enter the benchmark: the projector, the dressing, which abelian direction is Y, which right-handed axis, and the overall sign.

9.3 A model-independent layer

The following uses nothing from the arena. It calibrates how much of the hypercharge problem is generic.

Theorem 9.4 (The anomaly variety of one family)**.** *For one family with N_c colours and a right-handed neutrino, with left-handed hypercharges q, l and right-handed y_u, y_d, y_ν, y_e, the three linear conditions give $l = -N_c q$, $y_d = 2q - y_u$, $y_e = 2l - y_\nu$. With $h_q = y_u - q$ and $h_\ell = y_\nu - l$ the cubic condition factorises,*

$$\mathcal{A}_{Y^3} = 6 N_c q \, (h_\ell^2 - h_q^2),$$

using $2q^3-y_u^3-(2q-y_u)^3=-6qh_q^2$. *The solution set is the union of three planes:* Π_A *(*$h_\ell=h_q$*),* Π_B *(*$h_\ell=-h_q$*) and* Π_C *(*$q=l=0$*). On* Π_A *the four Yukawa relations* $y_u=q+h$*,* $y_d=q-h$*,* $y_\nu=l+h$*,* $y_e=l-h$ *hold, so they are consequences of anomaly cancellation rather than assumptions;* Π_B *is the image of* Π_A *under the exchange* $\nu_R\leftrightarrow e_R$*. At* $N_c=3$ *and* $h=\frac{1}{2}$*,* Π_A *is the line* $Y=Y_{\mathrm{SM}}+\lambda(B-L)$*.*

Anomaly cancellation does not supply the scalar, its hypercharge, or nonvanishing couplings. Quark Yukawa couplings alone do not exclude Π_C either: at $N_c=3$ the point $(q,l,y_u,y_d,y_\nu,y_e)=(0,0,\frac{1}{2},-\frac{1}{2},1,-1)$ with scalar hypercharge $\frac{1}{2}$ cancels all four anomalies and permits both quark Yukawas, yet $h_q=\frac{1}{2}\neq h_\ell=1$. What excludes it is integrality: with $q=0$ the left-handed quarks carry $\pm\frac{1}{2}$ and a three-quark singlet has half-integral charge.

Theorem 9.5 (Without ν_R hypercharge is quantised)**.** *Deleting* ν_R *gives* $y_e=2l$ *and* $\mathcal{A}_{Y^3}=6N_cq(N_c^2q^2-h_q^2)$*, so on the generic branch* $h_q=\pm N_cq$ *is locked and hypercharge is fixed up to scale and relabelling: at* $N_c=3$*,* $q=\frac{1}{6}$ *gives the Standard-Model values with no residual* $B-L$ *direction.*

This is the classical charge-quantisation result [39, 45]. Its bearing here is that the difficulty of selecting hypercharge on a family block exists because the block carries sixteen Weyl components rather than fifteen: the sterile neutrino is both the block's virtue and the source of the ambiguity.

Proposition 9.6 (Integrality, and what it does not fix)**.** *On* Π_A *with* $Q_{\mathrm{em}}=T_L^3+Y$*, the leptons carry* $Q_\nu=-\lambda$ *and* $Q_e=-1-\lambda$*, so integrality of the charged lepton forces* $\lambda\in\mathbb{Z}$*, and colour triality then gives integral charge to every colour singlet. It does not fix the fractional part of the quark charges, since* $Q_u=\frac{2}{3}+\lambda/3$*, and neutrality does not close the argument for composites:* $Q(ddd)=\lambda-1$ *and* $Q(uuu)=\lambda+2$*, so neutral colour-singlet operators of odd fermion parity exist at* $\lambda=1$ *and* $\lambda=-2$*. A neutral* elementary *colour-singlet fermion exists if and only if* $\lambda\in\{0,-1\}$*, and those two values are exchanged by* $\nu\leftrightarrow e$*.*

Three other routes fail to single out the Standard Model. The kernel of the representation is not a selection principle: its order depends on λ through the integer normalisation of the weights. Nor is a $\mathbb{Z}_6$ kernel evidence for three colours: with a neutral ν_R the kernel is $\mathbb{Z}_{2N_c}$ for odd N_c and $\mathbb{Z}_{N_c}$ for even N_c. And integrality does not select three either: on the standard branch every N_c-quark colour singlet has integral charge exactly for odd N_c, and the number N_c+1 of weak doublets per family, which the global $SU(2)$ anomaly requires to be even [148], is even exactly for odd N_c. All three facts hold at every odd N_c. With N_g generations the standard branch has $2N_g$ parameters, and the anomaly-free directions include L_e-L_μ and $L_\mu-L_\tau$ [61], so family universality is an input that grows with N_g. The global structure of the gauge group is a separate datum [133, 44].

9.4 Charge at one constituent

Complexifying, the spinor module of one constituent is $\Lambda^\bullet\mathbb{C}^3$, and its even half is spanned by $|0\rangle,|12\rangle,|13\rangle,|23\rangle$. The diagonal Fock charge N_1-N_2 has spectrum $\{0,0,+1,-1\}$ there, the charge skeleton of $(\nu,\bar\nu,e^+,e^-)$.

Theorem 9.7 (No unique charge at one constituent)**.** *The condition* $\operatorname{spec}Q=\{0,0,+1,-1\}$ *on the even half, for* $Q=\sum_i q_iN_i+c$*, has exactly twelve solutions, forming two orbits of* $S_3\times\{\pm1\}$*:* $Q=N_i-N_j$ *with* $c=0$*, and* $Q=N_i+N_j-1$ *with* $c=-1$*. The antiparticle condition* $CQC^{-1}=-Q$*, with* C *the Hodge conjugation, holds on both orbits and does not separate them.*

Proof. Summing the four values gives $4c+2\sum_i q_i=0$. If the neutral pair is $\{|0\rangle,|ij\rangle\}$ then $c=0$ and $q_i+q_j=0$, which forces the third coefficient to vanish; if it is $\{|ij\rangle,|ik\rangle\}$ then $q_j=q_k$, $c=-q_j$ and $q_i=0$. □

The further step, $Q|0\rangle = 0$ and hence $c = 0$, is the separate assumption that the state identified with the neutrino is the Fock vacuum. Choosing between the orbits is an identification, not a derivation, and it has the same shape as the hypercharge problem two rungs up, differing in size and not in kind.

10 Other Lorentz actions, and the vector bosons of the tower

Everything so far takes the Lorentz action from one constituent and treats the others as passengers. Theorem 6.1 is a statement about that choice, which we call the *spectator* action. The choice is adopted, not forced, and the alternatives are different mathematics with different verdicts.

10.1 The diagonal action

At three constituents let $\gamma_\mu^{(k)}$ be the interior four-plane of constituent k, of metric η. We call the two remaining generators $s_1^{(k)}, s_2^{(k)}$ of that constituent its surplus pair; both square to $+\mathbf{1}$. Write $\mathcal{S}$ for the six surplus directions and put

$$\Sigma_{\mu\nu}^{(k)} = \tfrac{1}{2}\gamma_\mu^{(k)}\gamma_\nu^{(k)}, \qquad \Delta_{\mu\nu} = \sum_{k=1}^{3} \Sigma_{\mu\nu}^{(k)}.$$

Bivectors of distinct constituents commute, so Δ satisfies the $\mathfrak{so}(1,3)$ brackets, and as a quadratic space

$$\mathbb{R}^{9,9} = \left(\mathbb{R}^3 \otimes \mathbb{R}^{1,3}\right) \oplus \mathbb{R}^{6,0}, \tag{10}$$

the first factor carrying a copy index with δ_{kl} and a Lorentz index with $\eta_{\mu\nu}$. The actions Δ and $\Sigma^{(1)}$ are not conjugate: their centralisers have different dimensions and their Casimirs different spectra.

Definition 10.1. $M_{kl} = \frac{1}{2}\sum_\mu \eta^{\mu\mu}\gamma_\mu^{(k)}\gamma_\mu^{(l)}$ for $k < l$ span the copy algebra $\mathfrak{k}_{\mathrm{cop}}$, and $\mathfrak{k}_{\mathcal{S}}$ is the span of the fifteen surplus bivectors.

Theorem 10.2 (The diagonal centraliser is compact)**.** $\mathfrak{z}(\Delta) = \mathfrak{k}_{\mathcal{S}} \oplus \mathfrak{k}_{\mathrm{cop}} \cong \mathfrak{so}(6) \oplus \mathfrak{so}(3)$*, of dimension* 18*, and every direction of it is compact.*

Proof. Under Δ the vector representation is $3V \oplus 6\cdot\mathbf{1}$ by (10), with V irreducible and nontrivial. By Schur an equivariant endomorphism acts as $A \otimes \mathbf{1}_V$ on the first summand and arbitrarily on the second, with no mixing since $\mathrm{Hom}_{\mathfrak{so}(1,3)}(V, \mathbf{1}) = 0$; antisymmetry for $\delta \otimes \eta$ makes $A \in \mathfrak{so}(3)$, and the surplus block gives $\mathfrak{so}(6)$. Both are realised: the surplus bivectors commute with Δ by (3), and M_{kl} is the spin lift of the copy rotation $E_{kl} - E_{lk}$ tensored with the identity on $\mathbb{R}^{1,3}$. Every surplus direction squares to $+\mathbf{1}$, and each term of M_{kl} pairs $\gamma_\mu^{(k)}$ with $\gamma_\mu^{(l)}$, two generators of the same square, so every generator is compact by Theorem 2.3. □

For comparison, the centraliser of $\Sigma^{(1)}$ is $\mathfrak{so}(8,6)$, of dimension 91 with 43 compact and 48 boost directions; that of all three $\Sigma^{(k)}$ together is $\mathfrak{k}_{\mathcal{S}}$ alone, of dimension 15; and that of $\Sigma^{(1)}$ and Δ together is $\mathfrak{k}_{\mathcal{S}} \oplus \mathbb{R}M_{23}$, of dimension 16. The diagonal choice exchanges 91 internal directions, only 43 of them compact, for 18 of which all are compact. By Theorem 11.3 below the noncompact directions are exactly the ghost-signed ones, so the diagonal internal algebra is free of ghosts as a whole while the spectator one is not.

10.2 A reductive dual pair

Put $C_\Delta = -\sum_{a<b}\eta^{aa}\eta^{bb}\Delta_{ab}^2$, $C_{\mathrm{cop}} = -\sum_{k<l} M_{kl}^2$ and $C_{\mathcal{S}} = -\sum_{s<s'}(\frac{1}{2}ss')^2$. For a single constituent $-\sum_{a<b}\eta^{aa}\eta^{bb}(\Sigma_{ab}^{(k)})^2 = \frac{3}{2}$ identically, the Dirac value, and $C_{\mathcal{S}} = \frac{15}{4}$ identically, so the surplus $\mathfrak{so}(6)$ acts isotypically on S_3 and its Casimir separates nothing.

Theorem 10.3 (Spectrum of the diagonal Casimir)**.** $\operatorname{spec} C_\Delta = \{\frac{3}{2}\,(160),\ \frac{11}{2}\,(288),\ \frac{15}{2}\,(64)\}$, *each eigenprojector being Lorentz invariant. The* $C_\Delta = \frac{3}{2}$ *block, the* Dirac block $\mathcal{D}$*, has rank* 160*.*

Proof. A quadratic Casimir of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ is $\alpha j_L(j_L+1) + \beta j_R(j_R+1)$, and the Dirac value $\frac{3}{2}$ together with the vector value 3 gives $\alpha = \beta = 2$. Since the constituent Casimir is $\frac{3}{2}$ identically, each $\mathbb{R}^8$ is $M \oplus M$ with M the real Majorana module, and $S_3 \cong M^{\otimes 3} \otimes \mathbb{R}^8$. Writing $A = (\frac{1}{2}, 0)$ and $B = (0, \frac{1}{2})$,

$$AAA = (\tfrac{3}{2},0) \oplus 2(\tfrac{1}{2},0), \quad AAB = (1,\tfrac{1}{2}) \oplus (0,\tfrac{1}{2}),$$
$$ABB = (\tfrac{1}{2},1) \oplus (\tfrac{1}{2},0), \quad BBB = (0,\tfrac{3}{2}) \oplus 2(0,\tfrac{1}{2}),$$

with three words of each mixed type. This gives multiplicities $1, 3, 5$ for $(\frac{3}{2},0) \oplus (0,\frac{3}{2})$, $(1,\frac{1}{2}) \oplus (\frac{1}{2},1)$ and $(\frac{1}{2},0) \oplus (0,\frac{1}{2})$, of complex dimensions $8, 12, 4$; the totals $8, 36, 20$ add up to 64, and the multiplicity factor 8 turns them into $64, 288, 160$. □

Theorem 10.4 (A Casimir identity)**.** *As operators on* S_3*,*

$$C_{\mathrm{cop}} + C_\Delta = \tfrac{15}{2}\,\mathbf{1}, \tag{11}$$

and for n *constituents with a* d*-dimensional interior,* $C_{\mathrm{cop}} + C_\Delta = \big[nd(d-1)/8 + \binom{n}{2}d/4\big]\mathbf{1}$*; for* $d = 4$ *this is* $\frac{1}{2}n(n+2)\,\mathbf{1}$*.*

Proof. Write $B^{kl}_{ab} = \gamma^{(k)}_a \gamma^{(k)}_b \gamma^{(l)}_a \gamma^{(l)}_b$, symmetric in $k \leftrightarrow l$ and in $a \leftrightarrow b$. Expanding,

$$\Delta^2_{ab} = \tfrac{1}{4}\Big[-n\,\eta_{aa}\eta_{bb} + 2\sum_{k<l} B^{kl}_{ab}\Big], \qquad M^2_{kl} = -\tfrac{d}{4} - \tfrac{1}{2}\sum_{\mu<\nu} \eta^{\mu\mu}\eta^{\nu\nu} B^{kl}_{\mu\nu},$$

the first from $\gamma_a\gamma_b\gamma_a\gamma_b = -\eta_{aa}\eta_{bb}$, the second from $\gamma^{(k)}_\mu \gamma^{(l)}_\mu \gamma^{(k)}_\nu \gamma^{(l)}_\nu = -B^{kl}_{\mu\nu}$ for $\mu \neq \nu$. Hence $C_\Delta = nd(d-1)/8 - \frac{1}{2}\mathsf{S}$ and $C_{\mathrm{cop}} = \binom{n}{2}d/4 + \frac{1}{2}\mathsf{S}$ with $\mathsf{S} = \sum_{k<l}\sum_{a<b} \eta^{aa}\eta^{bb} B^{kl}_{ab}$, the only term in which two constituents interact. It cancels in the sum. □

At $n = 3$, $d = 4$ the constant is $3 \cdot 4 \cdot 3/8 + 3 \cdot 4/4 = \frac{9}{2} + 3 = \frac{15}{2}$. The proof is half a page and the identity is exact; it is the cleanest result of the diagonal sector.

Corollary 10.5 (A dual pair)**.** $\mathfrak{so}(3)_{\mathrm{cop}}$ *and* $\mathfrak{so}(1,3)_\Delta$ *are mutual centralisers inside* $\mathfrak{so}(3,9)$*, and the joint spectrum of* $(C_{\mathrm{cop}}, C_\Delta)$ *has exactly three points:*

j_{cop}	dim	C_{cop}	C_Δ	Lorentz content	rank in S_3
0	1	0	$\frac{15}{2}$	$(\frac{3}{2},0) \oplus (0,\frac{3}{2})$	$8 \cdot 1 \cdot 8 = 64$
1	3	2	$\frac{11}{2}$	$(1,\frac{1}{2}) \oplus (\frac{1}{2},1)$	$12 \cdot 3 \cdot 8 = 288$
2	5	6	$\frac{3}{2}$	$(\frac{1}{2},0) \oplus (0,\frac{1}{2})$	$4 \cdot 5 \cdot 8 = 160$

The multiplicities $1, 3, 5$ *of Theorem 10.3 are the copy representations, so* $\mathcal{D} \cong L \otimes U$ *with* $L = (\frac{1}{2},0) \oplus (0,\frac{1}{2})$ *and* $U = [j_{\mathrm{cop}} = 2] \otimes \mathbb{R}^8$ *carrying the whole gauge action.*

A projector selected by a rotation condition alone need not be Lorentz invariant. The rotation Casimir $-\sum_{i<j} \Delta^2_{ij}$ has spectrum $\{\frac{3}{4}\,(256), \frac{15}{4}\,(256)\}$, and its rank-256 spin-$\frac{1}{2}$ projector commutes with the rotations and with no boost: inside the irreducible $(1,\frac{1}{2})$ a boost carries the $j = \frac{1}{2}$ piece into the $j = \frac{3}{2}$ piece. The Lorentz-invariant answer is rank 160.

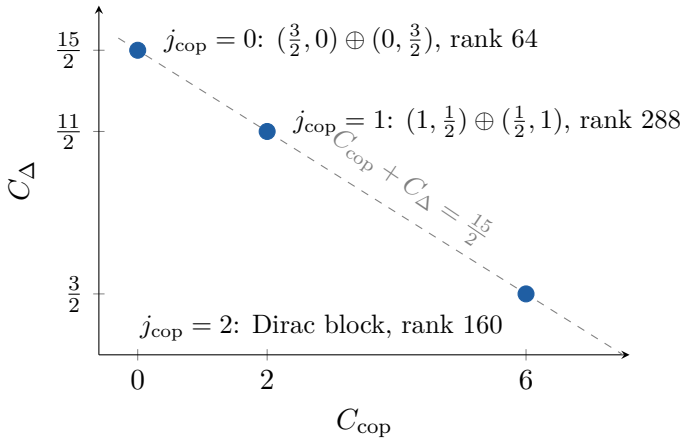


Figure 8: The diagonal sector at three constituents (Theorem 10.4 and Corollary 10.5). The copy Casimir and the Lorentz Casimir add up to a constant on the whole module, so copy spin is locked to Lorentz type: the Dirac block is exactly the copy quintet.

10.3 Gamma matrices on the Dirac block

The averaged gammas $V_\mu = 3^{-1/2}\sum_k \gamma_\mu^{(k)}$ satisfy $\{V_\mu, V_\nu\} = 2\eta_{\mu\nu}$ and transform as a Lorentz vector, but they are a copy vector, not a copy singlet: they map $\mathcal{D}$ into itself and into the rank-288 block with equal norms, and their compression to $\mathcal{D}$ degenerates on a rank-32 locus. The block does carry gamma matrices; they come from a copy-singlet cubic.

Theorem 10.6 (A gauge-invariant Cl(1, 3) on the Dirac block)**.** *Set* $\Psi_{abc} = \varepsilon_{klm}\gamma_a^{(k)}\gamma_b^{(l)}\gamma_c^{(m)}$,

$$Q_\mu = \tfrac{1}{2}\eta^{\nu\rho}\Psi_{\mu\nu\rho} = \sum_\nu \eta_{\nu\nu}(\gamma_\mu^{(1)}\gamma_\nu^{(2)}\gamma_\nu^{(3)} + \gamma_\nu^{(1)}\gamma_\mu^{(2)}\gamma_\nu^{(3)} + \gamma_\nu^{(1)}\gamma_\nu^{(2)}\gamma_\mu^{(3)}), \qquad G_\mu = \tfrac{1}{6}\,\omega_{\mathcal{S}} Q_\mu|_{\mathcal{D}},$$

with $\omega_{\mathcal{S}}$ *the product of the six surplus generators. Then* Ψ *is a copy singlet and totally symmetric in* (a, b, c)*; each* Q_μ *is a sum of ten grade-three blades, commutes with all eighteen gauge generators and with* C_Δ*, is a Lorentz vector, and satisfies* $\{Q_\mu, Q_\nu\}|_{\mathcal{D}} = -72\,\eta_{\mu\nu}$*. Consequently, on all* 160 *dimensions,*

$$\{G_\mu, G_\nu\} = 2\eta_{\mu\nu}\mathbf{1}_{\mathcal{D}}, \qquad [\Delta_{ab}, G_c] = \eta_{bc}G_a - \eta_{ac}G_b, \qquad [G_\mu, \mathfrak{z}(\Delta)] = 0, \qquad \tfrac{1}{4}[G_a, G_b] = \Delta_{ab}|_{\mathcal{D}}.$$

Proof. Reordering $\gamma^{(\sigma 1)}\gamma^{(\sigma 2)}\gamma^{(\sigma 3)}$ into copy order costs $\operatorname{sign}\sigma$, because generators of distinct constituents anticommute; against the factor ε this gives $(\operatorname{sign}\sigma)^2 = 1$, so antisymmetry in the copies is symmetry in the Lorentz labels. Block preservation follows because a copy singlet commutes with $C_{\rm cop}$ and hence, by (11), with C_Δ. Finally $\omega_{\mathcal{S}}$ commutes with Q_μ and $\omega_{\mathcal{S}}^2 = -\mathbf{1}$, so $\{G_\mu, G_\nu\} = \frac{1}{36}(-\mathbf{1})(-72\eta_{\mu\nu})$. □

The sign flip is not cosmetic. $\mathrm{Cl}(1,3) \cong M_2(\mathbb{H})$ has minimal real module of dimension eight, so the four-dimensional Majorana module carries no real Cl(1, 3); the Q_μ alone generate Cl(3, 1), and $\omega_{\mathcal{S}}$, with square $-\mathbf{1}$, converts them. The operator $\not{D} = G^\mu\nabla_\mu$ is Lorentz covariant and gauge invariant on $\mathcal{D}$; $G_5 = G_0G_1G_2G_3$ has $G_5^2 = -\mathbf{1}$, a complex structure rather than a chirality, as $M_2(\mathbb{H})$ requires.

10.4 The diagonal sector is vector-like

Theorem 10.7 (Vector-likeness)**.** *Let* $W_\mu = \gamma_\mu^{(1)}\gamma_\mu^{(2)}\gamma_\mu^{(3)}$*. Then* $W_\mu^2 = -\eta_{\mu\mu}$*, so* W_μ *is invertible;* $[W_\mu, \mathfrak{z}(\Delta)] = 0$*;* W_μ *anticommutes with* Δ_{ab} *when* $\mu \in \{a, b\}$ *and commutes otherwise, hence commutes with* C_Δ *and preserves every block; and* W_μ *anticommutes with the gauge-invariant chirality* $\omega = E_1 \cdots E_{18}$*. Therefore* W_0 *is an invertible* $\mathfrak{z}(\Delta)$*-equivariant map exchanging the two chiral halves of every* C_Δ *block, and all of* S_3 *is vector-like for* $\mathfrak{z}(\Delta)$*.*

Proof. Each property is (3) with $|I| = 3$: against a surplus bivector the sign is $(-1)^6$; against M_{kl}, the term with $\nu = \mu$ meets W_μ in two indices and the others in none; against Δ_{ab} it is $(-1)^{6-1}$ when $\mu \in \{a, b\}$; against ω it is $(-1)^{54-3}$. The intertwining conclusion is Schur's lemma. □

By Corollary 10.5 the reason is structural: $\mathcal{D} \cong L \otimes U$ with the gauge algebra acting on U only, so chirality and gauge charge are carried by different tensor factors. Vector-like means that the gauge representation permits a mass for every state, not that any mass is large; nothing here fixes a scale. A Weyl projection of the block is possible, and it keeps the kinetic operators and the gauge algebra, but it is then a choice of field content, and at three constituents it carries a colour anomaly (Proposition 10.10).

10.5 The candidate gauge algebra of every Lorentz action

A Lorentz action on the tower can be taken from each constituent in one of three ways: by the interior algebra $\mathfrak{l}_{\text{int}}$, by the algebra $\mathfrak{l}_G$ that centralises the binding operator (Section 12), or not at all.

Theorem 10.8 (Candidate algebras)**.** *Let the Lorentz algebra act on rung n by $\mathfrak{l}_{\text{int}}$ on k_I constituents, by $\mathfrak{l}_G$ on k_B constituents and trivially on k_S constituents, $k_I + k_B \geq 1$. The grade-two elements of* $\mathrm{Cl}(3n, 3n)$ *commuting with the action form*

$$\mathfrak{g}(k_I, k_B, k_S) \cong \mathfrak{so}(k_I) \oplus \mathfrak{so}(2k_I + 3k_S,\, 3k_S) \oplus \mathfrak{so}(k_B, \mathbb{C}),$$

a direct sum of ideals. Its maximal compact subalgebra is $\mathfrak{so}(k_I)\oplus\mathfrak{so}(2k_I+3k_S)\oplus\mathfrak{so}(3k_S)\oplus\mathfrak{so}(k_B)$, and the algebra is compact exactly when $k_S = 0$ and $k_B \leq 1$.

Proof. Grade two is $\mathfrak{so}(V_n)$, and $V_n = U \oplus W \oplus F$ with $U = \mathbb{R}^{k_I} \otimes \mathbb{R}^{1,3}$ the interior four-planes, W the Lorentz-trivial directions (the surplus pairs of the interior constituents and all directions of the spectators, of signature $(2k_I + 3k_S, 3k_S)$), and $F = \mathbb{R}^{k_B} \otimes V_f$ with $V_f = (1, 0) \oplus (0, 1)$ the binding frames. The modules $(\frac{1}{2}, \frac{1}{2})$, $\mathbf{1}$ and V_f are pairwise inequivalent and irreducible, so the mixed parts of $\Lambda^2 V_n$ contain no invariant. $\Lambda^2 W$ is invariant and gives $\mathfrak{so}(W)$. In $\Lambda^2 U = (\Lambda^2\mathbb{R}^{k_I} \otimes S^2\mathbb{R}^4) \oplus (S^2\mathbb{R}^{k_I} \otimes \Lambda^2\mathbb{R}^4)$ the invariants of $S^2\mathbb{R}^4$ are the multiples of η and those of $\Lambda^2\mathbb{R}^4$ vanish, leaving $\mathfrak{so}(k_I)$. On V_f the invariant symmetric forms are spanned by η and $\eta(\cdot, G\cdot)$ and there is no invariant antisymmetric form, leaving $\Lambda^2\mathbb{R}^{k_B} \otimes \mathrm{span}\{\mathbf{1}, G\} \cong \mathfrak{so}(k_B, \mathbb{C})$. □

Proposition 10.9 (Orthogonal against spin actions)**.** *A copy rotation of interior four-planes is a sum of four commuting terms of square $-\frac{1}{4}$, and one of binding frames a sum of six; their eigenvalues on S_n are i times integers, and a rotation through 2π is the identity. A rotation $\frac{1}{2}e_s e_t$ of two Lorentz-trivial directions of equal sign squares to $-\frac{1}{4}$ and its rotation through 2π is $-\mathbf{1}$. The copy algebras act through orthogonal groups, the algebra of the Lorentz-trivial directions through spin groups. Consequently the spin content of S_n is half-integer if k_I is odd and integer if k_I is even: fermions require an odd number of interior constituents.*

10.6 The diagonal tower

Under the diagonal interior action the $4n$ four-plane generators and the $2n$ surplus generators $e_2^{(k)}, e_3^{(k)}$ anticommute, so $S_n \otimes \mathbb{C} \cong S_{4n} \otimes S_{2n}$, with the Lorentz and copy algebras acting on the first factor and the surplus $\mathfrak{so}(2n)$ on the second. The connection algebra is the compact $\mathfrak{so}(n) \oplus \mathfrak{so}(2n)$, of dimension $n(5n-3)/2$.

Proposition 10.10 (The diagonal tower)**.** *(i) For $n \leq 7$ the factor S_{4n} decomposes under $\mathfrak{l}_{\text{int}} \oplus \mathfrak{so}(n)$ without multiplicity. Spin is half-integer exactly at odd n, and the Dirac components occur only at odd n, each once, tensored with the copy representation of highest weight* $(2, \ldots, 2)$*, of dimension* $1, 5, 35, 294$ *at $n = 1, 3, 5, 7$.*

(ii) At odd n the product $\mathcal{J}$ of the $2n$ surplus generators is a complex structure commuting with the whole connection algebra, and $\Gamma = \mathcal{J}\Omega$ is a chirality. With $\mathcal{J}$ as i the Dirac block is $[(\frac{1}{2},0)\oplus(0,\frac{1}{2})]\otimes V_{(2,\dots,2)}\otimes S^+$, with S^+ a half-spinor of the surplus $\mathfrak{so}(2n)$, and the charge $q=-\mathcal{J}\tilde{J}/2$ of $\tilde{J}=\sum_k J^{(k)}$ takes on S^+ the values

n	$\dim S^+$	values of q (multiplicity)
1	1	$\frac{1}{2}$
3	4	$-\frac{3}{2}$ (1), $\frac{1}{2}$ (3)
5	16	$-\frac{3}{2}$ (5), $\frac{1}{2}$ (10), $\frac{5}{2}$ (1)
7	64	$-\frac{7}{2}$ (1), $-\frac{3}{2}$ (21), $\frac{1}{2}$ (35), $\frac{5}{2}$ (7)

(iii) At $n=3$ the centraliser of $\tilde{J}$ in the surplus $\mathfrak{so}(6)$ is $\mathfrak{u}(3)$, whose derived algebra is a compact $\mathfrak{su}(3)$; on a Weyl half it acts as a triplet with $q=\frac{1}{2}$ and a singlet with $q=-\frac{3}{2}$, so that with $B-L=\frac{2}{3}q$ the triplet has $B-L=\frac{1}{3}$ and the singlet -1: lepton number as the fourth colour [102]. The eight gluons are neutral, the six remaining generators of $\mathfrak{so}(6)$ carry $q=\pm 2$ and turn the triplet into the singlet, as leptoquark bosons do. The copy $\mathfrak{so}(3)$ acts on the Dirac block with Casimir 6, that is isospin two: it is not weak isospin.

(iv) At $n=5$, with colour on three surplus planes and weak isospin on two, S^+ is one Standard-Model family with a right-handed neutrino, $(\mathbf{3},\mathbf{2})_{1/6}\oplus(\bar{\mathbf{3}},\mathbf{1})_{-2/3}\oplus(\bar{\mathbf{3}},\mathbf{1})_{1/3}\oplus(\mathbf{1},\mathbf{2})_{-1/2}\oplus(\mathbf{1},\mathbf{1})_{1}\oplus(\mathbf{1},\mathbf{1})_{0}$, the spinor $\mathbf{16}$ of $\mathfrak{so}(10)$ [41, 126], occurring in 35 copies. At $n=7$, S^+ under $\mathfrak{so}(10)\oplus\mathfrak{so}(4)$ is two families and two mirror families, the pattern found for the spinor of $\mathfrak{so}(18)$ [145].

(v) The Dirac block is vector-like at every odd n. Its Weyl projection admits no gauge-invariant mass, since no weight of S^+ has its negative in S^+; at $n=3$ it carries a colour anomaly, $\mathrm{Tr}(P_L\mathcal{J}H^3)=960$ for the colour Cartan element $H=J^{(1)}+J^{(2)}-2J^{(3)}$, and at $n=5$ every Standard-Model anomaly sum vanishes and $\mathfrak{so}(10)$ has no cubic anomaly [46].

Lemma 10.11 (No grade-two generator is chiral by itself)**.** *Let Γ be, up to sign, a product of $k\geq 5$ distinct generators with $\Gamma^2=\mathbf{1}$, and $P_{L,R}=\frac{1}{2}(\mathbf{1}\mp\Gamma)$. A grade-two element X with $XP_R=0$ is zero.*

Proof. $XP_R=0$ means $X\Gamma=-X$. For $a\neq b$, $e_ae_b\Gamma$ is a blade of grade $k-2$, k or $k+2$, so $X\Gamma$ has no grade-two component, and $X\Gamma=-X$ forces $X=0$. □

Under the diagonal and the spectator actions the chirality is, up to sign, the volume element of $\mathrm{Cl}(3n,3n)$, so a gauge interaction built from connection components can be chiral only through the representation the Weyl block carries.

10.7 The compact part of the spectator algebra

Under the spectator action the connection components form $\mathfrak{so}(3n-1,3n-3)$, noncompact for $n\geq 2$, and the gauge algebra is its compact part $\mathfrak{so}(3n-1)\oplus\mathfrak{so}(3n-3)$, acting through spinors. Let w be the product of the $6n-4$ Lorentz-trivial generators; it commutes with the whole connection algebra and serves as its complex structure, and $\Gamma=w\,\Omega^{(1)}$ is the chirality.

Proposition 10.12 (What the spectator gauge algebra acts on)**.** *At $n=2$ the Weyl block is $(\frac{1}{2},0)\otimes(\mathbf{4},\mathbf{2})$ under $\mathfrak{sp}(2)\oplus\mathfrak{su}(2)\cong\mathfrak{so}(5)\oplus\mathfrak{so}(3)$: isospin doublets without colour, in a real representation, so the weak interaction it carries is vector-like. At $n=3$ the Weyl block is*

$$(\tfrac{1}{2},0)\otimes[(\mathbf{8}_s,\mathbf{4})\oplus(\mathbf{8}_c,\bar{\mathbf{4}})]\quad\text{under }\mathfrak{so}(8)\oplus\mathfrak{so}(6).$$

With colour on three negative planes, weak isospin on two positive planes and a family $\mathfrak{so}(4)$ *on the remaining two, its* 64 *weights are two Standard-Model families with* ν^c *and two mirror families. The cubic* $\mathfrak{su}(4)$ *sum and the Standard-Model anomaly sums vanish, there are sixteen weak doublets, the weak roots carry* $Q = \pm 1$*, the gluons are neutral and the leptoquarks carry* $\pm\frac{2}{3}$*. The block is chiral and anomaly free under* $\mathfrak{so}(8)\oplus\mathfrak{so}(6)$ *and vector-like under the Standard-Model algebra.*

At $n = 3$ the compact spectator algebra therefore already contains the Pati–Salam algebra $\mathfrak{so}(4) \oplus \mathfrak{so}(6) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R \oplus \mathfrak{su}(4)$. The projector P_* of Section 8 is a way of cutting one chiral family out of this content.

10.8 Vector bosons rung by rung

n	diagonal connection algebra	bosons	fermions	action on the Dirac block
1	$\mathfrak{so}(2) = \mathfrak{u}(1)_J$	1	yes	one charge, also the complex structure
2	$\mathfrak{so}(2) \oplus \mathfrak{so}(4)$	7	no	integer spin only
3	$\mathfrak{so}(3) \oplus \mathfrak{so}(6) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(4)$	18	yes	$5 \times (\mathbf{3} \oplus \mathbf{1})$; the $\mathfrak{su}(2)$ is the copy algebra
4	$\mathfrak{so}(4) \oplus \mathfrak{so}(8)$	34	no	—
5	$\mathfrak{so}(5) \oplus \mathfrak{so}(10)$	55	yes	$35 \times \mathbf{16}$, one family each
7	$\mathfrak{so}(7) \oplus \mathfrak{so}(14)$	112	yes	$294 \times (2 \cdot \mathbf{16} \oplus 2 \cdot \overline{\mathbf{16}})$
n	spectator gauge algebra	bosons		action on the Weyl block
1	$\mathfrak{so}(2)$	1		one charge
2	$\mathfrak{so}(5) \oplus \mathfrak{so}(3)$	13		$(\mathbf{4}, \mathbf{2})$: doublets, no colour, real
3	$\mathfrak{so}(8) \oplus \mathfrak{so}(6)$	43		two families and two mirror families

The tempting assignment "$n = 1$: photon; $n = 2$: W and Z; $n = 3$: gluons" can now be tested. The photon at $n = 1$ is supported: the single connection component J gauges the Dirac spinor and is also its complex structure. Chiral W and Z at $n = 2$ are supported by no action: the diagonal action has no fermions there, the action that is interior on one constituent and binding on the other has the abelian algebra $\mathfrak{u}(1)$ only, and the spectator Weyl block is a real representation. Gluons at $n = 3$ are supported, without weak bosons under the diagonal action, where the $\mathfrak{su}(2)$ is the copy algebra, and with them under the spectator action. Rung three is the first at which some gauge algebra contains $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ acting on fermions in Standard-Model representations, there together with mirror families; among the actions with a compact connection algebra the first such rung is $n = 5$, through $\mathfrak{so}(10)$, with one family per copy and thirty-five copies. Nothing computed here reduces the copy multiplicity to three, and the rung index counts tensor factors, so none of this identifies particles with constituents.

Part III
Gauge fields and the electromagnetic reading

11 The projector as a field

Up to this point a projector has been a choice made once and for all. A choice of this kind becomes a field as soon as it is allowed to vary from point to point, and the selection problem then changes character: instead of asking which projector is right we ask which configurations an action prefers. Before any action can be written, three things must be known: the space in which the projector moves, the connection it induces on its own image, and the sign of the

natural metric on its variations. All three can be computed exactly, and each of them carries a surprise.

11.1 Configuration space

Let $\mathcal{P}_r$ be the set of orthogonal projectors P on $S_3 = \mathbb{R}^{512}$ of rank r that commute with the interior Lorentz action Σ. Differentiating $P^2 = P$ gives $dP\,P + P\,dP = dP$, hence

$$P\,dP\,P = 0, \qquad (\mathbf{1} - P)\,dP\,(\mathbf{1} - P) = 0 :$$

a variation of a projector is purely off-diagonal with respect to the splitting $\operatorname{im} P \oplus \ker P$. The tangent space is therefore

$$T_P\mathcal{P}_r = \{S :\ S^\top = S,\ PSP = 0,\ (\mathbf{1} - P)S(\mathbf{1} - P) = 0,\ [S, \Sigma] = 0\},$$

the tangent space of the real Grassmannian cut down by the Lorentz condition.

Proposition 11.1 (Configuration space)**.** *The Lorentz commutant $\mathcal{C} \subset \mathrm{End}_{\mathbb{R}}(S_3)$ is spanned by the* 32768 *blades that commute with the six interior bivectors. Its centre is $\{\mathbf{1}, \Omega\} \cong \mathbb{C}$ with $\Omega = \gamma^0\gamma^1\gamma^2\gamma^3$, $\Omega^2 = -\mathbf{1}$, and since a semisimple real algebra of dimension $2 \cdot 128^2$ with centre $\mathbb{C}$ is $M_{128}(\mathbb{C})$, we have $\mathcal{C} \cong M_{128}(\mathbb{C})$. Consequently*

$$\mathcal{P}_r \cong \mathrm{Gr}_{\mathbb{C}}(\tfrac{r}{4}, 128), \qquad \dim_{\mathbb{R}} \mathcal{P}_r = 2\,\frac{r}{4}\Big(128 - \frac{r}{4}\Big),$$

which is 3584 *at $r = 64$ and* 6144 *at $r = 128$, against* $\dim \mathrm{Gr}_{\mathbb{R}}(64, 512) = 28672$ *for the unconstrained real Grassmannian.*

The complex structure that appears here is the interior volume Ω, not the internal one ω_I of Theorem 4.1. That is not a contradiction. The commutant of the Lorentz action alone contains $e_2^{(1)}$, which commutes with the six interior bivectors but anticommutes with the fourteen-index volume ω_I, so ω_I is not central in $\mathcal{C}$; it becomes central only in the smaller algebra that also commutes with the internal bivectors, which is the algebra of Theorem 4.1. Each structure is central in its own ambient algebra, which is the pattern described after Proposition 2.7.

Remark 11.2 (No blade is tangent)**.** A count of tangent directions by blades fails, for a reason worth recording. Let $P = \prod_i \frac{1}{2}(\mathbf{1} + X_i)$ be one of the projectors of Section 7, built from k commuting involutions, and let B be a blade anticommuting with at least one X_i. Then $PBP = 0$, but $(\mathbf{1}-P)B(\mathbf{1}-P) = (\mathbf{1}-P-P_\varepsilon)B$ with $P_\varepsilon = \prod_i \frac{1}{2}(\mathbf{1}+\varepsilon_i X_i)$, which vanishes only when $P + P_\varepsilon = \mathbf{1}$, that is only for $k = 1$. There is also a rank argument: a blade is a signed permutation matrix, invertible of rank 512, while a tangent vector has the block form $\left(\begin{smallmatrix} 0 & A^\top \\ A & 0 \end{smallmatrix}\right)$ and rank at most $2r$. For $k \geq 2$ no single blade is tangent, the tangent space is not spanned by blades, and a blade count overestimates it, by a factor four at $r = 64$ and two at $r = 128$. The dimensions of Proposition 11.1 are obtained by imposing both conditions sector by sector.

11.2 The sign of a variation

The group that the geometry supplies is generated by the internal bivectors, and it moves a projector along an orbit inside $\mathcal{P}_r$. Whether a bivector moves P to another orthogonal projector, and with what sign the natural trace form measures the motion, is decided by the signature rule of Theorem 2.3.

Theorem 11.3 (The trace form splits by the Cartan involution)**.** *Let $P = P^\top = P^2$, let B be a bivector and $T = [B, P]$. Then $T^\top = PB^\top - B^\top P$, and*

(i) if B is compact, $B^\top = -B$, then T is symmetric and $\mathrm{Tr}(T^2) = \mathrm{Tr}(T^\top T) > 0$ unless $T = 0$;

(ii) if B is a boost, $B^\top = B$, then T is antisymmetric and $\operatorname{Tr}(T^2) = -\operatorname{Tr}(T^\top T) < 0$ unless $T = 0$.

Symmetric and antisymmetric matrices are trace-orthogonal, so on an orbit the trace form is an orthogonal sum of a positive definite piece on the directions generated by compact bivectors and a negative definite piece on those generated by boosts. Only the former are tangent to the manifold of orthogonal projectors.

Proof. $T^\top = (BP - PB)^\top = PB^\top - B^\top P$, which is T when $B^\top = -B$ and $-T$ when $B^\top = B$. The trace of the product of a symmetric and an antisymmetric matrix vanishes. A curve of orthogonal projectors has symmetric velocity, so the antisymmetric directions leave that manifold. □

Example 11.4 (The paired-mode projector at one constituent)**.** Take the rank-four projector Π of Example 2.6. Its stabiliser in $\mathfrak{so}(3,3)$ has dimension nine, so its orbit has dimension six. On the bivectors

$$B_I = \tfrac{1}{2}(e_1e_2,\ e_1e_3,\ e_2e_3,\ e_1e_5,\ e_1e_6,\ e_2e_6), \qquad T_I = [B_I, \Pi],$$

an exact 8×8 computation gives $\operatorname{Tr}(T_I T_J) = \operatorname{diag}(1,1,1,-1,-1,-1)$. The first three B_I are time–time rotations, the last three are boosts, exactly as Theorem 11.3 predicts. Only three of the fifteen basis bivectors commute with Π individually, while the stabiliser has dimension nine, so six of its directions are combinations of basis bivectors and no single one of them. Reading a stabiliser off basis elements overestimates the orbit.

Positive norm and tangency are nevertheless different conditions. For unit T_{sym} and T_{anti} of the two kinds, $\operatorname{Tr}\big((T_{\mathrm{sym}} + \tfrac{1}{2}T_{\mathrm{anti}})^2\big) = 1 - \tfrac{1}{4} > 0$, yet the direction is not symmetric and leaves the orthogonal projectors. What is exact is the orthogonal splitting, not an equivalence between the sign of the norm and tangency. On the manifold of orthogonal projectors itself every variation is symmetric and the trace form is positive definite, so the indefiniteness of the example is an artefact of moving Π along the full $\mathrm{Spin}(3,3)$ orbit. This is a statement about the target of a projector field and not about a field theory over a base: in a base of split signature the kinetic contraction $g^{\mu\nu}\operatorname{Tr}(D_\mu P\, D_\nu P)$ is indefinite in the spacetime index whatever the target metric is.

At three constituents the same reasoning counts the physical motions of P_{fam}. Of the 91 internal bivectors 43 are compact, forming the maximal compact $\mathfrak{so}(8) \oplus \mathfrak{so}(6)$, and the compact part of the stabiliser of P_{fam} has dimension nine (Theorem 7.5), so the compact orbit through P_{fam} has dimension

$$43 - 9 = 34,$$

which is also the rank of $B \mapsto [B, P_{\mathrm{fam}}]$ on the compact bivectors. Only seven of the 43 compact basis bivectors commute with P_{fam} individually, which repeats the warning of Example 11.4. Of the 3584 configuration directions at rank 64, thirty-four lie along the only group the geometry supplies. Whether these thirty-four are gauge redundancies or physical modes is a question about an action, which the algebra does not supply.

11.3 The induced connection over-generates

A varying projector carries its own connection. If the gauge field is not postulated but induced, $\nabla = P\,d$ on the bundle of images, then its curvature $F_{\mu\nu} = P[\partial_\mu P, \partial_\nu P]P$ is a composite, and the algebra in which it takes values is computed rather than chosen. To leading order in the Ambrose–Singer theorem that algebra is

$$\mathfrak{h}(P) = \operatorname{span}\big\{P[S, S']P :\ S, S' \in T_P\mathcal{P}_r\big\}.$$

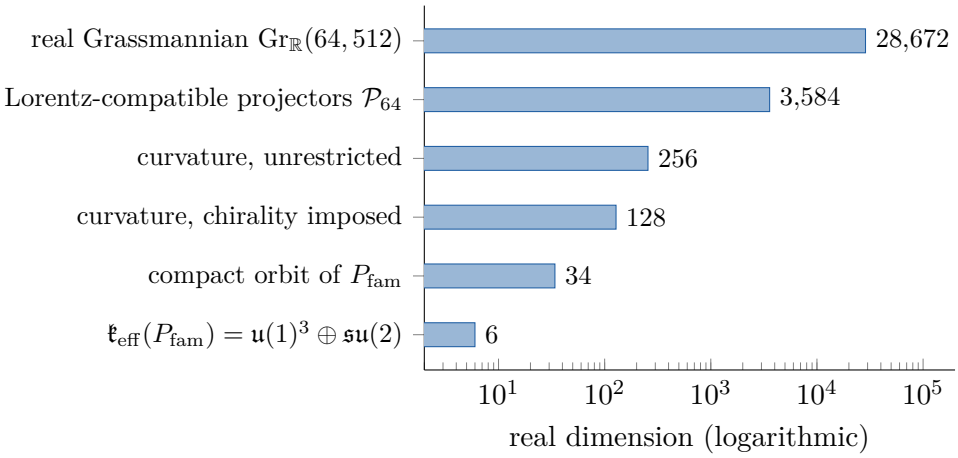


Figure 9: How much freedom a projector field has at rank 64, and how little of it is wanted. The configuration space of Lorentz-compatible projectors (Proposition 11.1) is far smaller than the real Grassmannian but still enormous; the group the geometry supplies moves P_{fam} along only 34 directions; and the curvature algebra the induced connection can reach (Theorem 11.5 and Proposition 11.6) exceeds the faithful gauge algebra of the block by more than an order of magnitude.

Theorem 11.5 (Over-generation)**.** *Let $P(x)$ range over all of $\mathcal{P}_{64}$. The curvature values $P[\partial_\mu P, \partial_\nu P]P$ are antisymmetric, hence compact, and lie in the block, and their span on* $\operatorname{im} P_{\mathrm{fam}}$ *has dimension*

$$\dim \mathfrak{h}(P_{\mathrm{fam}}) = 256 = \dim \mathfrak{u}(16),$$

the largest compact algebra the block admits at all. It exceeds the six dimensions of $\mathfrak{k}_{\mathrm{eff}}(P_{\mathrm{fam}}) = \mathfrak{u}(1)^3 \oplus \mathfrak{su}(2)$ by a factor of about 43*.*

The induced connection is not flat, every curvature value is compact, and every value lies in the block. The route fails by over-generating and by nothing else. In particular chirality is not among the casualties.

Proposition 11.6 (Chirality must be imposed, and costs nothing)**.** *Every projector of the family commutes with Γ. If a field $P(x)$ is required to satisfy $[P(x), \Gamma] = 0$ everywhere, then $[\partial_\mu P, \Gamma] = 0$ and*

$$\big[P[\partial_\mu P, \partial_\nu P]P,\ \Gamma\big] = 0,$$

so the induced curvature preserves chirality and the holonomy lies in $\mathfrak{u}(8) \oplus \mathfrak{u}(8)$. That algebra is still far larger than $\mathfrak{u}(1)^3 \oplus \mathfrak{su}(2)$.

The comparison of Theorem 11.5 is between what the construction permits and what is wanted, not an impossibility theorem about any particular field. A constant field has zero induced curvature, and a field compatible with Γ has holonomy inside $\mathfrak{u}(8) \oplus \mathfrak{u}(8)$. What the comparison does establish is the direction of the problem. A parent action must *cut* the freedom of the projector, by a constraint that shrinks $T_P\mathcal{P}$ before the curvature is formed or by a potential that freezes all but the intended holonomy directions, rather than build a gauge field up from nothing. This reverses the usual motivation for composite gauge fields. Figure 9 collects the numbers.

11.4 One explicit variational model

A list of fields does not determine field equations. What can be done without a derived action is to adopt one explicitly, derive its equation, and record what it can and cannot do. On a

four-dimensional Lorentzian base, with $D_\mu P = \partial_\mu P + [A_\mu, P]$ and with f and V as inputs, take

$$S_P = \int \sqrt{-g}\,\Big(\tfrac{f^2}{2}\,\mathrm{Tr}(D_\mu P\, D^\mu P) - V(P)\Big). \tag{12}$$

Varying along the Grassmannian, $\delta P = [\epsilon, P]$ with ϵ antisymmetric, gives

$$\big[P,\ f^2 D_\mu D^\mu P + V'(P)\big] = 0, \tag{13}$$

and for $V = 0$ this is the conservation law $D_\mu \mathcal{C}^\mu = 0$ of the matrix current $\mathcal{C}_\mu = [P, D_\mu P]$. The variation has to be projected onto the admissible tangent space whenever P is restricted further, for instance to commute with Γ.

Three consequences are immediate. First, a potential that depends only on the spectrum of a projector of fixed rank is constant, since $\mathrm{Tr}\, f(P) = r f(1) + (d-r) f(0)$; such a potential selects nothing, and for the pure kinetic term every constant admissible projector is a critical point. Selection needs couplings that distinguish orbits, and there is more than one orbit to distinguish: at rank 64 the orbits through P_{fam} and P_* have dimensions 34 and 20 (Proposition 8.6). Second, whether varying A_μ gives a Yang–Mills equation depends on how the action depends on A, not on A appearing in a list of fields; a composite A contributes through the chain rule to the equations of its constituents. Third, the pointwise stabiliser condition $[A_\mu, P] = 0$ is not the condition $D_\mu P = 0$ that a moving subbundle be preserved; the two agree only for constant P. Which connection acts on matter, $P(d + A)$, an ambient connection preserving P, or an independent connection in a moving frame of the block, is a further choice, and we make none of them here.

11.5 What Yang–Mills theory keeps open

Four conditions on a connection have been established as theorems: commutation with the Lorentz action (Theorem 6.1); commutation with the projector, with a faithful action on its image (Definition 7.3 and Theorem 7.5); antisymmetry, which for bivectors is Theorem 2.3; and preservation of the kinetic kernels K^μ. They are homogeneous and geometric, and together they determine the algebra in which A_μ takes its values, $\mathfrak{u}(1)^3 \oplus \mathfrak{su}(2)$ on the family block, $\mathfrak{so}(6) \oplus \mathfrak{su}(2)^{\oplus 3}$ at rank 256 and the Pati–Salam algebra on P_*, over the single frame of Proposition 6.3. They do not determine the field, since no parent action is derived; nor the couplings, since homogeneous conditions cancel every normalisation; nor which abelian direction is electromagnetic, by Theorem 9.1.

There is no representational obstruction behind this. By the theorem on universal connections [93], every connection on a bundle over a manifold of bounded dimension is the pullback of the canonical connection of a tautological bundle over a Grassmannian of sufficient rank, so the projector representation of a gauge field is always available in principle. The difficulty is dynamical, and power counting makes it concrete. A projector is dimensionless, so $\mathrm{Tr}(\partial P\, \partial P)$ carries a coupling of mass dimension $D - 2$ and $\mathrm{Tr}\, F^2$ one of dimension $D - 4$. In two dimensions the sigma model is renormalisable and the composite gauge field acquires a kinetic term at leading order in $1/N$, with linear confinement of the constituents [29, 146]. In four dimensions the sigma model is not power-counting renormalisable and that argument does not transfer.

Confinement is a separate question again. Invariant tensors permit meson-like and baryon-like operators, but they say nothing about masses or about the energy of separated sources, and the diagnostics with dynamical matter are flux-tube and string-breaking correlation matrices [9]. Moreover a block supplies a multiplicity index, not three constituents: three copies of a doublet form a direct sum, whereas three constituents require a tensor product with antisymmetrisation, so even the operator statement needs a Fock construction first.

12 The split space as a space of field strengths

The arena was introduced with an electromagnetic picture in mind: six directions, three of one sign and three of the other, paired one to one like the components of a magnetic and an electric field. This section turns the picture into a theorem and measures its price. The binding root G of Proposition 3.3 is the key, and its role in the interior dynamics, where it is the operator that makes bound motion possible, is established in Section 16. Here it is used only as a linear map with $G^2 = -\mathbf{1}$ and $G = G^\dagger$.

12.1 Arithmetic first

The space $\Lambda^2\mathbb{R}^{1,3}$ with the induced metric $\langle \theta^a \wedge \theta^b, \theta^a \wedge \theta^b \rangle = \eta^{aa}\eta^{bb}$ is negative on the three planes $\theta^0 \wedge \theta^i$ and positive on the three planes $\theta^j \wedge \theta^k$. It therefore has signature $(3,3)$, the Lorentz group acts on it by the adjoint representation, and the Hodge star satisfies $\star^2 = -\mathbf{1}$. This is counting, and it must be said before anything is built on it: every six-dimensional space of split signature is isometric to a Lorentzian field-strength space, so exhibiting one isometry predicts nothing. What is not arithmetic is which subalgebra $\mathfrak{so}(1,3) \subset \mathfrak{so}(3,3)$ realises the action, and whether the operator playing the Hodge star is one the arena already had.

12.2 The Lorentz algebra of the binding root

Theorem 12.1 (The centraliser of G)**.** *The centraliser $\mathfrak{l}_G$ of G in $\mathfrak{so}(3,3)$ is isomorphic to $\mathfrak{so}(3,\mathbb{C}) \cong \mathfrak{sl}(2,\mathbb{C}) \cong \mathfrak{so}(1,3)$. Its maximal compact subalgebra is the $\mathfrak{so}(3)$ of* diagonal *rotations, each acting identically on a temporal plane (e_i, e_j) and on its spatial partner (e_{i+3}, e_{j+3}).*

Proof. Since $G^2 = -\mathbf{1}$, the identification $z_j = x_j + ix_{j+3}$ makes $\mathbb{R}^6$ a complex space $\mathbb{C}^3$ on which G acts as i, and commuting with G is $\mathbb{C}$-linearity. Under this identification

$$\eta(X,X) = \sum_j x_j^2 - \sum_j x_{j+3}^2 = \operatorname{Re} q(z), \qquad q(z) := \sum_j z_j^2,$$

with q holomorphic. A $\mathbb{C}$-linear map A preserving $\operatorname{Re} q$ preserves it on every phase rotation $e^{i\vartheta}z$, and $q(e^{i\vartheta}z) = e^{2i\vartheta}q(z)$, so $\operatorname{Re}(e^{2i\vartheta}[q(Az) - q(z)]) = 0$ for all ϑ forces $q(Az) = q(z)$. The group is $O(3,\mathbb{C})$ and its algebra $\mathfrak{so}(3,\mathbb{C})$, whose underlying real Lie algebra is $\mathfrak{sl}(2,\mathbb{C}) \cong \mathfrak{so}(1,3)$ with maximal compact $\mathfrak{so}(3)$. A rotation of (e_i, e_j) alone does not commute with G, because G carries e_i to e_{i+3}; only the rotation performed identically on the partner plane does. □

Proposition 12.2 (Two Lorentz algebras, not conjugate)**.** *The interior algebra $\mathfrak{l}_{\text{int}}$ and $\mathfrak{l}_G$ are isomorphic as Lie algebras but not as subalgebras of $\mathfrak{so}(3,3)$. The interior algebra fixes the two-dimensional surplus plane pointwise, while $\mathfrak{l}_G$ contains simultaneous rotations of both triples and has no nonzero common fixed vector. Conjugation preserves the dimension of the common fixed subspace, so the two are not conjugate in $GL(6,\mathbb{R})$, let alone in $O(3,3)$.*

Write L_i for the diagonal rotations and put, for cyclic (i,j,k), $K_i = \frac{1}{2}(e_k e_{j+3} - e_j e_{k+3})$. The six operators L_i, K_i span $\mathfrak{l}_G$.

Theorem 12.3 (The field-strength reading)**.** *Write a vector of $\mathbb{R}^{3,3}$ as $X = \sum_i (B_i e_i + E_i e_{i+3})$. Then*

$$[L_i, L_j] = \epsilon_{ijk} L_k, \qquad [L_i, K_j] = \epsilon_{ijk} K_k, \qquad [K_i, K_j] = -\epsilon_{ijk} L_k,$$

both triples $\mathbf{B}$ and $\mathbf{E}$ transform as vectors under the L_i, and the K_i act by

$$\delta\mathbf{E} = \mathbf{v} \times \mathbf{B}, \qquad \delta\mathbf{B} = -\mathbf{v} \times \mathbf{E}. \tag{14}$$

Identify X with the two-form F_X of components $(F_X)_{0i} = E_i$ and $(F_X)_{jk} = \epsilon_{ijk}B_i$. Then G is the Hodge star, $F_{GX} = \star F_X$ with $(\star F)_{ab} = \frac{1}{2}\varepsilon_{abcd}F^{cd}$ and $\varepsilon_{0123} = +1$, and

$$\eta(X,X) = |\mathbf{B}|^2 - |\mathbf{E}|^2, \qquad \eta(X, GX) = -2\,\mathbf{E}\cdot\mathbf{B}$$

are the only invariant quadratic forms. With $\mathfrak{l}_G$ *as its Lorentz algebra,* $\mathbb{R}^{3,3}$ is $\Lambda^2\mathbb{R}^{1,3}$.

The orientation matters once, and we fix it here. With $F_{jk} = \epsilon_{ijk}B_i$ the vector $\mathbf{B}$ is minus the textbook magnetic field of the convention $F_{jk} = -\epsilon_{ijk}B_i$ [70], so we write $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$, as announced in Section 1.4; the electric field is the textbook one. Passing to a frame moving with velocity $\mathbf{u}$ changes the frame variables by $\delta\mathbf{B} = \mathbf{u} \times \mathbf{E}$ and $\delta\mathbf{E} = -\mathbf{u} \times \mathbf{B}$, which is (14) with $\mathbf{v} = -\mathbf{u}$; in textbook variables this is $\mathbf{E}' = \mathbf{E} + \mathbf{u} \times \mathbf{B}_{\mathrm{t}}$ and $\mathbf{B}'_{\mathrm{t}} = \mathbf{B}_{\mathrm{t}} - \mathbf{u} \times \mathbf{E}$, the familiar first-order transformation.

Example 12.4 (A boost along the third axis)**.** Take the pure magnetic field $X = e_1$, that is $\mathbf{B} = (1, 0, 0)$ and $\mathbf{E} = 0$, and the boost generator $K_3 = \frac{1}{2}(e_2e_4 - e_1e_5)$, acting on vectors by the commutator. Since e_1 anticommutes with e_2 and with e_4 it commutes with e_2e_4, while $[e_1e_5, e_1] = -2e_1^2e_5 = -2e_5$; hence $[K_3, e_1] = e_5$. Under $\delta X = \delta\,[K_3, X]$ the field acquires $\mathbf{E} = (0, \delta, 0)$, in accordance with $\delta\mathbf{E} = \mathbf{v} \times \mathbf{B}$ for $\mathbf{v} = (0, 0, \delta)$. In the same way $[K_3, e_5] = e_1$, so a pure electric field $\mathbf{E} = (0, 1, 0)$ acquires $\mathbf{B} = (\delta, 0, 0) = -\mathbf{v} \times \mathbf{E}$. The invariant $\eta(X, X) = |\mathbf{B}|^2 - |\mathbf{E}|^2$ changes only at second order, as it must.

Corollary 12.5 (The invariant complex structure)**.** $\mathbb{R}^6$ *is irreducible under* $\mathfrak{l}_G$*, and the commutant of* $\mathfrak{l}_G$ *in* $\mathrm{End}(\mathbb{R}^6)$ *is* $\mathrm{span}\{\mathbf{1}, G\} \cong \mathbb{C}$*. Hence, once* $\mathfrak{l}_G$ *is taken as the Lorentz algebra, its invariant complex structure is* $\pm G$ *and nothing else. The interior algebra, by contrast, has a commutant of dimension five,* $\mathbb{R}$ *on the four-plane and all of* $M_2(\mathbb{R})$ *on the surplus plane.*

The logic of this corollary runs from G to $\mathfrak{l}_G$ and back to $\pm G$, so the choice of G is recovered, not removed. It is a consistency statement, a uniqueness conditional on taking $\mathfrak{l}_G$ as the Lorentz algebra. Removing the choice would require a characterisation of $\mathfrak{l}_G$ that does not mention G, and we know none.

12.3 The two readings exclude one another

The price of the field-strength reading is not small.

Proposition 12.6 (Exclusion)**.** *Under* $\mathfrak{l}_{\mathrm{int}}$ *the six directions decompose as* $V \oplus 2\cdot\mathbf{1}$*, a four-vector and two scalars; under* $\mathfrak{l}_G$ *they are irreducible, with no fixed vector. The two representations are not similar, so a single theory cannot use both algebras as the Lorentz action on the same six directions. If the vector space* $\mathbb{R}^{3,3}$ *is read as the electromagnetic field strength, those six directions are not a four-plane plus a surplus pair, and the Dirac structure, the projector census,* P_* *and the diagonal sector all lose the arena on which they are built.*

What is excluded is reading one six-dimensional space in two ways, not a theory that contains both kinds of module. A theory may contain spacetime vectors and field strengths as different representations of one Lorentz algebra, which is the ordinary arrangement, and $\mathrm{Cl}(3, 3)$ realises it with the vectors at grade one and the field strength at grade two.

Proposition 12.7 (The field strength at grade two)**.** *Let* Λ^2E *be the span of the six bivectors of the interior four-plane. It is closed under the adjoint action of* $\mathfrak{l}_{\mathrm{int}}$*, on which it is the adjoint representation; the trace form* $\frac{1}{8}\mathrm{Tr}(XY)$ *on it has signature* $(3, 3)$*; and the interior volume* $\Omega = \gamma^0\gamma^1\gamma^2\gamma^3$*, with* $\Omega^2 = -\mathbf{1}$*, commutes with* $\mathfrak{l}_{\mathrm{int}}$*, preserves* Λ^2E *by left multiplication, exchanges* $\gamma^0\gamma^i$ *with the spatial bivectors and acts on* Λ^2E *as the Hodge star.*

Grade three behaves differently under the two algebras, and the difference sharpens the exclusion. Under $\mathfrak{l}_G$ one has $(\Lambda^3\mathbb{R}^6)_{\mathbb{C}} = 2\,(0, 0) \oplus 2\,(1, 1)$, of dimension $2 + 18$, so no field-strength module $(1, 0) \oplus (0, 1)$ occurs at grade three at all: trivectors cannot carry the field for this action. Under $\mathfrak{l}_{\mathrm{int}}$, $\mathbb{R}^6 = E \oplus \mathbb{R}^2$ gives $\Lambda^3\mathbb{R}^6 \cong 2E \oplus 2\Lambda^2E$, which contains two field-strength modules. The quadratic Casimir distinguishes the cases cleanly, with spectrum $\{0^{(2)}, 8^{(18)}\}$ on $\Lambda^3\mathbb{R}^6$ under $\mathfrak{l}_G$ and $\{3^{(8)}, 4^{(12)}\}$ under $\mathfrak{l}_{\mathrm{int}}$, the value 4 being that of $(1, 0) \oplus (0, 1)$ and 3

that of a four-vector. An explicit attempt confirms it: the six trivectors $C_i = \sum_{j,k} \epsilon_{ijk} e_j N_k$, $N_j = e_{k+3} e_{l+3}$ cyclically, together with $D_i = \mathcal{V} C_i$, transform as two vectors under the L_i, but all eighteen brackets $[K_i, C_j]$, $[K_i, D_j]$ leave their span, for instance $\frac{1}{2}[K_1, C_1] = e_2 e_3 e_4 + e_4 e_5 e_6$, and the Lorentz closure of the span has dimension 18.

12.4 How much of Maxwell's theory is forced

Theorem 12.3 supplies a field with the right transformation law and nothing that the field satisfies. How much of the dynamics does the transformation law fix by itself? The answer is: the form of the equations completely, and nothing else.

Theorem 12.8 (Exactly two first-order covariant operators)**.** *Consider linear first-order differential operators with constant coefficients on Minkowski space, acting on a field F valued in $\Lambda^2 \mathbb{R}^{1,3}$ and taking values in the vector representation. The derivative ∂F lies in $\Lambda^2 \otimes V$, of dimension* 24*, and under $\mathfrak{so}(1,3)$*

$$\Lambda^2 \otimes V = \underbrace{(\tfrac{3}{2}, \tfrac{1}{2}) \oplus (\tfrac{1}{2}, \tfrac{3}{2})}_{16} \;\oplus\; \underbrace{2\,(\tfrac{1}{2}, \tfrac{1}{2})}_{8},$$

so the vector module occurs with multiplicity exactly two, and no other module of dimension four occurs. The two copies are realised by

$$(dF)_{\mu\nu\rho} = 3\,\partial_{[\mu} F_{\nu\rho]} \quad (\text{through } \Lambda^3 \cong V), \qquad (\delta F)_\nu = \partial^\mu F_{\mu\nu},$$

and every covariant operator of the class is $\mathcal{D}F = a\,\delta F + b\,\delta(\star F)$, since $\delta\star = \pm \star d$.

The Casimir operator confirms the count: on $\Lambda^2 \otimes V$ its spectrum is $\{3^{(8)}, 9^{(16)}\}$, and the value 3 is the vector one. Nothing in the theorem constrains operators of higher order, operators with variable coefficients, or operators valued in the remaining $(\frac{3}{2}, \frac{1}{2}) \oplus (\frac{1}{2}, \frac{3}{2})$.

Corollary 12.9 (The form of Maxwell's equations)**.** *Within this class the two available equations are $dF = \mathcal{J}_m$ and $d \star F = \mathcal{J}_e$. Covariance does not rule out a magnetic current, which is a covariant field exactly as an electric current is; it identifies the two operators and leaves open which equations are imposed and which sources are admitted. The source-free system $dF = \delta F = 0$ is the only choice that needs no external data, and within this linear class it is the only one invariant under duality, because the Hodge star G exchanges the two operators. Finally, on a contractible region $dF = 0$ gives $F = dA$ by the Poincaré lemma, so the gauge potential is a consequence of the first equation rather than an assumption.*

In the variables of the frame the system takes a compact form, which we use repeatedly below.

Proposition 12.10 (Maxwell's equations on the frame)**.** *Read $\mathbb{R}^{3,3}$ as $\mathbb{C}^3$ through G, $z_j = B_j + iE_j$. Then $dF = 0$ and $d \star F = 0$ for the two-form F_X of Theorem 12.3 are*

$$\partial_t X = -G\,(\nabla \times X), \qquad \nabla \cdot X = 0, \tag{15}$$

that is $\partial_t z = -i\,\nabla \times z$ and $\nabla \cdot z = 0$, with the curl built from the rotation generators L_i. Splitting into the two triples gives $\partial_t \mathbf{B} = \nabla \times \mathbf{E}$, $\partial_t \mathbf{E} = -\nabla \times \mathbf{B}$ and $\nabla \cdot \mathbf{E} = \nabla \cdot \mathbf{B} = 0$; in textbook variables these are Faraday's law $\partial_t \mathbf{B}_{\mathrm{t}} = -\nabla \times \mathbf{E}$ and the Ampère–Maxwell law $\partial_t \mathbf{E} = \nabla \times \mathbf{B}_{\mathrm{t}}$ in vacuum. With a source, $\partial_\mu F^{\mu\nu} = (\rho, \mathbf{j})$ reads

$$\partial_t z = -i\,\nabla \times z - i\,\mathbf{j}, \qquad \nabla \cdot z = i\rho, \tag{16}$$

and taking the divergence of the first equation gives $\partial_t \rho + \nabla \cdot \mathbf{j} = 0$: continuity is a consequence of the system and the compatibility condition that any proposed current must satisfy.

Up to a factor i, z is the Riemann–Silberstein vector $\mathbf{E}+i\mathbf{B}_{\mathrm{t}}$, which obeys $i\partial_t(\mathbf{E}+i\mathbf{B}_{\mathrm{t}}) = \nabla\times(\mathbf{E}+i\mathbf{B}_{\mathrm{t}})$ [124, 14], and the holomorphic form of the proof of Theorem 12.1 has

$$\operatorname{Re} q(z) = |\mathbf{B}|^2 - |\mathbf{E}|^2, \qquad \operatorname{Im} q(z) = 2\,\mathbf{E}\cdot\mathbf{B},$$

so the two Maxwell invariants are the real and imaginary parts of one holomorphic quadratic form. Three classical facts fall into place. The duality rotation is the phase rotation $z \mapsto e^{i\vartheta}z$ that the proof of Theorem 12.1 already used, and it turns the pair of invariants by 2ϑ. The two quadratic forms that will compete in the interior dynamics of Sections 15 and 16 reappear with physical names: the η-form $\operatorname{Re} q$ is the Lorentz invariant, while the Hermitian form $|z|^2 = |\mathbf{B}|^2+|\mathbf{E}|^2$ is twice the energy density and depends on the frame. And the remaining sesquilinear covariant is the Poynting vector, $\frac{i}{2}\,\bar z\times z = \mathbf{E}\times\mathbf{B}$, so that $\mathbf{E}\times\mathbf{B}_{\mathrm{t}} = -\frac{i}{2}\,\bar z\times z$.

Example 12.11 (A circularly polarised wave on the frame)**.** The field $z = (1,i,0)\,e^{ik(x_3-t)}$ solves (15): $\nabla\cdot z = 0$ because z has no third component, and $\nabla\times z = ik\,(-i,1,0)e^{ik(x_3-t)} = k\,(1,i,0)e^{ik(x_3-t)}$, so $-i\nabla\times z = -ik\,z = \partial_t z$. Its invariant is $q(z) = (1+i^2)e^{2ik(x_3-t)} = 0$, so $|\mathbf{B}| = |\mathbf{E}|$ and $\mathbf{E}\cdot\mathbf{B} = 0$ at every point: the wave is null. Its Poynting vector $\frac{i}{2}\bar z\times z = \frac{i}{2}(1,-i,0)\times(1,i,0) = \frac{i}{2}(0,0,2i) = (0,0,-1)$ is $\mathbf{E}\times\mathbf{B}$, so $\mathbf{E}\times\mathbf{B}_{\mathrm{t}} = (0,0,1)$ points along the direction of propagation, as it should.

12.5 Frame vectors as spacetime planes

Each pairing plane (e_i, e_{i+3}) carries one complex component $z_i = B_i + iE_i$, and a frame vector is at the same time a bivector of spacetime. The two descriptions meet in the following way.

Proposition 12.12 (Frame vectors as planes)**.** *Let $X = \sum_i (B_i e_i + E_i e_{i+3})$ and F_X as above. Then $\operatorname{Pf} F_X = \mathbf{E}\cdot\mathbf{B}$ and $\det F_X = (\mathbf{E}\cdot\mathbf{B})^2$. A nonzero F_X is a single plane $u\wedge v$ exactly when $\mathbf{E}\cdot\mathbf{B} = 0$, that is when $\operatorname{Im} q = 0$. Otherwise there is a Lorentz frame in which $F_X = a\,\theta^0\wedge\theta^1 + b\,\theta^2\wedge\theta^3$ is a pair of orthogonal planes, and the duality rotation $e^{-\vartheta G}$ with $\tan\vartheta = a/b$ carries it to the single plane $\sqrt{a^2+b^2}\,\theta^2\wedge\theta^3$, the complexion of Rainich and of Misner and Wheeler [113, 86].*

Figure 10 shows the content of this proposition in the plane of the complex invariant. The real axis is the locus of single planes, the positive half of magnetic type and the negative half of electric type, the origin is the null cone, and the duality rotation moves a field around a circle at twice its own angle. In Section 16 the bound orbits of the interior dynamics will turn out to be exactly such rotations.

Everything in this section is a statement about a representation and its invariants. Theorem 12.8 is a multiplicity count that fixes the form of the equations and nothing about their content: it does not produce a source, does not say that a source is the current of anything, supplies no coupling constant and no Lagrangian. Nor is any of it special to this arena, since the count uses only $\mathfrak{so}(1,3)$ and Λ^2. What the arena adds is that the objects the count needs, the module, the duality and the invariants, were already present, so that the system can be written in the arena's own variables (15). The deflationary test of Section 1.2 applies to it in full.

13 Spin one in both readings

A photon is a massless particle of spin one with two helicities. Its covariant description is a field strength, a two-form, and its potential is a four-vector defined up to gauge. The arena offers two places for it. In the *interior reading* the potential is a grade-one element of the four-plane E on which the Dirac sector is built and the field strength is the grade-two module of Proposition 12.7. In the *frame reading* the field strength occupies the six directions themselves, under the algebra $\mathfrak{l}_G$ of Theorem 12.1. This section shows that each reading produces a free field

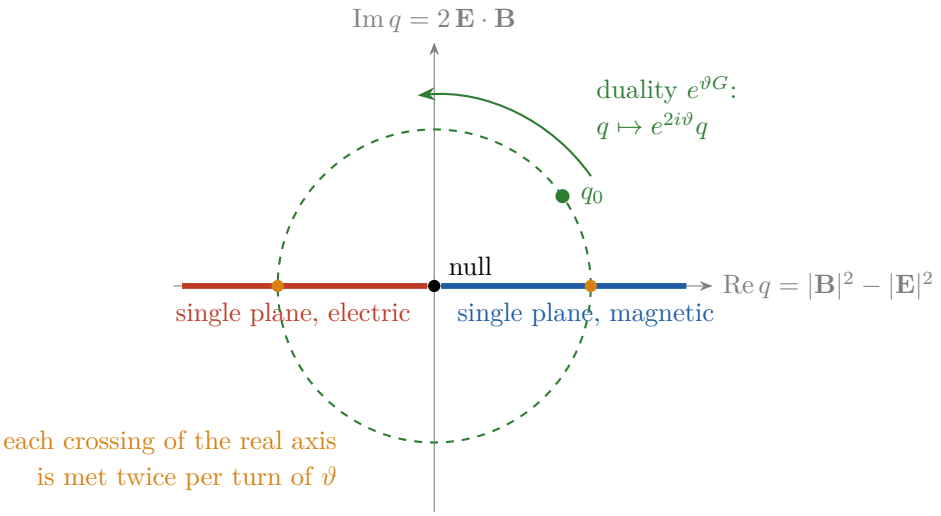


Figure 10: The plane of the complex invariant $q = z \cdot z$ of a frame vector (Theorem 12.3 and Proposition 12.12). The real axis is the locus $\mathbf{E}\cdot\mathbf{B} = 0$ on which the spacetime two-form is a single plane, magnetic on the right and electric on the left; the origin is the null cone. A duality rotation $e^{\vartheta G}$ preserves $|q|$ and turns q at twice the rate, so a non-null field becomes a single plane four times as ϑ runs once around.

with the spin content of the photon, that the two are related by the Klein correspondence, and that at the second rung both can be housed under one Lorentz action and coupled to a Dirac block. Figure 11 summarises the comparison.

13.1 The univectorial photon

Proposition 13.1 (Grade one under the interior action)**.** *Under* $\mathfrak{l}_{\mathrm{int}} \oplus \mathfrak{u}(1)_J$*, with* $J = e_2e_3$ *acting by* $\mathrm{ad}(J/2)$*,*

$$V = (\tfrac{1}{2},\tfrac{1}{2})_0 \oplus (0,0)_{+1} \oplus (0,0)_{-1}.$$

The Lorentz Casimir is $\mathrm{diag}(3,0,0,3,3,3)$ *on* $(e_1,\dots,e_6)$*, and* $\mathrm{ad}(J/2)$ *vanishes on the four-plane and acts on* (e_2,e_3) *by* $\left(\begin{smallmatrix}0 & \frac{1}{2}\\ -1 & 0\end{smallmatrix}\right)$*, with eigenvalues* $\pm i$*.*

The potential $A = A^\mu\gamma_\mu$ is therefore neutral under the charge, as a photon must be, while the surplus pair is a Lorentz scalar of charge ±1. The photon potential is the four-plane part of grade one, not the whole of it.

Proposition 13.2 (Spin one and spin one-half from the same rotors)**.** *The interior rotations* S_k *satisfy* $[S_i,S_j] = \epsilon_{ijk}S_k$*. On grade one* $\mathrm{ad}\,S_k$ *acts on* (e_4,e_5,e_6) *as the standard rotation generator* r_k *and annihilates* e_1,e_2,e_3*, so that* $-\sum_k(\mathrm{ad}\,S_k)^2 = \mathrm{diag}(0,0,0,2,2,2)$ *and* $\mathrm{spec}\,\mathrm{ad}\,S_3 = \{\pm i,0,0,0,0\}$*. On* $S = \mathbb{R}^8$*, by left multiplication,* $-\sum_k S_k^2 = \frac{3}{4}\mathbf{1}$ *and* S_3 *has eigenvalues* $\pm\frac{i}{2}$*. Consequently* $\exp(2\pi\,\mathrm{ad}\,S_3) = \mathbf{1}$ *on grade one while* $\exp(2\pi S_3) = -\mathbf{1}$ *on* S*.*

A vector transforms by conjugation, $A \mapsto RAR^{-1}$ with $R = e^{\theta S_3}$, and a spinor by left multiplication, $\psi \mapsto R\psi$. At $\theta = 2\pi$ the rotor is $-\mathbf{1}$; the sign cancels in the first case and survives in the second. This is the space-time-algebra account of the difference between vectors and spinors [64], realised here by the rotors of the Dirac sector: the same S_k give spin one to the potential and spin one-half to matter.

Theorem 13.3 (Maxwell's equations as one Clifford equation)**.** *Put* $\nabla = \gamma^\mu\partial_\mu$ *and* $\sigma_k = \gamma_k\gamma_0$*.*

(i) For $A = A^\mu\gamma_\mu$*,* $\nabla A = \partial\cdot A + \nabla\wedge A$*, and for* $F := \nabla\wedge A$ *the grade-one part of* ∇F *is* $(\Box A^\nu - \partial^\nu\partial\cdot A)\gamma_\nu$*, while its grade-three part vanishes identically.*

(ii) $F = \mathbf{E} + \Omega\,\mathbf{B}$ *with* $\mathbf{E} = \sum_k E_k\sigma_k$, $E_k = -\partial_k A^0 - \partial_t A^k$ *and* $\mathbf{B} = \nabla \times \mathbf{A}$.

(iii) For every $F = \sum_k (E_k\sigma_k + B_k\Omega\sigma_k)$,

$$\nabla F = (\nabla\cdot\mathbf{E})\gamma_0 + (\nabla\times\mathbf{B} - \partial_t\mathbf{E})^k\gamma_k + \Omega\big[-(\nabla\cdot\mathbf{B})\gamma_0 + (\partial_t\mathbf{B} + \nabla\times\mathbf{E})^k\gamma_k\big],$$

with no other grades.

Hence $\nabla F = \mathcal{J}$, *with* $\mathcal{J} = \rho\gamma_0 + \mathcal{J}^k\gamma_k$, *is Gauss's law, the Ampère–Maxwell law, the absence of magnetic charge and Faraday's law, and for* $F = \nabla \wedge A$ *the last two hold identically.*

This is Hestenes' single Maxwell equation [64], here on the four-plane of the Dirac sector inside the 8×8 matrices of $\mathrm{Cl}(3,3)$. Plane waves show how the gauge freedom removes the unphysical polarisations and how helicity appears.

Example 13.4 (A plane wave of the univectorial photon)**.** Let $k = \gamma_0 + \gamma_3$, so $k^2 = 0$, and $\varepsilon = \varepsilon^\mu\gamma_\mu$ with $k\cdot\varepsilon = 0$, that is $\varepsilon^0 = \varepsilon^3$. Then

$$F(\varepsilon) = k\varepsilon = \varepsilon^1 F_1 + \varepsilon^2 F_2, \qquad F_j = k\gamma_j,$$

because the component of ε along k drops out, $k^2 = 0$. The condition $k\cdot\varepsilon = 0$ has removed the component along the other null direction $\gamma_0 - \gamma_3$, so only the two transverse polarisations remain, and the gauge freedom $\varepsilon \mapsto \varepsilon + ck$ is exactly the helicity-zero direction. Moreover $F_j^2 = 0$ and $F_1F_2 + F_2F_1 = 0$, so every $F(\varepsilon)$ squares to zero: in the space-time algebra $F^2 = \mathbf{E}^2 - \mathbf{B}^2 + 2\Omega\,\mathbf{E}\cdot\mathbf{B}$, so this is the null condition $|\mathbf{E}| = |\mathbf{B}|$, $\mathbf{E}\perp\mathbf{B}$. On $\mathrm{span}\{F_1, F_2\}$,

$$\operatorname{ad} S_3 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -\operatorname{ad} S_3,$$

both with eigenvalues $\pm i$. For positive-frequency waves the eigenvectors of $\operatorname{ad} S_3$ with eigenvalues $\mp i$ carry helicity ± 1, and on them the Hodge star is i times the helicity: circular polarisation is self-duality or anti-self-duality of F.

Proposition 13.5 (Every field-strength module under the interior action)**.** *The Lorentz Casimir on the grades of* $\mathrm{Cl}(3,3)$, *the number of field-strength modules (Casimir value* 4*) and their charges under* $\mathrm{ad}(J/2)$ *are*

grade	spectrum of the Casimir (multiplicity)	field-strength modules	charges
$0, 6$	$0\,(1)$	0	—
$1, 5$	$3\,(4)$, $0\,(2)$	0	—
$2, 4$	$4\,(6)$, $3\,(8)$, $0\,(1)$	1	0
3	$4\,(12)$, $3\,(8)$	2	± 1

and on S *the Casimir is* $\frac{3}{2}$ *throughout.*

Because J commutes with the four-plane, the grade-four module is J times the grade-two one, and J is the complex structure at one constituent (Theorem 4.1). Up to that complex structure the neutral field strength is unique and sits at grade two; the two grade-three copies carry charge ± 1 and are not candidates for the photon.

13.2 The frame photon

In the frame reading the field strength is the vector X itself, and the boosts of $\mathfrak{l}_G$ turn out to be the spin matrices of the photon.

Proposition 13.6 (Spin matrices on the frame)**.** *$K_k = GL_k$ for $k = 1,2,3$, both families commute with G and preserve η, and the Casimir $\sum K_k^2 - \sum L_k^2$ equals $4\,\mathbf{1}$ on V, the field-strength value. On $z = \mathbf{B} + i\mathbf{E}$ the rotation L_k acts as r_k and the boost K_k as*

$$s_k = i\, r_k, \qquad (s_k)_{ab} = -i\epsilon_{kab},$$

which are Hermitian with $[s_i, s_j] = i\epsilon_{ijk}s_k$, $\sum_k s_k^2 = 2\,\mathbf{1}$ and $\operatorname{spec} s_3 = \{-1, 0, 1\}$. With $p = -i\nabla$ one has $(s \cdot p)z = \nabla \times z$ identically, so the frame Maxwell system (15) is the photon wave equation

$$i\partial_t z = (s \cdot p)\, z, \qquad p \cdot z = 0,$$

and the kernel of s_3 is spanned by $\hat{e}_3$: transversality removes exactly the helicity-zero component of a wave moving along $\hat{e}_3$.

These are the spin-one matrices that Oppenheimer and Bialynicki-Birula use to write the photon wave function [99, 13]. The rotations act on z as r_k, and their Hermitian generators $s_k = ir_k$ are the action of the boosts, because on this module a boost is G times a rotation.

Proposition 13.7 (Orthogonality on the frame)**.** *For $X = \sum_i (B_i e_i + E_i e_{i+3})$,*

$$z \cdot z = \eta(X, X) - i\,\eta(X, GX).$$

Hence $\mathbf{E} \perp \mathbf{B}$ exactly when $X \perp_\eta GX$, $|\mathbf{E}| = |\mathbf{B}|$ exactly when X is η-null, and both hold exactly when the plane $\operatorname{span}\{X, GX\}$ is totally η-null.

Theorem 13.8 (The axis of a null field)**.** *For real $a, b \in \mathbb{R}^3$,*

$$(s \cdot (a \times b))(a + ib) = |b|^2 a + i|a|^2 b - (a \cdot b)(b + ia).$$

Consequently a nonzero $z = \mathbf{B} + i\mathbf{E}$ satisfies $z \cdot z = 0$ if and only if $(s \cdot n)z = z$ for some real unit vector n, and then $n = \mathbf{B} \times \mathbf{E}/|\mathbf{B} \times \mathbf{E}|$ uniquely.

Proof. If $z \cdot z = 0$ then $\mathbf{B} \cdot \mathbf{E} = 0$ and $|\mathbf{B}| = |\mathbf{E}| \neq 0$; the identity gives $(s \cdot (\mathbf{B} \times \mathbf{E}))z = |\mathbf{B}|^2 z$ and $|\mathbf{B} \times \mathbf{E}| = |\mathbf{B}|^2$. Conversely $n \cdot ((s \cdot n)z) = 0$ and $z \cdot ((s \cdot n)z) = 0$ identically, and the characteristic polynomial of $s \cdot n$ is $-\lambda(\lambda^2 - |n|^2)$, so the eigenvalue 1 forces $n \cdot z = 0$ and $z \cdot z = 0$. For uniqueness, $(s \cdot n)z = in \times z = z$ splits into $\mathbf{B} = -n \times \mathbf{E}$ and $\mathbf{E} = n \times \mathbf{B}$, whence $n \perp \mathbf{B}$ and $\mathbf{B} \times \mathbf{E} = |\mathbf{B}|^2 n$. □

Since $\mathbf{B} \times \mathbf{E} = \mathbf{E} \times \mathbf{B}_\mathrm{t}$ is the Poynting vector, a null field is the $m = +1$ state of the photon spin along its own energy flux, the frame form of the classical fact that a null field determines a unique direction of propagation [130, 115]. Orthogonality with equal magnitudes therefore singles out the null fields and their axis. It does not single out a helicity.

Example 13.9 (Three null waves with one axis)**.** The fields

$$z_\pm = (1, i, 0)\, e^{\pm ik(x_3 - t)}, \qquad z_\mathrm{lin} = (1, i, 0) \cos k(x_3 - t)$$

all solve (15). In each, $\mathbf{B} \cdot \mathbf{E} = 0$, $|\mathbf{B}| = |\mathbf{E}|$ and $\mathbf{B} \times \mathbf{E}$ points along $+\hat{e}_3$ wherever $z \neq 0$, so each is an $m = +1$ state along $\hat{e}_3$. At $x_3 = 0$ the electric field turns according to $(\mathbf{E} \times \dot{\mathbf{E}}) \cdot \hat{e}_3 = +k,\ -k,\ 0$: helicity $+1$, helicity -1, and a linear superposition. The difference is the sign of the frequency, $z_+ \propto e^{-ikt}$ against $z_- \propto e^{+ikt}$, which is the Bialynicki-Birula assignment for the Riemann–Silberstein vector [13]. Not every radiation field is null either: $z = (1, i, 0) \cos k(x_3 - t) + (0, 1, i) \cos k(x_1 - t)$ solves the system and has $|\mathbf{B}| = |\mathbf{E}|$, but $\mathbf{B} \cdot \mathbf{E} = \cos k(x_1 - t) \cos k(x_3 - t)$.

Orthogonality alone is weaker still. The field $X = (\mathbf{B}; \mathbf{E}) = (1, 0, 0;\ 0, 2, 0)$ has $\eta(X, GX) = 0$, so $\mathbf{E} \perp \mathbf{B}$, but $\eta(X, X) = 1 - 4 = -3$; the boost $e^{\theta K_3}$ with $\theta = -\frac{1}{2}\log 3$ carries it to $(0, 0, 0;\ 0, \sqrt{3}, 0)$, a purely electric field with the same invariant. A field with $\mathbf{E} \perp \mathbf{B}$ and $|\mathbf{E}| \neq |\mathbf{B}|$ is purely electric or purely magnetic in a suitable frame [74]. On the null line $\mathbb{C}(1, i, 0)$, by contrast, $e^{\theta K_3}$ acts by multiplication by e^θ; with $\theta = -\operatorname{artanh}\beta$ for a frame moving with velocity $\beta\hat{e}_3$ this is the Doppler factor $\sqrt{(1-\beta)/(1+\beta)}$. The stabiliser of the axis acts on a null field by the helicity phase and by the Doppler factor.

13.3 Potentials on the frame, and helicity as a charge

Place a vector potential on each triple, $\mathbf{A}$ on e_1, e_2, e_3 and $\mathbf{C}$ on e_4, e_5, e_6, write $X_A = (\mathbf{A}; \mathbf{C})$ and $X_F = \nabla \times X_A$, the curl acting on each triple, and take

$$\mathcal{L} = \tfrac{1}{2}\,(\nabla \times X_A) \cdot G\dot{X}_A - \tfrac{1}{2}\,|\nabla \times X_A|^2 \tag{17}$$

with the Euclidean dot product of $\mathbb{R}^6$, for which G is orthogonal.

Theorem 13.10 (Duality-symmetric Maxwell theory on the frame)**.** *(i) The Euler–Lagrange expression of (17) is $G\nabla \times (\dot{X}_A + G\nabla \times X_A)$. Every solution of the first-order equation $\dot{X}_A = -G\nabla \times X_A$ solves it, and on a contractible region every solution satisfies the first-order equation up to a time-dependent gauge transformation.*

(ii) On solutions, X_F obeys the frame Maxwell system (15) with $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = \nabla \times \mathbf{C}$.

(iii) Under $X_A \mapsto X_A + (\nabla\lambda_1; \nabla\lambda_2)$, X_F is unchanged and $\mathcal{L}$ changes by a divergence.

(iv) The Hamiltonian is $\frac{1}{2}|\nabla \times X_A|^2 = \frac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2)$, the energy density.

(v) $\mathcal{L}(e^{\theta G} X_A) = \mathcal{L}(X_A)$ exactly, and the Noether current of $\delta X_A = GX_A$ has density $j^0 = -\frac{1}{2} X_A \cdot \nabla \times X_A$. For fields that decay or are periodic the conserved charge is $-\mathcal{K}$, where

$$\mathcal{K} = \tfrac{1}{2} \int X_A \cdot \nabla \times X_A \, d^3x = \tfrac{1}{2} \int (\mathbf{A} \cdot \mathbf{B} + \mathbf{C} \cdot \mathbf{E})\, d^3x$$

is the optical helicity.

This is the duality-symmetric first-order formulation of Schwarz and Sen [119], with two vector potentials in the temporal gauge and with the binding root G in the place of their $SO(2)$ generator; the conserved charge of the duality rotation is, up to sign, the optical helicity [18, 30, 19].

Example 13.11 (The G-charge is the helicity)**.** For waves $e^{i(kx_3 - \omega t)}$ the first-order equation reads $\omega a = G(k \times a)$, a 6×6 matrix with eigenvalues $\pm k$, each twice, and 0 twice (the longitudinal, pure-gauge pair). The eigenspace $\omega = +k$ is spanned by

$$\begin{aligned} a_1 &= (1, -i) \otimes (1, i, 0), \quad Ga_1 = ia_1, \qquad s_3 a_1 = a_1, \\ a_2 &= (1, i) \otimes (1, -i, 0), \quad Ga_2 = -ia_2, \quad s_3 a_2 = -a_2, \end{aligned}$$

with s_3 acting on each triple. For the real waves $X_A = \mathrm{Re}(a\, e^{ik(x_3 - t)})$ the helicity density $\frac{1}{2} X_A \cdot \nabla \times X_A$ and the energy density are (k, k^2) for a_1, $(-k, k^2)$ for a_2, and $(0, 4k^2 \sin^2 k(x_3 - t))$ for $a_1 + a_2$. On positive-frequency modes the eigenvalue of G is i times the helicity. The ratio of helicity to energy is $\pm 1/\omega$, which for a quantised field of energy $N\hbar\omega$ is $\pm N\hbar$: one unit of $\hbar$ per photon, the spin-one value.

The frame potentials are not components of a Lorentz tensor. Neither the constant boost $\delta X_A = K_3 X_A$ nor the boost of the coordinates is a symmetry of (17); the boost that is one is the Schwarz–Sen transformation $\delta X_A = t\,\partial_3 X_A - x_3\, G\nabla \times X_A$, whose curl is, identically,

$$\nabla \times (t\,\partial_3 X_A - x_3\, G\nabla \times X_A) = t\,\partial_3 X_F - x_3\, G\nabla \times X_F - K_3 X_F,$$

which on solutions is the boost of the field strength along $\hat{e}_3$ with its orbital part and its spin part K_3. Boosts reach the potentials only through their curl and only on shell [119]. Duality rotations, for their part, are not Lorentz transformations: $e^{\theta G}$ preserves the energy density and rotates $\eta(X, X)$ into $\eta(X, GX)$.

13.4 The bridge between the readings

Proposition 13.12 (Half-spinors under the two algebras)**.** *Under* $\mathfrak{l}_{\text{int}}$ *the Casimir on* $S^{\pm}$ *is* $\frac{3}{2}$*, and* S^{+} *carries a two-parameter family of invariant antisymmetric forms and no invariant symmetric one: it is a Weyl module. Under* $\mathfrak{l}_G$ *the Casimir on all of* S *is* 3*, the invariant symmetric forms on* S^{+} *form a one-parameter family of Lorentzian forms, and there is no invariant antisymmetric form:* S^{+} *is a Minkowski four-vector, and* $\exp(2\pi L_3) = +\mathbf{1}$ *on* S*. On* $\mathbb{C}^8$ *the two algebras give* $2\,(\frac{1}{2},0) \oplus 2\,(0,\frac{1}{2})$ *and* $2\,(\frac{1}{2},\frac{1}{2})$ *respectively, and they meet in zero.*

So under the binding algebra every spinor of $\mathrm{Cl}(3,3)$ has integer spin, in accordance with $\mathrm{Spin}(3,3) \cong SL(4,\mathbb{R})$, whose half-spinor modules are $\mathbb{R}^4$ and its dual [110]. This sharpens Proposition 12.6: with $\mathfrak{l}_G$ as the Lorentz algebra one constituent carries no spin one-half at all.

Theorem 13.13 (The Klein correspondence)**.** *The* $\mathrm{Spin}(3,3)$*-invariant bilinear forms on* S *form a two-dimensional space. For one of them,* β*, the map* $W(X)(\psi,\chi) = \beta(\psi, X\chi)$ *on* S^{+} *is antisymmetric and equivariant, and*

$$\operatorname{Pf} W(X) = \eta(X,X).$$

Hence $V \cong \Lambda^2(S^{+})^{*}$ *as* $\mathrm{Spin}(3,3)$*-modules, and the null cone of* η *is the set of decomposable two-forms.*

This is the Klein correspondence between the points of the Klein quadric and the lines of projective three-space [138]. It can be compared with the frame reading. Let $T : \mathbb{R}^{1,3} \to S^{+}$ be the $\mathfrak{l}_G$-intertwiner, unique up to scale. Then

$$T^{*}W(X) = -F_X + \star F_X = \sqrt{2}\, F_{e^{3\pi G/4}X} \qquad \text{for all } X,$$

so the canonical identification of V with two-forms on the half-spinor and the frame reading of the field strength differ by a duality rotation of an eighth of a turn, $-\pi/4$ modulo π. In the Klein frame $\eta(X,X)$ is the Pfaffian, the invariant of $\mathbf{E}\cdot\mathbf{B}$ type, and $\eta(X,GX)$ the Hodge pairing; in the frame reading the roles are the other way round. For free fields the two readings are equivalent, since duality rotations map solutions to solutions; a statement that singles out one invariant, such as which of them is the Lagrangian, holds in one reading and not in the other. A plane wave $\kappa \wedge \alpha$ built on S^{+}, with κ the lowered null momentum $(1,0,0,1)$ and α the polarisation $(0,1,0,0)$, lands on the doubly null vector $X = (-\frac{1}{2},-\frac{1}{2},0;\,\frac{1}{2},-\frac{1}{2},0)$, whose Poynting axis $\mathbf{B}\times\mathbf{E} = (0,0,\frac{1}{2})$ is the spatial momentum; the Klein planes of X and of GX meet in the null line $[\kappa]$, as two lines of projective space joined by a line of the quadric must.

13.5 The tower

On the tower the two readings multiply in different ways.

Proposition 13.14 (Commutants along the tower)**.** *On* $V_n = \mathbb{R}^{3n,3n}$ *the commutants have the following dimensions:*

commutant of	$n=1$	2	3	4	5
spectator $\mathfrak{l}_{\text{int}}$ in $\mathfrak{so}(3n,3n)$	1	28	91	190	325
diagonal $\mathfrak{l}_{\text{int}}$ in $\mathfrak{so}(3n,3n)$	1	7	18	34	55
diagonal $\mathfrak{l}_G$ in $\mathfrak{so}(3n,3n)$	0	2	6	12	20
diagonal rotations and G_n in $\mathfrak{so}(6n)$	1	4	9	16	25
the same inside $\mathfrak{so}(3n,3n)$	0	1	3	6	10

These are the dimensions of $\mathfrak{so}(3n-1,3n-3)$*,* $\mathfrak{so}(n)\oplus\mathfrak{so}(2n)$*,* $\mathfrak{so}(n,\mathbb{C})$*,* $\mathfrak{u}(n)$ *and* $\mathfrak{so}(n)$*.*

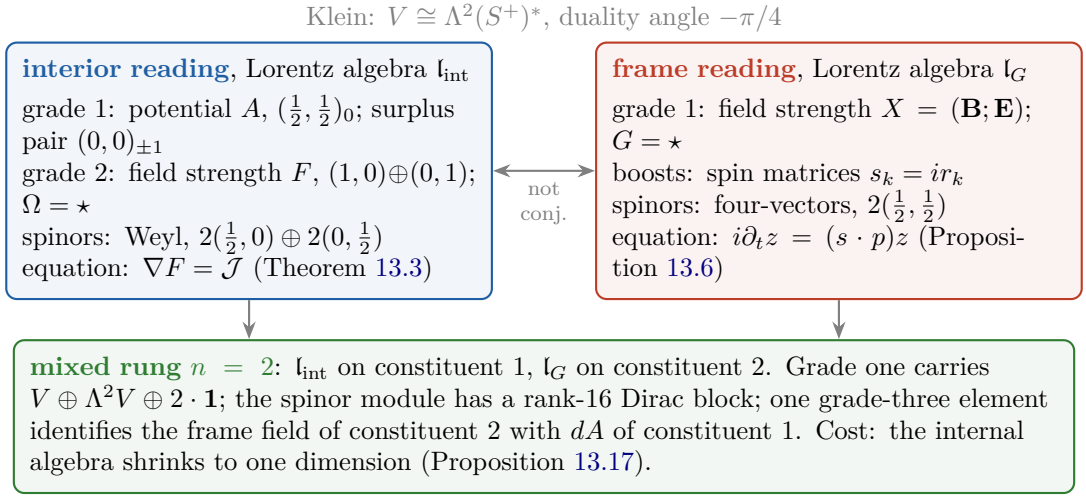


Figure 11: Two places for the photon in the arena (Section 13). Each reading gives a free field with the spin content of the photon; on one constituent they use non-conjugate Lorentz algebras and do not merge, and at the second rung they can be housed together at the price of the internal algebra.

Under the diagonal interior action $V_n = n\,(\frac{1}{2},\frac{1}{2})\oplus 2n\,(0,0)$, so there are n potentials forming the vector of the copy algebra $\mathfrak{so}(n)$, and none of them is copy-invariant at grade one for $n \geq 2$. A single univectorial potential in the diagonal sector therefore needs a choice of direction in copy space, and any such choice breaks $\mathfrak{so}(n)$ to $\mathfrak{so}(n-1)$. The obstruction belongs to grade one only.

Theorem 13.15 (Copy-invariant four-vectors)**.** *A copy-invariant four-vector in $\Lambda^g(\mathbb{R}^n\otimes\mathbb{R}^{1,3})$ requires n odd and g odd with $n \leq g \leq 3n$, and for every odd n there is one at $g = n$,*

$$Q_b^{(n)} = \epsilon_{k_1\cdots k_n}\,\eta^{a_1a_2}\cdots\eta^{a_{n-2}a_{n-1}}\,\gamma_{a_1}^{(k_1)}\wedge\cdots\wedge\gamma_{a_{n-1}}^{(k_{n-1})}\wedge\gamma_b^{(k_n)}.$$

At $n = 3$ there is exactly one at each of the grades $3,5,7,9$, and at $n = 2$ there is none at any grade.

Proof. An invariant element of $\Lambda^g(\mathbb{R}^n\otimes\mathbb{R}^{1,3})\otimes\mathbb{R}^{1,3}$ is an invariant tensor of $SO(n)\times SO^+(1,3)$ in $(\mathbb{R}^n)^{\otimes g}\otimes(\mathbb{R}^{1,3})^{\otimes(g+1)}$. By the first fundamental theorem for the special orthogonal groups [142], such tensors are spanned by products of metrics and at most one Levi-Civita symbol. On the Lorentz side every factor has an even number of indices, so $g+1$ is even. On the copy side $g = 2d+ne$ with $e\in\{0,1\}$; g odd forces $e = 1$ and n odd, so $g \geq n$. The Hodge star of $\mathbb{R}^n\otimes\mathbb{R}^{1,3}$ is an equivariant isomorphism $\Lambda^g \cong \Lambda^{4n-g}$, which gives $g \leq 3n$. The displayed $Q_b^{(n)}$ is invariant because ϵ is, and nonzero because the terms of $Q_0^{(n)}$ with all Lorentz indices equal to 0 add up to $n!\,\gamma_0^{(1)}\wedge\cdots\wedge\gamma_0^{(n)}$. □

At $n = 3$ the invariant four-vector of grade three is, up to a factor, the copy-singlet cubic Q_μ that builds the gamma matrices of the Dirac block in Theorem 10.6; the parity theorem explains why such an object exists at three constituents and could not exist at two. Under the diagonal $\mathfrak{l}_G$, finally, $V_n = n[(1,0)\oplus(0,1)]$: n field strengths mixed by the complex orthogonal flavour algebra $\mathfrak{so}(n,\mathbb{C})$. The energy-preserving duality group of n free Maxwell fields is $U(n)$, the maximal compact subgroup of the Gaillard–Zumino group $Sp(2n,\mathbb{R})$ [43]; its Lie algebra meets $\mathfrak{so}(3n,3n)$ only in $\mathfrak{so}(n)$, and its centre, generated by G_n and carrying the total helicity, lies outside $\mathfrak{so}(3n,3n)$, as G does at $n = 1$. A single photon would be one direction in this flavour space, and nothing selects one.

13.6 The mixed rung

Proposition 12.6 forbids reading one six-dimensional space in two ways. At $n \geq 2$ there is more than one constituent, and the obvious repair is to let one constituent carry the interior reading and another the frame reading, under one Lorentz action.

Proposition 13.16 (The mixed embedding)**.** *In* $\mathrm{Cl}(6,6)$ *let* $\mathfrak{l}_{\mathrm{mix}}$ *be spanned by* $S_k \otimes 1 + 1 \otimes L_k$ *and* $T_k \otimes 1 + 1 \otimes K_k$*: the interior algebra on constituent* 1 *and* $\mathfrak{l}_G$ *on constituent* 2*, paired generator by generator. It satisfies the* $\mathfrak{so}(1,3)$ *brackets, and*

$$\mathbb{R}^{6,6} = \underbrace{V}_{C=3} \oplus \underbrace{\Lambda^2 V}_{C=4} \oplus 2 \cdot \mathbf{1},$$

so one Lorentz action carries both a four-vector and a field strength at grade one. On $S_2 = \mathbb{R}^{64}$ *the Casimir satisfies* $(C_{\mathrm{mix}} - \frac{3}{2})(C_{\mathrm{mix}} - \frac{11}{2}) = 0$*, with the Dirac value* $\frac{3}{2}$ *on sixteen dimensions and* $\frac{11}{2}$ *on* 48*.*

Proposition 13.17 (The cost)**.** *The centraliser of* $\mathfrak{l}_{\mathrm{mix}}$ *in* $\mathfrak{so}(6,6)$ *is one-dimensional, spanned by the rotation of the surplus pair of constituent* 1*. At the same rung:*

Lorentz action at $n = 2$	centraliser	content of $\mathbb{R}^{6,6}$
interior on one constituent	28	$V \oplus 8 \cdot \mathbf{1}$
$\mathfrak{l}_G$ on one constituent	15	$\Lambda^2 V \oplus 6 \cdot \mathbf{1}$
mixed, across both	1	$V \oplus \Lambda^2 V \oplus 2 \cdot \mathbf{1}$

The mixed embedding buys the field-strength module at the price of the whole internal algebra: $\mathfrak{so}(5,3)$ *collapses to one* $\mathfrak{u}(1)$*.*

The count is for this pairing of the two Lorentz bases at $n = 2$; another pairing or a higher rung could do better and has not been examined. Within its scope, however, the mixed rung is an honest place to test whether the two readings can be coupled, and it turns out that they can.

Theorem 13.18 (The Dirac block of the mixed rung)**.** *Let* $P = \frac{1}{8}(11\,\mathbf{1} - 2C_{\mathrm{mix}})$ *and* $\hat{\gamma}_\mu = 2P\gamma_\mu^{(1)}P$*, with* $J_1 = e_2 e_3 \otimes 1$*.*

(i) *P is the orthogonal projector of rank* 16 *onto the block with the Dirac value, and commutes with the mixed generators.*

(ii) *$\{\hat{\gamma}_\mu, \hat{\gamma}_\nu\} = 2\eta_{\mu\nu}P$, $[\hat{\gamma}_\mu, J_1] = 0$, $J_1^2 = -\mathbf{1}$, and $P\mathcal{S}_k P = \frac{1}{2}\hat{\gamma}_i\hat{\gamma}_j$ (cyclic), $P\mathcal{T}_k P = \frac{1}{2}\hat{\gamma}_0\hat{\gamma}_k$: the compressed gammas generate the mixed Lorentz action on the block.*

(iii) *With J_1 as the imaginary unit the block is two complex Dirac modules, labelled by the half-spinors of constituent* 2*, with flavour algebra* $\mathfrak{u}(2)$*; $\mathcal{H}(p) = \hat{\gamma}^0\hat{\gamma}^a p_a + m\hat{\gamma}^0$ is symmetric and $\mathcal{H}(p)^2 = (|p|^2 + m^2)P$.*

(iv) *The block is the γ-trace part $\sum_a \gamma_a\chi \otimes \tau e_a$ of the vector-spinors (Dirac spinor of constituent* 1*) $\otimes$(four-vector of constituent* 2*), and the rank-*48 *block is their γ-traceless $(1, \frac{1}{2}) \oplus (\frac{1}{2}, 1)$ part.*

Item (iv) explains the factor 2 in $\hat{\gamma}_\mu$, since $P\gamma_\mu^{(1)}$ acts on the γ-trace part as $-\frac{1}{2}\gamma_\mu$, and it identifies the decomposition with the classical splitting of a vector-spinor into its trace and its traceless part [114].

A covariant identification of the field strength of constituent 1 with the frame field of constituent 2 is an invariant element of $\Lambda^2(\text{four-plane}_1) \otimes V_2$, that is, a grade-three element of the algebra commuting with the mixed generators. The invariant space is two-dimensional, and one invariant is

$$\Theta_F = \gamma_2^{(1)}\gamma_3^{(1)}e_1' + \gamma_3^{(1)}\gamma_1^{(1)}e_2' + \gamma_1^{(1)}\gamma_2^{(1)}e_3' - \gamma_0^{(1)}\gamma_1^{(1)}e_4' - \gamma_0^{(1)}\gamma_2^{(1)}e_5' - \gamma_0^{(1)}\gamma_3^{(1)}e_6',$$

with e'_a the generators of constituent 2; it pairs the spatial bivectors of constituent 1 with the magnetic directions of constituent 2 and the time–space bivectors with the electric ones. The other invariant is Θ_F followed by G, so up to scale the identifications form the family $X = e^{\theta G} X_F$.

Theorem 13.19 (A coupled model)**.** *On the block, in flavour coordinates $\mathbb{R}^8 \otimes \mathbb{R}^2$ with $\hat\gamma_\mu = -\gamma_\mu \otimes 1$, $J_1 = J \otimes 1$ and $U^\mu = \hat\gamma^0 \hat\gamma^\mu$, take for commuting real fields*

$$\mathcal{L} = -\tfrac{1}{4} F_{\mu\nu} F^{\mu\nu} + \psi^\top J_1 U^\mu D_\mu \psi - m\, \psi^\top \hat\gamma^0 \psi, \qquad D_\mu = \partial_\mu + q A_\mu J_1, \tag{18}$$

the real form of $-\frac{1}{4}F^2 + \bar\psi(i\gamma^\mu D_\mu - m)\psi$ with J_1 in the place of i. Then the field equations are Maxwell's equations $\partial_\mu F^{\mu\nu} = q\, j^\nu$ with $j^\nu = \psi^\top U^\nu \psi$ and $j^0 = |\psi|^2$, and the Dirac equation $J_1 \hat\gamma^\mu D_\mu \psi = m\psi$; the identity $\partial_\mu j^\mu + \psi^\top J_1 \mathcal{E}_\psi = 0$ holds with $\mathcal{E}_\psi$ the Dirac expression; the Lagrangian is exactly invariant under $\psi \mapsto e^{-q\lambda J_1}\psi$, $A \mapsto A + \partial\lambda$; and its flavour symmetry is the $\mathfrak{u}(2)$ spanned by $J \otimes 1$, $1 \otimes \epsilon$, $J \otimes \sigma_x$, $J \otimes \sigma_z$. The potential carries two polarisations of helicity ± 1 and no massive mode, and each momentum carries two flavours, two helicities and two signs of frequency.

In the frame reading the source is electric. For the frame field of an arbitrary potential one has, identically, $\partial_t z + i\nabla \times z + i\boldsymbol{\mathcal{J}} = 0$ and $\nabla \cdot z = i\rho$ with $(\rho, \boldsymbol{\mathcal{J}}) = \partial_\mu F^{\mu\nu}$, and the rotated field $e^{\theta G} X_F$ satisfies the same system with electric source $\cos\theta$ times the current and magnetic source $\sin\theta$ times it. Since the frame field takes no part in the dynamics of (18), which member of the family is called the frame field is a choice; requiring it to carry electric sources only fixes $\theta \equiv 0$ modulo π, as in the textbook treatment of duality with sources, where the angle is fixed by the convention that all charges are electric [70]. The duality rotation, a symmetry of the free system and of (17), is not a symmetry once electric sources are present.

The coupled model is classical two-flavour spinor electrodynamics with equal charges, written in the algebra of the second rung. It is a construction, not a selection, and it has a definite limitation that we record now and use later: the second-rung vacuum of Section 26 is defined for the spectator action, and under the mixed action it is preserved only by the rotations about one axis, so the coupled model lies outside that vacuum sector.

At even rungs, finally, the diagonal interior action places integer-spin modules inside the spinor module itself: at $n = 2$ the Casimir takes the values $0, 3, 4$ on $8, 32, 24$ dimensions, so S_2 contains scalars of copy charge ± 2, four-vectors of copy charge ± 1 and copy-neutral field strengths $(1,0) \oplus (0,1)$, and the copy and Lorentz Casimirs add up to 4, the $n = 2$ case of (11). These are modules of one spinor space, not bound states, and we do not propose them as the photon.

14 The orthogonality of the electric and magnetic fields

In the frame reading the orthogonality of $\mathbf{E}$ and $\mathbf{B}$ is the reality of the complex invariant, $\operatorname{Im} q = 2\,\mathbf{E} \cdot \mathbf{B} = 0$, and by Proposition 12.12 it is also the condition for the spacetime two-form to be a single plane. Since the arena puts $\mathbf{E}$ and $\mathbf{B}$ on two orthogonal triples, it is natural to ask what orthogonality means physically: what breaks it, whether its failure is a symmetry breaking, and whether it can produce mass. The answers are classical once the right questions are asked, and they come out sharply negative on the last point.

14.1 What orthogonality is

The field strength $F^\mu{}_\nu$ has eigenvalues $\pm\epsilon_E$ and $\pm i\epsilon_M$ with

$$\epsilon_E = |\operatorname{Im}\sqrt{q}|, \qquad \epsilon_M = |\operatorname{Re}\sqrt{q}|,$$

and $\mathbf{E}\cdot\mathbf{B} = \operatorname{Pf} F = 0$ exactly when F is one plane: a pure boost, a pure rotation or a null rotation. Otherwise F generates a screw, a boost and a rotation about a common axis. Two simple facts locate orthogonality among all fields.

Proposition 14.1 (Single modes are orthogonal)**.** *Let $F = k \wedge b$ with k, b real four-vectors. Then $\operatorname{Pf} F = 0$ and $q = (k\cdot k)(b\cdot b) - (k\cdot b)^2$ is real. The field of a mode $A_\mu = \operatorname{Re}(a_\mu e^{-ik\cdot x})$ with real k and $k\cdot a = 0$ is $k \wedge b(x)$ with $k\cdot b = 0$, for every polarisation. A Maxwell mode, $k\cdot k = 0$, is therefore null at every point, and a Proca mode, $k\cdot k = m^2$, has $q = m^2(b\cdot b) \le 0$: it is orthogonal and of electric type everywhere.*

A mass does not show in the angle between the fields of a single mode; it shows in the other invariant, which it makes nonzero. The null cone $q = 0$ is the massless locus, and orthogonality is a much larger set, containing every single mode of every mass.

Proposition 14.2 (Superposition breaks it)**.** $\operatorname{Pf}(k_1 \wedge b_1 + k_2 \wedge b_2) = \det(k_1^\flat, b_1^\flat, k_2^\flat, b_2^\flat)$. *At a point, two modes give $\mathbf{E}\cdot\mathbf{B} \neq 0$ exactly when their two planes together span spacetime. Waves propagating along one null direction ℓ never do: the total field is $\ell \wedge \sum_j \omega_j b_j$, a single null plane.*

Example 14.3 (Counter-propagating waves)**.** In textbook variables, with $\omega = k$, let a wave along $+z$ polarised along x meet a wave along $-z$. With crossed polarisations, $\mathbf{E} = (\cos(kz - kt), \cos(kz + kt), 0)$ and $\mathbf{B}_{\mathrm{t}} = (\cos(kz + kt), \cos(kz - kt), 0)$; with parallel ones, $\mathbf{E} = (\cos(kz - kt) + \cos(kz + kt), 0, 0)$ and $\mathbf{B}_{\mathrm{t}} = (0, \cos(kz - kt) - \cos(kz + kt), 0)$. Both solve Maxwell's equations, and

$$\begin{aligned} &\text{crossed:} && \mathbf{E}\cdot\mathbf{B}_{\mathrm{t}} = \cos 2kz + \cos 2kt, && |\mathbf{B}_{\mathrm{t}}|^2 - |\mathbf{E}|^2 = 0, \\ &\text{parallel:} && \mathbf{E}\cdot\mathbf{B}_{\mathrm{t}} = 0, && |\mathbf{B}_{\mathrm{t}}|^2 - |\mathbf{E}|^2 = -2(\cos 2kz + \cos 2kt). \end{aligned}$$

The two configurations have the same energy density and the same Poynting vector at every point. The reason is structural: the crossed configuration is the parallel one turned by $\pi/4$ about the z axis and then duality-rotated by $\pi/4$; the rotation preserves the energy density and the Poynting vector along z, the duality rotation preserves the entire energy–momentum tensor, and only the phase of q moves.

Nothing exotic is needed to break orthogonality. Near the ground the fair-weather electric field of the atmosphere, about 100 V/m and vertical [38], meets the geomagnetic field, which is inclined to the horizontal everywhere except on the magnetic equator.

14.2 How it changes in time

Proposition 14.4 (The local law)**.** *With $\partial_t \mathbf{B} = \nabla\times\mathbf{E}$ and $\partial_t \mathbf{E} = -\nabla\times\mathbf{B} - \mathbf{j}$,*

$$\partial_t q = -2i\, z\cdot(\nabla\times z) - 2i\, z\cdot\mathbf{j}, \qquad \partial_t(\mathbf{E}\cdot\mathbf{B}) = \mathbf{E}\cdot\nabla\times\mathbf{E} - \mathbf{B}\cdot\nabla\times\mathbf{B} - \mathbf{j}\cdot\mathbf{B},$$

in textbook variables $\partial_t(\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}) = \mathbf{B}_{\mathrm{t}}\cdot\nabla\times\mathbf{B}_{\mathrm{t}} - \mathbf{E}\cdot\nabla\times\mathbf{E} - \mathbf{j}\cdot\mathbf{B}_{\mathrm{t}}$. With potentials, $\mathbf{B}_{\mathrm{t}} = \nabla\times\mathbf{A}$, $\mathbf{E} = -\partial_t\mathbf{A} - \nabla\phi$, and in a source-free region $\mathbf{E} = -\nabla\times\mathbf{C}$, $\mathbf{B}_{\mathrm{t}} = -\partial_t\mathbf{C} + \nabla\chi$,

$$\partial_t(\mathbf{A}\cdot\mathbf{B}_{\mathrm{t}}) + \nabla\cdot(\phi\mathbf{B}_{\mathrm{t}} + \mathbf{E}\times\mathbf{A}) = -2\,\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}, \qquad \partial_t(\mathbf{C}\cdot\mathbf{E}) - \nabla\cdot(\mathbf{B}_{\mathrm{t}}\times\mathbf{C} + \chi\mathbf{E}) = -2\,\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}.$$

In vacuum, $\mathbf{E}\cdot\mathbf{B}$ changes at a point only through $\mathbf{E}\cdot\nabla\times\mathbf{E}$ and $\mathbf{B}\cdot\nabla\times\mathbf{B}$, which measure how much each field twists about itself. Integrated over a region with vanishing boundary fluxes, $\int \mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$ is minus half the rate of change of the magnetic helicity $\int \mathbf{A}\cdot\mathbf{B}_{\mathrm{t}}$ [149, 11], and equally of the electric helicity, so that their difference, the optical helicity of Theorem 13.10, is conserved in free space [18, 19]. Non-orthogonality is the agent that changes the helicity of each field while their combination is kept. In matter, charges moving along the magnetic field change $\mathbf{E}\cdot\mathbf{B}$ through $-\mathbf{j}\cdot\mathbf{B}$.

Conductors screen it. For the generalised Ohm law $\mathbf{E}+\mathbf{v}\times\mathbf{B}_{\mathrm{t}}=\eta_{\mathrm{R}}\mathbf{j}+(\mathbf{j}\times\mathbf{B}_{\mathrm{t}})/(ne)-(\nabla\cdot\mathsf{P}_e)/(ne)+(m_e/ne^2)\,d\mathbf{j}/dt$ [112],

$$\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}=\eta_{\mathrm{R}}\,\mathbf{j}\cdot\mathbf{B}_{\mathrm{t}}-\frac{\mathbf{B}_{\mathrm{t}}\cdot(\nabla\cdot\mathsf{P}_e)}{ne}+\frac{m_e}{ne^2}\,\mathbf{B}_{\mathrm{t}}\cdot\frac{d\mathbf{j}}{dt}\,:$$

the flow term and the Hall term drop out, ideal magnetohydrodynamics enforces orthogonality everywhere, and only resistivity, electron pressure and electron inertia allow $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}\neq0$. This is the criterion of Schindler, Hesse and Birn for magnetic reconnection, a nonvanishing integral of $E_{\parallel}$ along field lines [118]. Where charges are scarce the screening fails: a rotating magnetised neutron star in vacuum induces $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}\neq0$ at its surface, charges drawn from the surface fill the magnetosphere until $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}=0$ [52], and the regions left without enough charge keep $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}\neq0$ and accelerate particles [116].

14.3 Non-orthogonality breaks parity and time reversal, and is never spontaneous

Theorem 14.5 (Orthogonality is achirality)**.** *For a field strength F at a point the following are equivalent: (a) $\mathbf{E}\cdot\mathbf{B}=0$; (b) the mirror image PF equals ΛF for some proper orthochronous Lorentz transformation Λ; (c) the time-reversed field TF equals ΛF for some such Λ.*

Proof. If $\mathbf{E}\cdot\mathbf{B}=0$ and $\mathbf{B}\neq0$, the half-turn about $\mathbf{B}$ fixes $\mathbf{B}$ and reverses every vector perpendicular to it, sending $(\mathbf{B};\mathbf{E})$ to $(\mathbf{B};-\mathbf{E})=P(\mathbf{B};\mathbf{E})$; if $\mathbf{B}=0$ a half-turn about an axis perpendicular to $\mathbf{E}$ does the same. Conversely the Pfaffian is invariant under proper Lorentz transformations and changes sign under P, so $\operatorname{Pf}F=-\operatorname{Pf}F$. For (c), $T=CP$ on fields and $CF=-F$ has the same invariant q; since $\mathfrak{l}_G\cong\mathfrak{so}(3,\mathbb{C})$ acts transitively on each level set of q with the origin removed, $-F$ lies in the proper Lorentz orbit of F, and (c) is equivalent to (b). □

A non-orthogonal field is therefore chiral in Kelvin's sense, a screw of definite handedness, and the sign of $\mathbf{E}\cdot\mathbf{B}$ is that handedness; it is odd under P and T and even under C, PT and CPT. This is a statement about a field at a point. A configuration can be orthogonal everywhere and still chiral as a whole: a static Beltrami magnetic field with $\mathbf{E}=0$ carries a nonzero helicity density. The global chirality of a magnetic configuration is its helicity, P-odd and T-even, and $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$, P-odd and T-odd, is minus half its rate of change.

The breaking is never spontaneous. Since $u^2-|\mathbf{S}|^2=\frac14|q|^2$ for the energy density $u=\frac12(|\mathbf{E}|^2+|\mathbf{B}|^2)$ and the Poynting vector $\mathbf{S}$,

$$u\;\geq\;\tfrac12|q|\;\geq\;|\mathbf{E}\cdot\mathbf{B}|, \tag{19}$$

and the minimum of the energy is the zero field; there is no degenerate family of lowest-energy configurations among which $\mathbf{E}\cdot\mathbf{B}$ could choose a sign. In the quantum theory the corresponding statement is the theorem of Vafa and Witten: in a vector-like gauge theory at $\theta=0$, zero temperature and zero density, parity is not broken spontaneously through a Bose order parameter such as $F\tilde F$ [134]. When $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$ is imposed from outside the vacuum responds, for instance by rotating the chiral condensate of quantum chromodynamics through the electromagnetic triangle anomaly [21]; that is the response of a vacuum to an explicit breaking, not a breaking of its own.

Duality behaves differently. The rotation $X\mapsto e^{\vartheta G}X$ leaves the energy–momentum tensor invariant and multiplies q by $e^{2i\vartheta}$, so for $q\neq0$ only $\vartheta\in\pi\mathbb{Z}$ fixes the invariants, while for $q=0$ every rotation does. The fields whose invariants are unchanged by duality are the null ones, not the orthogonal ones, and orthogonality is not a symmetric point of duality. On $\mathbb{R}^{3,3}$, moreover, $(e^{\vartheta G})^{\top}\eta\,e^{\vartheta G}=\cos2\vartheta\,\eta+\sin2\vartheta\,\eta G$, so $e^{\vartheta G}$ is an isometry of η only for $\vartheta\in\pi\mathbb{Z}$. Duality is broken explicitly by electric sources, as Section 13.6 showed, and by the vacuum of a theory

with charged particles. For a Lagrangian $L(\mathcal{F},\mathcal{G})$ of the invariants $\mathcal{F}=\frac{1}{2}(|\mathbf{E}|^2-|\mathbf{B}_\mathrm{t}|^2)$ and $\mathcal{G}=\mathbf{E}\cdot\mathbf{B}_\mathrm{t}$, with $\mathbf{D}=\partial L/\partial\mathbf{E}$ and $\mathbf{H}=-\partial L/\partial\mathbf{B}_\mathrm{t}$,

$$\mathbf{D}\cdot\mathbf{H}-\mathbf{E}\cdot\mathbf{B}_\mathrm{t}=\mathcal{G}(L_\mathcal{F}^2-L_\mathcal{G}^2-1)-2\mathcal{F}L_\mathcal{F}L_\mathcal{G},$$

which vanishes for the Maxwell and Born–Infeld Lagrangians, the duality-invariant ones [43, 49]. For the weak-field Euler–Heisenberg Lagrangian [60, 36] it is $-\frac{8\alpha^2}{15m^4}\mathcal{F}\mathcal{G}+O(\alpha^3)$: the duality defect of the vacuum of quantum electrodynamics is proportional, at leading order, to $\mathbf{E}\cdot\mathbf{B}_\mathrm{t}$. For every parity-even Lagrangian $L=h(\mathcal{F},\mathcal{G}^2)$ it carries the explicit factor $\mathcal{G}$, so it vanishes on every orthogonal field.

14.4 Non-orthogonality does not generate mass

Mass can enter in four ways: as the rest energy of the field itself, as a photon mass, as a mass of charged matter in the background, and through an $\mathbf{E}\cdot\mathbf{B}$-type term in an action. None of them is produced by non-orthogonality.

Rest energy. For $q\neq 0$ the field has a local rest frame, moving with velocity $\mathbf{S}/(u+\frac{1}{2}|q|)$, in which the Poynting vector vanishes and the energy density is $\frac{1}{2}|q|=u\sqrt{1-|\mathbf{v}_E|^2}$, $\mathbf{v}_E=\mathbf{S}/u$; null fields transport their energy at the speed of light and have no rest frame. The rest-frame energy density is the modulus of q, orthogonality its phase, and duality turns the phase at fixed energy, momentum and stress. For a finite configuration with total energy U and momentum $\mathbf{P}$, $M^2=U^2-|\mathbf{P}|^2$ is positive whenever $q\neq 0$ on a set of positive measure, but M does not detect orthogonality: the Cauchy data $(\mathbf{E},\mathbf{B}_\mathrm{t})=(0,\mathbf{b})$ and $(\mathbf{b}\sin\vartheta,\mathbf{b}\cos\vartheta)$ for a compactly supported divergence-free $\mathbf{b}$ have the same energy density and zero Poynting vector at every point, hence the same M, while $\mathbf{E}\cdot\mathbf{B}_\mathrm{t}$ is 0 for the first and $\frac{1}{2}\sin 2\vartheta\,|\mathbf{b}|^2$ for the second.

Photon mass. Every Lorentz-covariant linear map from two-forms to vectors is zero; more strongly, a continuous covariant vector-valued function of the value of F at a point vanishes, because a non-null field is fixed by a rotation and a boost about a common axis that have no common fixed vector, and every tensor power of $(1,0)\oplus(0,1)$ contains only representations (j_1,j_2) with integer j_1,j_2. A mass term in $\partial_\mu F^{\mu\nu}=j^\nu$ must therefore involve the potential, other fields or derivatives of F; no function of the two invariants can serve. Derivatives give a massive mode that is a ghost, as Podolsky's Lagrangian shows. A constant nonlinear background moves the light cone without opening a gap, $\det W(s\omega,s\mathbf{k})=s^6\det W(\omega,\mathbf{k})$, and to first order the two phase velocities are $1-c_iT^{\mu\nu}k_\mu k_\nu$ for $L=\mathcal{F}+c_1\mathcal{F}^2+c_2\mathcal{G}^2$, which is duality invariant and blind to orthogonality. Written into the field equations, the binding rotation of Section 16 gives the frequencies $\kappa\pm|\mathbf{k}|$ to the two circular polarisations, a frequency offset that the time-dependent duality rotation $z=e^{-i\kappa t}w$ removes.

Charged matter. The invariant ϵ_E vanishes exactly when $\mathbf{E}\cdot\mathbf{B}=0$ and $|\mathbf{B}|\geq|\mathbf{E}|$, so every non-orthogonal field has $\epsilon_E>0$ and makes the vacuum of spinor electrodynamics decay into pairs [120], converting field energy into particles whose mass is a given parameter. For massless fermions the same field pumps chirality, $dN_5/dt=(e^2/2\pi^2)\int\mathbf{E}\cdot\mathbf{B}_\mathrm{t}$, the axial anomaly [1, 10, 95]; a fermion mass relaxes it, and none is produced. The background that does catalyse a fermion mass is a pure magnetic field, which is orthogonal [57].

Topological terms. In four dimensions the abelian term $\theta\,\mathbf{E}\cdot\mathbf{B}_\mathrm{t}$ is a total derivative and invisible to the field equations; a varying θ makes it a source, as in axion electrodynamics [144], and a constant gradient gives birefringence [22]. For a timelike gradient both branches remain gapless; for a spacelike gradient p^μ the dispersion relation $(k\cdot k)^2+(k\cdot k)(p\cdot p)-(k\cdot p)^2=0$ gives, at zero wave vector, $\omega=0$ on one branch and $\omega=|\mathbf{p}|$ on the other. That gap is a Lorentz-violating dispersion fixed by an external pseudoscalar, not a Proca mass, and it does not arise without the varying θ. In every known mechanism in which a term of this type makes mass, something is essential that a single abelian field on a topologically trivial space lacks: the Chern–Simons term in three dimensions [34], the anomaly in the two-dimensional Schwinger model [121], the topological susceptibility of a non-abelian theory with light fermions for the η'

Maxwell field	Dirac field
duality $e^{\vartheta G}$, turning q by 2ϑ	chiral rotation $e^{i\alpha\gamma_5}$, turning $\sigma + i\omega$ by 2α
energy–momentum tensor invariant	vector current J^μ invariant
$u^2 - \lvert\mathbf{S}\rvert^2 = \frac{1}{4}\lvert q\rvert^2$	$J^\mu J_\mu = \sigma^2 + \omega^2$
$\lvert\mathbf{B}\rvert^2 - \lvert\mathbf{E}\rvert^2$ and $2\,\mathbf{E}\cdot\mathbf{B}$	σ and ω
null field, $q = 0$	Weyl spinor, $\sigma = \omega = 0$
one mode: q real	one plane wave: $\omega = 0$
two modes whose planes span spacetime: $\operatorname{Im} q \neq 0$	both signs of frequency: $\omega \neq 0$
$\arg q$, twice the Rainich complexion	$\arg(\sigma + i\omega)$, the Yvon–Takabayasi angle

Table 1: The invariants of a field strength and the bilinears of a Dirac spinor. Each line is an identity or a worked example; the parallel is exact at the level of these identities and is not an identification of the two fields.

[147, 137], or a pseudoscalar partner for the axion [140, 143]. The frame field of this arena is a single abelian field strength with no fermions and no pseudoscalar partner, so no route from $\mathbf{E}\cdot\mathbf{B}$ to mass is open within what the arena has built.

The Dirac field supplies an exact counterpart, collected in Table 1. Its scalar and pseudoscalar densities $\sigma = \bar\psi\psi$ and $\omega = \bar\psi i\gamma_5\psi$ turn under a chiral rotation $e^{i\alpha\gamma_5}$ by the angle 2α, as $(\lvert\mathbf{B}\rvert^2 - \lvert\mathbf{E}\rvert^2, 2\mathbf{E}\cdot\mathbf{B})$ turn under duality; a plane wave has $\omega = 0$ as a single mode has $\operatorname{Im} q = 0$; and a pseudoscalar mass enters the mass shell only through $\sqrt{m_s^2 + m_p^2}$. The pseudoscalar density is a phase, not a mass, and $\mathbf{E}\cdot\mathbf{B}$ has exactly its place.

In the interior dynamics of the next part, orthogonality is broken by the bound motion itself. On an orbit without momentum the flow is the duality rotation $z \mapsto e^{-i\kappa\tau}z$, so $\mathbf{E}\cdot\mathbf{B}(\tau) = \frac{1}{2}\lvert q_0\rvert\sin(\phi_0 - 2\kappa\tau)$, which vanishes four times per period, and the rest-frame field turns from magnetic through screws to electric and back at the fixed rest-frame energy density $\frac{1}{2}\lvert q_0\rvert$. That is the content of Figure 10 read along a trajectory, and Section 16 derives it. It is a rotation of the phase of q at fixed modulus, and by everything in this section it moves no mass.

Part IV
Interior dynamics

15 The shifted Lagrangian in a positive form

Everything so far has been kinematics. The arena supplies one piece of interior dynamics without further input: a first-order shift operator built from the conversion between times and lengths (Section 3). This part asks what motion that operator generates. The answer depends on a single choice, whether the quadratic form in the Lagrangian is positive definite or the split form η of the arena itself, and the two answers are opposite. This section treats the positive form, where the verdict is negative and sharp; Section 16 treats the split form, where bound motion appears exactly in neutral signature.

15.1 Momentum and force

Let $X(\tau) \in \mathbb{R}^N$ be an interior configuration, g a constant real $N \times N$ matrix, and

$$\alpha_\pm := \partial_\tau \pm \kappa g, \qquad \mathcal{L} = \frac{m}{2}\left|\alpha_+ X\right|^2 = \frac{m}{2}\left|\dot X + \kappa g X\right|^2. \tag{20}$$

Only the product κg appears: $\kappa = c/\ell$ carries the conversion of (7) and g its direction. The Lagrangian is the square of a single first-order operator, which is what makes it the natural object to ask Clifford questions of.

Proposition 15.1 (Momentum and force)**.** *The canonical momentum and the Euler–Lagrange equation are*

$$p = \frac{\partial \mathcal{L}}{\partial \dot{X}} = m\,\alpha_+ X, \qquad \dot{p} = \kappa\, g^\top p, \tag{21}$$

and in second-order form, with $a := \frac{1}{2}(g - g^\top)$*,*

$$\mathcal{E}_g := m(\ddot{X} + 2\kappa a\dot{X} - \kappa^2 g^\top g X) = 0. \tag{22}$$

Proof. $\partial\mathcal{L}/\partial\dot{X} = m(\dot{X}+\kappa g X)$ and $\partial\mathcal{L}/\partial X = \kappa g^\top p$. Substituting $p = m(\dot{X}+\kappa g X)$ into $\dot{p} = \kappa g^\top p$ gives $m(\ddot{X} + \kappa g\dot{X}) = \kappa m g^\top \dot{X} + \kappa^2 m g^\top g X$, which rearranges to (22). □

The force law is $\dot{p} = \kappa g^\top p$. It takes the symmetric-looking form $\dot{p} = \kappa g p$ exactly when g is symmetric, and the next two results show that this is not a cosmetic condition.

Theorem 15.2 (Factorisation)**.** $\mathcal{E}_g = m\,\alpha_-\alpha_+ X$ *identically if and only if* $a = 0$*.*

Proof. For constant g, $\alpha_-\alpha_+ X = \ddot{X} - \kappa^2 g^2 X$, so $\mathcal{E}_g - m\alpha_-\alpha_+ X = m(2\kappa a\dot{X} + \kappa^2(g^2 - g^\top g)X)$. The two terms involve $\dot{X}$ and X independently, and both vanish for all trajectories if and only if $a = 0$, which makes g symmetric and $g^2 = g^\top g$. □

Proposition 15.3 (The factorised branch is an inverted oscillator)**.** *The Noether energy*

$$E = \langle p, \dot{X}\rangle - \mathcal{L} = \frac{m}{2}|\dot{X}|^2 - \frac{m\kappa^2}{2}\,|gX|^2 \tag{23}$$

is conserved for every g*. When* $a = 0$ *the motion is* $m\ddot{X} = -\nabla V$ *with* $V = -\frac{1}{2}m\kappa^2|gX|^2 \le 0$*, and the initial data whose solutions stay bounded for* $\tau \ge 0$ *form a linear subspace of dimension exactly* N *in the* $2N$*-dimensional phase space; for all real* τ *only the zero-frequency sector, of dimension* $\dim\ker g$*, stays bounded.*

Proof. Writing $u = \alpha_+ X$, $E = \frac{m}{2}|u|^2 - m\kappa\langle u, gX\rangle$, and expanding u gives (23). On shell $\dot{E} = \frac{\kappa}{m}\langle p, (g^\top - g)p\rangle = -\frac{2\kappa}{m}\langle p, ap\rangle = 0$ because a is antisymmetric. With $a = 0$ the equation is $\ddot{X} = \kappa^2 g^2 X$, and on the eigenplane of an eigenvalue $\lambda \neq 0$ of g the solutions are $Ae^{\kappa|\lambda|\tau} + Be^{-\kappa|\lambda|\tau}$, bounded for $\tau \ge 0$ iff $A = 0$; on $\lambda = 0$ they are $x_0 + v_0\tau$, bounded iff $v_0 = 0$. In each plane one condition is imposed on two dimensions. □

The factorisation condition and the inverted-potential condition are the same condition. Since $g^\top g \succeq 0$, the potential is never a well, and the bounded set has measure zero: no open set of initial data stays bounded. Only the velocity-coupled term $2\kappa a\dot{X}$ could bind, and it is exactly the term that obstructs factorisation.

Example 15.4 (The conversion operator as the shift)**.** Take $g = B_C$, the mixed bivector of Theorem 3.6 acting on S, which is symmetric. The equation of motion is $\ddot{X} = \kappa^2 B_C^2 X$, and by Theorem 3.6 the growth rates are $\kappa|\eta_1\sigma_1 + \eta_2\sigma_2 + \eta_3\sigma_3|$ for the four sign patterns η. For the symmetric clock, C has singular values $(1, 0, 0)$, every rate equals κ, and the dynamics cannot see the Clifford structure (Example 3.8). For C of rank two with $\sigma = (2, 1, 0)$ the rates are 3κ and κ, each on a four-dimensional subspace; for $\sigma = (1, 1, 0)$ they are 2κ and 0, and the second is a flat direction. Raising the rank makes the instability anisotropic and can create flat directions, never an oscillation.

15.2 No recurrence at nonzero momentum

It is tempting to stop here and blame the tension between factorisation and binding. That would be too generous to the unfactorised branch. A single identity settles the question for every g.

Theorem 15.5 (Virial identity)**.** *With the positive form of* (20)*, along every solution and for every constant g,*

$$\frac{d}{d\tau}\langle X,p\rangle = \frac{|p|^2}{m} \;\geq\; 0. \tag{24}$$

Consequently: (i) no solution with $p(0)\neq 0$ is periodic or recurrent; (ii) purely imaginary characteristic exponents may occur, but never semisimply, so the motion they carry is secular, and the Jordan blocks need not have size two; (iii) a trajectory bounded in phase space has $p\in L^2$, so either $p\equiv 0$ or p decays exponentially and the orbit is not recurrent.

Proof. $\frac{d}{d\tau}\langle X,p\rangle = \langle \dot X,p\rangle + \langle X,\kappa g^{\top}p\rangle = \langle \dot X+\kappa gX,p\rangle = |p|^2/m$. A recurrent orbit would return $\langle X,p\rangle$ to an earlier value, which (24) forbids unless $p\equiv 0$, and $p = e^{\kappa g^{\top}\tau}p(0)$ vanishes identically only if $p(0)=0$. A semisimple imaginary exponent would supply a periodic solution with $p\neq 0$. For (iii), boundedness makes $\langle X,p\rangle$ bounded and monotone, hence convergent, so $\int_0^\infty |p|^2<\infty$, and a bounded square-integrable solution of a constant-coefficient linear system is zero or exponentially decaying. □

Example 15.6 (A Jordan block of size four)**.** Let J_2 be the standard 2×2 rotation generator and $g=\left(\begin{smallmatrix} J_2 & \mathbf{1}_2 \\ 0 & J_2\end{smallmatrix}\right)$ on $\mathbb{R}^4$, with $\kappa=m=1$. The eight-dimensional phase-space generator M has its spectrum on $\pm i$, and $\operatorname{rank}\big((M^2+\mathbf{1})^k\big)=6,4,2,0$ for $k=1,2,3,4$. The blocks therefore have size four and the generic solution grows like τ^3, although every exponent is purely imaginary.

The zero-momentum sector is not empty, and it does oscillate. The general solution is

$$p(\tau)=e^{\kappa g^{\top}\tau}p_0, \qquad X(\tau)=e^{-\kappa g\tau}\Big[X_0+\frac{1}{m}\int_0^\tau e^{\kappa g s}e^{\kappa g^{\top}s}p_0\,ds\Big], \tag{25}$$

and the kernel of α_+, $X(\tau)=e^{-\kappa g\tau}X_0$, has $p\equiv 0$. The integral collapses to τp_0 only when $g^{\top}=-g$. For $g=J$ antisymmetric with $J^2=-\mathbf{1}$, the zero-momentum solutions are bounded rotations of period $2\pi/|\kappa|$, and any nonzero momentum adds the secular term $\tau\, e^{-\kappa J\tau}p_0/m$: the orbit becomes an outward spiral (Figure 12, left). In that marginal case $g^{\top}g=\mathbf{1}$, $a=J$, the equation reads $\ddot X+2\kappa J\dot X-\kappa^2X=0$, and in the eigenbasis of J the characteristic polynomial is $(\lambda+i\kappa)^2$, a double root on the imaginary axis; the corresponding Jordan block is exactly the instability just described.

So with a positive form, oscillation exists only on a measure-zero sector and is destroyed by any momentum. What is missing is neither curvature nor a dependence of g on τ. It is the positive form itself, and the next section removes it.

15.3 Towards a stress–energy tensor

The worldline statement lifts to a curved metric in two ways, which together indicate where a stress–energy tensor would come from. First, replacing ∂_τ by $D/D\tau$ along a worldline in (M,g) and promoting g to a section G of $\operatorname{End}(TM)$, the variation of $\int \frac{m}{2}g_{AB}(\alpha_+X)^A(\alpha_+X)^B\,d\tau$ gives $Dp_B/D\tau=\kappa G^A{}_Bp_A$, and the second-order form acquires one new term,

$$\frac{D^2X}{D\tau^2}+2\kappa a\frac{DX}{D\tau}+\kappa(D_\tau G)X-\kappa^2G^{\top}GX=0;$$

the factorisation computation goes through verbatim, since the extra term appears on both sides, so $a=0$ remains the factorisation condition. Second, for a field X in a bundle with fibre metric h and $D_M=\nabla_M+\kappa G_M$, the action $\int\sqrt{|g|}\,\frac{m}{2}h_{AB}g^{MN}(D_MX)^A(D_NX)^B$ has

$$T_{MN}=m\Big[h_{AB}(D_MX)^A(D_NX)^B-\tfrac12 g_{MN}\,h_{AB}\,g^{PQ}(D_PX)^A(D_QX)^B\Big],$$

whose trace $m(1-\frac{D}{2})|DX|^2$ vanishes only in $D=2$, the same dimension in which the projector sigma model of Section 11.5 is renormalisable. This is the stress tensor of the shifted kinetic

term only, with G a background, and Theorem 15.5 is a warning about it: a matter sector whose only oscillations sit at zero momentum, with an energy unbounded below, is not yet a source one can couple to gravity.

16 The split form and binding in neutral signature

The norm in (20) is positive definite, and that is what makes $g^\top g \succeq 0$ in Proposition 15.3 and $|p|^2 \geq 0$ in Theorem 15.5. But the arena does not hand us a positive form; it hands us η, of signature $(3,3)$. Redoing the derivation with η changes one inequality, and with it the verdict.

16.1 The split-form Lagrangian

Write $G^\dagger := \eta^{-1} G^\top \eta$ for the η-adjoint and take

$$\mathcal{L} = \frac{m}{2}\,\eta(\alpha_+ X, \alpha_+ X), \qquad p = m\,\eta\,(\alpha_+ X), \qquad \pi = \eta^{-1} p, \qquad \dot{p} = \kappa\, G^\top p, \tag{26}$$

with second-order form and energy

$$\ddot{X} + 2\kappa a \dot{X} - \kappa^2 G^\dagger G X = 0, \qquad a = \tfrac{1}{2}(G - G^\dagger), \qquad E = \frac{m}{2}\eta(\dot{X},\dot{X}) - \frac{m\kappa^2}{2}\eta(GX, GX). \tag{27}$$

Everything in Section 15 survives with the transpose replaced by the η-adjoint: factorisation holds exactly when $a = 0$, and E is conserved because a is η-antisymmetric. What does not survive is the sign of the virial identity.

Theorem 16.1 (The split virial identity)**.** *Let X be a section along a path in a manifold with metric of signature $(3,3)$ and let $D/D\tau$ come from any metric-compatible connection, torsion included. Then for every G,*

$$\frac{d}{d\tau}\,\eta(X,\pi) = \frac{\eta(\pi,\pi)}{m}. \tag{28}$$

Consequently every closed orbit satisfies $\oint \eta(\pi,\pi)\,d\tau = 0$: a bound orbit has null momentum on average.

Proof. Note that $\eta(X,\pi) = \langle p, X\rangle$. Metric compatibility gives $\frac{d}{d\tau}\eta(X,\pi) = \eta(DX/D\tau,\pi) + \eta(X, D\pi/D\tau)$, and torsion does not enter this Leibniz rule. Substituting $DX/D\tau = \pi/m - \kappa G X$ and $D\pi/D\tau = \kappa G^\dagger \pi$ gives $\eta(\pi,\pi)/m - \kappa\,\eta(GX,\pi) + \kappa\,\eta(X, G^\dagger\pi)$, and the last two terms cancel by the definition of the adjoint. □

Neither curvature nor torsion appears in (28); the identity sees only η. In definite signature $\eta(\pi,\pi)$ has one sign and the identity forbids recurrence, which is Theorem 15.5. In split signature it does not, and the question becomes whether solutions with null momentum on average exist. They do, and the operator that produces them is one we already have.

16.2 Binding exactly in neutral signature

Theorem 16.2 (Factorisation and binding coexist exactly in neutral signature)**.** *Let η have signature (P,Q) on $\mathbb{R}^N$. There is a linear map G with*

(i) $G = G^\dagger$, so that the factorisation of Theorem 15.2 holds, and

(ii) $G^2 = -\mathbf{1}$, so that (27) reads $\ddot{X} = -\kappa^2 X$ and every solution is periodic of period $2\pi/\kappa$,

if and only if $P = Q$. Such a G is an anti-isometry, $\eta(GX,GY) = -\eta(X,Y)$, and on $\mathbb{R}^{3,3}$ one of them, maximal in that it uses every paired direction, is the binding root of Proposition 3.3:

$$G:\ e_i \longmapsto e_{i+3}, \qquad e_{i+3} \longmapsto -e_i \qquad (i = 1,2,3). \tag{29}$$

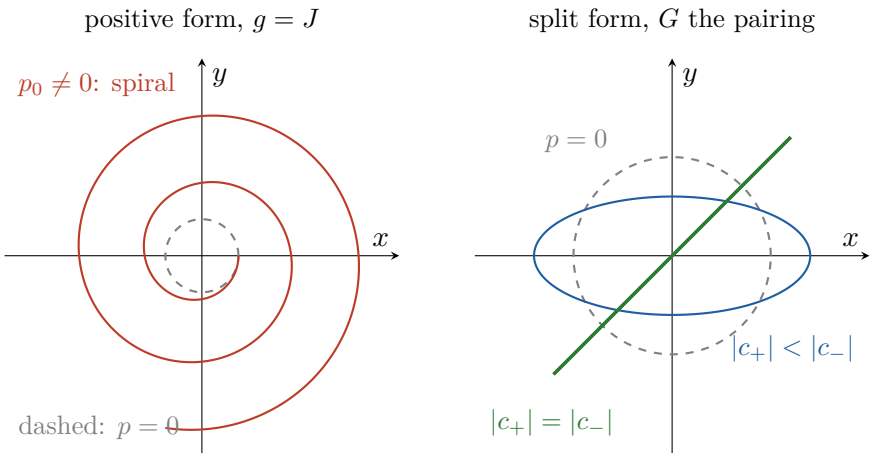


Figure 12: Interior motion in one pairing plane (Example 16.4). Left: with the positive form and an antisymmetric shift the zero-momentum orbits are circles, and any momentum turns them into outward spirals, as Theorem 15.5 requires. Right: with the split form and the binding root G every orbit is closed; the zero-momentum orbits are circles, the momentum-carrying ones ellipses, degenerating to a segment when the two rotating parts have equal size.

Proof. Given (i) and (ii), $\eta(GX, GY) = \eta(X, G^\dagger GY) = \eta(X, G^2 Y) = -\eta(X, Y)$, so G carries a maximal positive subspace isometrically onto a negative one and back, which forces $P = Q$. Conversely (29) satisfies $G^2 = -\mathbf{1}$, and $\eta(Ge_i, e_{i+3}) = \eta(e_{i+3}, e_{i+3}) = -1 = \eta(e_i, -e_i) = \eta(e_i, Ge_{i+3})$, so $G = G^\dagger$. With $G^\dagger G = G^2 = -\mathbf{1}$, (27) has $a = 0$ and becomes $\ddot{X} = -\kappa^2 X$. □

This reverses the verdict of Section 15 at its root. In definite signature the factorised branch is an inverted oscillator and nothing binds; in neutral signature the same factorisation condition $G = G^\dagger$ admits operators of purely imaginary spectrum, and the maximal one makes the whole interior a harmonic well of one frequency. The obstruction of Theorem 15.5 was a theorem about the positive form, not about the arena. Two qualifications keep the statement honest. The normalisation $G^2 = -\mathbf{1}$ imposes one common frequency; any η-self-adjoint shift built from independent paired blocks $\mu_i J_2$ with distinct $\mu_i > 0$ also factorises and binds, at unequal frequencies, so the construction is a family and $G^2 = -\mathbf{1}$ picks a member. And no equation of motion for G has been written: G is adopted, and calling its consequences a selection would be premature.

Proposition 16.3 (How much can bind)**.** *For $G = G^\dagger$ an eigenvector v with non-real eigenvalue satisfies $\eta(v, \bar{v}) = 0$, so each conjugate pair $\pm i\mu$ spans an η-neutral two-plane. The oscillating subspace has signature (k, k) with $k \leq \min(P, Q)$. On Minkowski space $\mathbb{R}^{1,3}$ at most a single two-plane can bind; on $\mathbb{R}^{3,3}$ all six directions can, simultaneously and at one frequency.*

Example 16.4 (One pairing plane)**.** In the plane (e_1, e_4) write $X = x\,e_1 + y\,e_4$, so that G acts as $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $z = x + iy$. The general solution of $\ddot{X} = -\kappa^2 X$ is $z(\tau) = c_- e^{-i\kappa\tau} + c_+ e^{i\kappa\tau}$, an ellipse with semi-axes $|c_-| + |c_+|$ and $||c_-| - |c_+||$. The momentum $\alpha_+ X = \dot{X} + \kappa G X$ corresponds to $\dot{z} + i\kappa z = 2i\kappa\, c_+ e^{i\kappa\tau}$, so it is carried entirely by the counter-rotating part, and the zero-momentum orbits are the circles $z = c_- e^{-i\kappa\tau}$, traversed at angular velocity $-\kappa$. The virial quantity $\eta(\pi, \pi) = m^2 \eta(\alpha_+ X, \alpha_+ X) = m^2 \operatorname{Re}((2i\kappa c_+ e^{i\kappa\tau})^2)$ oscillates with mean zero, as Theorem 16.1 requires of a closed orbit.

16.3 The orbits as duality rotations

Read through the field-strength dictionary of Theorem 12.3, an interior configuration X is a two-form and the binding root is the Hodge star. The bound motion then has a transparent

meaning.

Proposition 16.5 (Bound orbits in field-strength language)**.** *Let G be the pairing (29) and $z = \mathbf{B} + i\mathbf{E}$.*

(i) The zero-momentum orbits are $X(\tau) = e^{-\kappa G\tau}X_0$, that is $z(\tau) = e^{-i\kappa\tau}z_0$: continuous duality rotations, along which $q(z(\tau)) = e^{-2i\kappa\tau}q(z_0)$. A non-null orbit is a single spacetime plane at exactly four instants per period, alternately of magnetic and electric type, and a null orbit is a single plane at every instant.

(ii) Every solution is $z(\tau) = c_-e^{-i\kappa\tau} + c_+e^{i\kappa\tau}$ with $c_\pm \in \mathbb{C}^3$, the momentum corresponds to $2i\kappa c_+e^{i\kappa\tau}$, and the period averages are

$$\langle q\rangle = 2\,c_-\cdot c_+, \qquad \langle \mathbf{E}\cdot\mathbf{B}\rangle = \operatorname{Im}(c_-\cdot c_+), \qquad \langle |\mathbf{B}|^2 - |\mathbf{E}|^2\rangle = 2\operatorname{Re}(c_-\cdot c_+).$$

(iii) The energy is $\mathcal{E} = 2m\kappa^2\operatorname{Re}(c_-\cdot c_+)$, and the complex quantity $\frac{m}{2}(q(\dot z)+\kappa^2 q(z)) = 2m\kappa^2\, c_-\cdot c_+$ is conserved; its imaginary part is a second constant of motion.

(iv) On the orbits with $c_+ = \bar c_-$ the vector z is real at every instant: $\mathbf{E} = 0$ and $\mathbf{E}\cdot\mathbf{B} = 0$ throughout, while the momentum is not zero.

On a zero-momentum orbit $\mathbf{E}\cdot\mathbf{B}(\tau) = \frac{1}{2}|q_0|\sin(\phi_0 - 2\kappa\tau)$ with $q_0 = |q_0|e^{i\phi_0}$, and the rest-frame field of Section 14 turns from purely magnetic through screws to purely electric and back, at the fixed rest-frame energy density $\frac{1}{2}|q_0|$. So the binding flow is what breaks orthogonality on every non-null orbit without momentum, while orbits that carry momentum can stay orthogonal throughout. The mean non-orthogonality and the energy of a general orbit are the imaginary and real parts of one conserved complex number $c_-\cdot c_+$, and neither determines the other.

Theorem 16.6 (The parity-odd term is a total derivative)**.** *For the split-form Lagrangian with the pairing,*

$$\mathcal{L}_\kappa = \frac{m}{2}\Big(\eta(\dot X,\dot X) - \kappa^2\eta(X,X)\Big) - m\kappa\,\frac{d}{d\tau}(\mathbf{E}\cdot\mathbf{B}).$$

Consequently the cross term has identically vanishing Euler–Lagrange expression, the equation $\ddot X = -\kappa^2 X$ comes from the first bracket alone, and the momenta of $\mathcal{L}_\kappa$ and $\mathcal{L}_{-\kappa}$ differ by $2m\kappa\,\eta GX$. Parity maps $\mathcal{L}_\kappa$ to $\mathcal{L}_{-\kappa}$ and exchanges the orbits without momentum with those carried entirely by the counter-rotating part, while C and T leave $\mathcal{L}_\kappa$ invariant.

Proof. G is η-self-adjoint, so $\frac{d}{d\tau}\eta(X,GX) = 2\eta(\dot X, GX)$, and $\eta(X,GX) = -2\,\mathbf{E}\cdot\mathbf{B}$ by Theorem 12.3; $\eta(GX,GX) = -\eta(X,X)$ because G is an anti-isometry. Expanding $\eta(\dot X + \kappa GX, \dot X + \kappa GX)$ gives the formula. Parity reverses $\mathbf{E}$ and fixes $\mathbf{B}$, so $PGP^{-1} = -G$ and P acts on z as complex conjugation. □

The binding term carries the one parity-odd ingredient of the interior, the sign of κ, and it sits entirely in a total derivative. Since $\mathbf{E}\cdot\mathbf{B}$ is a single-valued function on the configuration space, the canonical transformation $p \mapsto p + 2m\kappa\,\eta GX$ removes it, classically and as a phase of wave functions. The structure is that of the θ-term of electrodynamics, a parity-odd coefficient times a total derivative, with the difference that here it has no topological sector in which to act. In the positive form the same cross term is gyroscopic, with Euler–Lagrange expression $2m\kappa G\dot X$, and the sign of κ enters the equations of motion; that is another way of seeing why the two forms behave so differently.

16.4 What the binding root selects, and what tears it

The binding root is not merely available; it carries structure. By Theorem 12.1 its centraliser is a Lorentz algebra $\mathfrak{l}_G$, not conjugate to the interior one (Proposition 12.2). The same operator limits what transport can do to the bound sector.

Proposition 16.7 (Transport and a homogeneous threshold)**.** *Let a connection along the path take values ω in $\mathfrak{so}(3,3)$. If ω lies in the compact part of $\mathfrak{l}_G$, transport and the shift commute and the orbit is carried rigidly. If ω has a boost component of strength c in a pairing plane (e_a, e_{a+3}), the homogeneous 2×2 block there has exponents $\pm\sqrt{c^2-\kappa^2}$: oscillatory for $c<\kappa$ and real for $c>\kappa$.*

Proof. In the basis (e_a, e_{a+3}) the boost generator is $\sigma = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and G is $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, so $-c\sigma + \kappa G = \left(\begin{smallmatrix} 0 & -c-\kappa \\ -c+\kappa & 0 \end{smallmatrix}\right)$, whose square is $(c^2-\kappa^2)\mathbf{1}$. □

The threshold classifies exponents of the homogeneous block, not boundedness of the full flow, and the difference is real. With the momentum included, the phase-space generator $M = \left(\begin{smallmatrix} -c\sigma-\kappa G & \mathbf{1}/m \\ 0 & -c\sigma+\kappa G \end{smallmatrix}\right)$ satisfies $M^2 - (c^2-\kappa^2)\mathbf{1} = \left(\begin{smallmatrix} 0 & -2c\sigma/m \\ 0 & 0 \end{smallmatrix}\right)$, nilpotent and nonzero for $c \neq 0$. At $\kappa = m = 1$ and $c = \frac{1}{2}$ one finds $\det(\lambda\mathbf{1} - M) = (\lambda^2 + \frac{3}{4})^2$ with $\operatorname{rank}(M^2 + \frac{3}{4}\mathbf{1}) = 2$: Jordan blocks, and generic secular growth strictly below the threshold. Along an open path, moreover, a connection coefficient can be changed by a frame rotation, so growth on one worldline proves nothing; the invariant is the holonomy around a closed path. Section 23 computes it in an explicit curved model and finds that normal curvature converts zero-momentum orbits into momentum-carrying ones, through exactly the failure of the normal rotation to commute with G.

What this part delivers is a bound sector, a frequency, a centralising Lorentz algebra and an exponent threshold, all from an adopted binding operator. It does not deliver a spectrum: κ is not fixed, the single frequency is degenerate across the three pairs, and the energy is indefinite, so these are bound orbits and not bound states in any quantum sense. Nor does it settle the relation between the two square roots of the pairing beyond Lemma 3.4: $\mathcal{K}$ is η-antisymmetric and grades, G is η-self-adjoint and binds, and they differ by the Cartan involution $\mathrm{Ad}_{e_1e_2e_3}$.

17 Closing the force law

The force law $\dot{p} = \kappa G^\top p$ carries its rate κ as a constant put in by hand. A force law of this shape is familiar from rigid bodies and ideal fluids, where the equation of motion is a geodesic equation on a Lie group and the rate is itself dynamical. This section asks whether the interior law can be closed in that sense: whether there is a Lie–Poisson system whose bracket and inertia are built from the same data that build G, and whose solutions contain the force law with κ as a dynamical variable. A bracket or an inertia is called *derived from the Clifford structure* when it is equivariant under the algebra $\mathfrak{l}_G$ of Theorem 12.1 and built from η, G and the bivectors. The answers, in brief: no closure lives on the frame alone; the closure on one extra direction is unique; its inertia is constrained by two requirements that exclude each other; and the dynamics it generates is a pendulum in the duality angle that drifts the frame except on null data.

Throughout, $V = \mathbb{R}^{3,3}$, a covector p is a column with $\langle p, X\rangle = p^\top X$, $\pi = \eta^{-1}p$, and $z(X) = x + iy$ for $X = \sum_j (x_j e_j + y_j e_{j+3})$, so that G acts as i. For a covector put

$$C(p) = p^\top \eta^{-1} p, \qquad D(p) = p^\top G \eta^{-1} p, \qquad w(p) := q(z(\pi)) = C(p) - iD(p). \tag{30}$$

In the field-strength reading, with $(\mathbf{B};\mathbf{E}) = \pi$, $w = |\mathbf{B}|^2 - |\mathbf{E}|^2 + 2i\,\mathbf{B}\cdot\mathbf{E}$ is the complex invariant of the field carried by the momentum.

17.1 The force law as a dual transport

Euler–Arnold equations are geodesic equations on a Lie group with a one-sided invariant metric [5, 6]; for a left-invariant metric they read $\dot{\mu} = \mathrm{ad}_w^\top \mu$ with $\mu = Aw$ for an inertia A. The rigid body is the case of $SO(3)$, where this is $\dot{\mathbf{M}} = \mathbf{M} \times \boldsymbol{\Omega}$, and incompressible Euler flow the case of the volume-preserving diffeomorphisms.

Proposition 17.1 (The dictionary). *(i) Since $\langle G^{\top}p, v\rangle = \langle p, Gv\rangle$, the law $\dot p = \kappa G^{\top}p$ transports the covector p by the transpose of κG. It is the Lie–Poisson equation $\dot p = \mathrm{ad}_w^{\top} p$ of a Lie bracket on the frame exactly when $\kappa G = \mathrm{ad}_w$ is an inner derivation.*

(ii) For a bracket with trivial centre on the $6n$-dimensional frame the inner derivations fill a $6n$-dimensional subspace of $\mathfrak{so}(3n, 3n)$, of dimension $3n(6n-1)$, when η is invariant: a fraction $2/(6n-1)$.

(iii) When $\kappa G = \mathrm{ad}_u$ and $p = Au$, the shift operators $\alpha_{\pm} = \partial_\tau \pm \kappa G$ are the constrained variations $\delta u = \dot\xi \pm [u, \xi]$ of Euler–Poincaré reduction [83, 66], in the body and in the spatial frame.

Proposition 17.2 (The binding root is inner for no bracket). *For every Lie bracket and every w, $\mathrm{ad}_w(w) = 0$, so every inner derivation is singular. The binding root satisfies $G^2 = -\mathbf{1}$ and is invertible, so κG is inner for no Lie bracket on $\mathbb{R}^{3n,3n}$, invariant or not, and the force law with that G is never a Lie–Poisson equation on the frame itself. It is one on the extension $\mathbb{R}\hat g \ltimes \mathbb{R}^{3n,3n}$, the frame taken abelian and $\hat g$ acting on it by G, where it is the Lie–Poisson equation of the fixed element $\kappa\hat g$ restricted to the ideal.*

Theorem 17.3 (Transport and factorisation are complementary). $\mathrm{End}(\mathbb{R}^{3n,3n}) = \mathfrak{so}(3n, 3n) \oplus \{G : G = G^{\dagger}\}$, *of dimensions $3n(6n-1) + 3n(6n+1) = 36n^2$. Along $\dot p = \kappa G^{\top}p$ the Casimir $C = \eta^{-1}(p, p)$ changes at the rate $dC/d\tau = 2\kappa\, p^{\top} G\eta^{-1}p$, which vanishes for every p exactly when G lies in the first summand, while factorisation in the split form requires the second. The binding root lies in the second summand and moves the Casimir at the rate $dC/d\tau = -4\kappa\, \mathbf{B}\cdot\mathbf{E}$ in the variables $(\mathbf{B}; \mathbf{E}) = \eta p$.*

At the first rung the frame itself is a Lie algebra. With $L_i = e_i$ and $K_i = e_{i+3}$ obeying the brackets of Theorem 12.3, $\mathbb{R}^{3,3}$ becomes $\mathfrak{so}(1, 3)$ with Killing form -4η; in the variable $z = \mathbf{B} + i\mathbf{E}$ the bracket is the complex cross product, G is multiplication by i and lies in the centroid, $[Gx, y] = G[x, y]$, and $z \cdot z$ is the form q. The inner derivations fill six of the fifteen dimensions of $\mathfrak{so}(3, 3)$. Closing the force law into an Euler–Arnold equation on the frame would need a bracket for which the shift is inner, and by Proposition 17.2 there is none. A closure must leave the frame.

17.2 No closure on the frame; the rotor is unique

Proposition 17.4 (Equivariant maps and forms). *For $\mathfrak{l}_G = \mathrm{span}\{L_i, K_i\}$: $\mathrm{Hom}_{\mathfrak{l}_G}(V, V) = \mathrm{span}\{\mathbf{1}, G\}$ and $\mathrm{Hom}_{\mathfrak{l}_G}(V, \mathbb{R}) = 0$; $\mathrm{Hom}_{\mathfrak{l}_G}(\Lambda^2 V, V) = \mathrm{span}\{\beta_0, G\beta_0\}$, where $\beta_0(X, Y)$ is the complex cross product $z(X) \times z(Y)$ returned to V, and $\mathrm{Hom}_{\mathfrak{l}_G}(\Lambda^2 V, \mathbb{R}) = 0$; and the $\mathfrak{l}_G$-invariant symmetric forms on V^* are $\mathrm{span}\{\eta^{-1}, G\eta^{-1}\}$.*

Theorem 17.5 (The brackets on the frame). *The $\mathfrak{l}_G$-equivariant Lie brackets on V are $\beta_L = L\beta_0$ with $L = a + bG$. The bracket β_0 makes $(V, \beta_0) \cong \mathfrak{l}_G$, its derivations are the six inner ones, its centroid is $\mathrm{span}\{\mathbf{1}, G\} \cong \mathbb{C}$, so G is not a derivation, and β_L has Killing form $-4\big[(a^2 - b^2)\eta + 2ab\,\eta G\big]$.*

Proof. By Proposition 17.4 every equivariant antisymmetric bilinear map is $(c_1 + c_2 G)\beta_0$. Since L lies in the centroid, $\beta_L(\beta_L(X, Y), Z) = L^2\beta_0(\beta_0(X, Y), Z)$, and β_L inherits the Jacobi identity. □

Theorem 17.6 (No closure on the frame alone). *Let $L = a + bG \neq 0$. For every symmetric N, with $\mathfrak{g}^*$ identified with $\mathfrak{g}$ through η, the Euler–Arnold field $\beta_0(M, NM)$ conserves $\eta(M, M)$ and $\eta(M, GM)$, whereas along the force law, in the form $\dot M = \kappa GM$, $\frac{d}{d\tau}\eta(M, M) = 2\kappa\,\eta(M, GM)$ and $\frac{d}{d\tau}\eta(M, GM) = -2\kappa\,\eta(M, M)$. No Hamiltonian flow on $(V, \beta_L)^*$ coincides with the force law on an open set, and this remains true with a state-dependent rate.*

Proof. The two quadratic forms are the Casimirs of $(V, \beta_L)^*$ and are conserved by every Hamiltonian flow, while the force law rotates them into each other. Equality on an open set would make both vanish there, that is $z \cdot z = 0$ on an open set, which no nonzero polynomial does. For a state-dependent rate, the equations $\beta_0(M_s, NM_s) = f_s\, GM_s$ at twelve rational null vectors already force every $f_s = 0$ and $N \in \operatorname{span}\{\mathbf{1}, G\}$, for which $\beta_0(M, NM) \equiv 0$. □

A closure therefore needs a direction beyond the frame. Adjoin one generator $\hat{g}$, the *rotor*, on which $\mathfrak{l}_G$ acts trivially; the rate κ is a Lorentz scalar, since G commutes with $\mathfrak{l}_G$.

Theorem 17.7 (Uniqueness of the closure)**.** *Let $\mathfrak{s} = \mathbb{R}\hat{g} \oplus V$ carry an $\mathfrak{l}_G$-equivariant Lie bracket for which $\operatorname{ad}_{\hat{g}}$ does not vanish on V. Then $[\hat{g}, X] = (a + bG)X$ and $[X, Y] = 0$ for $X, Y \in V$, with $(a, b) \neq (0, 0)$. If moreover the rotor rotates the frame covector without dilating it, as the force law does, then $a = 0$ and, after rescaling $\hat{g}$, $\mathfrak{s} \cong \mathbb{R} \ltimes_G V$: $[\hat{g}, X] = GX$, $[X, Y] = 0$.*

Proof. Equivariance gives $[\hat{g}, X] = AX + \alpha(X)\hat{g}$ with $A \in \operatorname{span}\{\mathbf{1}, G\}$ and $\alpha = 0$, and $[X, Y] = (c_1 + c_2 G)\beta_0(X, Y)$, the $\hat{g}$-component vanishing by Proposition 17.4. The Jacobi identity for $(\hat{g}, X, Y)$ reads $(a + bG)(c_1 + c_2 G)\beta_0(X, Y) = 0$, a linear system in (c_1, c_2) with determinant $a^2 + b^2 \neq 0$, so $c_1 = c_2 = 0$. On $\mathfrak{s}^*$ the rotor acts on p through $(a + bG)^\top$, and $\frac{d}{d\tau}|p|^2 = 2a\,(h/I_0)|p|^2$ because G is antisymmetric; a pure rotation needs $a = 0$. □

The extension of Proposition 17.2 is therefore not one choice among several: it is the only equivariant closure. The frame is abelian, all the dynamics sits in the rotor, and the rotor acts by the binding root itself. With several constituents the argument runs factor by factor, since equivariant maps between distinct constituents vanish, and a common rotation rate gives $\mathfrak{s}_n = \mathbb{R} \ltimes_{G_n} V_n$ with $G_n = \bigoplus_k G$.

17.3 The Lie–Poisson dynamics, and the two interiors as clamped reductions

Proposition 17.8 (Lie–Poisson equations)**.** *On $\mathfrak{s}^*$ with coordinates (h, p) let $H = h^2/(2I_0) + \frac{1}{2}p^\top Np$, with $I_0 > 0$ the rotor inertia and N a symmetric form on V^*, the inverse frame inertia. The Lie–Poisson equations are*

$$\dot{p} = \frac{h}{I_0}\, G^\top p, \qquad \dot{h} = -\,p^\top GNp, \tag{31}$$

so the force law holds with the dynamical rate $\kappa = h/I_0$, and $|p|^2$ is a Casimir. For $p \neq 0$ the coadjoint orbit through (h, p_0) is the cylinder $\mathbb{R} \times \{e^{\theta G} p_0\} \cong T^*S^1$, on which (θ, h) are canonical for $h^2/(2I_0) + U(\theta)$ with $U(\theta) = \frac{1}{2}p(\theta)^\top Np(\theta)$ and $p(\theta) = e^{-\theta G}p_0$: every reduced motion is a motion of the rotor angle alone.

Proposition 17.9 (The two interiors are clamped reductions)**.** *On the group S with product $(\theta, x)(\theta', x') = (\theta + \theta', x + e^{\theta G}x')$ let $L = \frac{1}{2}I_0\dot{\theta}^2 + \frac{m}{2}\eta(u, u)$ with body velocity $u = e^{-\theta G}\dot{x}$, and put $X = e^{-\theta G}x$. Then $u = \dot{X} + \dot{\theta}\, GX$, the Euler–Lagrange equations are (31) with $N = \eta^{-1}/m$, and clamping $\theta = \kappa\tau$ gives*

$$L = \tfrac{1}{2}I_0\kappa^2 + \tfrac{m}{2}\,\eta(\dot{X} + \kappa GX,\ \dot{X} + \kappa GX),$$

the split-form interior (26). With $\frac{m}{2}|u|^2$ in place of $\frac{m}{2}\eta(u, u)$ the clamped Lagrangian is the positive-form interior (20) with $g = G$.

So the two interiors of Sections 15 and 16 are the clamped reductions of one closure, with inverse inertias $N = \eta^{-1}/m$ and $N = \mathbf{1}/m$. Which inertias the Clifford data permit is the next question.

17.4 Two requirements on the inertia

Definition 17.10 (The two requirements)**.** A symmetric N on V^* satisfies the *Lorentz requirement* if it is $\mathfrak{l}_G$-invariant, $rN + Nr^\top = 0$ for all $r \in \mathfrak{l}_G$, and the *duality requirement* if the rate h/I_0 is conserved for all data. Write $N = N_c + N_a$ with $N_c = \frac{1}{2}(N - GNG)$ commuting and $N_a = \frac{1}{2}(N + GNG)$ anticommuting with G.

Theorem 17.11 (The two requirements exclude each other)**.** *(i) N_c and N_a are symmetric; the commuting forms have dimension 9 and the anticommuting ones 12.*

(ii) The Lorentz requirement gives $N = (a\eta^{-1} + bG\eta^{-1})/m$, a two-dimensional family inside the anticommuting forms, with frame energy $\frac{1}{2}p^\top Np = \mathrm{Re}((a+ib)w(p))/(2m)$.

(iii) The duality requirement holds exactly when $[N, G] = 0$. Among the forms invariant under the compact part of $\mathfrak{l}_G$ the only ones commuting with G are the multiples of $\mathbf{1}$: the positive form.

(iv) The two requirements meet in 0, and they do not exhaust the symmetric forms: $N = (2\,\mathbf{1} + \eta^{-1})/m$ is positive definite and satisfies neither.

In the field-strength reading the dichotomy is the classical one. The Lorentz class consists of the two invariants $|\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\mathbf{B}\cdot\mathbf{E}$; the duality class contains the energy density $|\mathbf{B}|^2 + |\mathbf{E}|^2$; and a duality rotation preserves the energy density while it rotates the Lagrangian density [18, 30, 43]. The split form lies in the Lorentz class and the positive form in the duality class.

Theorem 17.12 (The frame under the closure)**.** *Reconstruct the group curve from the body velocity,*

$$(\theta, x)^{-1}\tfrac{d}{d\tau}(\theta, x) = (h/I_0,\, Np), \qquad \theta(0) = 0.$$

Then $p = e^{-\theta G}p_0$ and

$$\dot{x} = (N_c + e^{2\theta G}N_a)p_0, \qquad \dot{h} = -\,p_0^\top G e^{2\theta G}N_a p_0.$$

With the rotor clamped at $\theta = \kappa\tau$, $\kappa \neq 0$,

$$x(\tau) = x_0 + \tau\, N_c p_0 + \big(e^{2\kappa\tau G} - \mathbf{1}\big)(2\kappa G)^{-1}N_a p_0,$$

so the frame stays bounded for all data exactly when $N_c = 0$ and drifts at the velocity $N_c p_0$ otherwise, while the rate is constant for all data exactly when $N_a = 0$.

A clamped rate that is also the dynamical one, and a bounded clamped frame, are therefore exclusive requirements: each is met by one summand of $\mathrm{Sym}(V^*) = \mathrm{Sym}_c \oplus \mathrm{Sym}_a$. For $N = \mathbf{1}/m$ the rate is conserved and the frame drifts at p_0/m, which is the non-recurrence of Theorem 15.5, $X = e^{-\kappa\tau G}(X_0 + \tau p_0/m)$; for $N = \eta^{-1}/m$ the clamped frame is bounded, as in Theorem 16.2, but the rate is conserved only on null data.

17.5 The free rotor: a pendulum in the duality angle

Theorem 17.13 (The duality pendulum)**.** *Let $N = (a\eta^{-1} + bG\eta^{-1})/m$ and write $(a+ib)w(p_0) = R_0 e^{i\varphi_0}$, $R_0 \geq 0$. Along the flow $z(\pi)$ turns as $e^{i\theta}$ and $w(p) = e^{2i\theta}w(p_0)$, the frame energy is $U(\theta) = \frac{R_0}{2m}\cos(2\theta + \varphi_0)$, and*

$$I_0\ddot{\theta} = \frac{R_0}{m}\sin(2\theta + \varphi_0), \qquad E_\theta = \tfrac{1}{2}I_0\dot{\theta}^2 + \frac{R_0}{2m}\cos(2\theta + \varphi_0) = H.$$

With $\psi := 2\theta + \varphi_0$ this is the pendulum $\ddot{\psi} = (2R_0/mI_0)\sin\psi$, stable at $\psi = \pi$. The rotor angle is the duality angle of the momentum field, and the potential is its Lorentz-invariant frame energy.

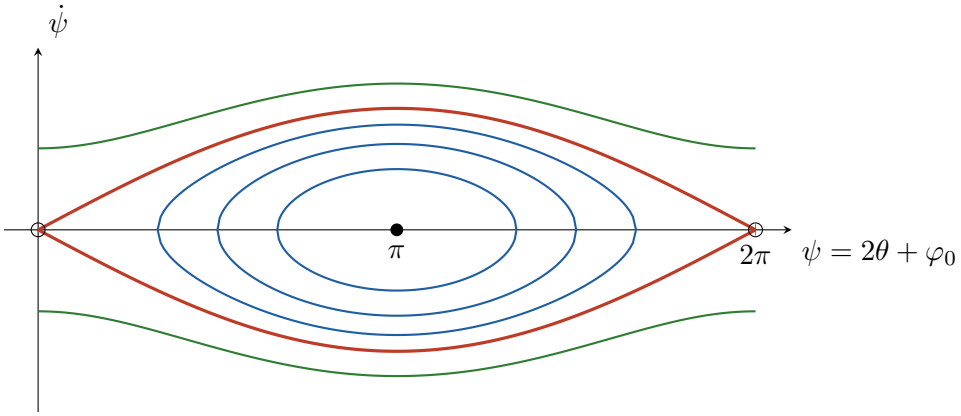


Figure 13: Phase portrait of the duality pendulum (Theorem 17.13), in units with $2R_0/(mI_0) = 1$; the filled dot is the stable point $\psi = \pi$ and the open dots the unstable one. Blue: librations, in which the rate $\kappa = \dot\theta$ changes sign periodically and the duality rotation reverses. Red: the separatrix. Green: rotations, in which the rotor turns for all time with a modulated rate. The constant-rate law of Section 16 is the limit of large rotor energy, far above the separatrix (Corollary 17.15).

Corollary 17.14 (When the rate is constant)**.** *With Lorentz-class inertia the rate is constant and nonzero along the flow exactly when $R_0 = 0$ and $h(0) \neq 0$; $R_0 = 0$ means $(a, b) = (0, 0)$ or $w(p_0) = 0$, that is $|\mathbf{B}| = |\mathbf{E}|$ and $\mathbf{B}\cdot\mathbf{E} = 0$ for the field of π_0: a null field, the class that contains plane waves [130, 115].*

Corollary 17.15 (The multiscale criterion)**.** *Let $R_0 > 0$, $\kappa_0 := \dot\theta(0) \neq 0$, $\psi_0 = 2\theta(0) + \varphi_0$ and $\varepsilon := R_0/(mI_0\kappa_0^2)$. Then $\kappa^2 = \kappa_0^2[1 + \varepsilon(\cos\psi_0 - \cos\psi)]$. The rotor turns for all time exactly when $\varepsilon(1 - \cos\psi_0) < 1$, and $\varepsilon < \frac{1}{2}$ suffices for every phase, in which case $|\kappa^2/\kappa_0^2 - 1| \leq 2\varepsilon$. At $\varepsilon(1 - \cos\psi_0) = 1$ the motion is the separatrix, and beyond it the rotor librates and the duality rotation reverses. The constant-rate law is the limit $\varepsilon \to 0$, and the multiscale requirement is*

$$I_0\,\kappa_0^2 \;\gg\; R_0/m. \tag{32}$$

A clamped rotor is therefore an idealisation of a heavy one. It is not a harmless idealisation, because the frame feels the modulation.

Theorem 17.16 (The free rotor drifts the frame)**.** *Let N be of Lorentz class with $R_0 > 0$ and $\kappa_0 \neq 0$, in the rotating regime. During one period of the rate, while θ changes by π, the frame is displaced by*

$$\Delta x = \lambda\, e^{-\varphi_0 G} N p_0, \qquad \lambda = \tfrac{1}{2}\int_0^{2\pi} \frac{\cos\psi\, d\psi}{|\kappa_0|\sqrt{1 + \varepsilon(\cos\psi_0 - \cos\psi)}} > 0,$$

and its mean velocity is $\frac{\varepsilon}{4} e^{-\varphi_0 G} N p_0 + O(\varepsilon^2)$. Every rotating solution with $R_0 \neq 0$ therefore has an unbounded frame.

Proof. For Lorentz-class N, $N = N_a$ and $\dot x = e^{2\theta G} N p_0$. In the rotating regime $\dot\theta$ never vanishes, so $d\tau = |d\theta|/|\dot\theta|$, and with $e^{2\theta G} = e^{-\varphi_0 G} e^{\psi G}$ and $e^{\psi G} = \cos\psi + G\sin\psi$ the $\sin\psi$ part integrates to zero because $1/|\dot\theta|$ depends on $\cos\psi$ only. Since $1/|\dot\theta|$ is a strictly increasing function of $\cos\psi$ when $R_0 > 0$, the integrand of λ exceeds its value at $\cos\psi = 0$ wherever $\cos\psi > 0$ and falls short of it where $\cos\psi < 0$, so $\lambda > 0$. Expanding to first order gives $\lambda = \pi\varepsilon/(4|\kappa_0|) + O(\varepsilon^2)$ over a period $\pi/|\kappa_0| + O(\varepsilon)$. □

Example 17.17 (A drift in closed form)**.** Take $p_0 = e^1$ and $m = I_0 = \kappa_0 = 1$, so that $w = 1$, $\varphi_0 = 0$, $\varepsilon = 1$ and $\dot\theta^2 = 1 + 2\sin^2\theta$. Writing $(1-2s)/\sqrt{1+2s} = 2/\sqrt{1+2s} - \sqrt{1+2s}$ with $s = \sin^2\theta$, the displacement per turn of the rotor is

$$\Delta x = (8K(-2) - 4E(-2))e_1 \approx 0.6336081022\, e_1,$$

with $K(m) = \int_0^{\pi/2}(1 - m\sin^2 t)^{-1/2}dt$ and $E(m) = \int_0^{\pi/2}(1 - m\sin^2 t)^{1/2}dt$ the complete elliptic integrals in the parameter m [35]. A numerical integration of the full system, frame included, reproduces this value to ten digits and returns h and p to their initial values after each turn.

The clamped frame of Theorem 17.12 moves on a circle of diameter $|Np_0|/|\kappa_0|$, while the free rotor adds $\frac{\pi\varepsilon}{2|\kappa_0|}|Np_0|$ per turn to first order, so the constant-rate picture tracks the bounded orbit for about $2/(\pi\varepsilon)$ turns. With Lorentz-class inertia and a free rotor, a constant rate and a bounded frame coexist exactly on null data.

Proposition 17.18 (Several constituents and one rotor)**.** *Let one rotor act on n constituents with $N_k = (a_k\eta^{-1} + b_kG\eta^{-1})/m_k$ and $w_k = w(p^{(k)})$. Then $\dot w_k = 2i(h/I_0)w_k$ for every k, all forces carry the common phase $e^{2i\theta}$, and with $R_{\mathrm{eff}}e^{i\varphi_{\mathrm{eff}}} := \sum_k(a_k + ib_k)w_k(0)/m_k$ the rotor obeys $I_0\ddot\theta = R_{\mathrm{eff}}\sin(2\theta + \varphi_{\mathrm{eff}})$. The* balance surface $R_{\mathrm{eff}} = 0$ *is preserved by the flow; on it the rate is exactly constant although no w_k need vanish, and every frame moves on a bounded orbit. With equal inertias, $p^{(2)} = -Gp^{(1)}$ gives $w_2 = -w_1$.*

The Clifford data η, G, $G\eta^{-1}$ and β_0 are integer matrices. The rescaling $(I_0, m) \mapsto (sI_0, tm)$ preserves admissibility and multiplies ε by $1/(st)$, so no value of ε, and no ratio among I_0, m and $|a + ib|$, is determined by the Clifford structure: the form of the inertia is derived, and (32) is an input.

17.6 Three constituents: the closure symmetry and the two projectors

At $n = 3$ write $e_A = e_i^{(k)}$ with $A = 6(k-1)+i$; the temporal indices are $\{1, 2, 3, 7, 8, 9, 13, 14, 15\}$, each with its partner $i' = i + 3$, and for temporal $i < j$ put

$$\mathrm{rot}(i,j) := e_{i'}e_{j'} - e_ie_j, \qquad \mathrm{bst}(i,j) := e_je_{i'} - e_ie_{j'}.$$

For one constituent these span $\mathfrak{l}_G$, with $\mathrm{rot}(2,3) = 2L_1$ and $\mathrm{bst}(2,3) = 2K_1$ and cyclically up to sign. In this numbering the weak directions of P_* are $W = \{7, 13, 14, 15\}$, the colour directions $\{10, 11, 12, 16, 17, 18\}$, and E denotes the span of the temporal directions $\{7, 8, 9, 13, 14, 15\}$ of constituents two and three; $G_3 = G \oplus G \oplus G$.

Theorem 17.19 (The closure symmetry is $\mathfrak{so}(9,\mathbb{C})$)**.** *$G_3$ is an anti-isometry of η_3 with $G_3^2 = -\mathbf{1}$, and its centraliser in $\mathfrak{so}(9,9)$ is 72-dimensional, spanned by the 36 paired rotations and the 36 paired boosts. In the basis of the nine temporal directions followed by their partners every element is $\left(\begin{smallmatrix} R & S \\ -S & R \end{smallmatrix}\right)$ with R, S antisymmetric, and $R - iS$ identifies the centraliser with $\mathfrak{so}(9,\mathbb{C})$, whose compact part consists of the paired rotations.*

In the field-strength reading $V_3 \cong \mathbb{C}^9$ carries three field strengths, and $\mathfrak{so}(9,\mathbb{C})$ consists of the complex orthogonal transformations of $\sum_k z^{(k)} \cdot z^{(k)}$.

Theorem 17.20 (No common Lorentz algebra)**.** *The centraliser of G_3 meets the interior $\mathfrak{so}(1,3)$ of the first constituent in 0; the elements commuting with both G_3 and that algebra form the 30-dimensional $\mathfrak{so}(6,\mathbb{C})$ of paired elements of constituents two and three. Under $\mathfrak{l}_G^{(1)}$ the complexified spinor module is $128 \cdot (\frac{1}{2}, \frac{1}{2})$, with no Weyl spinor.*

No nonzero Lorentz generator leaves both the closure and the spinor structure of matter invariant: the two Lorentz algebras are different subalgebras of $\mathfrak{so}(9,9)$ with zero intersection,

(q_1, q_2)	$(-1,-1)$	$(-1,0)$	$(-1,1)$	$(0,-1)$	$(0,0)$	$(0,1)$	$(1,-1)$	$(1,0)$	$(1,1)$
left, $\Gamma=-1$	1	0	1	1	2	1	0	2	0
right, $\Gamma=+1$	0	2	0	1	2	1	1	0	1

Table 2: Weyl content of the block of P_{fam} under the two surviving charges $Q_1 = -\frac{1}{2}J\,\mathrm{rot}(8,9)$ and $Q_2 = -\frac{1}{2}J\,\mathrm{rot}(14,15)$: the number of Weyl components with charges (q_1, q_2) on each chirality half.

which is Proposition 12.2 lifted to three constituents. A Pati–Salam generator acts on the closure variables only through its grade-two part, since ad_Z maps vectors to vectors exactly when Z has grades zero and two. The intersection with the Pati–Salam algebra can then be computed in general.

Theorem 17.21 (The Pati–Salam algebra and the binding root)**.** *Let U be any four-dimensional subspace of the span of the eight internal temporal directions, and $\mathfrak{ps}_U = \mathfrak{so}(U) \oplus \mathfrak{so}(6)_C$. Then neither summand meets the centraliser of G_3, and*

$$\mathfrak{ps}_U \cap \mathrm{Cent}(G_3) = \{A + G_3 A G_3^{-1} :\ A \in \mathfrak{so}(U \cap E)\} \cong \mathfrak{so}(U \cap E),$$

of dimension $\binom{d}{2} \in \{1, 3, 6\}$ with $d = \dim(U \cap E)$. For the Pati–Salam algebra of P_*, $U = \mathrm{span}\, W \subset E$, and the binding-compatible part is the diagonal $\mathfrak{so}(4)_\Delta = \mathrm{span}\{\mathrm{rot}(i,j) : i < j \in W\} \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, compact and non-abelian.

Proof. Let $A \in \mathfrak{so}(U)$ and $B \in \mathfrak{so}(6)_C$ with $[A + B, G_3] = 0$, and split the internal temporal span as $E \oplus F$ with $F = \mathrm{span}\{e_2, e_3\}$. G_3 maps F into the interior plane $\mathrm{span}\{e_5, e_6\}$, where A and B vanish, so $Av = 0$ for $v \in F$. For $v \in E$, $G_3 v$ is a colour direction and $Bv = 0$, so $G_3 A v = B G_3 v$ lies in the colour span and $Av \in E$. Hence $A \in \mathfrak{so}(U \cap E)$ and $B = G_3 A G_3^{-1}$ on the colour span, and conversely. Finally $d \geq 4 + 6 - 8 = 2$. □

Neither Pati–Salam factor contains an element commuting with the binding root; the compatible part pairs every rotated temporal direction with its spatial partner, which the colour $\mathfrak{su}(4)$, acting on spatial directions alone, never does.

Theorem 17.22 (What survives each projector)**.** *(i) The part of $\mathrm{Cent}(G_3)$ commuting with P_{fam} is $\mathrm{span}\{\mathrm{rot}(2,3), \mathrm{rot}(8,9), \mathrm{rot}(14,15), \mathrm{bst}(14,15)\}$, abelian of dimension four. The kinetic condition holds for the two internal rotations and fails for $\mathrm{bst}(14,15)$, so the gauge-admissible part is $\mathfrak{u}(1)^2 = \mathrm{span}\{\mathrm{rot}(8,9), \mathrm{rot}(14,15)\}$.*

(ii) The part commuting with P_* is $\mathrm{span}\{\mathrm{rot}(2,3), \mathrm{rot}(8,9)\} \oplus \mathfrak{so}(4)_\Delta$, compact of dimension eight, and its internal part $\mathfrak{so}(4)_\Delta \oplus \mathbb{R}\,\mathrm{rot}(8,9)$ acts faithfully and satisfies the kinetic condition.

(iii) Under either projector the frame Lorentz algebra $\mathfrak{l}_G^{(1)}$ keeps only $\mathrm{rot}(2,3) = e_5 e_6 - e_2 e_3$, a rotation of the interior plane (e_5, e_6) locked to the rotation of the surplus pair.

The rotor acts through G_3 and commutes with all of $\mathrm{Cent}(G_3)$, so a one-rotor closure is compatible with either projector. What differs is the symmetry each leaves: an abelian one for P_{fam}, and for P_* the non-abelian diagonal of weak isospin and a colour $\mathfrak{so}(4)$.

Theorem 17.23 (Chirality and anomalies of what survives)**.** *With anomaly coefficients $\mathcal{A}(f) = \frac{1}{4}[\mathrm{Tr}(fP_L) - \mathrm{Tr}(fP_R)]$ and charges $-JB$:*

(i) On P_{fam} the content is Table 2; $\mathcal{A}((\alpha Q_1 + \beta Q_2)^3) = -12\alpha\beta^2$ and $\mathcal{A}(\alpha Q_1 + \beta Q_2) = 0$, and the left and right charge multisets coincide exactly when $\alpha\beta = 0$. The anomaly-free subalgebras of $\mathfrak{u}(1)^2$ are 0, $\mathbb{R}Q_1$ and $\mathbb{R}Q_2$, each vector-like.

(ii) On P_*, with $\mathfrak{so}(4)_\Delta = \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$ and $Q = -\frac{1}{2}J\,\mathrm{rot}(8,9)$, the halves are $L = (0,0)_0 \oplus (\frac{1}{2},\frac{1}{2})_{+1} \oplus (1,0)_0$ and $R = (0,0)_0 \oplus (\frac{1}{2},\frac{1}{2})_{-1} \oplus (0,1)_0$. For $H = \alpha H_3^+ + \beta H_3^- + \gamma Q$, $\mathcal{A}(H^3) = 2\gamma(3\alpha^2 + 3\beta^2 + 4\gamma^2)$ and $\mathcal{A}(H) = 8\gamma$, so the anomaly-free subalgebras are exactly those of $\mathfrak{so}(4)_\Delta$, which has neither a perturbative nor a global anomaly.

(iii) The two halves of P_* carry inequivalent representations of $\mathfrak{so}(4)_\Delta$, so it acts chirally; but every summand is a real representation of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, so the left-handed content is real and gauge-invariant Majorana masses exist [129]. The chirality of $\mathfrak{so}(4)_\Delta$ on P_* forbids no mass.

The rotor has no continuous lift to spinors: $(e^{\theta G})^\top \eta\, e^{\theta G} = \cos 2\theta\,\eta + \sin 2\theta\,\eta G$, so $e^{\theta G}$ is an isometry of η only for $\theta \in \pi\mathbb{Z}$, and at $\theta = \pi$, $e^{\pi G_3} = -\mathbf{1}$ is implemented by Γ. This excludes a coupling of the rotor through the spin representation and says nothing about other couplings.

Theorem 17.24 (Inertia freedom at three constituents)**.** *The real dimensions of the symmetric forms on V_3^* in the Lorentz class, $\{N, G_3\} = 0$, and in the duality class, $[N, G_3] = 0$, that are invariant under a given symmetry are:*

symmetry	Lorentz class	duality class
$\mathrm{Cent}(G_3) = \mathfrak{so}(9,\mathbb{C})$	2	0
$\bigoplus_k \mathfrak{l}_G^{(k)}$	6	0
the P_{fam}-surviving algebra	18	13
the P_*-surviving algebra	8	6
none	90	81

The Lorentz-class inertia is fixed up to one complex scale under the full closure symmetry and up to three under the single-constituent Lorentz algebras, and imposing only what a projector leaves enlarges the freedom to nine complex scales for P_{fam} and four for P_*: neither projector selects the inertia.

17.7 Brackets supplied by hand

The closure above is derived; one may also supply a bracket on the frame by hand and ask what the arena's own inertias do with it. A quadratic Lie algebra whose invariant form is η must have a maximal compact subalgebra of exactly half its dimension, and such *balanced* simple real forms are rare: a complex simple algebra regarded as real, $\mathfrak{su}(p,q)$ with $|p-q| = 1$, or $\mathfrak{so}(p,q)$ with $(p,q) = (\frac{k(k+1)}{2}, \frac{k(k-1)}{2})$, and no exceptional form. At the first rung $\mathbb{R}^{3,3}$ admits exactly three semisimple quadratic structures, $\mathfrak{so}(1,3)$, $\mathfrak{so}(4)$ and $\mathfrak{so}(2,2)$, and only $\mathfrak{so}(1,3)$ if η is to be proportional to the Killing form. For $n \le 20$ a simple closure exists exactly at $n = 1, 4, 5, 6, 7, 8, 12, 15, 16, 20$.

Two observations about the arena's inertias hold across these closures. With the split-form inertia $A = m\eta$, which is an invariant form, every solution of the Euler–Arnold equation is a one-parameter subgroup and $\dot{p} \equiv 0$. With the positive form $A = m\,\mathbf{1}$ on $\mathfrak{so}(1,3)$ the system is $\dot{\mathbf{B}} = 0$, $\dot{\mathbf{E}} = \frac{2}{m}\mathbf{E} \times \mathbf{B}$: the magnetic half is frozen and the electric half precesses about it. The inertias with $G^\top A^{-1} G = -A^{-1}$ form a twelve-dimensional family of holomorphic Euler tops $\dot{Z} = Z \times MZ$ with M complex symmetric; they are Liouville integrable and never positive definite.

At the rungs $n = (N^2-1)/3$ with $3 \nmid N$, that is $(N,n) = (2,1), (4,5), (5,8), \ldots$, the compact half of the balanced form $\mathfrak{sl}(N,\mathbb{C})_{\mathbb{R}}$ is $\mathfrak{su}(N)$, which is Zeitlin's finite truncation of two-dimensional incompressible Euler flow [150]. Giving the 't Hooft clock–shift modes the discrete-Laplacian inertia, numerical integration at $N = 4$ and $N = 5$ conserves energy and enstrophy to 10^{-12} and moves up to 37% of the energy and 54% of the enstrophy between Laplacian shells: transfer between scales inside the frame. With either of the arena's own

name	plane	signature	position
T	$\langle e_1, e_2, e_3\rangle$	$(3,0)$	temporal triple: clock and surplus plane
S	$\langle e_4, e_5, e_6\rangle$	$(0,3)$	spatial triple, inside the interior four-plane
M_{21a}	$\langle e_1, e_2, e_4\rangle$	$(2,1)$	clock, one surplus direction, one space; one partner pair
M_{21b}	$\langle e_2, e_3, e_4\rangle$	$(2,1)$	surplus plane and one space; no partner pair
M_{12a}	$\langle e_1, e_4, e_5\rangle$	$(1,2)$	a 2+1 slice of the interior spacetime; one partner pair
M_{12b}	$\langle e_2, e_4, e_5\rangle$	$(1,2)$	a surplus direction and two spaces; one partner pair
M_{12c}	$\langle e_1, e_5, e_6\rangle$	$(1,2)$	a 2+1 slice of the interior spacetime; no partner pair

Table 3: Representatives of the coordinate three-planes of $\mathbb{R}^{3,3}$, classified by signature and by their position relative to the clock, the surplus plane, the interior four-plane and the pairing.

inertias the same data are exact equilibria, because both $\mathbf{1}$ and η restrict to multiples of the invariant form on $\mathfrak{su}(N)$. What the frame lacks for a fluid is the inertia. Finally, no closure dissipates: every Euler–Arnold flow stays on a coadjoint orbit and preserves every Casimir, which is exactly what viscosity destroys, so no closure of the force law can produce $\nu\Delta u$ on its own; and a constant generator that uniformly decreases the split form C has, by the inertia theorem of Ostrowski and Schneider [100], exactly $3n$ exponentially unstable directions.

Part V
The vacuum

18 The vacuum at one constituent

The word *vacuum* can mean several things, and the arena gives each of them a precise form. It can mean the structure an algebra selects on its module; the value of an order parameter chosen by a potential; the ground state of a classical field; or the Fock vacuum of a quantised one. At one constituent all four readings can be computed for every coordinate three-plane of $\mathbb{R}^{3,3}$ and for the full algebra. The exercise is worth doing for two reasons: it shows which vacuum properties belong to a part of the arena and which need all of it, and it contains, in miniature, the vacuum that will be selected at three constituents.

18.1 The subfields

A three-plane $W \subset \mathbb{R}^{3,3}$ generates a subalgebra $\mathrm{Cl}(W) \subset \mathrm{Cl}(3,3)$. Up to $O(3,3)$ only its signature matters, $(3,0)$, $(0,3)$, $(2,1)$ or $(1,2)$, but relative to the structures already built (the clock e_1, the surplus plane U, the interior four-plane E, and the pairing $e_i \leftrightarrow e_{i+3}$) the coordinate planes fall into the classes of Table 3.

18.2 The algebraic vacuum

Let ω_W be the ordered product of the three generators of W.

Theorem 18.1 (Complex structure or projector)**.** *For every coordinate three-plane W of signature (p,q), ω_W commutes with the generators in W and anticommutes with those outside it, and*

$$\omega_W^2 = -(-1)^q\,\mathbf{1}, \qquad \omega_W^\top = -(-1)^q\,\omega_W.$$

So ω_W is a complex structure for T and for the $(1,2)$ planes, and a symmetric involution, hence a rank-four orthogonal projector $\Pi_W = \frac{1}{2}(\mathbf{1} + \omega_W)$, for S and the $(2,1)$ planes. Three of these

volume elements are operators met before:

$$\omega_T = e_1e_2e_3 = \gamma^0 J, \qquad \omega_S = \gamma^1\gamma^2\gamma^3, \qquad \omega_T\,\omega_S = \Gamma,$$

the Dirac mass form, the time-reversal factor of Theorem 5.3, and the chirality, which is also the volume element of the full algebra.

Proof. A generator inside W shares one index with ω_W, so by (3) the sign is $(-1)^{3-1} = +1$; one outside shares none, and the sign is $(-1)^3 = -1$. The square and the transpose follow from (4) with $r = 3$: $(-1)^3 \prod \eta_{aa} = -(-1)^q$. □

The restriction to coordinate planes matters for the transpose, not for the square. For any nondegenerate three-plane W with an orthonormal basis, ω_W commutes with the vectors of W, anticommutes with those of $W^\perp$, and $\omega_W^2 = -(-1)^q\mathbf{1}$. The transpose rule, however, refers to the positive inner product of S, which is adapted to the splitting $V = V^+ \oplus V^-$: it holds for every plane spanned by vectors of V^+ and vectors of V^-, since rotations inside the two triples act on S by orthogonal matrices, and it fails for tilted planes. For the plane $W_t = \langle \cosh t\, e_4 + \sinh t\, e_1,\ e_5,\ e_6\rangle$, of signature $(0,3)$, $\omega_{W_t} = \cosh t\, e_4e_5e_6 + \sinh t\, e_1e_5e_6$ is the sum of a symmetric and an antisymmetric blade, so it is not symmetric for $t \neq 0$, although $\omega_{W_t}^2 = \mathbf{1}$. This is the mechanism behind the statement below that boost-generated variations leave the orthogonal projectors.

Under $\mathfrak{so}(3,3)$ the volume elements ω_T and ω_S have six-dimensional stabilisers that contain all six compact rotations, so no compact bivector moves them; only the nine boosts do, and by Theorem 11.3 boost-generated variations leave the orthogonal projectors. Each mixed volume element, by contrast, has a two-dimensional compact stabiliser and a four-dimensional compact orbit: the mixed planes are the only subfields whose vacuum projector or complex structure admits admissible fluctuations. Charge conjugation $C = e_2$ exchanges Π_S and Π_Γ with their complements and fixes the $(2,1)$ projectors, and the full volume element Γ commutes with all of $\mathfrak{so}(3,3)$: its projector is the rigid, Lorentz-invariant vacuum $\frac{1}{2}(\mathbf{1} \pm \Gamma)$ that will reappear in Section 26.

18.3 The order-parameter vacuum

Let $\mathfrak{so}(W)$ play the role of a Lorentz algebra, its antisymmetric bivectors acting as rotations and its symmetric ones as boosts.

Definition 18.2 (Mass kernels of a subfield)**.** The kernel space K_W is the space of antisymmetric $M \in \mathrm{End}(S)$ with $[M, \text{rotation}] = 0$ and $\{M, \text{boost}\} = 0$ for the generators of $\mathfrak{so}(W)$, and the internal algebra $\mathfrak{g}_W$ is the space of antisymmetric operators commuting with $\mathfrak{so}(W)$.

Antisymmetry makes $\psi^\top M\psi$ a nonzero Grassmann bilinear, and the two conditions make it Lorentz invariant, since a rotation is antisymmetric and a boost symmetric. Both conditions are diagonal on blades, so both spaces are spanned by blades.

Theorem 18.3 (Kernel spaces and invariant kernels)**.** *The kernel spaces, the internal algebras and the invariant kernels are:*

Lorentz algebra	$\dim \mathsf{K}_W$	$\dim \mathfrak{g}_W$	K_W under $\mathfrak{g}_W$	invariant kernel
$\mathfrak{so}(T)$, $\mathfrak{so}(S)$	10	10	adjoint of $\mathfrak{sp}(2)$, irreducible	none
each mixed $\mathfrak{so}(W)$	10	6	$9 \oplus 1$	the plane's own 2+1 mass form
full $\mathfrak{so}(3,3)$	1	0	—	$\gamma^0 J = e_1e_2e_3$
interior $\mathfrak{so}(1,3)$	6	4	irreducible	none

The invariant kernel of each mixed plane is $\pm\gamma_W^0\omega_W$, with γ_W^0 the generator of minority signature. For the interior algebra $\mathfrak{g}_I = \langle J, \Omega, e_2\Omega, e_3\Omega\rangle = \mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$ is the massless gauge algebra of Proposition 5.5.

Corollary 18.4 (The full algebra has one mass term)**.** *The only Grassmann mass kernel invariant under all of* $\mathfrak{so}(3,3)$ *is* $\gamma^0 J = \omega_T$*, and no antisymmetric operator commutes with* $\mathfrak{so}(3,3)$*. The order parameter of the full algebra is a single real number* r *in* $r\gamma^0 J$*; it breaks nothing and has no Goldstone direction, and its two signs are exchanged by conjugation with any spatial generator.*

The temporal triple therefore supplies more than a clock: its volume element is the Dirac mass form, the unique Lorentz-invariant mass term of the whole algebra. To see what a potential selects, take on K_W the trace potential

$$V = -\mu^2 N + \alpha N^2 + \beta Q, \qquad N = \operatorname{Tr} M^\top M, \qquad Q = \operatorname{Tr}(M^\top M)^2. \tag{33}$$

For $\beta > 0$ its minima minimise Q/N^2, a spread spectrum; for $\beta < 0$ they maximise it, a concentrated one. Every blade is a complex structure, so $\min Q/N^2 = \frac{1}{8}$ is attained on blades.

Theorem 18.5 (Two phases of the interior Lorentz algebra)**.** *For the interior* $\mathfrak{so}(1,3)$*, with* K *the irreducible six-dimensional module of* $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$*:*

(a) Dirac phase *($\beta > 0$). The mass form $\gamma^0 J$ is a minimum; its stabiliser in $\mathfrak{g}_I$ is $\langle J \rangle$, so $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$ breaks to $\mathfrak{u}(1)_J$ with three Goldstone directions. Fermion number is conserved and particle and antiparticle are distinct.*

(b) Majorana phase *($\beta < 0$). The kernels supported on the $\Gamma = +1$ half form a complex line; each anticommutes with J, has rank four, $Q/N^2 = \frac{1}{4}$ and $M^\top M \propto \frac{1}{2}(\mathbf{1} + \Gamma)$, and its stabiliser in $\mathfrak{g}_I$ is exactly $\langle J - J\Gamma \rangle$, the left-handed fermion number, again with three Goldstone directions.*

The Majorana phase is the one-constituent seed of the vacuum selected in Section 20. There, too, a concentrated phase of the trace potential selects a Majorana mass on one right-handed line, and the unbroken global charges include $1 - \Gamma$. At one constituent there is no gauge group and the three Goldstone directions are physical, classically; at three constituents the corresponding directions are gauge directions and are eaten.

18.4 The field-strength vacuum

Read through Theorem 12.3, a vector X restricted to W is a field strength of restricted type, and the interior dynamics of Section 16 restricts to it.

Theorem 18.6 (Field strengths on the subfields)**.** *For the coordinate three-planes and the full space:*

	Lorentz types	$\mathbf{E} \cdot \mathbf{B} \equiv 0$	partner pairs	Lorentz algebra kept	zero-point energy
T	magnetic only	yes	0	$\mathfrak{so}(3)$	$+\frac{3}{2}\hbar\kappa$
S	electric only	yes	0	$\mathfrak{so}(3)$	$-\frac{3}{2}\hbar\kappa$
M_{21b}	magnetic, electric, null	yes	0	$\mathfrak{so}(1,2)$	$+\frac{1}{2}\hbar\kappa$
M_{12c}	magnetic, electric, null	yes	0	$\mathfrak{so}(1,2)$	$-\frac{1}{2}\hbar\kappa$
M_{21a}	all four	no	1	none	$+\frac{1}{2}\hbar\kappa$
M_{12a}, M_{12b}	all four	no	1	none	$-\frac{1}{2}\hbar\kappa$
full	all four	no	3	$\mathfrak{so}(1,3)$	0

On every subfield the cross term of the split-form Lagrangian is a total derivative, the equation of motion is $\ddot{X} = -\kappa^2 X$, and the energy is $\frac{m}{2}(\eta(\dot{X},\dot{X}) + \kappa^2 \eta(X,X))$. The zero-momentum flow $\dot{X} = -\kappa G X$ stays in W exactly on $W \cap G^{-1}W$, of dimension 0 or 2 for the subfields and 6 for the full space.

On T the energy is positive definite and $X = 0$ is a true ground state: a pure magnetic field oscillates about zero with a positive quantum zero-point energy. On S the energy is negative definite: $X = 0$ is a maximum, dynamically stable, and its quantum oscillators are ghost-like, of negative energy or, if the energy is made positive, negative norm. On the mixed planes the energy is indefinite and there is no ground state; the zero-point energy is $\frac{1}{2}\hbar\kappa(p - q)$, and for the full $\mathbb{R}^{3,3}$ it vanishes. Neutral signature cancels the zero-point energy exactly. The mixed planes without a partner pair are the field strengths of electrodynamics in 2+1 dimensions, M_{12c} carrying one magnetic and two electric components and M_{21b} its dual, closed under an $\mathfrak{so}(1,2)$ and orthogonal by construction.

18.5 The Fock vacuum

A subfield with three anticommuting generators can carry a first-order Hamiltonian, and quantising it with Majorana operators as in Theorem 5.6 gives a Fock space.

Theorem 18.7 (A universal Fock vacuum)**.** *Let A be an antisymmetric first-order generator built from three anticommuting generators of a subfield, with $A^2 = -E^2$, and let Q be a commuting charge, antisymmetric with $Q^2 = -\mathbf{1}$ and balanced against A, $\operatorname{Tr}(AQ) = 0$. On the sixteen-dimensional Fock space the joint spectrum of energy and charge is*

$$(-2E, 0)^1,\ (-E, \pm 1)^{2+2},\ (0, 0)^4,\ (0, \pm 2)^{1+1},\ (E, \pm 1)^{2+2},\ (2E, 0)^1 :$$

a unique charge-zero ground state at $-2E$, the zero-point energy of four modes, and two particles and two antiparticles. The charge is the volume element ω_W for T and the $(1,2)$ planes; for S and the $(2,1)$ planes it must be borrowed from the complementary plane.

Proof. Commuting antisymmetric operators with $A^2 = -E^2\mathbf{1}$ and $Q^2 = -\mathbf{1}$ are simultaneously block diagonal on four orthogonal planes, on the k-th of which $A = Ea_kJ_2$ and $Q = s_kJ_2$ with $a_k, s_k \in \{\pm 1\}$. Each plane is one fermionic mode, and a Fock state is labelled by occupation signs $\epsilon_k = \pm 1$, with energy $\frac{E}{2}\sum_k \epsilon_k a_k$ and charge $\frac{1}{2}\sum_k \epsilon_k s_k$. Since $\operatorname{Tr}(AQ) = -2E\sum_k a_k s_k$, the balance condition says that $a_k s_k = +1$ on two planes and -1 on the other two, and counting the sixteen sign choices gives the table; in particular the ground state $\epsilon_k = -a_k$ has charge $-\frac{1}{2}\sum_k a_k s_k = 0$. □

The balance condition cannot be dropped. $A = Ee_4$ and $Q = e_4$ satisfy every other hypothesis, and then the Fock energy and charge are proportional, so the ground state at $-2E$ carries charge ± 2. The charges named in the theorem are balanced: $\operatorname{Tr}(\omega_T\, p \cdot e\, \omega_T) = -\operatorname{Tr}(p \cdot e) = 0$ on T, and $\operatorname{Tr}(AJ) = 0$ for $A = p \cdot (e_4, e_5, e_6)$ on S, because a product of distinct generators is traceless.

The subfields differ in the Hamiltonians they carry. T carries a massless Weyl-type Hamiltonian $\omega_T\, p \cdot e$ with an intrinsic charge; each $(1,2)$ plane a massive Dirac Hamiltonian in 2+1 dimensions with an intrinsic charge; S a real Majorana Hamiltonian $p \cdot (e_4, e_5, e_6)$ with no intrinsic charge; and the $(2,1)$ planes only a single antisymmetric generator, a mass term without momentum dependence.

Example 18.8 (Charge conjugation needs the whole structure)**.** Call charge conjugation of a charged subfield any operator anticommuting with its charge and commuting with every term of its Dirac generator, the condition of Theorem 5.1. For the interior 3+1 theory with charge J the solutions are exactly the surplus plane $\langle e_2, e_3\rangle$. For T, M_{12a}, M_{12b} and M_{12c} the solution spaces are eight-dimensional, and for S with charge J four-dimensional. For T the space contains e_4, e_5, e_6: the charge conjugation of the temporal Weyl theory lives in the spatial triple, as that of the four-dimensional theory lives in the surplus plane. Only the full four-dimensional Dirac structure fixes C up to the $U(1)_J$ rotation.

reading	temporal T	spatial S	full $\mathrm{Cl}(3,3)$
algebraic	$\omega_T = \gamma^0 J$: complex structure, rigid under rotations	$\omega_S = \gamma^1\gamma^2\gamma^3$: projector, rigid, swapped by C	$\Gamma = \omega_T\omega_S$: projector onto the half-spinors, invariant under $\mathfrak{so}(3,3)$
order parameter	adjoint of $\mathfrak{sp}(2)$; every vacuum breaks it, six Goldstone directions	the same	one invariant kernel $\gamma^0 J$; with the interior $\mathfrak{so}(1,3)$, a Dirac phase keeping J or a Majorana phase keeping $J - J\Gamma$
field strength	magnetic, positive energy, ground state	electric, negative energy	every type, indefinite energy, zero-point energy cancels
Fock	charged Weyl, C in the spatial triple	Majorana, charge only from outside	charged Dirac, $C = \langle e_2, e_3\rangle$ up to $U(1)_J$

Table 4: The vacuum at one constituent in four readings.

Table 4 summarises the four readings for the two triples and the full algebra. The temporal triple supplies the mass form and an intrinsic charge, the spatial triple the time-reversal factor and, as a field strength, the negative-energy half, and their product is the chirality. Charge conjugation, the alternative between a Dirac and a Majorana phase, and the cancellation of the zero-point energy all require the full algebra; none of them is settled inside a subfield.

19 The mass kernels of the chiral block

We now return to three constituents and to the block $V_* = \operatorname{im} P_*$ of Section 8, which carries one Pati–Salam family $(\mathbf{4},\mathbf{2},\mathbf{1})_{\Gamma=-1} \oplus (\mathbf{4},\mathbf{1},\mathbf{2})_{\Gamma=+1}$ with a right-handed neutrino. A vacuum for this matter is *dynamically selected* if it solves $\delta\Gamma_{\text{eff}}/\delta\Phi = 0$ with a second variation that is non-negative apart from the zero modes of broken symmetries, the equations being those of a stated theory. It is *compatible with matter* if, over it, the fermions carry a well-defined Dirac operator, the unbroken gauge algebra acts consistently on them, anomalies cancel, and the permitted couplings produce acceptable masses. This and the next three sections answer both questions inside the arena. Three ingredients are inputs, and we name them where they enter: the projector P_*, which is the field content; the promotion of the space of mass kernels to a dynamical field; and, for the radiative statements, the form of the tree potential at the matching scale.

19.1 Notation on the block

The colour algebra $\mathfrak{so}(6) \cong \mathfrak{su}(4)$ is spanned by the fifteen bivectors of the colour directions $C = \{10, 11, 12, 16, 17, 18\}$; with $w = (E_7, E_{13}, E_{14}, E_{15})$ and $\tau_i = -\frac{1}{2}\epsilon_{ijk} w_j w_k$, the two weak algebras are $t_i^L = \frac{1}{2}(\mathbf{1} - \chi_W)\tau_i$ and $t_i^R = \frac{1}{2}(\mathbf{1} + \chi_W)\tau_i$. These $15 + 3 + 3 = 21$ generators and the two central elements J and $J\Gamma$ commute with P_*, with the Lorentz action and with J, and on the block $q_A = E_2E_3$ acts as J and E_8E_9 as $J\Gamma$. With $H = -(E_{10}E_{16} + E_{11}E_{17} + E_{12}E_{18})$ the charges are

$$\hat{t} = -Jt, \qquad T_3^L = -Jt_3^L, \quad T_3^R = -Jt_3^R, \qquad B-L = -\tfrac{1}{3}JH, \qquad Y = T_3^R + \tfrac{1}{2}(B-L),$$

and the colour $\mathfrak{su}(3)$ is the part of $\mathfrak{so}(6)$ commuting with H. The charge table is that of Proposition 8.4. Finally γ^0 commutes with all 91 internal bivectors, with J and with the rotations, and anticommutes with Γ and the boosts, so it maps V_* onto an orthogonal rank-64 *mirror* on which the same internal algebra acts with Γ reversed (Remark 8.7).

19.2 The kernel space

Definition 19.1 (Mass kernels)**.** A Lorentz-invariant Grassmann mass kernel on the block is an $M \in \mathrm{End}(V_*)$ with

$$M^\top = -M, \qquad [M, P_* \Sigma^{ab} P_*] = 0, \qquad \{M, P_* K^a P_*\} = 0,$$

for the interior rotations Σ^{ab} and boosts $K^a = \frac{1}{2}\gamma^0\gamma^a$. Their real vector space is K, with inner product $\langle M, N\rangle = \mathrm{Tr}\, M^\top N$.

The definition is forced. For a Grassmann-valued $\psi \in V_* \otimes \Lambda$ the form $\psi^\top M \psi$ sees only the antisymmetric part of M, and it is invariant under $\delta\psi = X\psi$ iff $X^\top M + MX = 0$; the rotations are antisymmetric and the boosts symmetric, which gives the two conditions. A kernel is exactly a Lorentz-invariant mass term of the block.

Theorem 19.2 (The kernel space)**.** *(i)* $\dim_{\mathbb{R}} \mathsf{K} = 272$*. A basis is given by the compressions* $P_* B P_*$ *of the blades* $B = \gamma^0 e_I$ *and* $B = \gamma^1\gamma^2\gamma^3 e_I$*,* e_I *an internal blade, that are antisymmetric and commute with the three involutions defining* P_**: there are* 2176 *such blades and* 272 *distinct, mutually orthogonal compressions, and they span all solutions of Definition 19.1.*

(ii) Under $\mathfrak{su}(4) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$*,*

$$\mathsf{K} \cong \underbrace{(\mathbf{10},\mathbf{3},\mathbf{1})}_{60} \oplus \underbrace{(\mathbf{10},\mathbf{1},\mathbf{3})}_{60} \oplus \underbrace{(\mathbf{15},\mathbf{2},\mathbf{2})}_{120} \oplus \underbrace{2(\mathbf{6},\mathbf{1},\mathbf{1})}_{24} \oplus \underbrace{(\mathbf{1},\mathbf{2},\mathbf{2})}_{8},$$

the components being identified by their Casimir labels, $(-36,-2,0)$*,* $(-36,0,-2)$*,* $(-32,-\frac{3}{4},-\frac{3}{4})$*,* $(-20,0,0)$ *and* $(0,-\frac{3}{4},-\frac{3}{4})$ *respectively.*

(iii) Every basis kernel either anticommutes with J *and commutes with* Γ *(*Majorana type*, pairing fermions of equal chirality,* 144 *of them, spanning* $(\mathbf{10},\mathbf{3},\mathbf{1})\oplus(\mathbf{10},\mathbf{1},\mathbf{3})\oplus 2(\mathbf{6},\mathbf{1},\mathbf{1})$*), or commutes with* J *and anticommutes with* Γ *(*Dirac type*, pairing opposite chiralities,* 128 *of them, spanning* $(\mathbf{15},\mathbf{2},\mathbf{2}) \oplus (\mathbf{1},\mathbf{2},\mathbf{2})$*).*

(iv) No nonzero kernel is invariant under the 21 *Pati–Salam generators.*

(v) The kernels invariant under the twelve Standard-Model generators form the two-dimensional space spanned by Φ_0 *and* $J\Phi_0$*, where*

$$\Phi_0 = P_\nu\, P_*\, \gamma^0 E_3 E_9 E_{13} E_{14} E_{16} E_{17} E_{18}\, P_*\, P_\nu, \tag{34}$$

P_ν *being the projector onto* $\Gamma = +1$*,* $T_3^R = +\frac{1}{2}$*,* $B-L = -1$*, the right-handed neutrino. Moreover* $\Phi_0^\top \Phi_0 = P_\nu$*, of rank four,* J *anticommutes with* Φ_0*, and*

$$[t_3^R, \Phi_0] = J\Phi_0, \qquad [\tfrac{1}{3}H, \Phi_0] = -2J\Phi_0, \qquad [t_Y, \Phi_0] = 0,$$

so Φ_0 *carries* $T_3^R = 1$*,* $B-L = -2$ *and* $Y = 0$*. It lies in the* $(\mathbf{10},\mathbf{1},\mathbf{3})$ *component.*

Proof of the identification in (iii). A kernel commuting with Γ preserves the two chirality halves, which carry $(\mathbf{4},\mathbf{2},\mathbf{1})$ and $(\mathbf{4},\mathbf{1},\mathbf{2})$; one anticommuting with Γ exchanges them. A bilinear in two $(\mathbf{4},\mathbf{2},\mathbf{1})$ is an $\mathfrak{su}(2)_R$ singlet and one in two $(\mathbf{4},\mathbf{1},\mathbf{2})$ an $\mathfrak{su}(2)_L$ singlet, while a bilinear between the halves is a doublet of both. So $(\mathbf{15},\mathbf{2},\mathbf{2})$ and $(\mathbf{1},\mathbf{2},\mathbf{2})$ are of the second type and the others of the first, and $60 + 60 + 24 = 144$ and $120 + 8 = 128$ agree with the counts. □

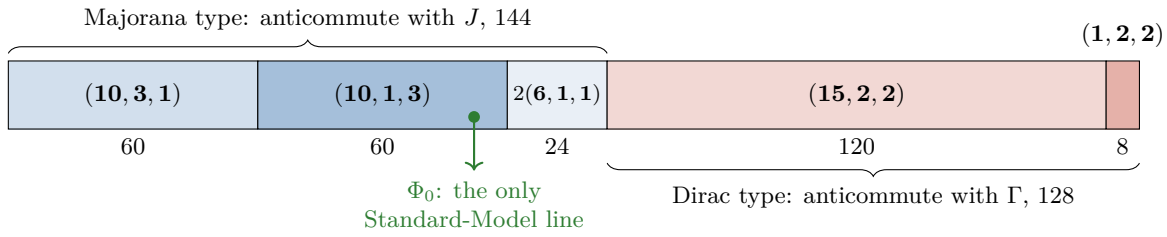


Figure 14: The 272-dimensional space of Lorentz-invariant mass kernels of the chiral block (Theorem 19.2), drawn to scale. The Standard-Model invariant kernels form a single complex line, the right-handed neutrino Majorana kernel Φ_0, inside $(\mathbf{10},\mathbf{1},\mathbf{3})$.

The complex dimension of K is $136 = \dim \mathrm{Sym}^2(\mathbb{C}^{16})$: a Lorentz-invariant mass term of sixteen Weyl fermions is a complex symmetric 16×16 matrix. In the language of $\mathfrak{so}(10)$, $\mathrm{Sym}^2(\mathbf{16}) = \mathbf{10}\oplus\mathbf{126}$ [126], and the decomposition (ii) is the Pati–Salam branching of $\mathbf{10}\oplus\mathbf{126}$; the $(\mathbf{10},\mathbf{1},\mathbf{3}) \subset \mathbf{126}$ is the field which, in minimal left–right and Pati–Salam models, gives the right-handed neutrino a Majorana mass [88, 89]. Here this content has been obtained from Definition 19.1, with only the Pati–Salam algebra acting. The only component containing a Standard-Model singlet is $(\mathbf{10},\mathbf{1},\mathbf{3})$: the colour-singlet parts of $(\mathbf{10},\mathbf{3},\mathbf{1})$, $(\mathbf{15},\mathbf{2},\mathbf{2})$ and $(\mathbf{1},\mathbf{2},\mathbf{2})$ carry $Y = \pm 1$ or $\pm\frac{1}{2}$, and $(\mathbf{6},\mathbf{1},\mathbf{1})$ has none.

Theorem 19.3 (Symmetric criticality)**.** *Let $V : \mathsf{K} \to \mathbb{R}$ be smooth and invariant under the Pati–Salam group acting by $M \mapsto gMg^{-1}$. Then for every $r \in \mathbb{R}$*

$$\nabla V(r\Phi_0) = \frac{1}{\langle \Phi_0, \Phi_0 \rangle} \frac{d}{dr} V(r\Phi_0)\, \Phi_0,$$

so $r\Phi_0$ is a critical point of V if and only if r is a critical point of $r \mapsto V(r\Phi_0)$.

Proof. The action is orthogonal. If h fixes $x = r\Phi_0$, invariance gives $h\nabla V(x) = \nabla V(x)$, so $\nabla V(x)$ is fixed by the Standard-Model group and lies in $\mathrm{span}\{\Phi_0, J\Phi_0\}$. Since $J\Phi_0 = [t_3^R, \Phi_0]$ is tangent to the orbit, $\langle \nabla V(x), J\Phi_0 \rangle = 0$, and only the radial component remains. □

This is Michel's theorem on critical orbits [84], the principle of symmetric criticality [101]. Its content here is that no invariant coupling on the whole kernel space can generate a tadpole for any other component along Φ_0. It says nothing about the second variation or about other minima, which are the subject of Section 20.

19.3 The kernel field and its action

Assumption 19.4. The kernels are the values of a scalar field $\Phi(x) \in \mathsf{K}$ with kinetic term $\frac{1}{2}\langle D_\mu\Phi, D^\mu\Phi\rangle$ and a potential $V(\Phi)$ invariant under the gauge algebra, and a single Yukawa coupling y in (35). The overall normalisation of the kinetic term is a convention; its use of one trace form on all of K, the single coupling y and the form of V are choices, restricted below only by symmetry.

With a real Grassmann field $\psi(x) \in V_*$, a gauge field A_μ with values in an algebra $\mathfrak{g}$ of antisymmetric operators, the kinetic kernels $K^\mu = P_*\gamma^0\gamma^\mu P_*$ (symmetric, $K^0 = \mathbf{1}$ on V_*) and the module metric $G_{ab} = \mathrm{Tr}(T_a^\top T_b)/512$, take

$$S = \int d^4x \Big[\tfrac{1}{2}\psi^\top K^\mu D_\mu \psi + \tfrac{y}{2}\psi^\top \Phi \psi + \tfrac{1}{2}\langle D_\mu\Phi, D^\mu\Phi\rangle - V(\Phi) - \tfrac{1}{4} G_{ab} F^a_{\mu\nu} F^{b\mu\nu} \Big], \tag{35}$$

with $D_\mu\psi = \partial_\mu\psi + A_\mu\psi$ and $D_\mu\Phi = \partial_\mu\Phi + [A_\mu, \Phi]$. Every term is Lorentz invariant by Definition 19.1, and gauge invariant for any $\mathfrak{g}$ inside the 23-dimensional faithful algebra of the block. The module metric is 1 on the colour bivectors and $\frac{1}{8}$ on t^L and t^R.

Proposition 19.5 (Fermion number)**.** *J commutes with the kinetic kernels and with the* 21 *Pati–Salam generators. The transformation $\psi \mapsto e^{\theta J}\psi$, $\Phi \mapsto e^{\theta J}\Phi e^{-\theta J}$, $A \mapsto A$ leaves every term of* (35) *except V invariant; on K it acts as $\Phi \mapsto e^{2\theta J}\Phi$ on Majorana-type kernels and trivially on Dirac-type ones. In the complex structure J each of the sixteen Weyl fermions has charge* 1*, so this is the fermion number $F = 3B + L$, and $4B = F + (B-L)$.*

By Proposition 8.5 the two central directions J and $J\Gamma$ are anomalous on this matter, so the gauge algebra can be any anomaly-free subalgebra of the 23, and $\mathfrak{su}(4) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ is the largest one containing Pati–Salam. A gauge field coupled to an anomalous current makes the quantum theory inconsistent [55, 16], which is why the central directions carry no gauge field; they remain global symmetries. In the vacuum $r\Phi_0$ the stabiliser among the 23 generators is fourteen-dimensional: the twelve Standard-Model generators together with $J + \frac{1}{3}H$ and $J - J\Gamma$, whose charges are $1 + (B-L) = 4B$ and $1 - \Gamma$, baryon number and twice the left-handed fermion number, both global.

Proposition 19.6 (The coupled field equations)**.** *The Euler–Lagrange equations of* (35) *are*

$$K^\mu D_\mu \psi + y\Phi\psi = 0, \qquad D_\mu D^\mu \Phi + \nabla V(\Phi) - \tfrac{y}{2}\Pi_{\mathsf{K}}(\psi\psi^\top) = 0, \qquad G_{ab} D_\nu F^{b\nu\mu} = -\big(j^\mu_a(\psi) + j^\mu_a(\Phi)\big),$$

with Π_{K} the orthogonal projection onto K. At constant $\Phi = r\Phi_0$, $\psi = 0$, $A = 0$ the first and third hold identically and the second reduces to $\frac{d}{dr}V(r\Phi_0) = 0$ by Theorem 19.3. The same holds with V replaced by the effective potential, which is again Pati–Salam invariant; at one loop

$$V_1 = \frac{1}{64\pi^2}\,\mathrm{STr}\Big[\mathcal{M}^4\Big(\ln\frac{\mathcal{M}^2}{\mu_R^2} - c\Big)\Big], \qquad \mathrm{STr} = \textstyle\sum_{\text{scalars}} + 3\sum_{\text{vectors}} - 2\sum_{\text{Weyl}}, \tag{36}$$

the fermion term being the determinant of the Dirac operator of the first equation [26, 69]; at stationary points it does not depend on the gauge fixing [94].

The fermions, classically zero, act on the vacuum only through the effective potential, and the matter content enters through the Yukawa coupling y. That there is one such coupling is part of Assumption 19.4, not a consequence of calling Φ one field: Φ is reducible under the Pati–Salam algebra, and symmetry alone allows an independent Yukawa coupling and an independent kinetic normalisation for each of its five irreducible components, with a mixing between the two copies of $(\mathbf{6},\mathbf{1},\mathbf{1})$. The same is true of the common gauge coupling of $\mathfrak{su}(4)$ and $\mathfrak{su}(2)_R$ used in Section 20. In the vacuum the block Hamiltonian is $H(p) = p_i K^i + iy\,r\,\Phi_0$; since $\{K^i, K^j\} = 2\delta^{ij}$, $\{K^i, \Phi_0\} = 0$ and $-\Phi_0^2 = P_\nu$,

$$H(p)^2 = |p|^2 + y^2 r^2 P_\nu :$$

one Majorana fermion, the right-handed neutrino, with four real components and mass $|y|r$, and sixty massless real components.

Theorem 19.3 settles stationarity for the whole coupled quantum system along Φ_0. Which orbit is the *minimum* is decided by the couplings, and that is the subject of the next section. A non-negative second variation on the $(\mathbf{10},\mathbf{1},\mathbf{3})$ directions alone does not imply it on all of K: fluctuations outside $(\mathbf{10},\mathbf{1},\mathbf{3})$ with the Standard-Model quantum numbers of $(\mathbf{10},\mathbf{1},\mathbf{3})$ fluctuations can mix with them, and Section 20.5 exhibits such a mixing.

20 Selecting the Standard-Model orbit

The Standard-Model line Φ_0 lies in the $(\mathbf{10},\mathbf{1},\mathbf{3})$ component, which is small enough to be analysed completely. We first classify the renormalizable potential on that component and give sufficient and necessary conditions for Φ_0 to be its global minimum; then we add the gauge and matter fluctuations; and finally we return to the whole of K.

20.1 The tensor model

Let $\mathfrak{su}(4)\oplus\mathfrak{su}(2)_R$ act on $\mathbb{C}^8=\mathbb{C}^4\otimes\mathbb{C}^2$ by Hermitian generators normalised by $\operatorname{Tr}T_aT_b=\tfrac12\delta_{ab}$ in the fundamental, and on complex symmetric 8×8 matrices by $S\mapsto USU^{\top}$. The subspace

$$\mathcal{T}=\operatorname{Sym}^2(\mathbb{C}^4)\otimes\operatorname{Sym}^2(\mathbb{C}^2)\subset\operatorname{Sym}^2(\mathbb{C}^8),\qquad \dim_{\mathbb{C}}\mathcal{T}=30,\qquad |S|^2=\operatorname{Tr}S^{\dagger}S,$$

is invariant and irreducible, isomorphic to $(\mathbf{10},\mathbf{1},\mathbf{3})$. With $e_1,\dots,e_4$ and $e_\pm$ the standard bases of $\mathbb{C}^4$ and $\mathbb{C}^2$, $B-L=\operatorname{diag}(\tfrac13,\tfrac13,\tfrac13,-1)$ and $T_3^R=\operatorname{diag}(\tfrac12,-\tfrac12)$, the vectors $e_i\otimes e_+$, $e_4\otimes e_+$, $e_i\otimes e_-$, $e_4\otimes e_-$ $(i\le3)$ carry $Y=\tfrac23,0,-\tfrac13,-1$: they are u_R, ν_R, d_R, e_R, and a component of S is a pair of them. Put $u_0=e_4\otimes e_+$ and $S_0=u_0u_0^{\top}$.

Proposition 20.1 (From kernels to tensors)**.** *The component $\mathsf{K}_{(\mathbf{10},\mathbf{1},\mathbf{3})}$ is J-invariant of real dimension 60, and with J as complex structure it is isomorphic to $\mathcal{T}$ by a Pati–Salam-equivariant ι with $\operatorname{Tr}\Phi^{\top}\Phi=4|\iota(\Phi)|^2$, unique up to a phase, with $\iota(\Phi_0)=e^{i\varphi}S_0$. For $S=\iota(\Phi)$, $\Phi^{\top}\Phi$ is supported on the $\Gamma=+1$ half and has the spectrum of $S^{\dagger}S$, each eigenvalue four times, so*

$$N:=\operatorname{Tr}\Phi^{\top}\Phi=4|S|^2,\qquad Q:=\operatorname{Tr}(\Phi^{\top}\Phi)^2=4\operatorname{Tr}(S^{\dagger}S)^2,\qquad \frac{Q}{N^2}=\frac14\,\frac{\operatorname{Tr}(S^{\dagger}S)^2}{|S|^4}.$$

Let $P_1,\dots,P_5$ be the orthogonal projectors of $\operatorname{Sym}^2\mathcal{T}$ onto its five irreducible components

$$(\mathbf{35},\mathbf{5})\oplus(\mathbf{35},\mathbf{1})\oplus(\mathbf{20'},\mathbf{5})\oplus(\mathbf{20'},\mathbf{1})\oplus(\mathbf{45},\mathbf{3}),\qquad 175+35+100+20+135=465,$$

and put $I_k(S)=\langle S\otimes S,P_k(S\otimes S)\rangle\ge0$, so that $\sum_k I_k=|S|^4$.

Theorem 20.2 (The renormalizable potential of $(\mathbf{10},\mathbf{1},\mathbf{3})$)**.** *The most general renormalizable invariant potential on $\mathcal{T}$ is*

$$V(S)=-\mu^2|S|^2+\sum_{k=1}^{5}\lambda_kI_k(S)+\kappa\,\mathcal{D}(S)+\bar\kappa\,\overline{\mathcal{D}(S)},\qquad \lambda_k\in\mathbb{R},\ \kappa\in\mathbb{C},\tag{37}$$

where the holomorphic quartic is

$$\mathcal{D}(S)=\tfrac{1}{24}\,\epsilon^{abcd}\epsilon^{efgh}P_{ab,ef}(S)P_{cd,gh}(S),\qquad P_{ab,ef}(S)=\varepsilon_{su}\varepsilon_{tv}S_{as,et}S_{bu,fv},$$

with colour indices $a,\dots,h$ and $\mathfrak{su}(2)_R$ indices s,t,u,v. There is no invariant cubic. Moreover:

(i) $\mathcal{D}(e^{i\theta}S)=e^{4i\theta}\mathcal{D}(S)$, every I_k is phase invariant, and $\mathcal{D}$ is not a combination of the I_k.

(ii) $\mathcal{D}$ depends on S only through the $(\mathbf{20'},\mathbf{1})$ component of $S\otimes S$, and $|\mathcal{D}(S)|\le\tfrac38 I_4(S)$, with equality at $S=\mathbf{1}_4\otimes\mathbf{1}_2$.

(iii) Under fermion number $S\mapsto e^{2i\theta}S$, so the I_k are invariant and $\mathcal{D}\mapsto e^{8i\theta}\mathcal{D}$: (37) is invariant under fermion number if and only if $\kappa=0$. $\mathcal{D}$ has $B-L=0$ and fermion number 8, hence baryon number 2.

The trivial representation occurs in $\mathcal{T}\otimes\bar{\mathcal{T}}$, $\operatorname{Sym}^2\mathcal{T}$, $\operatorname{Sym}^3\mathcal{T}$, $\operatorname{Sym}^2\mathcal{T}\otimes\bar{\mathcal{T}}$, $\operatorname{Sym}^4\mathcal{T}$, $\operatorname{Sym}^3\mathcal{T}\otimes\bar{\mathcal{T}}$ and $\operatorname{Sym}^2\mathcal{T}\otimes\operatorname{Sym}^2\bar{\mathcal{T}}$ with multiplicities $1,0,0,0,1,0,5$, which is the content of the theorem. The phase of κ can be removed by $S\mapsto e^{i\alpha}S$, so only $|\kappa|$ matters. The contraction of four $(\mathbf{10},\mathbf{1},\mathbf{3})$ fields with two $SU(4)$ ϵ tensors is the baryon-number-violating coupling of Pati–Salam models through which diquark components mediate neutron–antineutron oscillation [87, 24].

Two lemmas organise the quartics. The first is the recoupling identity $\operatorname{Tr}(S^{\dagger}S)^2=I_1-\tfrac12I_2-\tfrac12I_3+\tfrac14I_4+0\cdot I_5$, obtained from Schur–Weyl duality on $(\mathbb{C}^4)^{\otimes4}\otimes(\mathbb{C}^2)^{\otimes4}$. The second characterises the orbit of S_0.

Lemma 20.3 (The highest-weight orbit)**.** *For $S \in \mathcal{T} \setminus 0$ the following are equivalent: $S \otimes S$ lies in $(\mathbf{35}, \mathbf{5})$; $\operatorname{Tr}(S^\dagger S)^2 = |S|^4$; $\operatorname{rank} S = 1$; $S = (a \otimes b)(a \otimes b)^\top$ with $a \in \mathbb{C}^4$, $b \in \mathbb{C}^2$; S lies on the orbit $\mathcal{O}_{\mathrm{hw}}$ of $\mathbb{C}^\times S_0$.*

This is the Kostant–Lichtenstein description of the highest-weight orbit by quadrics [77]. On $\mathcal{O}_{\mathrm{hw}}$ the colour factor of S has rank one, so $\mathcal{D}$ vanishes there; more precisely $\mathcal{D}$ and its first and second derivatives vanish at S_0, and

$$\mathcal{D}(s_0 S_0 + \delta) = s_0\Big(\tfrac{1}{3}\epsilon_{ijk}\epsilon_{lmn}U_{il}D_{jm}D_{kn} - \tfrac{1}{12}\epsilon_{ijk}\epsilon_{lmn}M_{il}M_{jm}D_{kn}\Big) + \mathcal{D}(\delta),$$

with U, M, D the symmetric 3×3 matrices of the colour sextets $u_R u_R$, $u_R d_R$, $d_R d_R$. In the vacuum the holomorphic quartic contributes nothing to the masses and produces the cubic couplings $\kappa v\, \epsilon\epsilon\, \Delta_{uu}\Delta_{dd}\Delta_{dd}$ and $\kappa v\, \epsilon\epsilon\, \Delta_{ud}\Delta_{ud}\Delta_{dd}$ of diquarks of hypercharges $\frac{4}{3}, \frac{1}{3}, -\frac{2}{3}$, each of baryon number two. With the Yukawa couplings of the diquarks to quark pairs, here the single coupling y of (35), the vertex closes into a six-quark $\Delta B = 2$ operator, the Pati–Salam mechanism of neutron–antineutron oscillation; the present limits on the oscillation time are 8.6×10^7 s for free neutrons [8] and 4.7×10^8 s from neutrons bound in oxygen [128].

20.2 Where the Standard-Model orbit is the global minimum

Number the components $k = 1, \dots, 5$ as above, put $f_k(S) = I_k(S)/|S|^4$ (so $f_k \geq 0$, $\sum f_k = 1$), $d(S) = |\mathcal{D}(S)|/|S|^4$, and

$$\Delta_k = \lambda_k - \lambda_1 \ (k \geq 2), \qquad \Lambda(S) = \sum_k \lambda_k f_k(S) = \lambda_1 + \sum_{k \geq 2} \Delta_k f_k(S), \qquad \Lambda_\kappa(S) = \Lambda(S) - 2|\kappa|\, d(S).$$

For a unit vector $\hat{S}$, $\min_\varphi V(re^{i\varphi}\hat{S}) = -\mu^2 r^2 + \Lambda_\kappa(\hat{S}) r^4$, because the I_k do not depend on the phase and $2\operatorname{Re}(\kappa e^{4i\varphi}\mathcal{D}(\hat{S}))$ reaches $-2|\kappa||\mathcal{D}(\hat{S})|$. With $\mu^2 > 0$ the potential is bounded below iff $\Lambda_\kappa > 0$ on the unit sphere, the minimum along a ray is $-\mu^4/(4\Lambda_\kappa)$, and the vacuum is the configuration of smallest Λ_κ.

Theorem 20.4 (Selection without the holomorphic quartic)**.** *Let $\kappa = 0$, $\mu^2 > 0$, $\lambda_1 > 0$ and $\Delta_2, \dots, \Delta_5 > 0$. The global minima of (37) are exactly the points of $\mathcal{O}_{\mathrm{hw}}$ with $|S|^2 = \mu^2/(2\lambda_1)$, with $V_{\min} = -\mu^4/(4\lambda_1)$. They form one orbit of real dimension nine, and the stabiliser of each point is conjugate to $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_Y$ inside $\mathfrak{su}(4) \oplus \mathfrak{su}(2)_R$ with $Y = T_3^R + \frac{1}{2}(B - L)$; together with $\mathfrak{su}(2)_L$, which acts trivially on $(\mathbf{10}, \mathbf{1}, \mathbf{3})$, this is the Standard-Model algebra.*

Proof. $\Lambda \geq \lambda_1$, with equality iff $f_2 = \cdots = f_5 = 0$, that is iff $S \in \mathcal{O}_{\mathrm{hw}}$ by Lemma 20.3; the unit-norm part of $\mathcal{O}_{\mathrm{hw}}$ is a single orbit, of dimension $18 - 9$. □

The unbroken $\mathfrak{u}(1)$ is an output, computed as the stabiliser: the weights of the vacuum, not a choice of embedding, fix the hypercharge. This is the vacuum of Pati–Salam and left–right models with a $(\mathbf{10}, \mathbf{1}, \mathbf{3})$ field [102, 89]; the theorem says for which couplings the dynamics chooses it.

Theorem 20.5 (Selection with the holomorphic quartic)**.** *Let $\mu^2 > 0$ and $\kappa \in \mathbb{C}$.*

(a) If $\lambda_1, \lambda_2, \lambda_3, \lambda_5 > 0$ and $\lambda_4 > \frac{3}{4}|\kappa|$, the potential is bounded below; if it is bounded below then $\lambda_1 > 0$ and $2\lambda_1 + \lambda_2 + 2\lambda_3 + \lambda_4 > \frac{3}{4}|\kappa|$.

(b) If $\lambda_1 > 0$, $\Delta_2, \Delta_3, \Delta_5 > 0$ and $\Delta_4 > \frac{3}{4}|\kappa|$, the global minima are exactly the points of $\mathcal{O}_{\mathrm{hw}}$ with $|S|^2 = \mu^2/(2\lambda_1)$.

(c) If those points are global minima, then $\Delta_2 \geq 0$, $\Delta_3 \geq 0$, $2\Delta_2 + 2\Delta_3 + 4\Delta_4 + 9\Delta_5 \geq 0$ and $\Delta_2 + 2\Delta_3 + \Delta_4 \geq \frac{3}{4}|\kappa|$.

(d) The second variation at the Standard-Model orbit does not depend on κ.

At the benchmark couplings $\lambda = (1, 1.7, 2.3, 3.1, 2.9)$ the bounds read: $\frac{4}{3}\Delta_4 = 2.8$, sufficient for selection; $\frac{4}{3}(\Delta_2 + 2\Delta_3 + \Delta_4) = 7.2$, necessary for selection and equal to the computed threshold; $\frac{4}{3}\lambda_4 = 4.13$, sufficient for boundedness; and $\frac{4}{3}(2\lambda_1 + \lambda_2 + 2\lambda_3 + \lambda_4) = 15.2$, necessary for boundedness. Gradient flows confirm the transition at $|\kappa| = 7.2$: from random starts the flow reaches the Standard-Model orbit for every $|\kappa|$, as it must, since the orbit is always a local minimum, but for $6 \le |\kappa| \le 7.5$ one of two starts ends on the orbit of $\mathbf{1}_4 \otimes \mathbf{1}_2$, which is the higher of the two minima for $|\kappa| \le 7$ and the lower for $|\kappa| \ge 7.5$: at the benchmark, $f(\mathbf{1}_4 \otimes \mathbf{1}_2) = (\frac{1}{3}, \frac{1}{6}, \frac{1}{3}, \frac{1}{6}, 0)$ and $d(\mathbf{1}_4 \otimes \mathbf{1}_2) = \frac{1}{16}$, so $\Lambda_\kappa(\mathbf{1}_4 \otimes \mathbf{1}_2) - \Lambda_\kappa(S_0) = 0.9 - \frac{1}{8}|\kappa|$, which changes sign at $|\kappa| = 7.2$.

Theorem 20.6 (Second variation at the Standard-Model orbit)**.** *At $S_v = vS_0$ the sixty real fluctuations split into $SU(3) \times U(1)_Y$ multiplets, each with one mass:*

pair	$SU(3)_Y$	$\dim_{\mathbb{R}}$	m^2/v^2	role
$\nu_R\nu_R$ (modulus)	1_0	1	$4\lambda_1$	radial
$\nu_R\nu_R$ (phase), $\nu_R e_R$, $u_R\nu_R$	1_0, 1_{-1}, $3_{2/3}$	$1+2+6$	0	eaten
$e_R e_R$	1_{-2}	2	$\frac{4}{3}\Delta_2$	Δ_{ee}
$u_R e_R + d_R\nu_R$	$3_{-1/3}$	6	Δ_5	Δ_{ue}
$d_R e_R$	$3_{-4/3}$	6	$\frac{2}{3}\Delta_2 + \Delta_5$	Δ_{de}
$u_R u_R$	$6_{4/3}$	12	$\frac{4}{3}\Delta_3$	Δ_{uu}
$u_R d_R$	$6_{1/3}$	12	$\frac{2}{3}\Delta_3 + \Delta_5$	Δ_{ud}
$d_R d_R$	$6_{-2/3}$	12	$\frac{4}{9}\Delta_2 + \frac{2}{9}\Delta_3 + \frac{4}{9}\Delta_4 + \Delta_5$	Δ_{dd}

Consequently S_v is a non-degenerate local minimum on $\mathcal{T}$ iff $\lambda_1, \Delta_2, \Delta_3, \Delta_5 > 0$ and $2\Delta_2 + 2\Delta_3 + 4\Delta_4 + 9\Delta_5 > 0$, a region that strictly contains the cone of Theorem 20.4; and, independently of the couplings,

$$m^2_{de} = m^2_{ue} + \tfrac{1}{2}m^2_{ee}, \qquad m^2_{ud} = m^2_{ue} + \tfrac{1}{2}m^2_{uu}. \tag{38}$$

Example 20.7 (Checking a sum rule)**.** At the benchmark $\lambda = (1, 1.7, 2.3, 3.1, 2.9)$ one has $\Delta = (0.7, 1.3, 2.1, 1.9)$, so in units of v^2: $m^2_{ee} = \frac{4}{3}\cdot 0.7 = 0.9\overline{3}$, $m^2_{ue} = 1.9$, $m^2_{de} = \frac{2}{3}\cdot 0.7 + 1.9 = 2.3\overline{6}$, and indeed $1.9 + \frac{1}{2}\cdot 0.9\overline{3} = 2.3\overline{6}$. Likewise $m^2_{uu} = \frac{4}{3}\cdot 1.3 = 1.7\overline{3}$ and $m^2_{ud} = \frac{2}{3}\cdot 1.3 + 1.9 = 2.7\overline{6} = 1.9 + \frac{1}{2}\cdot 1.7\overline{3}$. The full 60×60 mass matrix at these couplings and $\mu^2 = 2$ has the predicted spectrum to 4×10^{-15}.

The colour-sextet components $\Delta_{uu}, \Delta_{ud}, \Delta_{dd}$ are the diquarks just met, and Δ_{ee} is the doubly charged scalar of left–right models. The sum rules (38) hold for the most general renormalizable potential of a single $(\mathbf{10}, \mathbf{1}, \mathbf{3})$, with or without κ. When one of the bounds $\Delta_2, \Delta_3, \Delta_5 > 0$ fails, gradient flows end on configurations of lower energy that break electric charge or colour: the configuration $(e_4 e_4)(e_+ e_-)$, for instance, keeps colour, but its unbroken abelian factor is T^R_3 instead of Y.

20.3 The trace potential

The kernel field carries two quartic invariants that do not refer to the Pati–Salam decomposition at all, N^2 and $Q = \mathrm{Tr}(\Phi^\top\Phi)^2$; they are invariant under every orthogonal change of basis of the block that preserves K, in particular under fermion number, so the trace potential has $\kappa = 0$.

Theorem 20.8 (The trace potential on $(\mathbf{10}, \mathbf{1}, \mathbf{3})$)**.** *Let $V_{\mathrm{tr}} = -\mu^2 N + \alpha N^2 + \beta Q$ with $\mu^2 > 0$. In the variables of Theorem 20.2, $\lambda_k = 16\alpha + 4\beta a_k$ with $a = (1, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{4}, 0)$, so $(\Delta_2, \Delta_3, \Delta_4, \Delta_5) = -4\beta\,(\frac{3}{2}, \frac{3}{2}, \frac{3}{4}, 1)$. If $\beta < 0$ and $\alpha + \frac{1}{4}\beta > 0$, the global minimum is the Standard-Model orbit; if $\beta > 0$ and $\alpha + \frac{1}{32}\beta > 0$, the minima are the S with $S^\dagger S = \frac{1}{8}|S|^2\mathbf{1}_8$, and every one of them breaks colour; if $\beta = 0$ the potential is $O(60)$-symmetric and the orbit is chosen by fluctuations.*

Q/N^2 measures how concentrated the Majorana mass matrix is: it equals $\frac{1}{4}$ when one fermion carries all of it and $\frac{1}{32}$ when the eight right-handed fermions share it equally. A negative β rewards concentration on a single line, and the only line on which the dynamics can concentrate without breaking colour is ν_R. The Standard-Model vacuum is the concentrated phase of the Majorana mass, which is the three-constituent form of the Majorana phase of Theorem 18.5.

20.4 The fluctuations decide: back-reaction on $(\mathbf{10},\mathbf{1},\mathbf{3})$

Take the tree potential degenerate on $(\mathbf{10},\mathbf{1},\mathbf{3})$, $\lambda_1 = \cdots = \lambda_5$ and $\kappa = 0$, the case $\beta = 0$. Every unit direction of $\mathcal{T}$ is then a tree-level minimum, and the orbit is chosen by the one-loop potential (36). Gauge and Yukawa interactions respect fermion number, so they do not generate κ in perturbation theory. Write $L = \ln(\Lambda_{\mathrm{UV}}^2/v^2)$. The orbit-dependent part of the leading logarithm is

$$\delta V = \frac{L}{64\pi^2}\Big[2\sum_{\mathrm{fam}} \mathrm{Tr}(M_F^\dagger M_F)^2 - 3\,\mathrm{Tr}\,M_V^4 - \mathrm{Tr}\,M_S^4\Big],$$

where for the degenerate tree potential $\mathrm{Tr}\,M_S^4$ depends on $|S|$ only. Here and in Section 20.5 the field is normalised canonically. By Proposition 20.1, $\mathrm{Tr}\,\Phi^\top\Phi = 4|\iota(\Phi)|^2$, so the kinetic term of Assumption 19.4 is $\mathrm{Tr}(DS)^\dagger(DS)$ for $S := \sqrt{2}\,\iota(\Phi)$, and from here on S denotes this field and $v = |S| = |\Phi|/\sqrt{2} = \sqrt{2}\,r$, the v of Section 20.5 and Section 21. In these variables the right-handed neutrino mass $|y|r$ of Section 19 reads $(|y|/\sqrt{2})\,v$, so the y of $M_F = yS$ below is the coupling of (35) divided by $\sqrt{2}$, while the coupling g is the same as in Theorem 21.1. The vector mass matrix is $(M_V^2)_{ab} = 2g^2\,\mathrm{Re}\,\mathrm{Tr}(T_aS + ST_a^\top)^\dagger(T_bS + ST_b^\top)$ with one coupling for $\mathfrak{su}(4)$ and $\mathfrak{su}(2)_R$, and the eight right-handed Weyl fermions get the Majorana matrix $M_F = yS$.

Theorem 20.9 (One-loop selection on $(\mathbf{10},\mathbf{1},\mathbf{3})$)**.** *At leading logarithm,* $\mathrm{Tr}\,M_V^4 = g^4\sum_k c_kI_k$ *with* $c = (33,3,-3,39,5)$ *and* $\mathrm{Tr}(M_F^\dagger M_F)^2 = y^4\sum_k a_kI_k$*, and the fluctuations generate the gaps*

$$\Delta_k = \frac{L}{64\pi^2}\,g^4(G_k - x\,F_k), \qquad G = (90,108,-18,84), \quad F = (3,3,\tfrac{3}{2},2), \qquad x = \frac{\sum_{\mathrm{fam}} y^4}{g^4}.$$

Gauge fluctuations favour the Standard-Model orbit in the directions $k = 2,3,5$*, and matter fluctuations disfavour it in every direction. The Standard-Model orbit is a non-degenerate local minimum on* $\mathcal{T}$ *iff* $x < 30$*, the binding conditions being those of* Δ_{ee} *and* Δ_{dd}*. The configuration* $\mathbf{1}_4 \otimes \mathbf{1}_2$ *has excess* $\frac{L}{64\pi^2}g^4(48 - \frac{7}{4}x)$ *over the Standard-Model orbit, negative for* $x > \frac{192}{7} \approx 27.43$*.*

At $x = 0$ the radiatively generated masses are, in units of $\frac{L}{64\pi^2}g^4v^2$, $m_{ee}^2 = 120$, $m_{ue}^2 = 84$, $m_{de}^2 = 144$, $m_{uu}^2 = 144$, $m_{ud}^2 = 156$ and $m_{dd}^2 = 120$, which satisfy (38); the fifty massive scalars are pseudo-Goldstone bosons of the accidental $O(60)$ symmetry of the tree potential [139, 51]. A projected gradient flow on the unit sphere of $\mathcal{T}$ from random starts finds no configuration below the Standard-Model orbit for $x \le 27$, finds the orbit of $\mathbf{1}_4 \otimes \mathbf{1}_2$ as the only competitor, and confirms that its energy crosses zero at $x = 192/7$; between $192/7$ and 30 the Standard-Model vacuum is metastable on $\mathcal{T}$. We conjecture that on $\mathcal{T}$ the Standard-Model orbit is the global minimum of the one-loop potential exactly for $x < 192/7$.

With $M_N = yv$ and $M_{W_R} = gv$, $x = \sum_{\mathrm{fam}}(M_N/M_{W_R})^4$; in the couplings of (35) and Theorem 21.1, $x = \sum_{\mathrm{fam}} y^4/(4g^4)$. Stated in terms of masses, x does not depend on the normalisation. The content of the bound is structural: the Majorana Yukawa coupling, matter's only handle on the vacuum, pushes towards configurations in which the Majorana mass is shared among several fermions, and the gauge fluctuations hold the Standard-Model orbit only while the heavy neutrinos are not much heavier than W_R.

20.5 The full kernel space

The general invariant potential on all of K has too many couplings to be classified usefully: the numbers of linearly independent Pati–Salam-invariant real polynomials of degrees $1, 2, 3, 4$ are $0, 18, 0, 686$, and with fermion number imposed $0, 12, 0, 402$. Two structures do not depend on these couplings.

Proposition 20.10 (A left–right involution)**.** *Let $O = E_1E_7E_8$ restricted to V_*. It commutes with P_* and with the rotations, satisfies $O^{\top}K^0O = K^0$, $O^{\top}K^iO = -K^i$, $O\Gamma O^{\top} = -\Gamma$ and $OJO^{\top} = J$, and maps t_i^L to $\pm t_i^R$ and each colour bivector to $\pm$ itself. Hence $\psi(t,\mathbf{x}) \mapsto O\psi(t,-\mathbf{x})$, $\Phi \mapsto O\Phi O^{\top}$, with the corresponding map of the gauge fields, is a symmetry of the kinetic, Yukawa and gauge terms of (35), and of the trace invariants. The image $\Phi_0' = O\Phi_0O^{\top}$ is a kernel in $(\mathbf{10},\mathbf{3},\mathbf{1})$ orthogonal to Φ_0, the mirror vacuum of left–right models, and along the great circle $\hat{u}(\theta) = \cos\theta\,\hat{u}_0 + \sin\theta\,\hat{u}_0'$, $Q = \frac{1}{4} - \frac{1}{8}\sin^2 2\theta$.*

The Standard-Model vacuum and its mirror are exchanged by a symmetry of the action, so an O-invariant potential cannot prefer one of them; the choice between breaking $\mathfrak{su}(2)_R$ and breaking $\mathfrak{su}(2)_L$ is spontaneous, as in models of spontaneous parity violation [122].

Now take the tree potential degenerate on all of K, $V_0 = -\mu^2 N + \alpha N^2$, invariant under the accidental $O(272)$. For a unit vector $\hat{u} \in \mathsf{K}$ put

$$Q(\hat{u}) = \operatorname{Tr}(\hat{u}^{\top}\hat{u})^2, \qquad T(\hat{u}) = \operatorname{Tr}\big(G^{-1}K(\hat{u})\big)^2, \qquad K(\hat{u})_{ab} = \langle [t_a,\hat{u}],[t_b,\hat{u}]\rangle,$$
$$E_x(\hat{u}) = 8x\,Q(\hat{u}) - \tfrac{3}{16}T(\hat{u}),$$

with the 21 Pati–Salam generators t_a and the module metric G. The orientation-dependent part of the leading logarithm is then $\delta V = \frac{L}{64\pi^2}g^4v^4E_x(\Phi/|\Phi|)$ with $v = |\Phi|/\sqrt{2}$, and at the Standard-Model point $E_x(\hat{u}_0) = 2x - 99$.

Theorem 20.11 (One-loop second variation on all of K)**.** *Let $H(x)$ be the second variation of E_x at $\hat{u}_0$ on the 271-dimensional tangent space of the unit sphere.*

(i) $H(x) = H_G + xH_F$ with $H_F = -8(\mathbf{1} - \Pi_\nu)$, where Π_ν is the orthogonal projector of rank 32 onto the kernels that pair ν_R with a Weyl fermion of the block. H_G preserves each Pati–Salam component of K.

(ii) With $m^2 = \frac{1}{2}\times$(eigenvalue of $H(x)$) in units of $\frac{L}{64\pi^2}g^4v^2$, the 271 angular modes have the masses of Table 5.

(iii) $H(x)$ is non-negative with kernel exactly the nine Goldstone directions if and only if $x < 27$. At $x = 27$ the six modes of $\ell_L\ell_L$, the left-handed triplet, become massless, and for $x > 27$ they are tachyonic.

Statement (i) is the mechanism by which matter opposes the Standard-Model orbit, now on all of K. At fixed $|\Phi|$ the fermion loop $2\sum m_f^4$ is largest when the whole Majorana mass sits on ν_R; it decreases quadratically along every direction that moves mass onto pairs not containing ν_R and is stationary along the directions that pair ν_R with another fermion. Every mode that does not involve ν_R therefore has slope -4, and the first of them to become massless is the one with the smallest gauge contribution, the left-handed triplet. The mixing in M_3, between the $(\mathbf{10},\mathbf{1},\mathbf{3})$ mode $u_Re_R + d_R\nu_R$ and the $(\mathbf{6},\mathbf{1},\mathbf{1})_R$ mode $u_Re_R - d_R\nu_R$, is an instance of the mixing announced at the end of Section 19: it moves the Δ_{ue} threshold from 42 to $315/11$. The thresholds are the zeros of the lowest lines,

$$x = 27\ (\ell_L\ell_L), \qquad x = 28.5\ (\ell_L\text{–}d_R), \qquad x = \tfrac{315}{11} \approx 28.64\ (M_3), \qquad x = 30\ (e_Re_R,\ d_Rd_R),$$

and Figure 15 shows them. Along the great circle of Proposition 20.10,

$$E_x(\hat{u}(\theta)) = 2x - 99 + (27 - x)\sin^2 2\theta,$$

fermion pair	multiplet	$\dim_{\mathbb{R}}$	component	m^2
$\nu_R\nu_R$ (phase), $\nu_R e_R$, $\nu_R u_R$	$(1,1)_0, (1,1)_1, (3,1)_{2/3}$	$1+2+6$	$(\mathbf{10},\mathbf{1},\mathbf{3})$	0 (Goldstone)
$e_R e_R$	$(1,1)_2$	2	$(\mathbf{10},\mathbf{1},\mathbf{3})$	$120-4x$
$d_R e_R$	$(3,1)_{4/3}$	6	$(\mathbf{10},\mathbf{1},\mathbf{3})$	$144-4x$
$u_R u_R$	$(6,1)_{4/3}$	12	$(\mathbf{10},\mathbf{1},\mathbf{3})$	$144-4x$
$u_R d_R$	$(6,1)_{1/3}$	12	$(\mathbf{10},\mathbf{1},\mathbf{3})$	$156-4x$
$d_R d_R$	$(6,1)_{2/3}$	12	$(\mathbf{10},\mathbf{1},\mathbf{3})$	$120-4x$
$u_R e_R \pm d_R \nu_R$	$(3,1)_{1/3}$	$6+6$	$(\mathbf{10},\mathbf{1},\mathbf{3})$, $(\mathbf{6},\mathbf{1},\mathbf{1})_R$	eigenvalues of M_3
$u_R d_R$ (antisymmetric)	$(3,1)_{1/3}$	6	$(\mathbf{6},\mathbf{1},\mathbf{1})_R$	$180-4x$
$u_L e_L - d_L \nu_L$, $u_L d_L$	$(3,1)_{1/3}$	$6+6$	$(\mathbf{6},\mathbf{1},\mathbf{1})_L$	$180-4x$
$\ell_L \ell_L$	$(1,3)_1$	6	$(\mathbf{10},\mathbf{3},\mathbf{1})$	$108-4x$
$q_L \ell_L$	$(3,3)_{1/3}$	18	$(\mathbf{10},\mathbf{3},\mathbf{1})$	$156-4x$
$q_L q_L$	$(6,3)_{1/3}$	36	$(\mathbf{10},\mathbf{3},\mathbf{1})$	$180-4x$
q_L–u_R, q_L–d_R (octet)	$(8,2)_{1/2}$	64	$(\mathbf{15},\mathbf{2},\mathbf{2})$	$174-4x$
ℓ_L–u_R, q_L–e_R	$(3,2)_{7/6}$	24	$(\mathbf{15},\mathbf{2},\mathbf{2})$	$162-4x$
ℓ_L–d_R	$(3,2)_{1/6}$	12	$(\mathbf{15},\mathbf{2},\mathbf{2})$	$114-4x$
q_L–ν_R	$(3,2)_{1/6}$	12	$(\mathbf{15},\mathbf{2},\mathbf{2})$	114
q_L–d_R, ℓ_L–e_R (singlets)	$(1,2)_{1/2}$	$4+4$	$(\mathbf{15},\mathbf{2},\mathbf{2})$, $(\mathbf{1},\mathbf{2},\mathbf{2})$	$138-4x$, $186-4x$
q_L–u_R, ℓ_L–ν_R (singlets)	$(1,2)_{1/2}$	$4+4$	$(\mathbf{15},\mathbf{2},\mathbf{2})$, $(\mathbf{1},\mathbf{2},\mathbf{2})$	eigenvalues of M_2

Table 5: The radiatively generated masses of the 271 angular modes of the kernel field at the Standard-Model vacuum, in units of $\frac{L}{64\pi^2} g^4 v^2$, with $x = \sum_{\text{fam}} (M_N/M_{W_R})^4$ (Theorem 20.11). Pairs written ff' are of Majorana type, pairs written f–f' of Dirac type; multiplets are given under $SU(3) \times SU(2)_L$ with $|Y|$. The two mixing matrices are $M_3 = \left(\begin{smallmatrix} 84-2x & 2x \\ 2x & 180-2x \end{smallmatrix}\right)$ and $M_2 = \left(\begin{smallmatrix} 138-x & \sqrt{3}x \\ \sqrt{3}x & 186-3x \end{smallmatrix}\right)$, with eigenvalues $132 - 2x \mp 2\sqrt{x^2+576}$ and $162 - 2x \mp 2\sqrt{x^2-12x+144}$.

so for $x > 27$ the Standard-Model vacuum descends towards the left–right symmetric configuration $\theta = \pi/4$, where ν_R and a left-handed lepton carry equal Majorana masses.

Proposition 20.12 (Random-start minimisation)**.** *E_x was minimised on the unit sphere of K by a quasi-Newton method from Gaussian random starts, with two independent implementations. For $x = 0, 20, 22$ every run ended at the energy $2x - 99$ of the Standard-Model orbit or of its mirror. At $x = 25$ a third of the runs ended below it, at a configuration with weight 0.968 on $(\mathbf{10},\mathbf{1},\mathbf{3})$ and 0.032 on $2(\mathbf{6},\mathbf{1},\mathbf{1})$, whose stabiliser of dimension nine contains $\mathfrak{su}(2)_L$ and projects onto colour with rank six, whereas any algebra containing $\mathfrak{su}(3)$ projects with rank at least eight: colour is broken. There four right-handed Weyl fermions carry Majorana masses, one of them about twice as heavy as the other three. Followed in x, this minimum crosses the energy of the Standard-Model orbit at $x_* = 24.88405$.*

We conjecture that in the radiative scenario on K the Standard-Model orbit and its mirror are the global minima exactly for $x < x_*$, and that for $x_* < x < 27$ the Standard-Model vacuum is metastable. The competitor again spreads the Majorana mass, now over four right-handed fermions instead of one, which is what the fermion loop rewards, at the price of breaking colour, which the gauge loop penalises. In terms of masses the local condition and the conjectured global one read

$$M_N < 27^{1/4} M_{W_R} = 2.280\, M_{W_R} \quad \text{and} \quad M_N < 2.233\, M_{W_R} \quad \text{(one family dominating)},$$
$$M_N < 9^{1/4} M_{W_R} = 1.732\, M_{W_R} \quad \text{and} \quad M_N < 1.697\, M_{W_R} \quad \text{(three degenerate families)}.$$

They hold under the following restrictions: a tree potential degenerate on K at the matching scale, one loop at leading logarithm, $\kappa = 0$, which fermion number makes automatic, a common gauge coupling for $\mathfrak{su}(4)$ and $\mathfrak{su}(2)_R$, and a single Yukawa coupling for all components of the kernel field (Assumption 19.4). The masses of Table 5 are curvatures of the leading-logarithm effective potential divided by the kinetic norm; pole masses would need the wave-function renormalisation and a gauge-consistent perturbative expansion, which are not attempted. The bounds

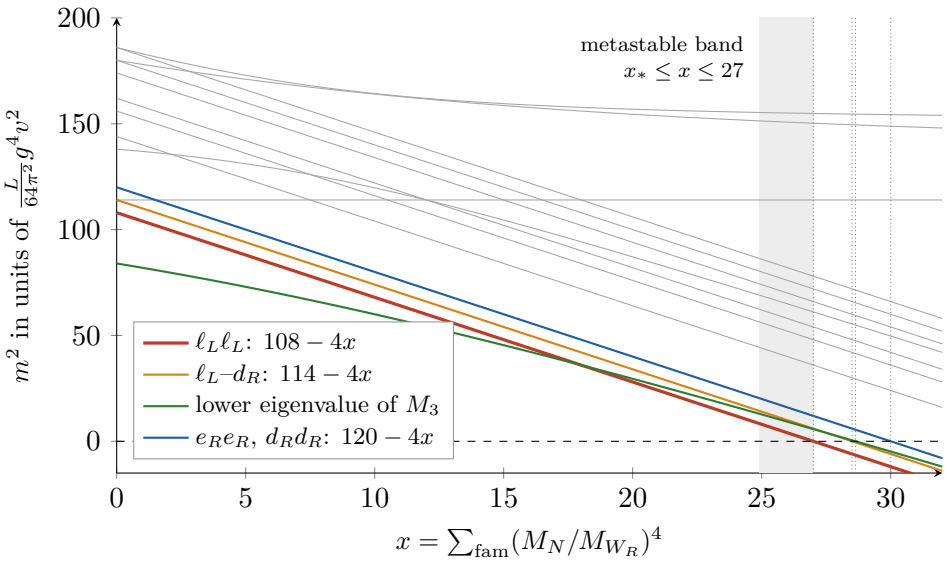


Figure 15: Radiatively generated masses of the angular modes of the kernel field at the Standard-Model vacuum (Theorem 20.11, Table 5). The coloured lines are the four lowest; they cross zero at $x = 27$, 28.5, 315/11 and 30 (dotted). The Standard-Model orbit is a non-degenerate local minimum for $x < 27$, and numerically it is the global minimum for $x < x_* \approx 24.884$; in the shaded band it is metastable.

are therefore statements about this model and approximation, not bounds on heavy-neutrino masses outside it. Within it, the result is that gauge fluctuations make the Standard-Model orientation of the Majorana condensate stable in every direction of the kernel space as long as the heavy neutrinos are not much heavier than W_R; beyond that, the local instability is the condensation of the left-handed triplet, and the configuration that overtakes the Standard-Model orbit globally, at the slightly smaller x_*, breaks colour.

21 Matter in the selected vacuum

We return to the vacuum $\Phi = r\Phi_0$, $r \neq 0$, with $v = |\Phi|/\sqrt{2} = \sqrt{2}\,r$, and ask whether the matter of the block is compatible with it in the sense stated at the beginning of Section 19.

Theorem 21.1 (Compatibility with matter)**.** *(a)* Unbroken algebra. *Among the* 21 *Pati–Salam generators the stabiliser of* Φ_0 *has dimension* 12 *and closes under brackets; its part orthogonal to* $\mathfrak{su}(3) \oplus \mathfrak{su}(2)_L$ *is spanned by* $t_Y = t_3^R + \frac{1}{6}H$*. Among the* 23 *generators it has dimension* 14*; the two further directions* $J + \frac{1}{3}H$ *and* $J - J\Gamma$ *carry the charges* $4B$ *and* $1 - \Gamma$ *and are global.*

(b) Vector bosons. *The matrix* $M_{ab} = \langle [t_a, \Phi_0], [t_b, \Phi_0] \rangle / |\Phi_0|^2$*, taken relative to the module metric, has eigenvalues* 0 *(twelve times),* 4 *(eight times) and* 20 *(once), so the squared vector masses are* 0, g^2v^2 *and* $5g^2v^2$*. The eight carry* $Y = \pm\frac{2}{3}$ *(six leptoquarks* X*) and* ± 1 *(the two* $W_R^{\pm}$*); the one at* $5g^2v^2$ *is neutral, a* Z'*.*

(c) Couplings. *In the module trace* $|t_3^L|^2 = \frac{1}{8}$ *and* $|t_Y|^2 = \frac{5}{24}$*, so*

$$\sin^2\theta_W = \frac{|t_3^L|^2}{|t_3^L|^2 + |t_Y|^2} = \frac{3}{8}$$

at the matching scale, the value of the Pati–Salam and $\mathfrak{so}(10)$ *embeddings [48].*

(d) Dirac operator. *The kinetic kernels are symmetric with $K^0 = \mathbf{1}$ and are preserved by the twelve unbroken generators, and $H(p)^2 = |p|^2 + y^2 r^2 P_\nu$.*

(e) Chirality. *The massless fermions are $(3,2)_{1/6} \oplus (1,2)_{-1/2}$ with $\Gamma = -1$ and $(3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus (1,1)_{-1}$ with $\Gamma = +1$: the unbroken $\mathfrak{su}(2)_L$ acts on one Lorentz chirality only.*

(f) Anomalies. *On the block the Y^3, mixed gravitational, $SU(2)^2Y$, $SU(3)^2Y$ and $SU(3)^3$ anomalies vanish, with and without ν_R, and there are four left-handed $SU(2)$ doublets, an even number [148].*

The formula in (c) deserves a line. With one coupling for the whole Pati–Salam algebra, the effective couplings of t_3^L and t_Y are in the ratio of their norms in the module, $g'^2/g^2 = |t_3^L|^2/|t_Y|^2 = \frac{3/24}{5/24} = \frac{3}{5}$, the familiar normalisation of hypercharge in unified models, and $\sin^2\theta_W = g'^2/(g^2+g'^2) = \frac{3}{8}$. Every requirement of a matter-compatible vacuum is met at this stage except mass generation for the charged fermions, which needs the electroweak stage.

21.1 The electroweak stage and the fermion masses

Everything in this subsection assumes, in addition, a vacuum value of the bidoublet kernels; their dynamics and their scale are not derived. In the radiative scenario of Section 20.5 the colour-singlet doublets have squared masses $138 - 4x$, $186 - 4x$ and the eigenvalues of M_2, so they sit at the Pati–Salam scale and a light doublet requires a tuning that we do not derive.

Proposition 21.2 (Electroweak stage)**.** *(i) The kernels commuting with $\mathfrak{su}(3)$ and with $Q = T_3^L + Y$ form a 12-dimensional space: two each in $(\mathbf{10},\mathbf{1},\mathbf{3})$ and $(\mathbf{10},\mathbf{3},\mathbf{1})$ and four each in $(\mathbf{15},\mathbf{2},\mathbf{2})$ and $(\mathbf{1},\mathbf{2},\mathbf{2})$. In each bidoublet the up-type kernels (no d, e entries) and the down-type kernels (no u, ν entries) form two-dimensional spaces.*

(ii) On the charge sectors u ($Q = \frac{2}{3}$), d ($-\frac{1}{3}$), ν (0) and e (-1), the $(\mathbf{1},\mathbf{2},\mathbf{2})$ kernels give $m_u = m_\nu^D$ and $m_d = m_e$; the $(\mathbf{15},\mathbf{2},\mathbf{2})$ kernels give lepton masses three times the quark masses, with relative sign -3 once aligned with a $(\mathbf{1},\mathbf{2},\mathbf{2})$ kernel. For $\Phi = \kappa_1 K_1 + \kappa_{15} K_{15}$ of each type,

$$m_{\text{quark}} = |\kappa_1 + \kappa_{15}|, \qquad m_{\text{lepton}} = |\kappa_1 - 3\kappa_{15}|,$$

the Georgi–Jarlskog factor [47] used in fits of fermion masses with $\mathbf{10} \oplus \mathbf{126}$ [7].

(iii) With $\Phi = v_R \Phi_0 + \kappa K_{1u}$ the neutral sector has singular values $\frac{1}{2}(\sqrt{v_R^2 + 4\kappa^2} \mp v_R)$, each four times: the seesaw pair [85, 88], with light mass $\approx \kappa^2/v_R$.

(iv) For $\Phi = v_R\Phi_0 + \kappa_u K_{1u} + \kappa_d K_{1d}$ the stabiliser among the 21 generators is nine-dimensional, $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_Q$, and Q annihilates Φ.

Example 21.3 (An electroweak spectrum)**.** At $(v_R, \kappa_u, \kappa_d) = (20, 1, \frac{1}{2})$ the squared vector masses, for $g = 1$ in the units of Theorem 21.1(b), are 0 (nine times), 79.360 (twice, $W^\pm$), 127.872 (once, Z), 6400 (six times), 6480.64 (twice) and 32032.1 (once). The ratio $m_W^2/m_Z^2 = 0.62062$ is close to $\cos^2\theta_W = \frac{5}{8}$, and at $v_R = 200$ it differs from $\frac{5}{8}$ by -4.4×10^{-5}: $\rho = 1$ at tree level up to corrections of order κ^2/v_R^2. In the same vacuum at $(v_R, \kappa_u, \kappa_d) = (20, 3, 1)$ the fermion mass eigenspaces are a light Majorana neutrino of mass $(\sqrt{436} - 20)/2 \approx 0.440$ and a heavy one of mass $(\sqrt{436} + 20)/2 \approx 20.440$, each of real dimension four, the d quarks and the electron with $m_d = m_e = 1$, and the u quarks with mass 3. The chirality content of the neutral pair is $\langle\Gamma\rangle = \mp\cos 2\vartheta$ with $\tan 2\vartheta = 2\kappa_u/v_R$, which is their mixing, and the Majorana-type part of the full mass operator is exactly $v_R\Phi_0$: ν_L acquires its Majorana mass only through the mixing.

What is not obtained here is equally definite: the values of κ_u and κ_d, the potential that would select them together with v_R, and the hierarchy $\kappa \ll v_R$, which in Pati–Salam and left–right models is a tuning of the scalar potential [50].

21.2 What the vacuum does not supply

The statements of this subsection are impossibility results within explicitly stated algebras of operators. They say where an input enters, not that no mechanism outside those algebras could replace it.

Proposition 21.4 (Chirality is field content)**.** *Let $\mathcal{A}$ be the algebra of operators on S_3 generated by the* 91 *internal bivectors and J. Then γ^0 commutes with every element of $\mathcal{A}$ and with the rotations, anticommutes with Γ and the boosts, and maps V_* isometrically onto the orthogonal rank-*64 *subspace $\gamma^0 V_*$, on which $\mathcal{A}$ acts by the same matrices and Γ has the opposite sign. Hence every eigenspace of an element of $\mathcal{A}$ has a mirror image with the same eigenvalue and the opposite chirality, and no vacuum value of a field with values in $\mathcal{A}$, nor any function of such values, separates V_* from $\gamma^0 V_*$.*

The block is selected by $X = \Gamma\chi_W$, which contains the spacetime orientation through $\Gamma = J\gamma^0\gamma^1\gamma^2\gamma^3$ and is not in $\mathcal{A}$. The kernel field is defined on V_* and presupposes it, so it does not escape the proposition either. The Standard-Model vacuum of Theorem 20.4 is chiral because the matter on which it acts already is; this is the arena's form of the fact that a vector-like gauge theory does not generate chirality through its vacuum [135].

Proposition 21.5 (Hermitian order parameters keep rank five)**.** *On the spin-up part of the block, the operators commuting with J, Γ and the twelve Standard-Model generators form a twelve-dimensional algebra whose symmetric part is six-dimensional: the commutative algebra of the six multiplet projectors (Q, L, u, d, e, ν), which are pairwise inequivalent. A Hermitian vacuum value in it commutes with T_3^R and B–L, so its stabiliser contains $\mathfrak{su}(3)\oplus\mathfrak{su}(2)_L\oplus\mathfrak{u}(1)_{T_3^R}\oplus\mathfrak{u}(1)_{B-L}$, of rank five; the Standard-Model algebra, of rank four, is not the unbroken algebra of such an order parameter.*

This is the block version of the statement that order parameters of adjoint type do not lower the rank [76]. The kernel field escapes it because a Grassmann bilinear carries charge: $[t_3^R, \Phi_0] = J\Phi_0 \neq 0$.

Proposition 21.6 (Gauge alignment does not see chirality)**.** *Consider rank-*64 *projectors P commuting with Lorentz and J, with the alignment energy $V(P) = \frac{1}{512}\sum_T \|[T, P]\|^2$ over the* 43 *compact internal bivectors. For P_*, P_{fam}, the colour-singlet lepton projector P_{lep} and the vector-like $Q_v = P_L + \gamma^0 P_L \gamma^0$ with $P_L = P_* \frac{1}{2}(\mathbf{1} - \Gamma)$:*

	P_*	P_{fam}	P_{lep}	Q_v
$V(P)$	5	9	3	5
stabiliser among the 43	23	9	37	25
axial, $P\gamma^0 P = 0$	yes	—	—	no

Alignment prefers P_{lep} to P_*, and $V(P_*) = V(P_L) + V(P_R)$ with $V(P_L) = V(P_R) = \frac{5}{2}$ while $V(Q_v) = 2V(P_L) = V(P_*)$: the alignment energy does not see chirality.

Vacuum alignment [109, 111] selects among projectors only by the amount of gauge symmetry they keep; it neither selects the Pati–Salam block nor distinguishes a chiral block from a vector-like one of the same rank. Finally, with the single Yukawa coupling of Assumption 19.4, matter opposes the Standard-Model vacuum: on $(\mathbf{10}, \mathbf{1}, \mathbf{3})$ every F_k of Theorem 20.9 is positive, and on all of K the Yukawa part of the second variation is $-8x(\mathbf{1} - \Pi_\nu) \leq 0$. The fermion loop lowers the energy along every direction that moves Majorana mass away from ν_R and raises it along none; on its own it would select configurations in which the mass is spread, which break colour. The matter sector is not the agent that selects the Standard-Model orbit. At one loop the gauge fluctuations are, and they succeed for $x < 27$ locally and, numerically, for $x < x_*$.

22 Goldstone bosons, antiparticles, baryon number and topology

The selected vacuum can be examined from four further angles: which of its zero modes are Goldstone bosons and which are eaten; how the discrete operations that exchange particles and antiparticles act on it; whether the potential protects baryon number; and what topological defects its orbit supports. Each answer depends on counting the two global central generators J and $J\Gamma$ alongside the 21 gauged ones.

22.1 Goldstone bosons

Proposition 22.1 (Goldstone directions attached to the symmetries)**.** *J and $J\Gamma$ are antisymmetric, commute with the kinetic kernels and the 21 Pati–Salam generators, span the centre of the 23-dimensional faithful algebra, are anomalous, and carry no gauge fields. If V is invariant under all 23 generators and Φ is stationary, the tangent space of the 23-orbit lies in the kernel of the Hessian; the gauge bosons eat the tangents of the 21-orbit, whose dimension is the rank of the vector mass matrix, and*

$$\#\{\textit{symmetry-generated Goldstone directions not eaten}\} = \dim(23\text{-}\textit{orbit}) - \dim(21\text{-}\textit{orbit}).$$

The count is classical, and it counts the Goldstone directions attached to the 23 specified generators, not all massless scalars: a potential with an accidental symmetry has more zero modes, as the degenerate tree potential of Section 20.5 has 271 against a gauge orbit of dimension nine. Since J and $J\Gamma$ are anomalous, an uneaten direction is a pseudo-Goldstone boson, massive once the anomaly is accounted for.

Theorem 22.2 (No Majoron)**.** *At $r\Phi_0$ the 21-orbit and the 23-orbit both have dimension nine, so the global symmetries add no Goldstone direction beyond those eaten. The reason is*

$$[J, \Phi_0] = [J\Gamma, \Phi_0] = 2\,[t_3^R, \Phi_0]\,:$$

the fermion-number rotation of the $\nu_R\nu_R$ condensate is a gauge rotation, and $J + \frac{1}{3}H$, of charge $4B$, stays unbroken. The nine eaten modes are the $\nu_R\nu_R$ phase (Z'), $\nu_R e_R$ ($W_R^{\pm}$) and $u_R\nu_R$ (six leptoquarks), matching the vector masses of Theorem 21.1.

The would-be Majoron of a spontaneously broken lepton number is eaten by the Z'. At the electroweak stage the situation changes.

Theorem 22.3 (One uneaten direction at the electroweak vacuum)**.** *For $\Phi = v_R\Phi_0 + \kappa_u K_{1u} + \kappa_d K_{1d}$:*

vacuum	stabiliser in 21	stabiliser in 23	massive vectors	23-orbit	Goldstone directions
generic, $\kappa_u\kappa_d \neq 0$	9	10	12	13	**1**
$\kappa_d = 0$ or $\kappa_u = 0$	9	11	12	12	0
$v_R = 0$	16	17	5	6	1

At the electroweak vacuum the tangents of J and $J + \frac{1}{3}H$ lie in the gauge orbit and those of $J\Gamma$ and $J - J\Gamma$ do not; $J + \frac{1}{3}H$ annihilates Φ and $J - J\Gamma$, of charge $1 - \Gamma$, does not.

The mechanism is simple. $J\Gamma$ acts on every Dirac-type kernel as the complex structure $2J\Gamma$, so it rotates the up-type and the down-type lines by the same phase, while the only Standard-Model generator that moves within these lines, the Z direction, rotates them by opposite phases. When both values are nonzero the $J\Gamma$ direction cannot be absorbed. The broken global charge is $1 - \Gamma$, whose anomalies per Weyl fermion in the all-left convention are

charge	$SU(3)^2$	$SU(2)_L^2$	Y^2	Q^2	gravitational	electroweak vacuum
$1-\Gamma$	$\mathbf{2}$	4	$\frac{4}{3}$	$\frac{16}{3}$	16	broken
$4B = 1 + (B-L)$	0	2	-2	—	0	unbroken

The current of the broken charge has a colour anomaly, so its classical Goldstone direction is a pseudo-Goldstone boson of Peccei–Quinn–Weinberg–Wilczek type [105, 140, 143]: an axion. Since its only order parameter is the bidoublet, its decay constant is of the electroweak scale, and such an axion is excluded experimentally [104]. Within this field content, leaving $U(1)_{1-\Gamma}$ intact is therefore a phenomenological obstruction, and the electroweak-stage potential must break it explicitly. The trace potential and the one-loop gauge and Yukawa potential of Section 20 are invariant under every orthogonal conjugation commuting with the Pati–Salam algebra, hence under $\exp\theta(J - J\Gamma)$, and the holomorphic quartic lives on the $\Gamma = +1$ half, where $J - J\Gamma = 0$; none of them lifts the axion.

Proposition 22.4 (Bidoublet mass terms and their CP parity)**.** *On* $(\mathbf{1},\mathbf{2},\mathbf{2})$ *there are three Pati–Salam- and J-invariant symmetric forms, one of them $J\Gamma$-invariant. The CP operator of Theorem 22.6 acts on them with eigenvalues* $(+1,+1,-1)$*: the invariant form B_{inv} is CP-even, and the two-dimensional breaking space splits into a CP-even B_e and a CP-odd B_o. Along the $J\Gamma$-orbit,*

$$B_e\big(e^{\theta\,\mathrm{ad}\,J\Gamma}\Phi\big) = -\kappa_u\kappa_d\cos 4\theta, \qquad B_o\big(e^{\theta\,\mathrm{ad}\,J\Gamma}\Phi\big) = -\kappa_u\kappa_d\sin 4\theta, \qquad B_{\mathrm{inv}} = \mathit{const},$$

the analogues of Re *and* $\mathrm{Im}\,\mathrm{Tr}\,\tilde\Phi^\dagger\Phi$ *in left–right models.*

Theorem 22.5 (The kinetic norm of the axion direction)**.** *Let $a = J\Phi_0$, $e_u = J\Gamma K_{1u}$, $e_d = J\Gamma K_{1d}$; these are mutually orthogonal with $|a|^2 = 4$ and $|e_u|^2 = |e_d|^2 = n_K = 32$, and $t = [J\Gamma,\Phi] = 2v_R a + 2\kappa_u e_u + 2\kappa_d e_d$. The gauge tangent space splits orthogonally into its part in* $\mathrm{span}\{a,e_u,e_d\}$*, spanned by a and $z = \kappa_u e_u - \kappa_d e_d$, and its part orthogonal to that span, and for all v_R,κ_u,κ_d*

$$|P_\perp t|^2 = |t|^2 - \frac{\langle t,a\rangle^2}{|a|^2} - \frac{\langle t,z\rangle^2}{|z|^2} = \frac{16\,n_K\,\kappa_u^2\kappa_d^2}{\kappa_u^2+\kappa_d^2}.$$

The heavy-condensate component of t is a gauge direction, which is why v_R drops out. If $\mu_e^2 B_e$ is the only term breaking $U(1)_{1-\Gamma}$, with canonical amplitudes $h_{u,d} = \sqrt{32}\,\kappa_{u,d}$ and $m_{12}^2 = \mu_e^2/32$, then to leading order in the explicit breaking

$$m_A^2 = \frac{\partial_\theta^2 V}{|P_\perp[J\Gamma,\Phi]|^2} = \frac{m_{12}^2}{\sin\beta\cos\beta}, \qquad \tan\beta = \frac{\kappa_u}{\kappa_d},$$

the pseudoscalar mass of the two-Higgs-doublet model with a soft m_{12}^2 term [17]. Phase-dependent quartics add to it, and whether this curvature is a mass eigenvalue once mixing with the other neutral scalars is included requires the complete Hessian of a specified potential. With CP imposed ($\mu_o^2 = 0$) every extremum of the breaking mass terms has $\sin 4\theta = 0$, so no quadratic term can break CP spontaneously; that needs terms of charge ± 8 under $J\Gamma$, and the potential has $31+31$ of them among its quartics, the analogues of the phase-dependent quartics of left–right models [32].

22.2 Particles and antiparticles

At one constituent charge conjugation is unique up to a $U(1)_J$ rotation, lies in the surplus plane and is the Majorana real structure (Theorem 5.1), and the Fock space of one momentum has a unique charge-zero ground state of energy $-2E$ (Theorem 5.7). That value is the zero-point energy of two particle and two antiparticle modes; summed over momenta it is the fermion term $-2\sum_{\mathrm{Weyl}}$ of the one-loop potential (36), the very term through which matter opposes

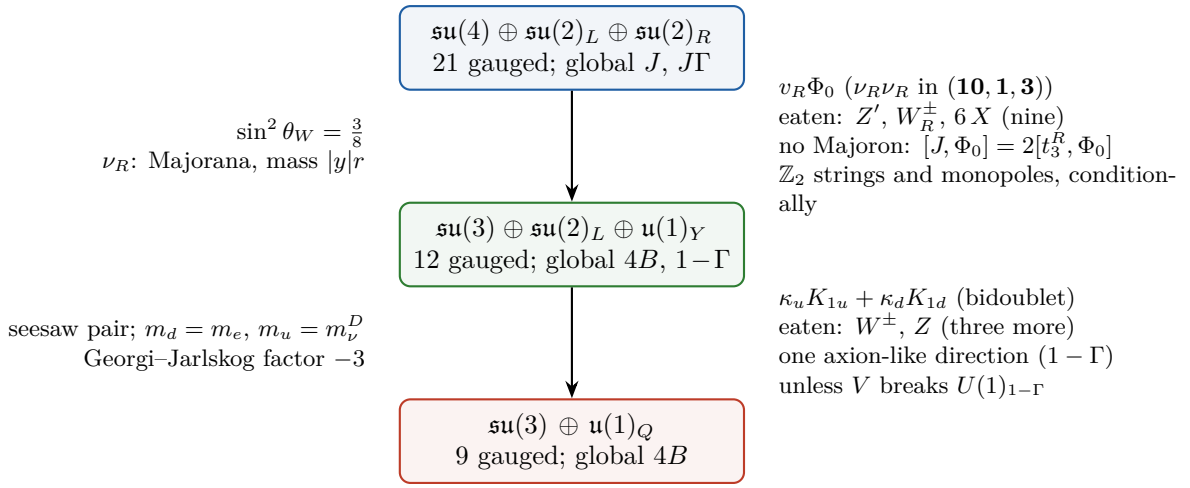


Figure 16: The two stages of symmetry breaking in the selected vacuum (Sections 21 and 22). Each arrow lists the order parameter, the vectors that eat the Goldstone directions, and the uneaten directions attached to the global central charges.

the Standard-Model vacuum. At three constituents the discrete operations are classified as follows. A map $\psi \mapsto S\psi$ with S orthogonal on V_* exchanges particles and antiparticles when $SJ = -JS$; it keeps or reverses chirality according to $S\Gamma S^{-1} = \pm\Gamma$; it goes with $\mathbf{x} \mapsto -\mathbf{x}$ when $S^{\top} K^i S = -K^i$.

Theorem 22.6 (Four classes of discrete operators)**.** *The blades of* $\mathrm{Cl}(9,9)$ *that commute with the three involutions of* P_* *and with the rotations number* 8192*, and they fall into exactly four classes of* 2048*: internal (*J*-linear,* Γ *kept, boosts kept),* P *(*J*-linear,* Γ *reversed, boosts reversed),* C *(*J*-antilinear,* Γ *reversed, boosts kept) and* CP *(*J*-antilinear,* Γ *kept, boosts reversed). The two missing types are forbidden for every operator, since an antilinear map commuting with the Lorentz action and keeping* Γ *would identify* $(\frac{1}{2},0)$ *with its conjugate* $(0,\frac{1}{2})$*. Representatives are*

$$C = E_2E_7E_{10}E_{11}E_{12}, \qquad P = O = E_1E_7E_8, \qquad CP = E_1E_2E_7E_8E_{10}E_{11}E_{12}E_{14},$$

and all three normalise the Pati–Salam algebra and preserve the kinetic, gauge and Yukawa terms.

The factors of C are instructive: E_2 is the surplus generator of the first constituent, where the one-constituent charge conjugation lives; E_7 exchanges left and right; and $E_{10}E_{11}E_{12}$, an odd number of colour generators, induces the outer automorphism of $\mathfrak{su}(4)$ that exchanges $\mathbf{4}$ and $\bar{\mathbf{4}}$. $P = O$ is parity composed with an $SU(2)$ Weyl reflection, and $CP = \pm(C\,O)\,E_7E_{14}$ with E_7E_{14} a gauge element.

Theorem 22.7 (CP fixes the order parameter; C and P move it)**.** $CP\,\Phi_0\,CP^{\top} = \Phi_0$. $C\Phi_0C^{\top}$ *is the* $\nu_L\nu_L$ *kernel, which commutes with* $\mathfrak{su}(3)$*,* $\mathfrak{su}(2)_R$ *and* Q *but not with* $\mathfrak{su}(2)_L$*, and* $P\Phi_0P^{\top}$ *is the* e_Le_L *kernel. Both images lie on the mirror orbit in* $(\mathbf{10},\mathbf{3},\mathbf{1})$*: the degeneracy of Proposition 20.10 is realised by two distinct discrete symmetries.*

Invariance of the order parameter is not invariance of the theory. The kinetic, gauge and Yukawa terms and the trace invariants are CP-invariant; the general potential on K is invariant only for couplings satisfying reality conditions; and on $(\mathbf{10},\mathbf{1},\mathbf{3})$ the holomorphic quartic needs a generalised CP, composed with a baryon-number rotation $e^{\theta(J+H/3)}$, which is a classical symmetry of the renormalisable theory for every κ but is anomalous, since $4B$ has $SU(2)_L^2$ anomaly 2. Whether a CP survives in the quantum theory is not settled.

Which fermions are their own antiparticles follows from a pairing rule. If M is a mass kernel and t an unbroken generator with $[t,M] = 0$, then a Majorana-type M satisfies $M\hat{t} = -\hat{t}M$,

so within one chirality it pairs a representation of the unbroken group with its conjugate, while a Dirac-type M pairs equal representations of opposite chirality. A single irreducible multiplet can therefore carry a Majorana mass only if its representation of the unbroken group is real: a nonzero charge under an unbroken $U(1)$ excludes this, while a nontrivial real representation of a non-abelian group does not. In the block, every multiplet other than ν_R carries a nonzero hypercharge at the Pati–Salam stage, and every multiplet other than ν_L and ν_R is a colour triplet or electrically charged at the electroweak stage. So ν_R is the only Majorana fermion at the first stage, the light ν and heavy N of Example 21.3 are the Majorana fermions at the second, and the massive ν_R is mapped to itself by CP with eigenvalues $\pm i$, since $CP^2 = -\mathbf{1}$ on the block.

22.3 Baryon number in the potential

For a Pati–Salam invariant $B{-}L = 0$, so $4B = F$ and $\Delta B = F/4$ in terms of the fermion number F.

Theorem 22.8 (Invariant polynomials by fermion number)**.** *The Pati–Salam-invariant polynomials on* K *of degrees two and four, graded by* F*, number*

degree	$F = 0$	$F = \pm 4$ ($\lvert\Delta B\rvert = 1$)	$F = \pm 8$ ($\lvert\Delta B\rvert = 2$)	total
2	12	$3 + 3$, all on $2(\mathbf{6}, \mathbf{1}, \mathbf{1})$	—	18
4	402	$127 + 127$	$15 + 15$, including $\mathcal{D}$	686

With $U(1)_F$ *imposed the couplings number* $12 + 402$*; with* $U(1)_{1-\Gamma}$*,* $8 + 228$*; with both* F *and* $J\Gamma$*-charge,* $6 + 162$*.*

Pati–Salam gauge invariance alone does not forbid baryon-number violation. The six $F = \pm 4$ quadratic invariants are holomorphic bilinears in the complex sextets of $2(\mathbf{6}, \mathbf{1}, \mathbf{1})$; under $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{B-L}$ each sextet is $\bar{\mathbf{3}}_{2/3} \oplus \mathbf{3}_{-2/3}$, diquark components coupling to quark pairs and leptoquark components coupling to quark–lepton pairs, and $\epsilon^{abcd}\Delta_{ab}\Delta'_{cd}$ pairs one with the other. Through the Yukawa coupling such a mass mixing generates tree-level $\Delta B = \Delta L = 1$ four-fermion operators, of the type that mediates proton decay. Two protections are available and they are not equivalent: continuous $U(1)_F$ forbids every $F \neq 0$ term and conserves baryon number perturbatively (it is violated by $SU(2)$ instantons [132]); the $\mathbb{Z}_4$ generated on K by conjugation with $e^{\pi J/4}$ forbids the $\lvert\Delta B\rvert = 1$ terms but allows the $\lvert\Delta B\rvert = 2$ ones that mediate neutron–antineutron oscillation. $U(1)_F$, which the radiative scenario of Section 20 has, is sufficient, not necessary.

22.4 Topology of the vacuum orbit

Theorem 22.9 ($\mathbb{Z}_2$ strings)**.** *On the block* $H = 3J(B{-}L)$*, so* $e^{\theta H}$ *acts on quarks as* $\cos\theta + J\sin\theta$ *and on leptons as* $\cos 3\theta - J\sin 3\theta$*. The colour element* $g_* = e^{\pi H/3}$ *fixes* Φ_0*, acts as* -1 *on the* ν_R *line and on every lepton, and the path* $\theta \mapsto e^{\theta H}\Phi_0 e^{-\theta H} = \cos 6\theta\,\Phi_0 - \sin 6\theta\, J\Phi_0$*,* $\theta \in [0, \pi/3]$*, is a closed loop on which the condensate phase winds once. Every generator of the stabiliser annihilates the* ν_R *line, so the identity component of the stabiliser acts trivially there and* g_* *is not in it. Lifting to* $SU(4) \times SU(2)_L \times SU(2)_R$*, the stabiliser has exactly two components, so the gauge orbit of the vacuum has* $\pi_1 = \mathbb{Z}_2$*, and* $\pi_2 \supseteq \mathbb{Z}$ *from* $U(1)_Y$*.*

If the gauge orbit is the complete set of minima of the chosen potential, as it is on $(\mathbf{10}, \mathbf{1}, \mathbf{3})$ in the region of Theorem 20.4, the vacuum supports topologically stable $\mathbb{Z}_2$ strings around which ν_R changes sign, the Pati–Salam form of the $\mathbb{Z}_2$ strings of $\mathfrak{so}(10)$ broken by a **126** [71], and Pati–Salam monopoles. If the minimum set is larger, as in the degenerate limit with accidental $O(272)$ symmetry, the relevant fundamental group is that of the larger manifold, and a string may unwind through flat directions. The Standard-Model orbit and its mirror lie in

different Pati–Salam components, so no gauge transformation connects them; whether they are disconnected components of the full minimum set, which would give domain walls [151], depends on paths through configurations outside both orbits. Along the great circle of Proposition 20.10 the radiative energy has a barrier between them for $x < 27$, but other paths are not controlled.

Part VI
Curvature, transport and the photon

23 A curved arena

Nothing in the algebra requires the ambient metric to be flat. Promoting the metric to a field of signature $(3,3)$, the generators to an orthonormal frame with a spin connection, and the pairing to a field of endomorphisms, one obtains a curved arena. This section describes its geometry, builds one explicit curved model against which statements can be tested, and computes the six-dimensional Einstein tensor of that model exactly. The two readings of $\mathbb{R}^{3,3}$ then each produce the Einstein tensor of a four-metric, in quite different ways, and neither determines the metric.

23.1 The pairing as a para-complex structure

Proposition 23.1 (The arena is almost para-Hermitian)**.** *Let (M, g) have signature $(3,3)$ and let $\mathcal{K} \in \mathrm{End}(TM)$ satisfy $\mathcal{K}^2 = \mathbf{1}$ and $g(\mathcal{K}\cdot, \mathcal{K}\cdot) = -g$. Then $(M, g, \mathcal{K})$ is almost para-Hermitian: the eigendistributions $T^{\pm}M$ are isotropic of rank three, since $g(X,Y) = -g(\mathcal{K}X, \mathcal{K}Y) = -g(X,Y)$ on $T^{+}M$. It is para-Kähler, equivalently bi-Lagrangian, if and only if $\nabla^{\mathrm{LC}}\mathcal{K} = 0$ for the Levi-Civita connection [28, 37]; then each eigendistribution is parallel, involutive and totally geodesic, and $\Omega(X,Y) = g(\mathcal{K}X, Y)$ is skew and closed.*

The para-complex structure need not be posited: it is the grading root of Proposition 3.3, $\mathcal{K}(v) = \frac{1}{2}[\mathsf{P}, v]$ with $\mathsf{P} = \sum_i e_i e_{i+3}$, whose eigenspaces are the two totally null three-planes spanned by $e_i \mp e_{i+3}$. The null 3+3 splitting it induces is a different decomposition from the physical 4+2 splitting into the interior four-plane and the surplus plane, and conflating them is the first thing to avoid. A connection with torsion can preserve $\mathcal{K}$ without making the eigendistributions integrable or Ω closed, so a $\mathcal{K}$-preserving connection does not deliver the para-Kähler conclusion [82, 79]. Para-Kähler geometry is the world of $\mathcal{K}^2 = +\mathbf{1}$, while chirality is comfortable in the world of $J^2 = -\mathbf{1}$; this is one way of naming the difficulty that runs through the whole construction.

23.2 An explicit model with a normal $U(1)$

Split the tangent space as $TM = E \oplus N$ with $E = \mathrm{span}\{\widehat{W}, e_4, e_5, e_6\}$ of signature $(1,3)$, the symmetric clock of Proposition 3.5 playing the role of γ^0, and $N = \mathrm{span}\{u_1, u_2\}$ with

$$u_1 = \tfrac{1}{\sqrt{2}}(e_1 - e_2), \qquad u_2 = \tfrac{1}{\sqrt{6}}(e_1 + e_2 - 2e_3),$$

which complete the clock inside the positive triple: the relative times of Section 3.3. The metric does not select this split; it is extra data.

Lemma 23.2 (The normal plane carries a commuting rotation)**.** $J_N := u_1 u_2 = \frac{1}{\sqrt{3}}(e_2e_3 + e_3e_1 + e_1e_2)$ *satisfies $J_N^2 = -\mathbf{1}$ and commutes with $\widehat{W}$, e_4, e_5 and e_6.*

Proof. J_N is a sum of time–time bivectors and has no spatial index, so it commutes with e_4, e_5, e_6 by (3); it is the unit bivector of the plane orthogonal to $\widehat{W}$ inside the positive triple, so it commutes with $\widehat{W}$; and u_1, u_2 are orthonormal of square $+\mathbf{1}$, so $(u_1u_2)^2 = -u_1^2u_2^2 = -\mathbf{1}$. □

The normal plane therefore carries a compact $\mathfrak{u}(1)$ commuting with the physical generators. Give it a connection, $\nabla^\perp u_1 = -A u_2$, $\nabla^\perp u_2 = A u_1$, with spin connection $\Omega_N = \frac{1}{2} A J_N$. The model metric is

$$ds_6^2 = g^{(4)}_{\mu\nu}(x)\, dx^\mu dx^\nu + (du + vA)^2 + (dv - uA)^2, \qquad A = A_\mu(x)\, dx^\mu. \tag{39}$$

Proposition 23.3 (The model)**.** *The metric* (39) *has signature* $(3,3)$*; its restriction to* $\Sigma = \{u = v = 0\}$ *is* $g^{(4)}$*; the induced normal connection is* A*; and* $R^u{}_{v\mu\nu}|_\Sigma = (dA)_{\mu\nu}$*. For* $g^{(4)}$ *flat and* $A = Bx\, dz$ *one finds* $R^u{}_{vxz}|_\Sigma = B$ *while every component* $R^\mu{}_{\nu\rho\sigma}|_\Sigma$ *vanishes.*

This is the Kaluza–Klein construction for a rank-two real bundle with structure group $SO(2)$, and $R^u{}_{v\mu\nu} = F_{\mu\nu}$ is the standard formula for the curvature of a normal bundle [4, 98]. Two consequences are easy to misread in either direction: ambient curvature does not imply observable curvature, since the last example has a curved six-geometry with a flat induced four-metric; and a four-dimensional distribution is not a spacetime until an integral submanifold or a quotient is produced, which here Σ supplies.

Proposition 23.4 (The surplus constraints are first class)**.** *For a geodesic test particle in* (39) *put* $\ell = v p_u - u p_v$*,* $k_\mu = p_\mu - A_\mu \ell$*, so that* $H = \frac{1}{2}(g^{\mu\nu}_{(4)} k_\mu k_\nu + p_u^2 + p_v^2)$*. With* $\phi_1 = p_u$*,* $\phi_2 = p_v$ *and* $C := A_\mu g^{\mu\nu}_{(4)} k_\nu$*,*

$$\{\phi_1, \phi_2\} = 0, \qquad \{\phi_1, H\} = -C\phi_2, \qquad \{\phi_2, H\} = +C\phi_1,$$

so the constraints are first class for every $A_\mu(x)$ *and every* $g^{(4)}$*. The reduced phase space has dimension* $12 - 2 \times 2 = 8$*, that of* T^*M^4*, and* $H_{\mathrm{red}} = \frac{1}{2} g^{\mu\nu}_{(4)} p_\mu p_\nu$*.*

The potential has cancelled from the reduced Hamiltonian, and not by accident: the constraints generate translations in (u, v), so the normal $U(1)$ acts trivially on the reduced phase space and a spinless test particle cannot see it. Whatever makes the normal connection physical must come from the spinor sector, which Section 25 examines. This is a first instance of the constraint structure that Section 30 derives from a worldline gauge symmetry.

23.3 The six-dimensional Einstein tensor

In polar normal coordinates $u + iv = re^{i\phi}$ the model reads $ds_6^2 = h_{\mu\nu} dx^\mu dx^\nu + dr^2 + r^2(d\phi - A)^2$ with $h = g^{(4)}$. Write $W_{\mu\nu} = F_{\mu\rho} F_\nu{}^\rho - \frac{1}{4} h_{\mu\nu} F^2$. In signature $(+,-,-,-)$, $-W_{00} = \frac{1}{2}(|\mathbf{E}|^2 + |\mathbf{B}|^2)$, so a physical Maxwell field enters Einstein's equation as $\mathcal{G} = -2W$ in units where the coupling is absorbed.

Theorem 23.5 (The ambient Einstein tensor)**.** *For every* $r > 0$*, and not as a series in* r*,*

$$\mathcal{G}^{(6)}_{\mu\nu} = \mathcal{G}_{\mu\nu}[h] - \tfrac{r^2}{2} W_{\mu\nu}, \qquad \mathcal{G}^{(6)}_{rr} = -\tfrac{1}{2} R[h] + \tfrac{r^2}{8} F^2, \qquad \mathcal{G}^{(6)}_{\hat\phi\hat\phi} = -\tfrac{1}{2} R[h] + \tfrac{3r^2}{8} F^2,$$

$$\mathcal{G}^{(6)}_{\mu\hat\phi} = \tfrac{r}{2} \nabla^\rho F_{\rho\mu}, \qquad \mathcal{G}^{(6)}_{\mu r} = \mathcal{G}^{(6)}_{r\hat\phi} = 0, \qquad R^{(6)} = R[h] - \tfrac{r^2}{4} F^2,$$

with indices and ∇ *referring to* h*. On the slice* Σ *the Einstein tensor is*

$$\mathcal{G}^{(6)}\big|_\Sigma = \mathrm{diag}\big(\mathcal{G}_{\mu\nu}[h],\, -\tfrac{1}{2} R[h],\, -\tfrac{1}{2} R[h]\big)\,:$$

its tangential block is the four-dimensional Einstein tensor exactly, with no stress term from F*.*

Proof. The projection onto (M, h) is a semi-Riemannian submersion with flat, totally geodesic fibres: for $U, V \in \{\partial_r, \partial_\phi\}$ and the horizontal lifts $E_\mu = \partial_\mu + A_\mu \partial_\phi$ one has $[U, E_\mu] = 0$ and $E_\mu\, g(U, V) = 0$, so the Koszul formula gives $g(\nabla_U V, E_\mu) = 0$. The integrability tensor is $\frac{1}{2}[E_\mu, E_\nu]^{\mathrm{vert}} = \frac{1}{2} F_{\mu\nu} \partial_\phi$ with $|\partial_\phi|^2 = r^2$. O'Neill's formulas, valid for semi-Riemannian submersions with nondegenerate fibres [97, 98, 12], then give $R^{(6)}_{\mu\nu} = R_{\mu\nu}[h] - \frac{r^2}{2} F_{\mu\rho} F_\nu{}^\rho$, $R^{(6)}_{\mu\hat\phi} = \frac{r}{2} \nabla^\rho F_{\rho\mu}$,

$R^{(6)}_{\hat\phi\hat\phi} = \frac{r^2}{4}F^2$ and $R^{(6)}_{rr} = 0$, from which the Einstein components follow. They have also been checked symbolically on a Friedmann base $h = \mathrm{diag}(1, -t^2, -t^2, -t^2)$ with $A = z\,dt + \frac{1}{2}x^2 dy$, which carries both electric and magnetic fields. □

Corollary 23.6 (Sign, reduction and rigidity)**.** *(i) The tangential equation $\mathcal{G}[h] = +\frac{r^2}{2}W$ has the opposite sign to $\mathcal{G} = -2W$: the normal field strength gravitates with negative energy. The sign is forced: a Lorentzian four-plane in $\mathbb{R}^{3,3}$ uses one positive and three negative directions, so its normal plane has signature $(2, 0)$, the sign of time.*

(ii) Over a normal disc of radius r_c, $\iint \sqrt{|g_6|}\, R^{(6)}\, d\phi\, dr = \sqrt{|h|}(\pi r_c^2 R[h] - \frac{\pi r_c^4}{8}F^2)$. The Euler–Lagrange equation of $\int \sqrt{|h|}(a_1 R + a_2 F^2)$ is $a_1\mathcal{G} + 2a_2 W = 0$, so Einstein–Maxwell theory has $a_2/a_1 = +1$ while the reduced action has $a_2/a_1 = -r_c^2/8$: the wrong relative sign.

(iii) The vacuum is rigid: $\mathcal{G}^{(6)} = 0$ if and only if $F = 0$ and h is Ricci-flat. The internal equations differ by $\frac{r^2}{4}F^2$, which forces $F^2 = 0$ and then $R[h] = 0$; the tangential equation must hold at every r while $\mathcal{G}[h]$ and W do not depend on r, so $\mathcal{G}[h] = 0$ and $W = 0$ separately, and $W = 0$ forces $F = 0$ because $-W_{00}$ is positive definite.

Reduction on directions of timelike sign is known to give fields whose kinetic terms have the wrong sign [68], and the negative-energy vector found here is an instance. The two mechanisms of rigidity can be separated by changing the fibre. With a normal fibre of constant size, $ds_6^2 = h + \ell^2(d\psi - A)^2 + m^2 d\chi^2$, still of signature $(3, 3)$, the internal equations force only $F^2 = 0$, and null fields survive.

Example 23.7 (Negative-energy null radiation)**.** Take $h = 2\,dU\,dV - H\,dU^2 - dX^2 - dY^2$ and $A = a(U)X\,dU$, so $F = a(U)\,dX\wedge dU$ is null. The only nonzero component of the six-dimensional Einstein tensor is $\mathcal{G}^{(6)}_{UU} = -\frac{1}{2}(H_{XX} + H_{YY} - \ell^2 a^2)$, so

$$H = \tfrac{1}{4}\ell^2 a(U)^2(X^2 + Y^2)$$

solves the six-dimensional vacuum with $F \neq 0$. In four-dimensional terms $\mathcal{G}_{UU}[h] = -\frac{1}{2}\ell^2 a^2 < 0$, whereas a physical null Maxwell wave has $\mathcal{G}_{UU} = -2W_{UU} = +2a^2 > 0$: the wave is negative-energy radiation. With the fibre size the coordinate r, as in (39), the same field gives a tangential component graded in r that forces $a = 0$.

More generally, with $ds_6^2 = h + e^{2\alpha(x)}(d\psi - A)^2 + e^{2\beta(x)}d\chi^2$,

$$\mathcal{G}^{(6)}_{\hat\psi\hat\psi} - \mathcal{G}^{(6)}_{\hat\chi\hat\chi} = \tfrac{1}{4}e^{2\alpha}F^2 - \Box(\alpha - \beta) - \nabla(\alpha - \beta)\cdot\nabla(\alpha + \beta).$$

For a breathing mode $\alpha = \beta$ the difference is a pure constraint and only null F survives; for a relative modulus it is a field equation for $\alpha - \beta$, and $F^2 \neq 0$ becomes admissible. Non-null F therefore needs the ratio of the two normal sizes to be a dynamical field. None of these escapes repairs the sign.

23.4 Off the slice

On Σ the F-couplings, which all carry r, switch off. Taking the four-geometry off the zero section requires care, because the level set $\{r = r_0\}$ is five-dimensional. Its quotient by the $U(1)$ is four-dimensional and carries h, and the ambient Einstein tensor in the horizontal frame is $\mathcal{G}[h] - \frac{r_0^2}{2}W$, with the sign above. A section $\phi = \chi(x)$ of $\{r = r_0\}$ is a four-dimensional submanifold with induced metric

$$h_\chi = h + r_0^2\, B\otimes B, \qquad B := d\chi - A,$$

where B is the Stückelberg combination, invariant under $A \mapsto A + d\lambda$, $\chi \mapsto \chi + \lambda$; $\det h_\chi = \det h\,(1 + r_0^2|B|^2)$, so h_χ is Lorentzian only where $1 + r_0^2|B|^2 > 0$, and the section is totally geodesic on an open set only where $B = 0$, which requires $F = 0$.

Example 23.8 (A uniform magnetic field)**.** Take $h = \eta$ and $F = b\,dx \wedge dz$. On a static section the circulation of B around a circle of radius R_c in an x–z plane is $-\pi b R_c^2$, whatever χ, so $\sum_i B_i^2 \geq b^2 R_c^2/4$ somewhere on it, and the section metric has signature $(2,2)$ there as soon as $R_c > 2/(r_0 b)$. In the gauge $A = bx\,dz$ with $\chi = 0$,

$$h_\chi = dt^2 - dx^2 - dy^2 - (1 - k^2x^2)\,dz^2, \qquad R[h_\chi] = -\frac{2k^2}{(1-k^2x^2)^2}, \qquad k = r_0 b,$$

which degenerates at $|x| = 1/k$, where its curvature diverges, and has signature $(2,2)$ beyond. The Lorentzian region of a static section in a uniform field contains no circle of radius larger than $2/(r_0 b)$.

23.5 Holonomy acting on the bound orbits

The interior dynamics of Section 16 can now be transported around a loop of the model. A circuit enclosing normal flux Φ rotates the frame by $h = \exp(\Phi R_N)$, with R_N the rotation of the normal plane N. R_N preserves η, but it rotates a temporal plane without its spatial partner, so it is not among the diagonal rotations of Theorem 12.1, and exactly

$$h = \mathbf{1} + \sin\Phi\, R_N + (1 - \cos\Phi) R_N^2, \qquad hGh^{-1} - G = \sin\Phi\, A_1 + (1 - \cos\Phi)\, A_2$$

with A_1, A_2 linearly independent, so $hGh^{-1} = G$ if and only if $\Phi \in 2\pi\mathbb{Z}$. The second-order equation $\ddot{X} = -\kappa^2 X$ does not see this, but the momentum does: a zero-momentum orbit $X(\tau) = e^{-\kappa G\tau} X_0$ carried once around becomes $hX(\tau)$, and

$$\alpha_+(hX) = \kappa\,(G - hGh^{-1})\,hX,$$

which vanishes for every X_0 only when $\Phi \in 2\pi\mathbb{Z}$. Normal curvature therefore converts bound orbits of zero momentum into orbits that carry momentum, through exactly the failure of the normal $U(1)$ to centralise the binding root.

23.6 Flows do not transfer

In DeTurck gauge Ricci flow reads $\partial_t g_{MN} = \Delta_g g_{MN} + Q(g, \partial g)$, and short-time existence requires the principal symbol $g^{PQ}\xi_P\xi_Q$ to be definite [59, 31]. In signature $(3,3)$ it has signature $(3,3)$: the operator is ultrahyperbolic, the gauge-fixed flow is not parabolic, and the standard Ricci–DeTurck theory does not apply. In the para-Kähler case the metric comes locally from a potential depending on coordinates along both foliations, $g_{i\bar{\jmath}} = \partial^2\Phi/\partial x^i \partial y^j$ [123], but suppressing one set of variables degenerates the metric rather than producing a Hessian one; a real Monge–Ampère flow on a convex locus [20] would need its own domain, data and a proof that it preserves the arena's structures, and is a programme rather than a result.

24 Curvature in the field-strength reading

In the field-strength reading the six directions are $\Lambda^2\mathbb{R}^{1,3}$, η is the induced metric and G the Hodge star (Theorem 12.3). A tensor R_{abcd} antisymmetric in each pair is then an operator $\mathcal{R}$ on the frame, $(\mathcal{R}\omega)_{ab} = \frac{1}{2} R_{ab}{}^{cd}\omega_{cd}$ [125, 12], and the conditions that make it a Riemann tensor become conditions stated with η and G. Write $\mathrm{Ric}(\mathcal{R})_{bd} = R^a{}_{bad}$.

Proposition 24.1 (The curvature operator lives on the frame)**.** *$\mathcal{R}$ comes from a pair-symmetric tensor, $R_{abcd} = R_{cdab}$, if and only if $\mathcal{R}$ is η-self-adjoint, a 21-dimensional space $\mathcal{S}$; the η-antisymmetric operators give $R_{abcd} = -R_{cdab}$ instead. On $\mathcal{S}$, $\operatorname{Tr}\mathcal{R} = \frac{1}{2}R$ and*

$$R_{abcd} + R_{acdb} + R_{adbc} = -\tfrac{1}{2}\operatorname{Tr}(G\mathcal{R})\,\varepsilon_{abcd}, \qquad \varepsilon_{0123} = +1,$$

so the first Bianchi identity is the single linear condition $\operatorname{Tr}(G\mathcal{R}) = 0$*, and the algebraic curvature operators form a* 20*-dimensional subspace* $\mathcal{V}$*, the dimension of the space of Riemann tensors in four dimensions.*

Pair symmetry is the same condition $G = G^\dagger$ that makes the interior shift factorise (Theorem 16.2), imposed on an operator of another type: the frame is shared, not the object. The binding root, which is η-self-adjoint, is itself in $\mathcal{S}$ but not in $\mathcal{V}$, since $\operatorname{Tr}(G \cdot G) = -6$; it is the one direction of $\mathcal{S}$ that violates the Bianchi identity.

24.1 A phase family that ends at the Einstein tensor

Theorem 24.2 (The phase family)**.** *For every* $\mathcal{R} \in \mathcal{S}$*, on or off the Bianchi shell, and every* θ*, the operator* $\mathcal{R}_\theta := e^{\theta G}\mathcal{R}\, e^{\theta G}$ *lies in* $\mathcal{S}$ *and*

$$\operatorname{Ric}(\mathcal{R}_\theta)_{ab} = R_{ab} - \tfrac{1}{2}\sin^2\theta\, R\, g_{ab} + \tfrac{1}{2}\sin 2\theta\, \operatorname{Tr}(G\mathcal{R})\, g_{ab}.$$

At $\theta = \pi/2$ *the operator is the double dual* $G\mathcal{R}G$ *of Lanczos [73, 92], and its contraction is the Einstein tensor* $R_{ab} - \frac{1}{2}Rg_{ab}$*.*

Proof. Two lemmas hold on all of $\mathcal{S}$: $\operatorname{Ric}(G\mathcal{R}G) = \operatorname{Ric}(\mathcal{R}) - \frac{1}{2}Rg$, and $\operatorname{Ric}(G\mathcal{R}) + \operatorname{Ric}(\mathcal{R}G) = \operatorname{Tr}(G\mathcal{R})\, g$. With $e^{\theta G} = \cos\theta + \sin\theta\, G$,

$$\mathcal{R}_\theta = \cos^2\theta\, \mathcal{R} + \sin\theta\cos\theta\, (G\mathcal{R} + \mathcal{R}G) + \sin^2\theta\, G\mathcal{R}G,$$

and the lemmas contract this to the formula. It has also been checked as a polynomial identity in the 21 entries, modulo $\cos^2\theta + \sin^2\theta = 1$. Since G is η-self-adjoint, so is $e^{\theta G}$, and $\mathcal{R}_\theta \in \mathcal{S}$. □

Example 24.3 (A maximally symmetric curvature)**.** For de Sitter space of curvature k, $R_{abcd} = k(g_{ac}g_{bd} - g_{ad}g_{bc})$, the curvature operator is $\mathcal{R} = k\,\mathbf{1}$ on the frame up to the sign convention of the induced metric, so $\operatorname{Tr}\mathcal{R} = 6k = \frac{1}{2}R$ gives $R = 12k$ and $\operatorname{Tr}(G\mathcal{R}) = k\operatorname{Tr}G = 0$. Then $G\mathcal{R}G = kG^2 = -k\,\mathbf{1}$, and the formula gives $\operatorname{Ric}(\mathcal{R}_{\pi/2}) = R_{ab} - \frac{1}{2}Rg_{ab} = 3kg_{ab} - 6kg_{ab} = -3kg_{ab}$, the Einstein tensor of de Sitter space, which is $-\Lambda g_{ab}$ with $\Lambda = 3k$. The double dual of a constant-curvature operator is minus itself, and the trace reversal that turns Ricci into Einstein is the sign change $G^2 = -\mathbf{1}$.

Corollary 24.4 (What selects the endpoint)**.** *(i) For the curvature operator of a metric and a constant phase, the contracted Bianchi identity gives* $\nabla^a H_{ab}(\theta) = \frac{1}{2}\cos^2\theta\, \nabla_b R$ *for* $H(\theta) := \operatorname{Ric}(\mathcal{R}_\theta)$*, so divergence-freedom for every metric selects* $\theta \equiv \pi/2 \pmod{\pi}$*.*

(ii) $\operatorname{Tr}(G\mathcal{R}_\theta) = \cos 2\theta\, \operatorname{Tr}(G\mathcal{R}) - \frac{1}{2}\sin 2\theta\, R$*, so on the Bianchi shell and for* $R \neq 0$*,* $\mathcal{R}_\theta$ *is itself a curvature operator only at* $\sin 2\theta = 0$*, compatible with (i) and narrowing nothing further.*

(iii) $Q(\mathcal{R}) := \frac{1}{2}\operatorname{Tr}(\mathcal{R}G\mathcal{R}G) = -\frac{1}{8}(R^2 - 4R_{ab}R^{ab} + R_{abcd}R^{abcd})$ *for every pair-symmetric* $\mathcal{R}$*, and* $dQ[\mathcal{D}] = \operatorname{Tr}(G\mathcal{R}G\mathcal{D})$*: the double dual is the algebraic gradient of the Gauss–Bonnet density, exactly so on* $\mathcal{V}$*, where* $G\mathcal{R}G = \operatorname{grad}_{\mathcal{V}} Q$*.*

The gradient in (iii) does not lift to a field equation. The spacetime integral of the Gauss–Bonnet density has identically vanishing metric variation in four dimensions, which is Lanczos's identity [73, 80]: the Euler–Lagrange tensor $P_a{}^{cde}R_{bcde} - \frac{1}{2}g_{ab}\mathcal{E}_{\mathrm{GB}} - 2\nabla^c\nabla^d P_{acdb}$ with $P = -2\,{*R*}$ vanishes because the double dual is divergence-free and $({*R*})_a{}^{cde}R_{bcde} = -\frac{1}{4}g_{ab}\mathcal{E}_{\mathrm{GB}}$ in four dimensions. The two facts sit one Lovelock degree apart: the algebraic gradient of the quadratic density contracts to the metric gradient of the linear density $\operatorname{Tr}\mathcal{R} = \frac{1}{2}R$, and the quadratic density itself is inert. Which members of the family are variational at all is a sharper question, and it has a sharp answer.

Theorem 24.5 (Only the Einstein phase is variational)**.** *Consider the operators* $g \mapsto H(\theta)[g]_{ab} = R_{ab} - \frac{1}{2}\sin^2\theta\, R g_{ab}$*. There is a Lagrangian* $L(g, \partial g, \partial^2 g, \dots)$*, covariant or not, defined on a neighbourhood of flat space, whose Euler–Lagrange expression is* $H(\theta)$ *if and only if* $\cos\theta = 0$*; then* $R\sqrt{|g|}$ *is one.*

Proof. An Euler–Lagrange operator has a formally self-adjoint Fréchet derivative at every point of its domain, the necessity half of the Helmholtz conditions [136, 96]. At $g = \eta$, $H(\theta)[\eta] = 0$, so the density factor does not contribute and the linearisation acts on plane waves $h_{ab}e^{ik\cdot x}$ with a symbol quadratic in k. Pairing it with a second perturbation q_{ab}, the antisymmetric part of the bilinear form is

$$q^{ab}\,\delta H_{ab}[h] - h^{ab}\,\delta H_{ab}[q] = \tfrac{1}{2}\cos^2\theta\big[(k\cdot q\cdot k)\ \mathrm{tr}\, h - (k\cdot h\cdot k)\ \mathrm{tr}\, q\big],$$

which vanishes identically only when $\cos\theta = 0$. There the operator is the Einstein tensor, and $\delta(R\sqrt{|g|}) = \sqrt{|g|}\,\mathcal{G}_{ab}\,\delta g^{ab}$ up to a divergence. □

For a diffeomorphism-invariant action Noether's second theorem makes the Euler–Lagrange tensor divergence-free, so Corollary 24.4(i) is the covariant necessary condition and Theorem 24.5 the full condition, obtained without assuming covariance. Both select the same phase.

24.2 The first-order form: the phase is a Barbero–Immirzi angle

Let e^a be a coframe, ω a Lorentz connection with curvature F^{ab}, and $\Sigma^{ab} = e^a \wedge e^b$. Both Σ and F are two-forms valued in $\Lambda^2\mathbb{R}^{1,3}$, which is the frame by Theorem 12.3. Pair them by $\langle \Sigma \wedge X\rangle = \frac{1}{2}\Sigma_{ab}\wedge X^{ab}$, orient by $\mathrm{vol} = e^0\wedge e^1\wedge e^2\wedge e^3$, and let $\mathcal{R}$ be the curvature operator of ω in the frame e.

Proposition 24.6 (Palatini and Holst on the frame)**.** *(i)* $\frac{1}{4}\varepsilon_{abcd}\,e^a\wedge e^b\wedge F^{cd} = \langle G\Sigma\wedge F\rangle = \mathrm{Tr}(\mathcal{R})\,\mathrm{vol}$*: the Palatini Lagrangian, which on the torsion-free locus is* $\frac{1}{2}R\,\mathrm{vol}$*.*

(ii) $e_a\wedge e_b\wedge F^{ab} = 2\langle\Sigma\wedge F\rangle = -2\,\mathrm{Tr}(G\mathcal{R})\,\mathrm{vol}$*: the Holst term is the first-Bianchi datum of Proposition 24.1.*

(iii) For the phase family, $\langle e^{\theta G}\Sigma\wedge e^{\theta G}F\rangle = \cos 2\theta\,\langle\Sigma\wedge F\rangle + \sin 2\theta\,\langle G\Sigma\wedge F\rangle = -\,\mathrm{Tr}(G\mathcal{R}_\theta)\,\mathrm{vol}$*. In the normalisation* $\frac{1}{2}\varepsilon_{abcd}e^a e^b F^{cd} - \gamma^{-1}e_a e_b F^{ab}$ *this is the Holst action [67] with*

$$\gamma = -\tan 2\theta,$$

the Barbero–Immirzi parameter.

(iv) The connection equation of $\int\langle e^{\theta G}\Sigma\wedge e^{\theta G}F\rangle$ *is* $e^{2\theta G}D\Sigma = 0$*, so the torsion vanishes, and the coframe equation is* $-2\sin 2\theta\,\mathcal{G}^{af} = 0$ *on the torsion-free locus. The vacuum equations are Einstein's for every* $\theta \notin \frac{\pi}{2}\mathbb{Z}$ *and empty for* $\theta \in \frac{\pi}{2}\mathbb{Z}$*.*

The roles of the phase differ between the two formalisms, as Figure 17 shows. In the second-order family the variational point is $\theta = \pi/2$; in the first-order family Einstein's equations hold for every $\theta \notin \frac{\pi}{2}\mathbb{Z}$, the Palatini point is $\theta = \pi/4$, and the endpoint $\theta = \pi/2$, where $\mathcal{R}_\theta = G\mathcal{R}G$, is exactly where the first-order Lagrangian degenerates to the pure Holst term. These are two uses of one rotation and must not be conflated. The self-dual value $\gamma = \pm i$ of Ashtekar's variables is not attained for any complex θ; it is approached as $\mathrm{Im}\,\theta \to \mp\infty$, where $e^{2\theta G}$ becomes proportional to a projector onto an eigenspace of G. With torsion sources such as fermions the Holst term is no longer inert and the phase would enter an effective four-fermion interaction [108, 40]; nothing here fixes it.

Both readings of $\mathbb{R}^{3,3}$ therefore construct the Einstein tensor of a given four-metric: the interior reading by restricting the six-dimensional Einstein tensor to the slice (Theorem 23.5), the

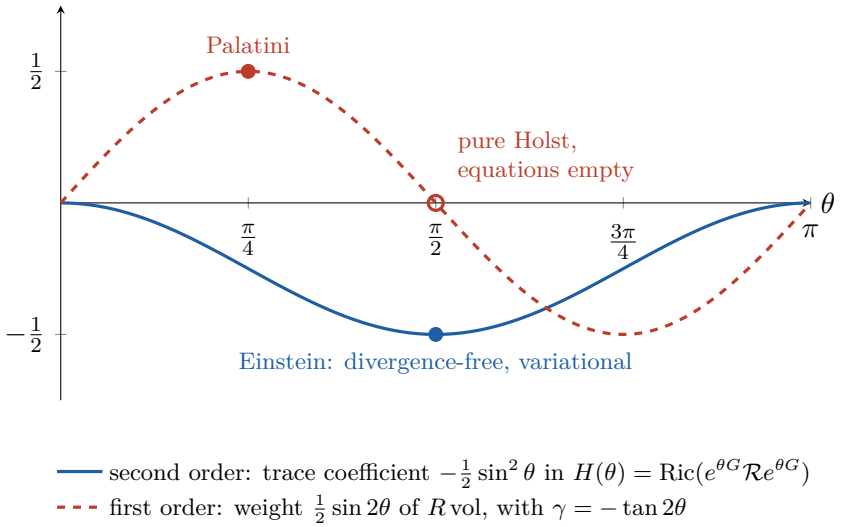


Figure 17: Two uses of one duality rotation of the curvature (Theorems 24.2 and 24.5 and Proposition 24.6). In the second-order family $H(\theta) = \mathrm{Ric}(e^{\theta G}\mathcal{R}e^{\theta G})$ the endpoint $\theta = \pi/2$ is the Einstein tensor, the only divergence-free and the only variational member. In the first-order family the same rotation weights the Einstein–Hilbert term by $\frac{1}{2}\sin 2\theta$: Einstein's equations follow for every $\theta \notin \frac{\pi}{2}\mathbb{Z}$, the Palatini point is $\pi/4$, and at $\theta = \pi/2$ the Lagrangian degenerates to the pure Holst term, with $\gamma = -\tan 2\theta$.

field-strength reading as the contraction of $G\mathcal{R}G$ (Theorem 24.2). Neither determines the metric. No gravitational coupling, source or equation of motion for G follows, the selection criteria come from Riemannian geometry rather than from the Clifford structure, and the coefficient $\frac{1}{2}$ of the trace reversal comes from the algebra of the Hodge star.

25 The projector under transport

In a curved arena an observation projector becomes a field $\Pi(x) = \Pi(x)^2$ that is carried around by the spin connection, $D\Pi = d\Pi + [\Omega, \Pi]$. Two curvatures then compete on its image, and their competition decides whether the normal $U(1)$ of the model of Section 23.2 can be electromagnetism.

25.1 Two curvatures on the selected bundle

Differentiating $\Pi^2 = \Pi$ gives $\Pi(D\Pi)\Pi = 0$ and $(\mathbf{1}-\Pi)(D\Pi)(\mathbf{1}-\Pi) = 0$: the covariant derivative of a projector is purely off-diagonal, which is the configuration-space statement of Section 11 in covariant form. The induced connection $D^{\mathrm{obs}}s = \Pi Ds$ on the selected bundle has curvature

$$\mathcal{F}_{\mathrm{obs}} = \Pi\,\mathcal{R}\,\Pi + \Pi\,(D\Pi \wedge D\Pi)\,\Pi, \tag{40}$$

with $\mathcal{R}_{MN} = \frac{1}{4}R_{MNAB}\gamma^A\gamma^B$ the ambient spin curvature [75]. The two terms have distinct origins: ambient curvature, and the motion of the selected subspace relative to ambient transport. The flat case of Section 11 keeps only the second.

Proposition 25.1 (Transport obstruction)**.** *If ambient transport preserves the projector, $D\Pi = 0$, then $[\mathcal{R}_{MN}, \Pi] = 0$. For the paired-mode projector $\Pi = \frac{1}{2}\mathbf{1} - \frac{1}{4}(\mathsf{P} + \mathcal{V})$ of Example 2.6,* $\mathrm{rank}[J_N, \Pi] = 4$*, and in the model (39) with $g^{(4)}$ flat, $\mathcal{R}_{\mu\nu} = \frac{1}{2}(F_N)_{\mu\nu}J_N$ on Σ, so $[\mathcal{R}_{\mu\nu}, \Pi] = \frac{1}{2}(F_N)_{\mu\nu}[J_N, \Pi] \neq 0$ wherever the normal curvature $F_N = dA$ is nonzero.*

Proof. $D^2\Pi = [\mathcal{R}, \Pi]$ for any section of the endomorphism bundle, so $D\Pi = 0$ forces the condition. The rank is a direct computation in the 8×8 realisation. □

This statement is about the spinor projector and the spin curvature, and the corresponding tangent-level statement is false; the two must not be run together. At the tangent level J_N commutes with every generator of E (Lemma 23.2), so the rotation it generates fixes E pointwise, and in (39) every mixed Christoffel symbol vanishes on Σ: the splitting $TM = E \oplus N$ is Levi-Civita parallel there although $F_N \neq 0$. At the spinor level the same rotation is not trivial, since $\operatorname{rank}[J_N, \Pi] = 4$. Both statements hold simultaneously, and together they are the content of the remark after Proposition 23.4: a spinless test particle cannot see the normal connection, and whatever makes it physical has to come from the spinor sector. The general principle that a normal-bundle curvature commutes with the projector is a statement about TM and does not transfer to End of the spinor bundle. Curvature cannot be chosen independently of the selection.

Proposition 25.2 (The adapted connection)**.** *Set* $\mathcal{C} := [\Pi, D\Pi]$ *and* $\widehat{D} := D + \mathcal{C}$*. Then* $\widehat{D}\Pi = 0$*. For the paired-mode projector, with the Levi-Civita spin connection,* $D\mathcal{V} = 0$ *and* $\mathcal{V}$ *commutes with bivectors, so* $\mathcal{C} = \frac{1}{16}[\mathsf{P}, D\mathsf{P}]$*, again a bivector: the correction stays inside* $\mathfrak{spin}(3,3)$*. The corresponding tangent connection is* $\widehat{\nabla}_X Y = \nabla_X Y + \frac{1}{2}\mathcal{K}(\nabla_X \mathcal{K})Y$*, metric-compatible and* $\mathcal{K}$*-preserving, hence generically with torsion.*

Proof. Write $\delta = D\Pi$, which is off-diagonal. Direct expansion gives $[[\Pi, \delta], \Pi] = -\delta$ for such δ, so $\widehat{D}\Pi = \delta + [\mathcal{C}, \Pi] = \delta + [[\Pi, \delta], \Pi] = 0$. The form of $\mathcal{C}$ follows from $D\Pi = -\frac{1}{4}D\mathsf{P}$ and $[\mathcal{V}, D\mathsf{P}] = 0$. On tangent vectors $\mathcal{K}^2 = \mathbf{1}$ gives $\mathcal{K}(\nabla\mathcal{K})\mathcal{K} = -\nabla\mathcal{K}$, so the correction $\frac{1}{2}\mathcal{K}\nabla\mathcal{K}$ satisfies $\nabla\mathcal{K} + [\frac{1}{2}\mathcal{K}\nabla\mathcal{K}, \mathcal{K}] = 0$, and it is η-antisymmetric because $\mathcal{K}$ and $\nabla\mathcal{K}$ are η-antisymmetric and anticommute. The Levi-Civita connection is the unique torsion-free metric connection, so a metric connection preserving a structure that Levi-Civita transport does not preserve carries torsion. □

The adapted generator is not an internal charge. In the model, with $\Omega = \frac{1}{2}AJ_+$ and $J_+ = J_N$, the adapted connection on Σ is $\widehat{\Omega} = \frac{A}{4}(J_+ - J_-)$ with $J_- = \frac{1}{\sqrt{3}}(e_5e_6 + e_6e_4 + e_4e_5)$, and indeed $[J_+ - J_-, \Pi] = 0$; but $[\frac{1}{2}(J_+ - J_-), e_4] = \frac{1}{\sqrt{3}}(e_6 - e_5) \neq 0$, and likewise for e_5 and e_6. Preserving the projector has bought commutation with Π at the cost of commutation with the physical generators, which is the first of the four conditions of Section 11.5. The identification of the normal $U(1)$ with electromagnetism is not available on this route.

Nor is the observed curvature the field-strength module of Proposition 12.7. $\mathcal{F}_{\text{obs}}$ takes values in $\operatorname{End}(\operatorname{im}\Pi)$, and the interior bivectors are not elements of that space: all six of them fail to commute with Π, for instance $[e_4e_5, \Pi] = \frac{1}{2}(e_2e_4 - e_1e_5)$. Their compressions $\Pi B \Pi$ are endomorphisms of the image, but compression is not an algebra map and carries no Lie brackets. The model supplies a direct counterexample: with Π constant and $\Omega = \frac{1}{2}AJ_N$, $\mathcal{F}_{\text{obs}} = \frac{1}{2}\,dA\,\Pi J_N \Pi$, and exact expansion gives

$$4\sqrt{3}\,\Pi J_N \Pi = e_1e_2 - e_1e_3 + e_2e_3 - e_4e_5 + e_4e_6 - e_5e_6 \\ - (e_1e_2e_3e_4 + e_1e_2e_3e_5 + e_1e_2e_3e_6) + (e_1e_4e_5e_6 + e_2e_4e_5e_6 + e_3e_4e_5e_6),$$

with grade-four terms of the same size as the bivector terms. An electromagnetic two-form and a Lorentz-algebra-valued curvature two-form carry different kinds of index, and identifying them needs structure that is not supplied.

25.2 Projector dynamics is adapted transport

With the kinetic term $S_\Pi = \frac{f^2}{2}\int \sqrt{|g|}\,\operatorname{Tr}(D_\mu\Pi\, D^\mu\Pi)$ varied along the spin orbit, the Euler–Lagrange equation is (13) in curved form,

$$[\Pi, \Box_D\Pi] = 0 \qquad \Longleftrightarrow \qquad D_\mu\mathcal{C}^\mu = 0, \qquad \mathcal{C}_\mu = [\Pi, D_\mu\Pi],$$

so the conserved matrix current of the projector *is* the connection correction of Proposition 25.2: projector dynamics and adapted transport are the same data. Built with $\widehat{D}$ instead of D the

functional vanishes identically, which is the same statement read backwards. By Theorem 11.3 and Example 11.4, the trace form on the spin orbit of the paired-mode projector has three positive and three negative directions, the compact and the boost generators, and on the manifold of orthogonal projectors it is positive definite.

Example 25.3 (The angle mode is a Stückelberg field)**.** The $U(1)$ direction is one of the healthy three: $T = [\frac{1}{2}J_N, \Pi]$ is symmetric with $\mathrm{Tr}(T^2) = +1$. But T is a tangent vector, not a generator, and conjugating by $e^{\theta T}$ with T symmetric leaves the orthogonal projectors; the correct one-angle ansatz conjugates by the compact generator itself,

$$\Pi(\theta) = e^{\theta J_N/2}\,\Pi_0\,e^{-\theta J_N/2}.$$

The one-angle geodesic condition $[\Pi,[B,[B,\Pi]]] = 0$ with $B = \frac{1}{2}J_N$ holds at Π_0 and hence along the orbit, so the ansatz is consistent with the projector equation, which reduces to an equation for $b_\mu = \partial_\mu\theta + A_\mu$. The action reduces to $\mathcal{L}_\theta = \frac{f^2}{2}b_\mu b^\mu$, invariant under $\theta \mapsto \theta - \lambda$, $A \mapsto A + d\lambda$: the projector angle is a Stückelberg field for the normal frame rotation [117]. Adjoining a Maxwell term by hand gives a Proca field with $m_A^2 = g_A^2 f^2$.

Read strictly, this is a constraint and not a prediction: neither the Maxwell term nor the scales g_A and f is derived, and a massive normal vector is not the photon. Together with the failure of the adapted generator to commute with the physical generators it leaves a dichotomy: either the normal $U(1)$ is not electromagnetic, or the paired-mode projector is not the observation projector. The next section resolves it in one direction.

25.3 Is the flat obstruction a torsion statement?

The factorisation condition $a = \frac{1}{2}(G - G^\top) = 0$ of Theorem 15.2 survives covariantisation, and requiring transport to preserve the observation projector adds to the Levi-Civita connection a frame-antisymmetric correction, which is contorsion. One may conjecture that these are the same object: that in a first-order formulation with an independent connection [106] G is not a background but the connection's own shift. The types do not yet match, and that is part of the work: a is an endomorphism while contorsion is a one-form valued in metric-skew endomorphisms, so a relation such as $a = K_\mu\dot{x}^\mu$ needs a stated contraction and normalisation before it is even a proposition. And the flat exclusion cannot be borrowed for it: by Theorem 15.5 and Theorem 16.2, what repairs binding is the split form, not curvature or torsion, so the antisymmetric part a is not what binding was waiting for.

26 Sources, and whether the photon can be massless

Theorem 12.8 fixes the form of Maxwell's equations and leaves the source entirely open. The geometry of the curved arena supplies two candidate currents, and testing them gives a sharp negative result about the photon, together with the criterion that repairs it and a vacuum, at the second rung, that satisfies the criterion.

26.1 Two currents

On the slice Σ the oriented normal two-plane bundle of Section 23.2 has an $SO(2)$ connection A with curvature $F_N = dA$, an ordinary two-form on spacetime whose six components sit in the module of Theorem 12.3, and which is distinct from the compressed spinor curvature $\mathcal{F}_{\mathrm{obs}}$. Because the structure group is abelian, $dF_N = 0$ identically: the homogeneous half of Maxwell's equations is the Bianchi identity of the normal bundle, not an assumption. Setting $j^\nu_{\mathrm{geom}} := \lambda\,\nabla_\mu F_N^{\mu\nu}$ gives a current conserved identically, by the antisymmetry of F_N. On a flat base the connection $A = Bx\,dz$ has constant curvature and zero current, while $A = a(x)\,dz$ gives

$F_N = a'(x)\,dx\wedge dz$ and $j^z_{\text{geom}} = \lambda\, a''(x)$. This current is defined from F_N, so it is the current of a geometry that has been specified, not an independent source; its value is to show that continuity can be met by a geometric object.

The stronger candidate is independent of F. Fix a compact $\mathfrak{u}(1)$ generator Q, let $D_\mu\Pi = \nabla_\mu\Pi + A_\mu[Q,\Pi]$, and add a Maxwell term $-F_{\mu\nu}F^{\mu\nu}/4g_A^2$. Varying the kinetic term $\frac{f^2}{2}\operatorname{Tr}(D_\mu\Pi D^\mu\Pi)$ with respect to A_μ gives

$$j^\mu_\Pi = -f^2\operatorname{Tr}(Q\,[\Pi, D^\mu\Pi]) = -f^2\operatorname{Tr}([Q,\Pi]\,D^\mu\Pi), \tag{41}$$

the component of projector motion along the chosen charge rotation. It is conserved on the projector equation of motion, $D_\mu\mathcal{C}^\mu = 0$, rather than identically.

Theorem 26.1 (The paired-mode projector sources a massive vector)**.** *For $Q = J_N/2$ and the angle ansatz of Example 25.3, $\operatorname{Tr}([Q,\Pi]^2) = 1$ exactly, so $j^\mu_\Pi = -f^2(\partial^\mu\theta + A^\mu)$, and coupling it to the Maxwell term reproduces the Stückelberg mass $m_A^2 = g_A^2 f^2$. The mass and the current come from the same trace, so no choice of f gives a massless photon with this source.*

Corollary 26.2 (The massless criterion)**.** *A massless vector requires a vacuum Π_0 with $[Q,\Pi_0] = 0$, so that the kinetic term generates no mass, and fluctuations π with $[Q,\pi]\neq 0$, so that charged modes still carry the current $-f^2\operatorname{Tr}(Q[\pi, D^\mu\pi]) + \cdots$. If $[Q,\Pi(x)] = 0$ is imposed everywhere, the current vanishes identically and nothing is sourced.*

26.2 The first rung

Theorem 26.3 (The criterion can be met)**.** *In $\mathrm{Cl}(3,3)$ with $Q = J_N/2$ the rank-four orthogonal projectors commuting with Q form $\mathrm{Gr}_{\mathbb{C}}(2,4)$, of real dimension eight, since $2Q$ is a complex structure. For a generic member Π_0,*

$$\dim\{[B,\Pi_0] : B\in\mathfrak{so}(3,3)\} = 14, \qquad \dim\{\pi \text{ in that span with } [Q,\pi]\neq 0\} = 8,$$

so the vacuum is massless and not rigid; the mechanism is the Jacobi identity $[Q,[B,\Pi_0]] = [[Q,B],\Pi_0]$, with ad_Q of rank eight on $\mathfrak{so}(3,3)$. But only two of the eight charged directions are admissible: by Theorem 11.3 a variation $[B,\Pi_0]$ stays among the orthogonal projectors exactly when B is compact, and of the fourteen orbit directions only five are compact-generated, of the eight charged ones only two.

Proposition 26.4 (Lorentz invariance makes the first rung rigid)**.** *The commutant of $\{\mathfrak{l}_{\text{int}}, Q\}$ in $\mathrm{End}(\mathbb{R}^8)$ has dimension four, with symmetric part spanned by $\mathbf{1}$ and $\mathcal{V}$. So the only rank-four orthogonal projectors commuting with both the interior Lorentz action and Q are $\frac{1}{2}(\mathbf{1}\pm\mathcal{V})$, the chirality projectors of Theorem 18.1. Every bivector commutes with $\mathcal{V}$, so these have a spin-orbit tangent space of dimension zero: they are rigid, and by Corollary 26.2 their current vanishes identically.*

The paired-mode projector fails the same Lorentz condition, since all six interior bivectors fail to commute with it; non-invariance of the observation projector is a standing feature of the one-constituent construction. What Proposition 26.4 adds is that imposing Lorentz invariance together with $[Q,\Pi_0] = 0$ forces rigidity at the first rung. The tower relaxes this: a surplus rotation commuting with the interior generators of every constituent is unique at $n = 1$, while at $n = 2$ there are six, and ad_Q has rank 8 of 15 at $n = 1$ and 20 of 66 at $n = 2$ on the bivectors, so the higher rungs offer a genuine choice of which $\mathfrak{u}(1)$ to gauge.

26.3 A Lorentz-invariant charged vacuum at the second rung

Definition 26.5 (The second-rung vacuum)**.** In $\mathrm{Cl}(6,6)$ with the spectator interior action, whose internal complement has signature $(5,3)$, let a,b,c,d,e be orthonormal positive internal

generators: the surplus pair of the first constituent and the three positive directions of the second. Set

$$Q := \tfrac{1}{2}ab, \qquad H := abcd, \qquad \Pi_0 := \tfrac{1}{2}(\mathbf{1} + H).$$

Theorem 26.6 (Lorentz invariant, uncharged and not rigid)**.** *$H^2 = \mathbf{1}$, $H^\top = H$ and $\operatorname{Tr} H = 0$, so Π_0 is an orthogonal projector of rank 32 on $\mathbb{R}^{64}$. Moreover:*

(i) *$[Q, \Pi_0] = 0$, so the kinetic term gives the vacuum no gauge-boson mass;*

(ii) *Π_0 commutes with every interior gamma matrix, hence with the interior Lorentz algebra, and with the internal volume that supplies the complex structure;*

(iii) *it is moved by a compact internal bivector: for $B = \tfrac{1}{2}be$, $\pi = [B, \Pi_0] = -\tfrac{1}{2}acde$ and $[Q, \pi] = \tfrac{1}{2}bcde \neq 0$, with π symmetric and $\operatorname{Tr}(\pi^2) = 16$.*

Proof. H is a product of four positive generators, so $H^2 = (-1)^{4\cdot 3/2}\mathbf{1} = \mathbf{1}$ and $H^\top = H^2 H = H$, and it is a traceless blade. $Q = \tfrac{1}{2}ab$ shares two indices with H and commutes with it by (3). Each interior gamma shares no index with H, and $4 \cdot 1$ is even, so it commutes. $B = \tfrac{1}{2}be$ shares one index with H and anticommutes with it, so $[B, \Pi_0] = \tfrac{1}{2}[B, H] = BH = \tfrac{1}{2}\, be\, abcd$; since $b\, e\, a\, b = b^2\, e\, a = e\, a$, this is $\tfrac{1}{2}\, eacd = -\tfrac{1}{2}\, acde$. The blade $acde$ has four positive generators and is symmetric, and $\operatorname{Tr}(\pi^2) = \tfrac{1}{4}\operatorname{Tr}\mathbf{1} = 16$. Finally Q shares one index with π, so $[Q, \pi] = 2Q\pi = -\tfrac{1}{2}\, ab\, acde = \tfrac{1}{2}\, bcde$. □

Theorem 26.7 (The compact family and its current)**.** *Let $I_5 := abcde$ and $Z_i := I_5 v_i$ for $v_i \in \{a, b, c, d, e\}$. The Z_i are symmetric of grade four with $Z_i Z_j + Z_j Z_i = 2\delta_{ij}$, so for every unit $n \in \mathbb{R}^5$*

$$\Pi(n) := \tfrac{1}{2}(\mathbf{1} + I_5 n)$$

is a rank-32 orthogonal projector commuting with every interior gamma matrix. These form a compact family $S^4 = \mathrm{Spin}(5)/\mathrm{Spin}(4)$ with $\Pi_0 = \Pi(e)$. At the vacuum the tangents $T_i = [\tfrac{1}{2}v_i e, \Pi_0] = \tfrac{1}{2}I_5 v_i$ satisfy $\operatorname{Tr}(T_i T_j) = 16\delta_{ij}$ and

$$[Q, T_a] = -T_b, \qquad [Q, T_b] = T_a, \qquad [Q, T_c] = [Q, T_d] = 0 :$$

one charged complex scalar $\phi = n_a + i n_b$ and two neutral real scalars. With $D_\mu \phi = (\partial_\mu - iA_\mu)\phi$ the kinetic term and its current are

$$\mathcal{L}_\Pi = 8f^2\, D_\mu n \cdot D^\mu n, \qquad j^\mu_\Pi = 16 f^2 (n_a D^\mu n_b - n_b D^\mu n_a) = 16 f^2 \operatorname{Im}(\bar\phi D^\mu \phi),$$

the factor 16 being the real trace on the 64-dimensional module. At $n = e$, $\phi = 0$, so no $A_\mu A^\mu$ term is generated, while charged fluctuations carry current.

Theorem 26.8 (The curved extension)**.** *Reduce the internal connection so that a, b carry the normal $SO(2)$ connection and c, d, e are parallel, $\Omega = \Omega_{\text{Lorentz}} + AQ$. Both summands commute with Π_0, so*

$$D\Pi_0 = 0, \qquad [\mathcal{R}, \Pi_0] = 0$$

for arbitrary spacetime curvature and arbitrary normal curvature, the two conditions that the transport obstruction of Proposition 25.1 separates, here satisfied together by construction. Throughout the family $\mathcal{C}_\mu = [\Pi, D_\mu \Pi] = \tfrac{1}{4}[n, D_\mu n] = \tfrac{1}{2}\, n \wedge D_\mu n$, a compact internal bivector at every point, so the adapted connection stays spin-valued away from the vacuum as well.

For a general projector $[\Pi, D\Pi]$ need not be a bivector: in a normal frame with unit clock w and negative s, t, the projector $\Pi = \tfrac{1}{2}(\mathbf{1} + X)$ with $X = (w + ws)/\sqrt{2}$ and $B = st$ gives $[\Pi, [B, \Pi]] = \tfrac{1}{2}(t - st)$, which has a grade-one part. The S^4 family avoids this by construction.

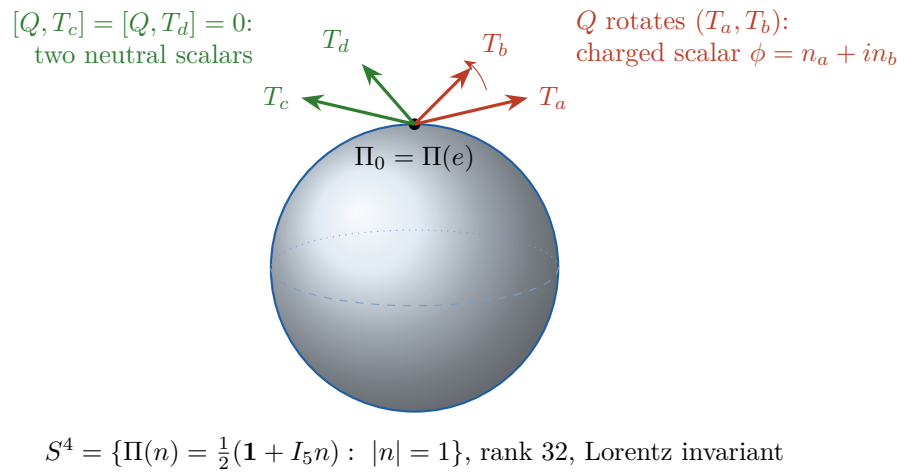


Figure 18: The compact family of second-rung vacua (Theorem 26.7). Every point is a Lorentz-invariant projector; at the vacuum Π_0, which commutes with the charge Q, the four tangent directions are generated by compact bivectors, two of them rotated into each other by Q and two neutral.

Proposition 26.9 (The vacuum and the matter)**.** *The stabiliser of* Π_0 *in the internal* $\mathfrak{so}(5,3)$ *is* $\mathfrak{so}(4) \oplus \mathfrak{so}(1,3)$*, of dimension* 12 *with compact part* $\mathfrak{su}(2) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(2)$*, which does not contain* $\mathfrak{su}(3)$*. The construction is therefore incompatible with colour at* $n = 2$*; it is compatible from the third rung on:*

rung	internal algebra	stabiliser of H	compact part
$n = 2$	$\mathfrak{so}(5,3)$, dim 28	$\mathfrak{so}(4) \oplus \mathfrak{so}(1,3)$, dim 12	9, no $\mathfrak{su}(3)$
$n = 3$	$\mathfrak{so}(8,6)$, dim 91	$\mathfrak{so}(4) \oplus \mathfrak{so}(4,6)$, dim 51	27, containing $\mathfrak{so}(6) \supset \mathfrak{su}(3)$
$n = 4$	$\mathfrak{so}(11,9)$, dim 190	$\mathfrak{so}(4) \oplus \mathfrak{so}(7,9)$, dim 126	63

At $n = 3$*, the rung of* P_**, the unbroken compact algebra is* $\mathfrak{so}(4) \oplus \mathfrak{so}(4) \oplus \mathfrak{so}(6)$ *and contains colour.*

The second-rung vacuum settles that a vacuum which is simultaneously Lorentz invariant, uncharged, not rigid and moved by a compact generator exists, so the first-rung rigidity of Proposition 26.4 is a property of that rung and not a law; its fluctuation spectrum is one charged complex scalar and two neutral ones, its current is an ordinary charged-scalar current, and its curved extension has $D\Pi_0 = 0$ with a spin-valued adapted connection across the whole family. It does not settle which point of S^4 is selected, since a spectrum-only potential is constant on the family; whether j_Π is the electric current of a fermion, with which charge normalisation; the values of f and g_A; stability; or a gravitational action. It must be done at $n \geq 3$ to leave room for colour. And it uses the spectator action throughout: under the mixed action of Section 13.6, Π_0 is preserved only by the rotations about one axis, its commutators with the six mixed generators span a five-dimensional space, and the rank-16 Dirac block of the coupled model meets the range of Π_0 only in zero, with $P\Pi_0 P$ having eigenvalues $\frac{1}{4}$ and $\frac{3}{4}$ on it. What the coupled model and the second-rung vacuum share is the charge: $Q = \frac{1}{2}J_1$.

The dichotomy left at the end of Section 25 is thereby resolved in one direction. If the normal $U(1)$ is to be electromagnetism, the paired-mode projector must go, and Theorem 26.6 shows what can replace it. At one constituent, the paired-mode projector of Example 2.6 fails the photon test.

Part VII
Causal structure

27 Causal structure: the six-dimensional baseline

Every construction so far has used a Lorentzian four-plane chosen inside $\mathbb{R}^{3,3}$: the interior plane E of the Dirac structure, or the slice Σ of the curved model. This part asks whether that choice, and with it the light cone and the distinction between past and future, can be obtained from the split geometry rather than postulated. Two levels must be distinguished. *Weak emergence* means that an already selected 3+1 sector carries a consistent quantum causal structure. *Strong emergence* means that the dynamics determine why that sector, with that causal structure, arises from the six-dimensional theory. The light cone must not be put into the admissibility conditions and then be said to emerge. We fix the failure conditions in advance: uncontrolled growing modes in the six-dimensional theory; a projector onto the four-dimensional sector imposed by hand; ghosts or negative spectral weights in the effective theory; propagation outside the derived cone; extra modes or imposed equations in the Maxwell sector.

27.1 The unrestricted scalar

Proposition 27.1 (A free field on $\mathbb{R}^{3,3}$)**.** *For the action* $S_6 = \frac{1}{2}\int d^6X\,(\eta^{AB}\partial_A\phi\,\partial_B\phi - m^2\phi^2)$ *in coordinates* (t,u,v,x,y,z)*, with* t,u,v *temporal:*

(i) plane waves $e^{-i\omega t+i(q_1u+q_2v)+i\mathbf{k}\cdot\mathbf{x}}$ *obey*

$$\omega^2 = |\mathbf{k}|^2 + m^2 - q^2, \qquad q^2 = q_1^2 + q_2^2; \tag{42}$$

(ii) for $q^2 > |\mathbf{k}|^2 + m^2$ *the growth rate is* $\gamma = \sqrt{q^2 - |\mathbf{k}|^2 - m^2}$*, and* $(1+q^2)^{-s/2}e^{\gamma} \to \infty$ *for every* s*: data small in any Sobolev norm produce arbitrarily large solutions at* $t = 1$*, so the Cauchy problem with respect to* t *is ill-posed in the sense of Hadamard [58];*

(iii) the canonical energy density $\frac{1}{2}(\pi^2 - \phi_u^2 - \phi_v^2 + |\nabla_{\mathbf{x}}\phi|^2 + m^2\phi^2)$ *averages to* $\frac{1}{4}a^2(m^2 - q^2)$ *on* $\phi = a\sin qu$*: the energy is unbounded below.*

Proposition 27.2 (No time orientation in signature $(3,3)$)**.** *The timelike set* $\{\eta(X,X) > 0\}$ *of* $\mathbb{R}^{p,q}$ *is connected for* $p \geq 2$*: the path* $X(s) = (\cos s, \sin s, 0;\, 0,0,0)$ *joins* X *to* $-X$ *with* $\eta = 1$ *throughout. Only for* $p = 1$ *does it split into a past and a future component.*

Causal order is therefore not merely unproven in $\mathbb{R}^{3,3}$; it is absent. Anything called a light cone with a future has to come from a reduction to a single time. The arena already contains such a reduction, as an input. For the geodesic test particle of Proposition 23.4, $H = \frac{1}{2}(h^{\mu\nu}k_\mu k_\nu + p_u^2 + p_v^2)$ with $h = \mathrm{diag}(1,-1,-1,-1)$, the mass shell $2H = m^2$ is $p_0^2 = |\mathbf{p}|^2 + m^2 - p_u^2 - p_v^2$, the relation (42) with $q = (p_u, p_v)$. The first-class constraints $p_u = p_v = 0$ remove exactly the unstable band, as Figure 19 shows.

Example 27.3 (Compact surplus times give tachyons)**.** If instead u and v are made periodic with period $2\pi R$, the surplus momenta become n/R with $n \in \mathbb{Z}^2$, and the four-dimensional masses are

$$m_n^2 = m^2 - \frac{|n|^2}{R^2}.$$

Every mode with $|n| > mR$ is a tachyon, and there are infinitely many: for $mR = 3$ there are 92, 412 and 1652 of them within $|n_i| \leq 5$, 10 and 20. Compactification, the usual cure for extra space dimensions, is no cure for extra times.

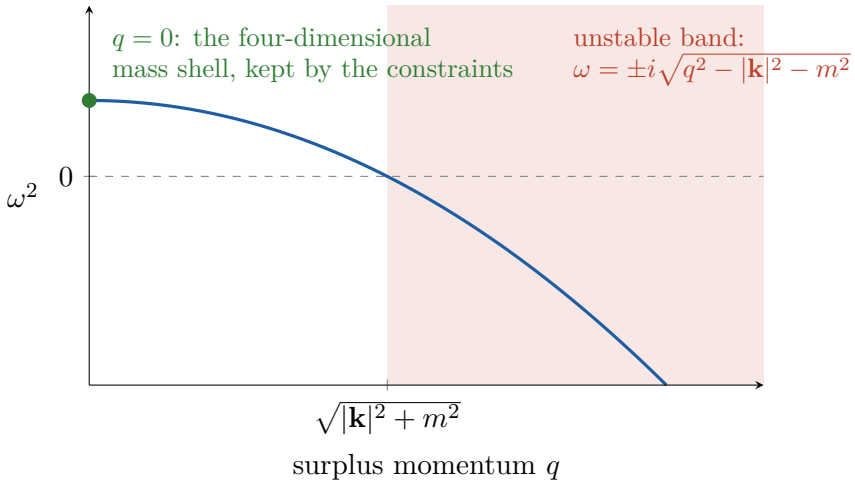


Figure 19: The dispersion relation of a free scalar on $\mathbb{R}^{3,3}$ at fixed spatial momentum (Proposition 27.1). Every surplus momentum beyond $\sqrt{|\mathbf{k}|^2 + m^2}$ gives an exponentially growing mode, with a rate that grows without bound. The constraints of Proposition 23.4 keep exactly the point $q = 0$.

An ultrahyperbolic problem can be made well posed by a nonlocal constraint on the data [27], and the dimensionality of spacetime has been argued from such considerations [131]. The constraints of the arena, $\partial_u \phi = \partial_v \phi = 0$, are local and stronger: they keep only the zero modes, which is dimensional reduction. Either way the cure is a restriction, and at this stage it is an input. Section 30 asks where it can come from; the next two sections ask whether the four-plane it presupposes can be selected.

28 The kinematic selector

Choosing a Lorentzian four-plane inside $\mathbb{R}^{3,3}$ is equivalent to choosing its orthogonal complement, a two-plane, and a two-plane is described by its unit bivector. The question is which bivectors describe a Lorentzian complement, and what the arena does with them.

28.1 One constituent

Theorem 28.1 (The selector is a complex structure)**.** *Let $C = a \wedge b$ be a simple bivector of $\mathbb{R}^{3,3}$ and $N = \langle a, b \rangle$.*

(i) $C^2 = -\det \mathrm{Gram}(a, b)\, \mathbf{1}$.

(ii) For a nondegenerate N with C normalised: $C^2 = -\mathbf{1}$ exactly when N is definite, that is exactly when $N^\perp$ is Lorentzian, of signature $(1,3)$ when N is positive and $(3,1)$ when N is negative; $C^2 = +\mathbf{1}$ exactly when $N^\perp$ has signature $(2,2)$; and $C^2 = 0$ exactly when N is degenerate.

(iii) The centraliser of C in $\mathfrak{so}(3,3)$ is $\mathfrak{so}(N) \oplus \mathfrak{so}(N^\perp)$, of dimension seven, so the choices of N of a given type form an eight-dimensional homogeneous space.

(iv) The surplus plane is such an N, with $C = J = e_2 e_3$ and $N^\perp$ the interior four-plane.

Proof. (i) is the Clifford identity $(a \wedge b)^2 = (a \cdot b)^2 - a^2 b^2$. (ii) For an orthonormal pair, $(ab)^2 = -a^2 b^2$, which is -1 when $a^2 = b^2$ and $+1$ otherwise, and the signature of $N^\perp$ is the complement of that of N in $(3,3)$. (iii) holds for coordinate planes by (3), and every

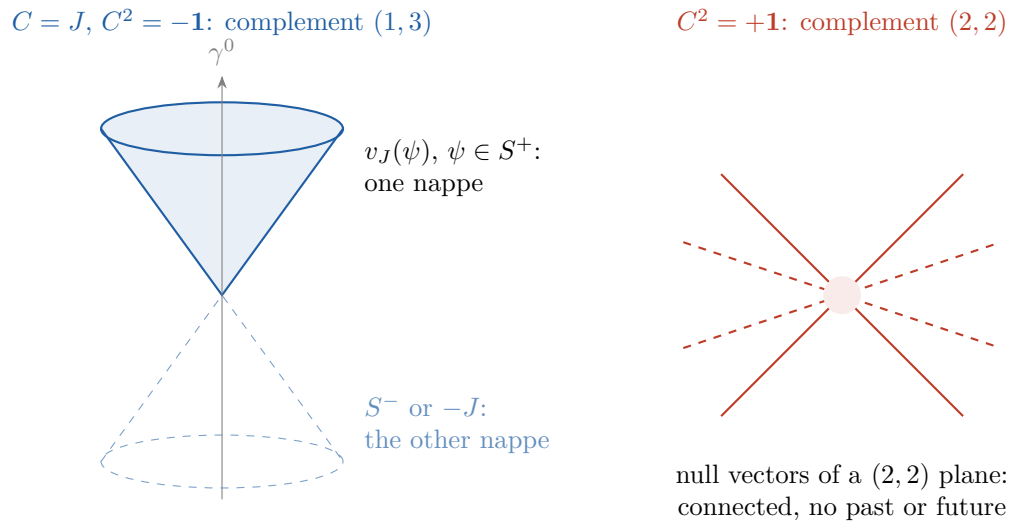


Figure 20: The selector at one constituent (Theorems 28.1 and 28.2). A unit bivector C with $C^2 = -\mathbf{1}$ selects a Lorentzian complement, and the C-twisted squares of chiral spinors fill one nappe of its light cone, the other nappe belonging to the other chirality or to $-C$. A bivector with $C^2 = +\mathbf{1}$ selects a $(2,2)$ complement, whose null set has no time orientation.

nondegenerate plane is an $SO(3,3)$-image of a coordinate plane of the same signature, by Witt's theorem. □

For coordinate planes C is moreover antisymmetric on S when $C^2 = -\mathbf{1}$, an orthogonal complex structure; for tilted planes it is not. Orthogonality refers to the Euclidean structure of S, which singles out the temporal triple, while $C^2 = -\mathbf{1}$ is invariant. The distinction will matter.

Spinors give the light cone. The volume element $\omega_S = e_4e_5e_6$ satisfies $\omega_S e_A \omega_S^{-1} = -e_A^\top$ for every generator, so $\langle\psi,\varphi\rangle = \psi^\top\omega_S\varphi$ is a Spin$(3,3)$-invariant pairing, and it pairs S^+ with S^-.

Theorem 28.2 (The light cone as twisted squares)**.** *Let ψ range over the chiral half S^+.*

(i) The forms $m_A = \omega_S e_A$ are antisymmetric and satisfy $X^\top m_A + m_A X = -\omega_S[X, e_A]$ for every bivector X; their restrictions to S^+ span $\Lambda^2(S^+)^*$, so $\psi\wedge\varphi \mapsto (\psi^\top\omega_S e_A\varphi)_A$ is an equivariant isomorphism $\Lambda^2 S^+ \cong \mathbb{R}^{3,3}$, the Klein correspondence of Theorem 13.13.

(ii) On S^+, $J^2 = -\mathbf{1}$, so $S^+ \cong \mathbb{C}^2$, and the interior $\mathfrak{so}(1,3)$ acts $\mathbb{C}$-linearly, as $\mathfrak{sl}(2,\mathbb{C})$.

(iii) For a unit simple bivector C with plane N define $v_C(\psi)_A = \psi^\top\omega_S e_A C\psi$. Then $v_C(\psi) \in N^\perp$ and $\eta(v_C, v_C) = 0$ identically, and the image has dimension three.

(iv) If $C^2 = -\mathbf{1}$, the component of v_C along the timelike direction of $N^\perp$ is definite: v_C fills one nappe of the light cone of $N^\perp$. If $C^2 = +\mathbf{1}$, no component is definite, and v_C fills the null vectors of a $(2,2)$ plane, which has no time orientation.

(v) $v_{-C} = -v_C$; on S^- the J-twisted square lies in the other nappe; and on S^+, $v_J = -j$, where $j_A = \psi^\top\gamma^0 e_A\psi$ is the Dirac current.

The selector of the reduction is therefore not a new field. Choosing the spacetime $N^\perp$ and choosing the complex structure C are one choice, and in the arena that complex structure is the charge J, the same J that is the imaginary unit of the Dirac theory (Theorem 4.1). Null directions are the squares of Weyl spinors twisted by J, and the time orientation is fixed by the sign of J together with the chirality. The mathematics is the Klein–Penrose correspondence, in

which a null vector is $\psi\bar{\psi}$ [107, 138]; what is specific to the arena is that J already plays this role. All of this is kinematics: nothing yet says why J, rather than an indefinite or degenerate bivector, is chosen. For degenerate planes v_C is still null with a three-dimensional image, and for the totally null plane $\langle e_1 + e_4, e_2 + e_5\rangle$ it lies in the totally null three-plane $\langle e_i + e_{i+3}\rangle$, an eigenspace of the grading root $\mathcal{K}$.

28.2 Three constituents

At three constituents J is the product of the fourteen generators of the Lorentz complement W_3, of signature $(8, 6)$, and Ω is the volume element of E.

Theorem 28.3 (The selector at three constituents)**.** *(i) $J^2 = \Omega^2 = -\mathbf{1}$, $[J, \Omega] = 0$ and $\Gamma = J\Omega$ with $\Gamma^2 = \mathbf{1}$; $\Omega = -J$ on $\Gamma = +1$. Since $\Gamma^2 = \mathbf{1}$ at every rung, any coordinate four-plane E' with volume element $\Omega_{E'}$ and complementary volume element $J_{E'}$ has $J_{E'}^2 = \Omega_{E'}^2 = \prod_{i\in E'} e_i^2$: the volume element of the complement is a complex structure exactly when E' is Lorentzian, at every n.*

(ii) The centraliser of J among the 153 bivectors has dimension $97 = 6 + 91$, so the choices form a space of dimension 56.

(iii) With either $\mathrm{Spin}(9, 9)$-invariant pairing, the product β_{odd} of the nine negative generators, which is antisymmetric, or the product β_{even} of the nine positive ones, which is symmetric, $\beta e_A J$ is symmetric exactly for the fourteen directions of W_3: the J-twisted square lives on the complement, not on E. The untwisted $\beta_{\mathrm{odd}} e_A$ is symmetric for all eighteen directions, so the vector sits in $\mathrm{Sym}^2 S^+$, and at random spinors the untwisted square is not null.

The identification of the selector with J survives at three constituents, but the covariant construction of the cone from spinors is a feature of dimension six. At $n = 3$ the cone of E is recovered by the Dirac current $j_\mu = \psi^\top \gamma^0 \gamma_\mu \psi$, which is future causal, being a sum of one-constituent currents, and null on product states; but it uses the clock γ^0 and the transpose, that is, the positive inner product of the spinor module.

29 Type selection

Kinematics identifies the selector with a complex structure. Dynamics would have to explain why the field that plays this role takes a value of the complex-structure type at all, rather than an indefinite or degenerate one. The natural test is a potential: take the order parameter to be a bivector field $B \in \mathfrak{so}(3, 3)$, ask that the potential be invariant under $SO(3, 3)$, so that nothing is put in by hand, and require that its minimum set be exactly the orbit of J, over an open set of couplings, with healthy fluctuations. Restriction to S^+ is the isomorphism $\mathfrak{so}(3, 3) \cong \mathfrak{sl}(4, \mathbb{R})$, and the invariant polynomials are the polynomials in

$$t_k(B) = \mathrm{tr}\big((B|_{S^+})^k\big), \qquad k = 2, 3, 4,$$

the coefficients of the characteristic polynomial; a continuous invariant function is constant on orbit closures.

Proposition 29.1 (Invariants of the three types)**.** *Rotations of a positive plane and of a negative plane both have $(t_2, t_3, t_4) = (-4, 0, 4)$ and characteristic polynomial $(y^2 + 1)^2$; boosts of an indefinite plane have $(4, 0, 4)$ and $(y - 1)^2(y + 1)^2$. The positive and the negative planes induce opposite orientations of $S^+ = \mathbb{R}^4$: they are distinct $SL(4, \mathbb{R})$-orbits with identical invariants.*

So no invariant potential can prefer $(1, 3)$ to $(3, 1)$. The two are exchanged by the anti-isometries $\mathcal{K}$ and G and correspond to the two sign conventions for a Lorentzian metric; physically this is harmless, but it means the selection is never unique. The type itself is harder.

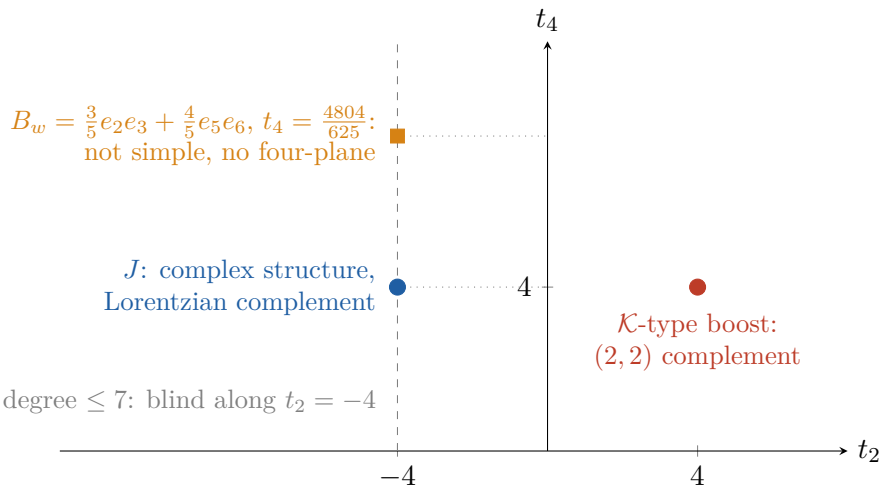


Figure 21: Type selection in the plane of the invariants (t_2, t_4) at $t_3 = 0$ (Theorem 29.2). A bounded-below invariant potential of degree at most seven depends on t_2 alone along $t_3 = 0$, so it cannot distinguish the complex structure J from the non-simple element B_w on the same vertical line. Degree eight is the first that can.

Theorem 29.2 (No clean selection below degree eight)**.** *Let V be an $SO(3,3)$-invariant polynomial on $\mathfrak{so}(3,3)$, bounded below.*

(i) If $\deg V \leq 4$, then $V = m\,t_2 + \lambda\,t_2^2 + \text{const}$.

(ii) If $\deg V \leq 7$, then V restricted to $\{t_3 = 0\}$ is a polynomial of degree at most two in t_2 alone.

(iii) Hence if J minimises V, so does $B_w = \frac{3}{5}e_2e_3 + \frac{4}{5}e_5e_6$, which has $t_2 = -4$, $t_3 = 0$ and $t_4 = \frac{4804}{625} \neq 4$, is not simple, and selects no four-plane.

(iv) In degree eight, $f = (t_2+4)^2 + t_3^2 + (t_4-4)^2$ vanishes exactly on the elements with characteristic polynomial $(y^2+1)^2$, and its Hessian at J has rank one.

Proof. Use test elements of $\mathfrak{sl}(4,\mathbb{R})$ with rational entries. $A_0 = \mathrm{diag}(1,-1) \oplus \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $B_0 = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right) \oplus \left(\begin{smallmatrix} -1 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ have $t_2 = t_3 = 0$ and $t_4 = 4, -16$; $C_0 = \mathrm{diag}(-1,-3) \oplus \left(\begin{smallmatrix} 2 & -3 \\ 3 & 2 \end{smallmatrix}\right)$ has $t_2 = 0$ and $t_3 = -120$; the complex and para-complex structures have $(t_2, t_4) = (\mp 4, 4)$; $D_0 = \mathrm{diag}(1,-1,2,-2)$ and two rotations of speeds $1, 2$ have $(t_2, t_4) = (\pm 10, 34)$, all with $t_3 = 0$. (i) Along sA_0 and sB_0 a quartic term $\lambda_2 t_4$ would give $\lambda_2 t_4 s^4$ with both signs of t_4, so $\lambda_2 = 0$ for boundedness; along sC_0 a cubic term gives $-120\kappa s^3$, so $\kappa = 0$. (ii) The top homogeneous part must be non-negative, so an odd top degree vanishes, and in degree six the top part $e\,t_2^3 + f\,t_3^2 + g\,t_2 t_4$ takes the values $\mp(64e + 16g)$ on the complex and para-complex structures and $\pm(1000e + 340g)$ on the last two elements; non-negativity forces $g = -4e$ and then $e = g = 0$, with $f \geq 0$ from C_0. The quartic coefficient of t_4 must again vanish, leaving $a\,t_2 + b\,t_3 + c\,t_2^2 + h\,t_2 t_3 + f\,t_3^2$, which on $t_3 = 0$ is $a\,t_2 + c\,t_2^2$. (iii) follows. (iv) $f \geq 0$ with equality exactly when $(t_2, t_3, t_4) = (-4, 0, 4)$, which fixes the characteristic polynomial by Newton's identities; at J the gradients of t_3 vanish and those of t_2 and t_4 are proportional, so the Hessian has rank one. □

Even where a potential does isolate the type, the minimum cannot be clean, and the reason holds at every degree.

Theorem 29.3 (Degeneracy and flat directions at every degree)**.** *Let V be any $SO(3,3)$-invariant function on $\mathfrak{so}(3,3)$.*

(i) Let $n = (e_1 + e_4)e_5$, a null rotation of the interior four-plane. Then $[J, n] = 0$ and $n^2 = 0$, so $J+n$ has the invariants of J, but its centraliser has dimension three and does not contain $\mathfrak{so}(1,3)$: it breaks Lorentz invariance. With $K = \frac{1}{2}e_1e_4$, $[K, n] = -n$ and $[K, J] = 0$, so conjugation by e^{tK} maps $J + n$ to $J + e^{-t}n$. Hence J lies in the closure of the orbit of $J + n$, and $V(J + n) = V(J)$ whenever V is continuous.

(ii) If V is smooth and has a local minimum at J, its Hessian vanishes on the six $\mathfrak{so}(1,3)$ directions. The $\mathrm{Stab}(J)$-invariant symmetric forms on $\mathfrak{so}(1,3)$ are the real and imaginary parts of the complex Killing form of $\mathfrak{sl}(2,\mathbb{C})$, and each nonzero one has signature $(3,3)$, so a positive semidefinite one is zero.

At best the fluctuations about J are one massive radial mode, eight Goldstone modes and six massless modes that no symmetry protects.

Proposition 29.4 (Kinetic terms)**.** *$\mathfrak{so}(3,3)$ is simple, so every $SO(3,3)$-invariant quadratic kinetic term is a multiple of the Killing form, which on the fifteen bivectors has signature $(9,6)$, boosts with one sign and rotations with the other. At J the radial direction is a rotation, the eight Goldstone directions are two rotations and six boosts, and the six $\mathfrak{so}(1,3)$ directions are three of each. Whatever the overall sign, at least six directions are ghosts, and the induced metric on the eight-dimensional space of choices has signature $(6,2)$.*

In covariant terms the Goldstone directions form a charged four-vector and the $\mathfrak{so}(1,3)$ directions a two-form, and healthy vectors and two-forms owe their positivity to a gauge symmetry that removes the wrong-sign components; a scalar order parameter has none. This is Theorem 11.3 again, in a new setting. There is a positive alternative, and it shows what is missing.

Proposition 29.5 (The positive alternative)**.** *On the fifteen bivectors the positive form $\langle X, Y\rangle = \mathrm{tr}(XY^{\top})/8$ is the identity, and its invariance algebra is the six compact rotations $\mathfrak{so}(3)_T \oplus \mathfrak{so}(3)_S$. Split $B = B_T + B_S + B_p$ into these two parts and the nine boosts. The potential*

$$V_+ = \left(|B_T|^2 - 1\right)^2 + |B_S|^2 + |B_p|^2$$

has as minimum set the unit sphere of $\mathfrak{so}(3)_T$. Every element there is a rotation of a plane N in the temporal triple, the clock is the axis of the rotation, the spacetime is the clock line plus the spatial triple, and $J = \gamma^0\omega_T$. The Hessian at J has eigenvalues 8, 0, 0 and 2 (twelve times): two Goldstone modes and everything else massive with positive kinetic energy. But the boosts e_1e_4, e_1e_5, e_1e_6 of the selected spacetime are not symmetries of this sector.

The requirement was a clean selection of the Lorentzian type over an open set of couplings, with every mode healthy and accounted for. An $SO(3,3)$-invariant scalar order parameter cannot deliver it: below degree eight it does not isolate the type; at every degree the minimum is degenerate with Lorentz-violating configurations and has six unprotected massless modes; and its kinetic term has ghosts. The positive route selects cleanly only by giving up the boosts it was meant to produce.

30 Where the surplus constraints come from

The baseline showed that the constraints $p_u = p_v = 0$ remove exactly the unstable band, and the curved model showed that they are first class. A first-class constraint generates a gauge symmetry, so the question is which symmetry, and whether it is a natural one.

Theorem 30.1 (A worldline gauge symmetry)**.** *For arbitrary $A_\mu(x)$ and flat h, with $C = A_\mu h^{\mu\nu} k_\nu$, the constraints satisfy $\{p_u, H\} = -Cp_v$, $\{p_v, H\} = Cp_u$, $\{p_u, p_v\} = 0$ and $\{\ell, H\} = 0$. The extended action*

$$\int d\tau \left(p \cdot \dot{x} - NH - \lambda_1 p_u - \lambda_2 p_v\right)$$

is invariant under

$$\delta u = \varepsilon_1(\tau), \qquad \delta v = \varepsilon_2(\tau), \qquad \delta\lambda_1 = \dot\varepsilon_1 + NC\varepsilon_2, \qquad \delta\lambda_2 = \dot\varepsilon_2 - NC\varepsilon_1,$$

all other variables fixed: local translations along the normal plane, which are gauge symmetries although they are not isometries when $F \neq 0$, the algebra closing with structure functions [63]. In Dirac quantisation the physical states are the functions $\Psi(x)$ on spacetime, and on them $\hat H\Psi = -\frac{1}{2}\Box_h\Psi$; the potential A drops out.

Proposition 30.2 (Gauging the isometry is not enough)**.** *In polar normal coordinates $\ell = -p_\phi$ and $p_u^2 + p_v^2 = p_r^2 + p_\phi^2/r^2$. Gauging only the rotation ℓ, the isometry of the normal plane, leaves $p_0^2 = |\mathbf{p}|^2 + m^2 - p_r^2$: the radial surplus momentum keeps the unstable band. For fields, the rotation-invariant modes $J_0(qr)e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}$ obey $\omega^2 = |\mathbf{k}|^2 + m^2 - q^2$.*

The symmetry that removes the instability is therefore the full group of local translations along the normal plane, not its rotation subgroup.

Corollary 30.3 (Weak emergence for scalar matter)**.** *If the normal translations are gauged, the scalar sector reduces to the Klein–Gordon field on (M^4, h): its energy is bounded below, its evolution is unitary, and $[\phi(x), \phi(y)] = 0$ at spacelike separation for the cone of h [141]. In the free theory the zero modes ψ and the surplus modes χ do not mix, since translation invariance along the normal plane makes the mixed kernel $K_{\psi\chi}$ vanish, so the Schur complement is trivial and $K_{\text{eff}} = K_{\psi\psi}$; the content is not in the kernel but in whether χ is physical, and under the gauge principle it is not. The causal cone is that of h, the metric of the selected four-plane: it is inherited from the reduction, not derived from the six-dimensional sum.*

What the reduction leaves over is the geometry of the normal plane itself, and it is not healthy.

Theorem 30.4 (Every normal-plane vector is a ghost)**.** *For $ds^2 = h + \sigma\big[(du + B^u)^2 + (dv + B^v)^2\big]$ with flat h, $B^u = b_1 x^1 dx^2$ and $B^v = b_2 x^0 dx^3$, one finds $R_6 = -\frac{\sigma}{4}\sum F_{\mu\nu}F^{\mu\nu}$, and for the rotation ansatz $B^u = vA$, $B^v = -uA$, $R_6 = -\sigma\, r^2 F^2/4$, which at $\sigma = +1$ is the ambient scalar curvature of Theorem 23.5. The coefficient is odd in σ, so the healthy sign is that of a normal plane of the sign of space, $\sigma = -1$, the ordinary Kaluza–Klein case. By Theorem 28.1(ii) the normal plane of a Lorentzian four-plane in $\mathbb{R}^{3,3}$ always has the sign of time. Every Kaluza–Klein vector from the normal plane, from its translations or its rotation, therefore gravitates with the wrong sign, and by Theorem 30.1 none of them couples to spinless matter.*

The failure condition "ghosts or negative spectral weights" is therefore met by the normal-plane vectors, and the photon cannot come from them. If the Maxwell field emerges, it must come from elsewhere: from the univectorial potential of the interior reading, or from the field-strength reading $\mathbb{R}^{3,3} = \Lambda^2\mathbb{R}^{1,3}$, where the energy is the positive Hermitian form $|\mathbf{B}|^2 + |\mathbf{E}|^2$ (Section 13).

30.1 What the causal analysis establishes

Table 6 collects the outcome against the failure conditions fixed at the start of Section 27.

Weak emergence holds for scalar matter, conditionally on a gauge principle, local translations along the normal plane, that is consistent but chosen rather than derived. Strong emergence through a scalar order parameter is excluded. What survives is a sharper statement of the question. The selector is the complex structure J, and every construction here that produced a healthy Lorentzian sector also used one extra datum: a positive inner product, which is the transpose of the realisation, equivalently the temporal triple. Within these computations the totality of $SO(3,3)$-invariant dynamics fixes at most the type of the structure, never the instance, and it fixes even the type only imperfectly. The datum that singles out time, wherever things

stage	result	verdict
baseline	growing modes and no ground state; exactly the band the surplus constraints remove; compactification makes tachyons	obstruction confirmed, cure located
reduction	kinematics: selector $= J$, cone $=$ twisted squares at one constituent; dynamics: no $SO(3,3)$-invariant scalar mechanism; the positive route breaks the boosts	kinematics exact; dynamical selection fails as posed
effective theory	scalar matter: no mixing, $K_{\text{eff}} = K_{\psi\psi}$; normal-plane vectors are ghosts	failure condition met by the vectors
causality	the scalar sector is microcausal for the cone of h	conditional on gauged translations; inherited
Maxwell	normal-plane vectors excluded as photons	blocked on this route

Table 6: The causal analysis against the failure conditions fixed in advance (Sections 27 to 30).

worked, is the positivity of an inner product; in quantum mechanics it is the positivity of the Hilbert-space norm. That is where a derivation of causal order from the split geometry would have to look.

Several routes remain open. A gauge mechanism for type selection, an $SO(3,3)$ analogue of the MacDowell–Mansouri and Stelle–West breaking [81, 127], $SO(3,3) \to SO(2) \times SO(1,3)$, would make the would-be ghosts gauge, solder the eight broken directions to the base and remove the six flat ones. A principle that makes the normal translations gauge, rather than adding the multipliers by hand, would turn weak emergence from a conditional statement into a derived one. At three constituents, a $\mathrm{Spin}(9,9)$-covariant construction of the cone of E, for which pure spinors are the natural candidates, or a proof that none exists without the positive inner product, would settle whether the cone survives the tower.

Part VIII
Assessment

31 What the algebra decides

We sort the results of this paper into the four classes announced in Section 1.2, and then name the choices that remain. The test applied throughout is the deflationary one of Corollary 4.4: a statement about the arena carries information only if it could have come out otherwise, and compatible representation data is not a prediction.

31.1 Fixed by the algebra

At one constituent the algebra fixes more than one might expect. The signature rule separates the bivectors that can gauge from those that can only grade, and the pole operators grade the algebra by $\mathbb{Z}_2^3$ through a maximally split Cartan subalgebra. The pairing of the two triples has exactly two square roots, the grading root $\mathcal{K}$ and the binding root G, related by the Cartan involution $\mathrm{Ad}_{e_1e_2e_3}$. Under permutation symmetry the clock is the barycentre of the Fisher–Rao octant, and exactly that clock makes the conversion invisible in the equations of motion. The imaginary unit is selected, not adjoined: inside the joint commutant of the interior Lorentz action and the internal bivectors the only complex structures commuting with the gamma matrices are $\pm J$. Charge conjugation is the surplus plane, time reversal squares to $-\mathbf{1}$, the kinetic forms alone fix the massless gauge algebra $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$, the mass form cuts it to $\mathfrak{u}(1)_J$, and that $\mathfrak{u}(1)$ is the whole internal centraliser. Along the tower the internal centraliser is $\mathfrak{so}(3n-1, 3n-3)$, nested and

over one Lorentz frame, and the diagonal Lorentz action gives a compact internal algebra forming a reductive dual pair with the Lorentz algebra, with the exact identity $C_{\mathrm{cop}} + C_{\Delta} = \frac{1}{2}n(n+2)$.

In the field-strength reading the centraliser of G is a Lorentz algebra under which $\mathbb{R}^{3,3}$ *is* $\Lambda^2\mathbb{R}^{1,3}$, G is the Hodge star, the two Maxwell invariants are the real and imaginary parts of one holomorphic quadratic form, and covariance fixes the form of Maxwell's equations through a multiplicity count. In the interior dynamics, factorisation and binding coexist exactly in neutral signature, and the force law closes in one way only, into a rotor acting by G whose free motion is a pendulum in the duality angle. In the curved arena the curvature operator lives on the frame, the first Bianchi identity is the single trace $\mathrm{Tr}(G\mathcal{R}) = 0$, and the duality rotation of the curvature ends, where conservation and variationality both select it, at the double dual contracting to the Einstein tensor; in first-order form the same rotation is a Barbero–Immirzi angle. Kinematically, choosing a Lorentzian four-plane is choosing a complex structure, and at one constituent the light cone is the set of chiral spinors squared with the help of J.

31.2 Constructed

The constructions are existence results; in each case the algebra admits the object and nothing selects it among alternatives. At three constituents the axial projector P_* of rank 64 carries the Pati–Salam algebra and one chiral family with a right-handed neutrino, and settles the question of colour with chirality. The diagonal Dirac block carries a gauge-invariant real $\mathrm{Cl}(1,3)$ built from a copy-singlet cubic. At the mixed second rung the univectorial potential, a Dirac block and the frame field can be coupled into two-flavour spinor electrodynamics. On the 272 mass kernels of the chiral block the renormalizable potential of the $(\mathbf{10},\mathbf{1},\mathbf{3})$ component is classified, sufficient conditions and separate necessary conditions under which the Standard-Model orbit is its global minimum are found (with a gap between them that numerics, not a proof, resolve), and the radiative spectrum of all 271 angular modes is computed, with the electroweak stage, the seesaw and the Georgi–Jarlskog factor as consequences of a bidoublet vacuum value. At the second rung a Lorentz-invariant, uncharged, non-rigid vacuum with a charged-scalar current exists on a compact S^4 of projectors, with a curved extension in which it is parallel. Both readings of $\mathbb{R}^{3,3}$ construct the Einstein tensor of a given four-metric.

31.3 Obstructed

Each obstruction holds under hypotheses stated where it was proved, and those hypotheses are part of the result. Hypercharge cannot be read off anomaly cancellation for charges built from three commuting involutions; $\mathfrak{su}(3)$ fits in the effective algebra of no axial projector of the five-projector family; the two central $\mathfrak{u}(1)$s of P_* cannot be gauged on its matter; the diagonal sector is vector-like on its complete blocks. The two readings of the same six directions exclude one another, and the obvious repair at the second rung costs the whole internal algebra. The positive-form interior oscillates only at zero momentum. No Lie–Poisson closure lives on the frame alone, and no inertia is both Lorentz invariant and rate-conserving. Non-orthogonality of $\mathbf{E}$ and $\mathbf{B}$ breaks P and T, is never spontaneous, and generates no mass. The paired-mode projector sources only a massive vector, and at the first rung a vacuum that is both massless and Lorentz invariant is rigid. The normal $U(1)$ of the curved model fails as electromagnetism on three independent counts: a spinless particle cannot see it, its adapted generator does not commute with the physical generators, and it gravitates with the wrong sign. Ricci flow does not transfer to split signature. No vacuum value of a field built from the internal bivectors and J makes the matter chiral, Hermitian order parameters keep rank five, and vacuum alignment does not see chirality. In the causal analysis, the six-dimensional free field is ill-posed and has no time orientation, no $SO(3,3)$-invariant scalar potential of degree below eight isolates the Lorentzian type, every such potential is degenerate with a Lorentz-violating orbit and has six unprotected flat directions, every invariant kinetic term has ghosts, and every vector from the

choice	status
the projector	a field with a computable configuration space; at rank 64 at least two admissible orbits, of dimensions 34 and 20; a spectrum-only potential is constant on them, so selection needs couplings that distinguish orbits
the Lorentz action	spectator and diagonal actions are not conjugate and trade against each other: a large internal algebra with ghost-signed directions and a chiral block, or a small compact algebra on vector-like matter
the hypercharge direction	a compatible embedding exists on P_* and is output by the vacuum of the kernel field, but the vacuum itself is selected only on an open set of couplings
the binding operator and its reading	G is adopted; conditional uniqueness once its Lorentz algebra is taken, not selection; the field-strength reading and the interior reading of the same six directions exclude one another
scales and inertias	no ratio among the rotor inertia, the frame inertia and the coupling scales is fixed; the inertia of the closure has 21 free directions satisfying neither requirement
the action	no parent action is derived; the gauge field, the metric and the kernel field are given dynamics by adopted actions whose equations follow
the four-plane and positivity	the selector is J; the type is not selected by invariant dynamics, and every healthy construction used a positive inner product
families	the block carries one family; three copies of a module are a multiplicity index, not three constituents, and nothing reduces the copy multiplicity to three

Table 7: The choices the algebra leaves open.

normal plane is a ghost.

31.4 Open, and the choices that remain

Table 7 lists the choices that the algebra leaves free, with the place where each enters. Most of the open problems are downstream of them.

Several problems are sharply posed by the results. A coupling that distinguishes the points of the compact family of second-rung vacua, done at $n \geq 3$ where colour survives, would turn the photon construction into a selection. A finite search over non-diagonal pairings of the two Lorentz bases at the second rung and above would settle whether the two readings of $\mathbb{R}^{3,3}$ can share one theory at an acceptable price. The spectrum and the index of the gauge-invariant Dirac operator of the diagonal block on a curved base are ordinary index-theorem calculations, and a nonvanishing index would not contradict vector-likeness. The global structure of the potential on all 272 mass kernels, the choice between the Standard-Model vacuum and its mirror, the electroweak hierarchy, and the explicit breaking of the left-handed fermion number that the axion bound requires are all open. A mechanism placing the interior momenta on the null cone or on the balance surface, the only places where a constant rate and a bounded frame coexist, and the quantisation of the rotor, whose angle would then have to be related to an external time, are open. Whether the antisymmetric part of the shift is the contorsion of an independent connection is a conjecture that is not yet type-correct. A gravitational action relating metric, pairing, connection and projector, from which the metric would follow rather than be given, is the largest open problem of the curved arena; and in the causal analysis a gauge mechanism for type selection and a principle that makes the normal translations gauge are the two steps from which a derivation of causal order would have to begin.

31.5 The test, applied

The largest gap is the same in every sector, and it is one of selection rather than absence. The gauge algebra is derived and the gauge field is not; the arena is derived and its dynamics is not; the Einstein tensor of a given metric is constructed and the metric is not determined; the projector is a field whose equation follows once an action is adopted, but no action is derived and none of its admissible orbits is singled out; the vacuum of the kernel field is selected, but only on a stated open set of couplings and with field content given. Recovering the hypercharge table, the anomaly conditions or $\sin^2\theta_W = \frac{3}{8}$ is a consistency result, not evidence, since any construction with that charge content gives them. New mathematics would be a theorem that removes one of the choices of Table 7 without adding another; new physics would require a selected spectrum, a coupling or scale relation, a mass restriction, or a deviation that survives re-expression in the complex language of Theorem 4.3. None of these is claimed here. What the paper does claim is a precise map of where the split-signature Clifford algebra decides and where it leaves the decision to the physicist, and on that map the decided territory is larger, and its borders sharper, than one might have guessed from the size of the arena.

A The explicit realisation

Every finite-dimensional statement of this paper is a statement about integer matrices, and this appendix collects the data needed to check any of them by hand.

One constituent. Label the basis of $S = \mathbb{R}^8 = (\mathbb{R}^2)^{\otimes 3}$ by bit strings $b = b_1b_2b_3$, ordered as the binary numbers $0, \dots, 7$, and write δ_i for the string with a single 1 in position i. The generators (1) act by

$$e_i\,|b\rangle = (-1)^{b_1+\cdots+b_{i-1}}\,|b\oplus\delta_i\rangle, \qquad e_{i+3}\,|b\rangle = (-1)^{b_1+\cdots+b_i}\,|b\oplus\delta_i\rangle \qquad (i=1,2,3):$$

both e_i and e_{i+3} flip the i-th bit, and they differ only by the sign carried by that bit. As matrices, with $\bar{1} = -1$ and dots for zeros,

$$e_1 = \begin{pmatrix} \cdot&\cdot&\cdot&\cdot&1&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&1&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&1&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&1\\ 1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&1&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&1&\cdot&\cdot&\cdot&\cdot \end{pmatrix}, \quad e_2 = \begin{pmatrix} \cdot&\cdot&1&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&1&\cdot&\cdot&\cdot&\cdot\\ 1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\bar{1}\\ \cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot \end{pmatrix}, \quad e_3 = \begin{pmatrix} \cdot&1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ 1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\bar{1}&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&1\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&1&\cdot \end{pmatrix},$$

$$e_4 = \begin{pmatrix} \cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\bar{1}\\ 1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&1&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&1&\cdot&\cdot&\cdot&\cdot \end{pmatrix}, \quad e_5 = \begin{pmatrix} \cdot&\cdot&\bar{1}&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot&\cdot&\cdot\\ 1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&1&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&1\\ \cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot \end{pmatrix}, \quad e_6 = \begin{pmatrix} \cdot&\bar{1}&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ 1&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&1&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\bar{1}&\cdot&\cdot&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&1&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\bar{1}&\cdot&\cdot&\cdot\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\bar{1}\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&1&\cdot \end{pmatrix}.$$

The first three are symmetric and the last three antisymmetric, as (4) requires, and $e_ae_b + e_be_a = 2\eta_{ab}\mathbf{1}$ is checked entry by entry.

Derived operators at one constituent. The pole operators are diagonal, $\omega_i\,|b\rangle = (-1)^{b_i}|b\rangle$, so their joint eigenlines are the basis lines themselves, and the volume element is $\omega = e_1\cdots e_6 = \Gamma$ with $\Gamma|b\rangle = -(-1)^{b_1+b_2+b_3}|b\rangle$; the half-spinors S^+ and S^- are spanned by the basis vectors with an odd, respectively even, number of ones. Further,

$$J = e_2e_3 :\ |b\rangle \mapsto (-1)^{b_2}\,|b\oplus\delta_2\oplus\delta_3\rangle, \qquad \Omega = e_1e_4e_5e_6, \qquad T = \gamma^0 J = e_1e_2e_3, \qquad \Gamma = J\Omega.$$

On $V = \mathbb{R}^{3,3}$ the two square roots of the pairing are

$$\mathcal{K}e_i = -e_{i+3}, \quad \mathcal{K}e_{i+3} = -e_i, \qquad G\,e_i = e_{i+3}, \quad G\,e_{i+3} = -e_i,$$

with $\mathcal{K} = \frac{1}{2}\,\mathrm{ad}_{\mathsf{P}}$, $\mathsf{P} = \sum_i e_i e_{i+3}$, and $G = -\mathcal{K} \circ \mathrm{Ad}_{e_1e_2e_3}$. The interior four-plane is $E = \langle e_1, e_4, e_5, e_6\rangle$ with $\gamma^0 = e_1$ and $\gamma^a = e_{a+3}$; the surplus plane is $U = \langle e_2, e_3\rangle$; the paired-mode projector is $\Pi = \frac{1}{2}\mathbf{1} - \frac{1}{4}(\mathsf{P} + \mathcal{V})$, which in the basis above is the diagonal projector onto the strings with at least two ones.

The tower. At n constituents $S_n = (\mathbb{R}^8)^{\otimes n}$ and

$$E_{6(k-1)+a} = \omega \otimes \cdots \otimes \omega \otimes e_a \otimes \mathbf{1} \otimes \cdots \otimes \mathbf{1},$$

with $k-1$ factors ω. Since ω is diagonal and every e_a is a signed permutation, every E_A and every blade is a signed permutation matrix of size 8^n, and products, commutators, traces and ranks reduce to integer arithmetic. At three constituents the indices are grouped as follows:

temporal directions	$\{1,2,3,7,8,9,13,14,15\}$, partners $i+3$
interior four-plane E	$\{1,4,5,6\}$, $\gamma^0 = E_1$, $\gamma^a = E_{3+a}$
surplus pair of the first constituent	$\{2,3\}$, $q_A = E_2E_3$
weak directions W	$\{7,13,14,15\}$, $\chi_W = E_7E_{13}E_{14}E_{15}$
colour directions C	$\{10,11,12,16,17,18\}$, $\chi_C = \prod_{i\in C} E_i$
auxiliary directions	$\{2,3,8,9\}$

J is the ordered product of the fourteen generators outside E, $J^2 = -\mathbf{1}$, and $\Gamma = J\gamma^0\gamma^1\gamma^2\gamma^3$. The projectors used in the text are

$$P_* = \tfrac{1}{8}(\mathbf{1} + \Gamma\chi_W)(\mathbf{1} - J\chi_C)(\mathbf{1} - Jq_A), \qquad P_{\mathrm{fam}} = \tfrac{1}{8}(\mathbf{1} + X_1)(\mathbf{1} + X_2)(\mathbf{1} + X_3),$$

with $X_1 = J^{(1)}J^{(2)}$, $X_2 = \omega^{(3)}$ and $X_3 = \omega^{(1)}e_1^{(3)}e_4^{(3)}$, where $J^{(k)} = e_2^{(k)}e_3^{(k)}$ and $\omega^{(k)}$ is the volume element of constituent k. The colour Cartan element is $H = -(E_{10}E_{16} + E_{11}E_{17} + E_{12}E_{18})$, the weak generators are $t_i^{L,R} = \frac{1}{2}(\mathbf{1} \mp \chi_W)\tau_i$ with $\tau_i = -\frac{1}{2}\epsilon_{ijk}w_jw_k$, $w = (E_7, E_{13}, E_{14}, E_{15})$, and the Standard-Model line of mass kernels is

$$\Phi_0 = P_\nu\, P_*\, \gamma^0 E_3E_9E_{13}E_{14}E_{16}E_{17}E_{18}\, P_*\, P_\nu,$$

with P_ν the projector onto $\Gamma = +1$, $T_3^R = +\frac{1}{2}$, $B - L = -1$. The left–right involution is $O = E_1E_7E_8$, and the charge conjugation and CP blades are $C = E_2E_7E_{10}E_{11}E_{12}$ and $CP = E_1E_2E_7E_8E_{10}E_{11}E_{12}E_{14}$.

Second rung. At $n = 2$ the twelve generators are $e_a \otimes \mathbf{1}$ and $\omega \otimes e_a$ on $\mathbb{R}^{64}$. The mixed Lorentz action pairs the interior rotations $S_k = \frac{1}{2}\gamma^i\gamma^j$ and boosts $T_k = \frac{1}{2}\gamma^0\gamma^k$ of the first factor with the diagonal rotations L_k and boosts K_k of $\mathfrak{l}_G$ on the second, and the second-rung vacuum uses $a, b = e_2 \otimes \mathbf{1},\ e_3 \otimes \mathbf{1}$ and $c, d, e = \omega \otimes e_1,\ \omega \otimes e_2,\ \omega \otimes e_3$.

B How the statements were verified

The finite-dimensional statements of this paper fall into three kinds, and they were verified in different ways.

Exact statements. Most results are identities, ranks, dimensions, spectra and decompositions of explicitly given matrices. Since every generator of the realisation of Appendix A is a signed permutation matrix, every blade at every rung is one as well, and products, commutators and traces are computed in integer arithmetic without rounding. Ranks and kernel dimensions, such as the dimension 272 of the space of mass kernels, the dimensions of centralisers and stabilisers, and the orbit dimensions 34 and 20, were obtained by Gaussian elimination over the rationals. Decompositions into irreducible modules were identified by Casimir operators, computed exactly and diagonalised on rational or algebraic eigenvalues; a statement such as $\operatorname{spec} C_\Delta = \{\frac{3}{2}, \frac{11}{2}, \frac{15}{2}\}$ with multiplicities $160, 288, 64$ is the exact characteristic polynomial of an integer matrix. Polynomial identities, such as the phase family of the curvature operator, the Casimir identity of the diagonal sector, the closed form of the drift of the frame, or the Einstein tensor of the curved model on a Friedmann base, were checked as identities in the symbolic variables, modulo the stated relations.

The exact statements were obtained by symbolic and exact computation in the Wolfram Language, and the central ones were recomputed by a second, independent instrument: a program in pure Python using exact rational arithmetic (the `Fraction` type) and sparse representations of signed permutations, written without reference to the first. Where possible the second instrument uses different methods: the kernel space by solving the linear conditions on all antisymmetric matrices rather than by enumerating blades, invariant counts by enumerating weight multisets rather than by Brauer's formula, and the holomorphic quartic in an independent tensor model. Where the two instruments use different normalisations, as for the invariant pairing in the Klein correspondence, the comparison was made on normalisation-independent quantities such as the duality angle.

Derived statements. Some statements follow from exact ones by an argument given in the text, for instance the identification of an unbroken $\mathfrak{u}(1)$ with hypercharge, the physical reading of the second-rung vacuum, or the statement that a non-orthogonal field carries no mass. These are proved in the text and do not rest on computation beyond the exact statements they cite.

Numerical statements. A small number of statements rest on floating-point computation, and they are identified as such where they occur: the random-start minimisations of the one-loop energy on $\mathcal{T}$ and on the full kernel space, which locate the competing configuration and the crossing value $x_* = 24.88405$; the gradient flows at the benchmark couplings, which illustrate the transition at $|\kappa| = 7.2$; the numerical integrations of the free rotor, which reproduce the exact drift of Example 17.17 to ten digits; the integrations of a Zeitlin truncation in Section 17; and a few spot checks at random spinors in Section 28. In every case the numerical result is either a confirmation of an exact formula or a statement explicitly labelled as numerical, and the two conjectures of Section 20, the global thresholds $192/7$ on $\mathcal{T}$ and x_* on all of K, rest on such minimisations and are stated as conjectures.

Inherited statements. Results of the established literature, such as the anomaly cancellation of the Standard Model, the Vafa–Witten theorem, Michel's theorem on critical orbits, the Helmholtz conditions or Lanczos's identity, are cited and not recomputed; where the paper uses them, it says so.

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